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Master Thesis

Detection, Evaluation and Mitigation of Language Artefacts in the Competition On Legal Information Extraction and Entailment Dataset

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Abstract

The increasing prevalence of transformer models in Natural Language Processing (NLP) has revolutionized various domains, including the legal sector. However, the limited availability of domain-specific training data can lead to models exploiting superficial patterns or 'language artefacts' in the data, rather than genuinely understanding the underlying semantics. This reliance on artefacts can significantly impact the robustness and generalizability of these models, particularly in high-stakes applications like legal text analysis. This thesis investigates the presence and impact of language artefacts in the context of the Competition on Legal Information Extraction/Entailment (COLIEE) Task 4, which focuses on statute law textual entailment. The research employs a three-pronged approach: detection, evaluation, and mitigation. The detection phase involves a comprehensive analysis of the COLIEE dataset to identify various types of language artefacts, including lexical, syntactic, and semantic patterns that might mislead models. The evaluation phase assesses the robustness of several BERT-based models against these artefacts, employing quantitative and qualitative methods to understand their impact on model performance. The mitigation phase explores data augmentation techniques to reduce the influence of specific artefacts, aiming to enhance model robustness and generalizability. The findings reveal a significant presence of language artefacts in the COLIEE dataset and their detrimental impact on model performance, particularly in adversarial scenarios. The research also demonstrates the effectiveness of data augmentation in mitigating certain types of artefacts, leading to improved model robustness and generalization. The insights gained from this study contribute to a deeper understanding of the challenges posed by language artefacts in legal NLP tasks and offer potential solutions to enhance the reliability and trustworthiness of transformer models in the legal domain.

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1. Introduction

Natural Language Processing (NLP) is a domain within artificial intelligence (AI) that focuses on enabling machines to understand, interpret, and generate human language. NLP technologies underpin a wide range of applications, from conversational agents and translation services to sentiment analysis and information retrieval systems. These applications are increasingly integrated into various aspects of our daily lives, providing significant enhancements in efficiency and accessibility. As the field progresses, the challenge of ensuring the robustness and reliability of these systems becomes increasingly critical, especially in specialized domains such as legal text analysis.

Among the many tasks within NLP, Natural Language Inference (NLI) stands out as particularly challenging and significant [60]. Natural Language Inference (NLI) is the task of determining the relationship between two texts: a premise and a hypothesis. Specifically, it involves assessing whether the hypothesis is true based on the premise (entailment), false (contradiction), or whether the two texts are unrelated (neutral) [45]. This task is essential for various applications, including legal text analysis, where accurate inference is crucial for tasks such as case law and statute law analysis, legal research, and decision-making support. The ability of machines to accurately infer relationships within legal texts can drastically enhance the efficiency and effectiveness of legal professionals.

In the context of NLI, a premise is a statement or a reference document, and the hypothesis is a statement whose truth is being tested. The goal is to determine whether the hypothesis can be logically inferred from the premise. For instance, if the premise states "All humans are mortal," and the hypothesis is "Jack is mortal," the relationship is "entailment" because Jack is a human. Conversely, if the hypothesis were "Jack is immortal," the relationship would be a contradiction. However, one of the significant challenges in developing a robust NLI system is the presence of language artefacts. These are simple patterns or heuristics in the training data that the model can capture during training, allowing it to make predictions based on these patterns without fully considering the context of the text.

1.1 Motivation

Deep neural network models are widely used in natural language inference (NLI) tasks, and recently, the usage of transformer models has increased due to their capability to process large amounts of textual data and superior performance. However, in specific domains, these models often lack access to substantial amounts of training data. Consequently, domain-specific datasets are typically limited in size, which may not be preferable for deep neural network models that require more data for better feature representation and generalization. Therefore, models "trained on unrepresentative or limited samples may develop biases and exhibit poor generalization capabilities" [56].

One notable weakness of these models is their tendency to exploit weak features present in the dataset, which do not necessarily reflect the complete context of the text when making predictions [45]. These weak features are superficial patterns implicitly embedded in the training dataset and are referred to as language artefacts in the NLP domain.

Language artefacts can significantly distort the evaluation of model performance. A model might appear to perform well on a given dataset because it has learned to exploit these artefacts rather than genuinely capturing the semantic relationships between the premise and hypothesis. This issue is particularly concerning in specialized domains, such as legal text analysis, where high-stakes decisions necessitate accurate inference.

Despite the importance of addressing language artefacts, limited work has been done in the legal domain to detect their presence and understand their impact on models. The Competition on Legal Information Extraction/Entailment (COLIEE) provides a valuable benchmark for evaluating the performance of legal NLI systems. Various transformer models, including BERT, LegalBERT, RoBERTa, Electra, and DeBERTa, are employed in the competition to tackle statute law entailment tasks. However, no study has focused on whether these models perform well by understanding the context of the text or merely exploiting the language artefacts present in the training dataset. Thus, the presence of language artefacts in these datasets can lead to misleading conclusions about the actual capabilities of these models. Addressing language artefacts can also uncover implicit biases present in the dataset. Biases in training data can propagate through models, leading to biased predictions with significant implications in legal contexts.

Understanding the different types of language artefacts present in statute law text entailment tasks and their impact on models is crucial. Investigating the true performance capabilities of these BERT-based encoder-only transformer models will provide information on their strengths and limitations. This understanding can inform the creation of better evaluation test datasets and help develop more robust and reliable legal NLI systems.

By detecting, evaluating and mitigating the impact of language artefacts, we can contribute to developing more robust and transparent AI systems.

1.2 Research Questions

This section introduces the research questions and describes the evaluation phases designed to address these questions. The thesis poses the following three research questions.

Research Question 1: Is there evidence of language artefacts within the COLIEE Task 4 statute law textual entailment datasets?

Language artefacts in NLP datasets are patterns that allow models to make predictions based on superficial cues rather than considering the context of the text. The different types of language artefacts we aim to detect include:

- **Lexical artefacts**: These include annotation artefacts created during dataset creation, contradiction word artefacts (presence of contradiction words in the hypothesis), and word overlap artefacts between the premise and hypothesis [56].
- **Syntactic artefacts**: These include sentence length artefacts (irregular distribution in hypothesis sentences) and subsequence heuristic artefacts (presence of the hypothesis completely in the premise).

We analyze the COLIEE Task 4 statute law entailment datasets from 2018 to 2022 to identify the presence and types of these artefacts.

Research Question 2: What is the extent of the impact of these language artefacts on the performance of BERT-based models in statute law entailment tasks?

We evaluated the impact of detected language artefacts on the performance of various BERT-based models include BERT [11], LegalBERT [7], RoBERTa [25], Electra [9], and DeBERTa [18].

The evaluation for robustness involves quantitative methods using standard test sets, adversarial test sets and a hypothesis-only baseline, and qualitative method using a model interpretability technique named **Integrated Gradients**.

Research Question 3: Can data-level debiasing through data augmentation techniques effectively reduce these artefacts within the dataset?

This research explores data-level debiasing techniques, focusing on data augmentation methods to mitigate the presence of contradiction word artefacts and word overlap artefacts. Techniques such as synonym replacement and paraphrasing were implemented to create augmented datasets. The effectiveness of these techniques is evaluated by training BERT-based models on the augmented datasets and comparing their performance with models trained on the original datasets.

1.3 Structure of the Thesis

This thesis is organized around three central themes: (i) detection of language artefacts, (ii) evaluation of the impact of these artefacts on BERT-based models, and (iii) mitigation of the artefacts through data augmentation techniques.

Chapter 2 reviews the background and related work on language artefacts in natural language inference (NLI) datasets and the transformer models utilized in this research. It sets the foundation for understanding the challenges and existing solutions in the field. Chapter 3 provides an overview of the Competition on Legal Information Extraction/Entailment (COLIEE), focusing on the statute law textual entailment task and the specific dataset used in this study.

Chapter 4 offers a detailed explanation of the methodologies used to detect language artefacts, evaluate model performance against these artefacts, and implement strategies to mitigate their effects. It includes an in-depth discussion of the techniques and approaches used to address the research objectives. Chapter 5 describes the experimental setup and implementation details for model fine-tuning, evaluation, and data augmentation. This chapter outlines the practical aspects of the research, including the tools and processes used to conduct the experiments.

Chapter 6 presents the results for each research question, along with insights derived from the experiments. It provides a comprehensive analysis of the findings, highlighting the effectiveness of the proposed methods and their implications. Chapter 7 discusses the interpretation of the results and the limitations of this research. It critically examines the outcomes, considering potential biases and the broader context of the findings. Finally, chapter 8 concludes the thesis by summarizing the key findings and offering a brief overview of potential future research directions. This chapter reflects on the contributions of the thesis and suggests avenues for further investigation.

2. Background and Related Work

2.1 Language Artefacts

The objective of a Natural Language Inference (NLI) task is to determine the relationship between two sentences: the premise and the hypothesis. The premise is an initial statement that serves as a factual foundation. The hypothesis is a secondary statement that is evaluated in relation to the premise. The model is required to categorize this relationship into one of three categories: entailment (the hypothesis logically follows from the premise), contradiction (the hypothesis is logically inconsistent with the premise), or neutral (the hypothesis is neither supported nor contradicted by the premise).

NLI models are designed to learn and determine the relationship between a premise and a hypothesis. However, these models can be affected by various task-agnostic heuristics or spurious correlations present in NLI datasets, which can mislead the models into relying on superficial patterns rather than stronger relevant features in the text to predict the entailment relationship. In the literature, this phenomenon has been addressed using different terminologies, such as annotation artefacts, lexical overlap, spurious correlations and task-agnostic heuristics. For consistency, this research will refer to these hidden biases in the dataset as language artefacts, as they are specific to language datasets.

This section examines the types of language artefacts observed in NLI datasets and reviews the related work on detecting, evaluating, and mitigating these artefacts.

Lexical Artefacts

Lexical artefacts refer to patterns within the lexical features (words and phrases) of the dataset that correlate with specific output labels, leading models to make erroneous predictions by exploiting this weak feature. The primary types of lexical artefacts include annotation artefacts and lexical overlap.

Annotation Artefacts emerge when annotators, while labeling sentences for specific classes or labels, follow a consistent pattern. This pattern, when generalized across a dataset, leads to implicit biases. For example, if a unigram feature ‘A’ is strongly correlated with a label 'X', the model might predict "X' whenever it encounters 'A', neglecting other relevant features or the overall semantic context. This type of bias is referred to as annotation artefacts [56]. Table 2.1 provides an example of the annotation artefacts taken from examples in Stanford Natural Language Inference (SNLI) [5] corpus dataset.

Annotation artefacts are superficial patterns that arise during the data annotation process or during curation of certain datasets from limited sources which increases the possibility of these spurious correlations. Gururangan et al. [15] examined the co-occurrence of hypothesis unigram features and class labels in the SNLI [5] and Multi-Genre Natural Language Inference (MNLI)[57] datasets. They discovered a significant number of unigram features that exhibited a considerable correlation with class labels. These features, while not directly influencing the prediction of class labels, can bias the model. Similarly, Tan et al. [49] investigated the influence of bigram features in the SNLI dataset, finding similar correlations. Additionally, Zhou and Bansal [60] identified a specific type of lexical co-occurrence known as Contradiction-words Bias, where certain words in the hypothesis are strongly correlated with the contradiction label in both SNLI and MNLI datasets.

Lexical Overlap occurs when there is a significant overlap between the words in the premise and the hypothesis [56]. McCoy et al. [26] found that models often use lexical overlap as a heuristic to predict the output label in the MNLI dataset, assuming that a premise entails all hypotheses generated from its words. Zhou and Bansal [60] further explored this phenomenon by examining high word overlap between the premise and hypothesis in the SNLI and MNLI datasets, suggesting that this correlation is often associated with entailment. In this research, the lexical overlap is referred to as word overlap. Table 2.1 provides the examples for word overlap artefacts.

Syntactic Artefacts

Syntactic artefacts arise from specific syntactic patterns within the data. These patterns can lead models to make predictions based on superficial syntactic cues rather than deeper semantic understanding. The primary types of syntactic artefacts include sentence length, subsequence heuristics, and constituent heuristics.

Sentence Length Artefact as noted by Gururangan et al. [15], there is often an unequal distribution of sentence lengths in the hypotheses for each inference class in the SNLI and MNLI datasets. This discrepancy can cause models to rely on sentence length as a heuristic, leading to an overestimation of their performance, as mentioned by Gururangan et al. [15].

Subsequence Heuristic The subsequence heuristic, proposed by McCoy et al. [26], suggests that every part of a sentence (contiguous subsequence) in the hypothesis is considered included in the premise. This heuristic can mislead models to predict entailment when the hypothesis is a subsequence of the premise, regardless of the actual logical relationship. Table 2.1 provides examples for subsequence heuristic.

Constituent Heuristic The constituent heuristic, also suggested by McCoy et al. [26], assumes that a premise entails all complete subtrees present in its parse tree. This heuristic can cause models to incorrectly predict entailment based on syntactic structure rather than true semantic understanding. According to studies by Gururangan et al. [15] and McCoy et al. [26], NLI models often rely on these syntactic heuristics, which leads to an overestimation of their performance. Table 2.1 provides examples for the constituent heuristic artefacts.

Table 2.1: Examples of language artefacts. The label column shows the correct label for the sentence pair. E stands for Entailment and N stands for Non-entailment [3, 26].

Artefact Type	**Premise**	**Hypothesis**	**Label**
Annotation Artefact	The cat is sitting on the mat indoors.	The cat is **not** on the mat.	N
	She is attending the party indoors.	She is **not** going to the party.	N
	He loves ice cream indoors.	He does **not** like ice cream.	N
	The event is happening in the open air.	The event was held **outdoors**.	E
	They are playing outside.	They are playing **outdoors**.	E
	He enjoys activities in nature.	He enjoys activities **outdoors**.	E
Lexical Overlap	The banker near the judge saw the actor.	The banker saw the actor.	E
	The lawyer was advised by the actor.	The actor advised the lawyer.	E
	The judge by the actor stopped the banker.	The banker stopped the actor.	N
Subsequence Heuristic	The artist and the student called the judge.	The student called the judge.	E
	Angry tourists helped the lawyer.	Tourists helped the lawyer.	E
	The senator near the lawyer danced.	The lawyer danced.	N
Constituent Heuristic	Before the actor slept, the senator ran.	The actor slept.	E
	If the actor slept, the judge saw the artist.	The actor slept.	N

2.1.1 Language Artefacts Detection

Detecting language artefacts involves identifying superficial patterns in the training dataset that may mislead models into exploiting those weak features to make predictions. The detecting methods can be widely classified into two broader categories of data-centric and model-centric approaches.

Data-Centric Detection

Lexical Artefacts Detection A study by Gururangan et al. [15] utilized point-wise mutual information (PMI) calculations to detect annotation artefacts, identifying unigram features in hypothesis and class label pairs with significant PMI values as artefacts within the datasets. This approach was further refined by Gardner et al. [12], who introduced the use of a z-statistic test to provide a more quantitative analysis of spurious correlations.

This advanced method accounted for both the frequency of feature appearances and their correlation with a single class label, whereas PMI only considers the association of a unigram feature with a class label.

Additionally, McCoy et al. [26] developed a technique to identify instances with lexical overlaps between the premise and hypothesis. This technique involved creating a template to detect lexical overlap heuristics, ensuring that the hypothesis in these instances was neither a direct subsequence nor a constituent of the premise. A subsequence is a sequence that appears in the same order within another sequence but not necessarily consecutively, while a constituent is a complete subtree within the syntactic parse tree of the premise. By excluding these conditions, the technique allowed for a clearer identification of lexical overlap artefacts independent of other syntactic heuristics.

Syntactic Artefacts Detection In the context of detecting syntactic artefacts, McCoy et al. [26] identified subsequence and constituent heuristics by developing specific templates for each heuristic. These templates were designed to both support and contradict the heuristics, ensuring that in cases of subsequence, the hypothesis was not a constituent of the premise. This approach enabled a more precise identification of syntactic artefacts by isolating the influence of each heuristic.

Model-Centric Detection

A model-centric approach for detecting language artefacts involves analyzing the behavior of the model itself. Pezeshkpour et al. [33] proposed a method that integrates training feature attribution and instance attribution techniques. Training feature attribution refers to identifying which features (words, phrases, or other input elements) the model considers most important when making predictions. By analyzing these features, researchers can determine if certain patterns are disproportionately influencing the model's decisions. Instance attribution, on the other hand, involves examining specific instances (individual data points) to understand how the model's predictions are influenced by particular examples in the training dataset. This helps in identifying whether certain instances contribute to biased or erroneous predictions. The combination of training feature attribution and instance attribution allows for the comprehensive discovery of language artefacts in the training dataset that may influence the prediction of class labels.

2.1.2 Robustness Evaluation

The performance of numerous Natural Language Inference (NLI) models has been shown to be overestimated [45]. This overestimation stems from the learning algorithms' ability to concentrate on weaker, task-agnostic heuristics that facilitate correct class label prediction without adequately considering the semantics underlying the task. This section will outline various strategies used to assess the robustness of NLI models against language artefacts.

Hypothesis-Only Baseline

This technique involves training the model using only the hypothesis as input, excluding the premise, to predict the class label. If a model's performance using this method exceeds the majority baseline, it suggests the presence of language artefacts or spurious correlations in the training dataset as this indicates that the hypothesis alone contains generalizable features that enable the model to predict the entailment relationship, independent of the premise. This phenomenon has been extensively investigated across various NLI models and datasets [1, 2, 21, 34, 49, 59]. Alternatively this method is recommended to be a better baseline than the majority baseline.

Challenging Datasets

Another approach to examining model robustness involves constructing controlled evaluation datasets. For instance, McCoy et al. [26] proposed the HANS (Heuristic Analysis for NLI Systems) dataset, specifically designed to test whether models have incorporated syntactic heuristics. HANS includes examples that both support and contradict these heuristics, thus challenging the model's reliance on them. Similarly, Swayamdipta et al. [48] introduced a technique to categorize training instances into 'easy', 'hard', or 'ambiguous' categories to assess the difficulty level for the models during training. Instances categorized as ambiguous or hard to learn are particularly informative for evaluating model robustness. These specially created datasets are designed to prevent models from predicting the correct answer using only the hypothesis or simple heuristics, thereby providing a rigorous test of the model's robustness.

Adversarial Attacks

An alternative method for evaluating robustness involves the use of adversarial attacks. Glockner et al. [14] proposed a simple adversarial attack by creating a test set with minor lexical alterations that change the ground truth. Poor performance on these specific instances suggests that the model lacks certain lexical knowledge not captured during training and instead relies on spurious correlations. Similarly, Wang et al. [54] proposed a strategy of interchanging the premise and hypothesis within the NLI dataset. The rationale is that performance should diminish for entailment pairs after such a swap, as the reordered pairs will lack an entailment relationship, while the performance for neutral and contradiction pairs should remain steady. Failure to meet this expectation would indicate that the model's predictions are potentially influenced by language artefacts.

2.1.3 Language Artefacts Mitigation

The observed language artefacts can be mitigated using data-centric and model-centric approaches. This section will provide an outline of these methods used for mitigating language artefacts.

2.1.3.1 Data-centric approach

Local Edits to Instances: Gardner et al. [12] proposed a method to make edit-based data augmentation that makes small changes to the biased dataset, reducing the single feature artefacts associated with the class label. The edits made were based on the sensitivity-based model [16].

Data Augmentation : Gururangan et al. [15] suggests a method to distribute the artefacts across the labels using an adversarial system that flags the over-represented heuristics. Similarly Zhou and Bansal [60] proposed data enhancement by repeating the biased data to distribute the bias over all the labels ensuring artefact distribution. Likewise Belinkov et al.

[1] suggested a method to augment the training dataset by randomly swapping premises to reduce the impact of weak heuristics adapted by the model by creating noisy examples. Additionally Wang et al. [54] also proposed augmenting the training dataset with swapped instances of premise and hypothesis to reduce the influence of spurious correlations.

Adversarial Filtering: Data filtering is one of the techniques proposed by Gardner et al. [12] and Bhargava et al. [3] to filter ambiguous datasets using dataset cartography as described by Swayamdipta et al. [48]. This approach aims to enhance generalization and reduce language artefacts. Bras et al. [6] introduced AFLITE, an adversarial filtering approach designed to counteract the overestimation of machine performance due to spurious correlations. AFLITE identifies and removes examples that are most predictable by a given feature representation. The selection process is guided by a predictability score, which measures the strength of the feature-label correlation for each example. By filtering the dataset to retain only examples with predictability scores above a certain threshold, models trained on this refined dataset exhibit improved generalization to out-of-distribution tasks, providing a practical approximation for bias reduction.

More Annotators: To minimize the effect of language artefacts, Geva et al. [13] suggested that the test set should not include examples from the same annotators who contributed to the training set. This approach can help to ensure that the model generalizes well to new annotators and reduces the influence of language artefacts on model performance. Gardner et al. [12] also suggested having a diverse set of annotators during the creation of a dataset to avoid lexical and syntactic artefacts.

Model-centric approach

Adversarial Classifiers: Belinkov et al. [1] introduced an adversarial learning strategy that employs an adversarial classifier specifically designed to compel the hypothesis encoder to disregard hypothesis-only biases. This approach helps in mitigating the influence of biases derived solely from the hypothesis. Additionally, Clark et al. [8] proposed an ensemble method to address dataset biases. This method involves training a naive model exclusively on the biased dataset and, as part of the ensemble, concurrently training a robust model. The robust model is encouraged to focus on other patterns in the data that are more likely to generalize, thereby reducing the impact of the biases present in the dataset [23].

End to End bias mitigation: Likewise Karimi Mahabadi et al. [21] proposed two end-to-end learning strategies using Product of Experts and Debiased Focal Loss that focus on reducing the reliance on dataset biases by concentrating training on the hard examples. Similarly Sanh et al. [43] also proposed the products of experts method by using a combination of the weak and main model without any knowledge of the available dataset biases.

Mitigation in Language Models: Bhargava et al. [3] conducted a study for generalisation in BERT-based NLI models using Siamese and adapter networks, explicit debiasing using HEX projections and finally increasing the model size with the normal fine-tuning approaches. Tu et al. [51] discusses various methods to mitigate the spurious correlations in pretrained language models. Based on their study fine-tuning the language models on challenging datasets and using Multitask Learning objectives were found to be effective in mitigating the spurious correlations. Finally, Utama et al. [52] applied a regularization method for prompt-based finetuning in pretrained BERT-based language models to mitigate the impact of spurious correlations.

The existing literature predominantly addresses the detection, evaluation, and mitigation of language artefactss within general NLI datasets such as SNLI [5] and MNLI [57]. In contrast, this research focuses specifically on these aspects within the legal domain, particularly in the context of the COLIEE statute law textual entailment task. The study applies existing robustness evaluation methods, including challenging dataset creation [26] and the hypothesis-only baseline approach [1, 2, 21, 34, 49, 59], to assess the impact of language artefactss on BERT-based encoder-only models. Additionally, data augmentation techniques [15] are employed to mitigate these artefactss in the COLIEE dataset. The following section will discuss the BERT-based transformer models that will be evaluated in this research.

2.2 Transformers

Transformer models have revolutionized the field of deep learning, particularly in Natural Language Processing (NLP). Vaswani et al. [53] introduced the Transformer model in 2017 to address the limitations of recurrent neural networks (RNNs) [42] and long short-term memory (LSTM) networks [19], which were the primary architectures used for sequence modeling tasks prior to the advent of Transformers.

RNNs were initially developed to handle sequential data by maintaining a hidden state that captures information from previous time steps. This architecture allows RNNs to process input sequences of variable length, making them suitable for tasks such as language modeling, speech recognition, and machine translation. However, RNNs suffer from several significant drawbacks, including the problem of vanishing and exploding gradients, which makes training deep RNNs challenging. Additionally, RNNs struggle with capturing long-range dependencies due to their sequential nature, leading to the loss of important contextual information over extended sequences.

To mitigate some of these issues, the LSTM model was introduced by Hochreiter and Schmidhuber [19]. LSTMs extend RNNs by incorporating a memory cell and gating mechanisms (input, forget, and output gates) that regulate the flow of information and help maintain long-term dependencies. While LSTMs improved upon RNNs in handling long-range dependencies, they still inherit the sequential processing nature, making them computationally inefficient for long sequences.

The Transformer model, proposed by Vaswani et al. [53], fundamentally departed from the sequential processing approach of RNNs and LSTMs. Instead, it leverages a fully attention-based mechanism to model dependencies between tokens in a sequence. The key innovation of the Transformer is the self-attention mechanism, which allows the model to weigh the importance of different tokens in a sequence simultaneously, regardless of their positions.

The architecture of the Transformer consists of an encoder-decoder structure, where both the encoder and decoder are composed of multiple layers of self-attention and feed-forward neural networks. Each layer in the encoder has a self-attention mechanism followed by a position-wise feed-forward network, while the decoder layers include an additional cross-attention mechanism to attend to the encoder's output. This design enables the Transformer to capture complex dependencies and relationships within the input data effectively.

Self-Attention Mechanism

The self-attention mechanism is the cornerstone of the Transformer's architecture. This mechanism helps the transformer models process the input sequence in parallel. It operates by computing a set of attention weights that determine the relevance of each token in the sequence concerning every other token. These attention weights are used to create a weighted sum of the token representations, which is then passed through a feed-forward network to produce the output for each position.

Mathematically, self-attention can be described as follows:

1. **Input Representation:** Given an input sequence $X = [x_1, x_2, \ldots, x_n]$, each token x_i is first transformed into three vectors: a query q_i, a key k_i, and a value v_i. These are obtained through learned linear projections:

$$q_i = W_q x_i, \quad k_i = W_k x_i, \quad v_i = W_v x_i \tag{2.1}$$

   where W_q, W_k, and W_v are weight matrices for the query, key, and value projections, respectively.

2. **Scaled Dot-Product Attention:** The attention score for each token pair (i, j) is computed using the dot product of the query and key vectors, scaled by the square root of the dimensionality of the key vectors d_k:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{2.2}$$

   The scaled dot-production attention helps to stabilize the gradients during training and allows for better convergence. These scores are then passed through a softmax function to obtain the attention weights:

$$\alpha_{ij} = \text{softmax}\left(\frac{q_i \cdot k_j}{\sqrt{d_k}}\right) \tag{2.3}$$

3. **Weighted Sum of Values:** The output for each token x_i is computed as the weighted sum of the value vectors v_j, where the weights are the attention scores α_{ij}:

$$z_i = \sum_{j=1}^{n} \alpha_{ij} v_j \tag{2.4}$$

This mechanism enables the model to focus on different parts of the input text to varying degrees when encoding a particular token, capturing long-range dependencies and relationships in the text.

In a bid to harness information from multiple representation subspaces, the Transformer model employs multi-head attention, which consists of several parallel attention layers (or "heads"). Each head operates on different learned linear projections of the input vectors, thereby capturing different aspects of the relationships between tokens. The outputs of these heads are concatenated and linearly transformed to produce the final representation. The formulation of multi-head attention is given by:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_h) W_O \tag{2.5}$$

where $\text{head}_i = \text{Attention}(QW_{Qi}, KW_{Ki}, VW_{Vi})$, and W_{Qi}, W_{Ki}, W_{Vi}, and W_O are parameter matrices learned during training. This process can be parallelized, allowing for efficient computation and making the Transformer highly scalable.

Since their introduction, Transformer models have become the foundation for most downstream tasks in NLP. Models such as Bidirectional Encoder Representations from Transformers (BERT) [11], Generative Pre-trained Transformer (GPT) [38] , and Text-to-Text Transfer Transformer (T5) [40] have demonstrated state-of-the-art performance across a wide range of applications, including text classification, translation, summarization, and question answering.

In recent years, researchers have proposed various modifications to the original Transformer architecture to further enhance its performance and adapt it to specific tasks. For instance, BERT introduced bidirectional self-attention, enabling the model to consider both left and right context simultaneously, which significantly improved its performance on tasks requiring deep understanding of the context [11]. Robustly Optimized BERT Pretraining Approach (RoBERTa) [25] modified the pre-training process of BERT, leading to further improvements in downstream task performance. Other models like ALBERT (A Lite BERT) [24] and DistilBERT [44] have focused on reducing the model size and computational requirements while maintaining high performance.

This research work focuses on five Transformer architectures, all of which are primarily based on BERT. Detailed descriptions of these architectures and their pre-training methodologies are provided in the following sections.

2.2.1 BERT

The Bidirectional Encoder Representations from Transformers (BERT) model, introduced by researchers at Google AI Language in 2018, marks a significant milestone in the evolution of Natural Language Processing (NLP) techniques [11].

Architecture

BERT is founded on the Transformer architecture, leveraging attention mechanisms such as scaled dot-product attention and multi-head attention. BERT's bidirectional training distinguishes it from preceding models like ELMo [31] and GPT [38]. Unlike ELMo, which conditions each word solely on its preceding words, and GPT, which conditions on following words, BERT conditions each word on both its preceding and following words. This bidirectional context is achieved through a masking strategy, where a portion of the input data is obscured during training.

The implementation of BERT involves pre-training on a large corpus of text data followed by fine-tuning for specific NLP tasks. The pre-training phase consists of two tasks: masked language modeling (MLM) and next sentence prediction (NSP). The MLM task helps in learning robust representations by predicting the masked words, while the NSP task enables the understanding of relationships between different sentences. After pre-training, BERT can be fine-tuned for various NLP tasks, such as question answering, named entity recognition, and sentiment analysis, by adding a task-specific layer on top of the pre-trained BERT model.

Pre-training Data

The original BERT model was pre-trained on two corpora: the BooksCorpus (800M words) [61] and English Wikipedia (2,500M words). The BooksCorpus consists of text from 11,038 books, covering a broad spectrum of genres and topics. The natural, narrative style of the text in this corpus aids in training the model to understand more colloquial and informal aspects of language. In contrast, the text from Wikipedia articles provides a more formal and informative style, covering an extensive array of topics. This combination of datasets offers a diverse and comprehensive linguistic foundation for pre-training, encompassing a wide range of topics and writing styles.

2.2.2 LegalBERT

LegalBERT by Chalkidis et al. [7] is an adaptation of the original BERT model, fine-tuned specifically for handling legal texts. This section delves into the data utilized for fine-tuning LegalBERT and its implementation details, aligning it with the unique characteristics of the legal domain.

Architecture

LegalBERT retains the core architecture of BERT, leveraging the powerful Transformer-based bidirectional self-attention mechanism to capture complex dependencies within legal texts. The implementation of LegalBERT involved pre-training on a substantial corpus of legal documents from scratch, which enabled the model to adapt to the specific language patterns and contexts prevalent in legal texts.

Pre-training Data

The pre-training data for LegalBERT includes a variety of legal documents from multiple sources. The dataset consists of 61,826 documents from EU Legislation, totaling 1.9 GB (16.5%), and 19,867 documents from UK Legislation, amounting to 1.4 GB (12.2%). It also incorporates 19,867 documents from European Court of Justice (ECJ) cases, comprising 0.6 GB (5.2%), and 12,554 documents from European Court of Human Rights (ECHR) cases, which total 0.5 GB (4.3%). From the United States, the dataset includes 164,141 US Court cases, totaling 3.2 GB (27.8%), and 76,366 US Contracts, which comprise 3.9 GB (34.0%). This dataset laid the foundation for training various versions of LegalBERT, ensuring that the model is finely tuned to understand and process legal language and contexts effectively.

2.2.3 RoBERTa

The RoBERTa model, denoting a Robustly Optimized BERT Pretraining Approach, is an enhancement of the original BERT model designed to improve performance on a range of Natural Language Processing (NLP) tasks [25]. RoBERTa was developed to address the limitations observed in BERT's pretraining phase, specifically targeting the efficiency and effectiveness of the pretraining regimen. While the architectural blueprint of RoBERTa remains largely consistent with that of BERT, inheriting the Transformer architecture, it introduces several optimizations to enhance its performance.

Architecture

RoBERTa refines BERT's architecture through the following modifications.

1. **Tokenizer**
   - Transitioning from BERT's WordPiece tokenizer [11], RoBERTa employs a byte-level Byte-Pair Encoding (BPE) tokenizer [46], facilitating a more granular tokenization scheme.
2. **Pretraining Scheme**
   - The Next Sentence Prediction (NSP) task, a constituent of BERT's pretraining objectives, is omitted in RoBERTa's pretraining regimen.
   - An innovation in RoBERTa is the advent of dynamic masking, allowing distinct masking patterns for each epoch during pretraining, contrary to the static masking paradigm observed in BERT. In each epoch, the positions of the masked tokens are randomly selected, introducing variability in the masking pattern. This dynamic aspect produces a change in the mathematical computation of the Masked Language Model (MLM) objective, enhancing the model's exposure to different contextual relationships in the text during pretraining.

The objective function in RoBERTa primarily revolves around the Masked Language Model (MLM) task, akin to BERT, but with dynamic masking. The MLM task is formulated as a prediction problem, where the model aims to predict the original tokens of the masked positions in a text sequence, given the remaining unmasked tokens. Mathematically, the objective function $\mathcal{L}$ for the MLM task can be written as:

$$\mathcal{L} = -\sum_{i=1}^{N} \log P(x_i | x_{-i}; \theta) \tag{2.6}$$

where:

- N is the total number of masked positions in the training corpus.
- x_i is the original token at the masked position i.
- x_{-i} denotes the unmasked tokens in the text sequence.
- θ represents the model parameters.

This objective function aims to minimize the negative log-likelihood of the original tokens at the masked positions, guiding the model to learn robust representations of text that capture contextual relationships among tokens.

Pre-training Data

RoBERTa significantly increases the volume of training data compared to the original BERT model, leveraging a much larger and diverse dataset.

- Common Crawl News: 63 million articles.
- OpenWebText: A dataset similar to the one used for training GPT.
- Stories from Common Crawl: Filtered to retain relevant narrative texts.
- Wikipedia (English): An extensive collection of English Wikipedia articles.
- BooksCorpus: The same dataset utilized in training BERT.

The model was pretrained on a corpus encompassing approximately 160GB of text, which is a substantial enlargement compared to the corpus used for BERT. This expanded dataset allows RoBERTa to cover a broader linguistic spectrum and enhances its generalization capabilities, contributing to its superior performance on various natural language processing benchmarks [25].

2.2.4 ELECTRA

ELECTRA, which stands for "Efficiently Learning an Encoder that Classifies Token Replacements Accurately," was introduced by Clark et al. [9]. Unlike traditional Masked Language Modeling (MLM) pre-training methods such as BERT, which corrupts input data by replacing some tokens with a [MASK] token and then trains a model to reconstruct the original tokens, ELECTRA introduces a more sample-efficient pre-training task called Replaced Token Detection (RTD).

Architecture

ELECTRA employs a novel approach where, instead of masking tokens, it replaces some tokens with credible alternatives sampled from a small generator network. The framework comprises two transformer models: a generator and a discriminator. The generator's role is to replace tokens in a sequence, and it is trained similarly to a masked language model. Conversely, the discriminator is trained to predict whether each token in the corrupted input was replaced by a generator sample or not. This bifurcated structure facilitates learning from all input tokens, rendering ELECTRA more computationally efficient compared to MLM pre-training methods like BERT, which learn only from a small masked-out subset of tokens.

In the ELECTRA model, both the generator and the discriminator are trained through separate objectives. The generator employs a masked language modeling (MLM) objective, while the discriminator employs a binary classification objective. The detailed mathematical exposition of these training objectives is provided below.

Generator Training Objective The generator in ELECTRA is trained to perform masked language modeling (MLM), akin to the objective used in BERT. Given an input sequence $x = \{x_1, x_2, \ldots, x_N\}$, a subset of the tokens are masked out, and the generator is trained to predict the original tokens at the masked positions. Mathematically, the training objective $\mathcal{L}_{\text{MLM}}$ for the generator can be written as:

$$\mathcal{L}_{\text{MLM}} = -\sum_{i \in \mathcal{M}} \log P(x_i | x_{-i}; \theta_{\text{gen}}) \tag{2.7}$$

where:

- $\mathcal{M}$ is the set of masked positions,
- x_i is the original token at position i,
- x_{-i} denotes the unmasked tokens in the sequence,
- θ_{gen} represents the parameters of the generator.

Discriminator Training Objective The discriminator is trained to distinguish between the original tokens and the tokens replaced by the generator. For each token position i, the discriminator predicts whether the token x_i is original or replaced. The training objective $\mathcal{L}_{\text{disc}}$ for the discriminator can be written as a binary classification loss:

$$\mathcal{L}_{\text{disc}} = -\sum_{i=1}^{N} \left(y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right) \tag{2.8}$$

where:

- y_i is the true label (1 for original, 0 for replaced),
- p_i is the predicted probability that the token x_i is original,
- N is the total number of tokens in the sequence.

Combined Training Objective The combined loss $\mathcal{L}$ for training ELECTRA is a weighted sum of the generator loss $\mathcal{L}_{\text{MLM}}$ and the discriminator loss $\mathcal{L}_{\text{disc}}$, which can be written as:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{MLM}} + \beta \cdot \mathcal{L}_{\text{disc}} \tag{2.9}$$

where α and β are hyperparameters controlling the relative importance of the two losses.

This combination of masked language modeling by the generator and token classification by the discriminator is called the Replaced Token Detection (RTD) task used in the pre-training of ELECTRA. After pre-training, the generator is discarded, and only the discriminator is fine-tuned on downstream tasks [9].

Pre-training Data

ELECTRA's pre-training utilizes a dataset similar to those used for BERT and other transformer models, ensuring a comprehensive linguistic representation. The key datasets used are mentioned below.

- Wikipedia (English): An extensive dataset of English Wikipedia articles.
- BooksCorpus: A large dataset of books, similar to the one used for BERT.
- OpenWebText: A dataset designed to be similar to the WebText dataset used for training Generative Pre-trained Transformer.
- Common Crawl: A vast repository of web data.

The ELECTRA model is trained on these datasets, leveraging the RTD task to learn robust language representations.

2.2.5 DeBERTa V3

DeBERTa, which stands for Decoding-enhanced BERT with Disentangled Attention, is an advanced transformer-based architecture designed to improve upon the BERT model's natural language understanding capabilities [18]. DeBERTa V3 introduces two key techniques: disentangled attention and ELECTRA-style pre-training with the Replaced Token Detection task.

Architecture

DeBERTa V3 introduces significant improvements over its predecessors by integrating techniques from ELECTRA and incorporating new gradient-disentangled embedding sharing methods.

Disentangled Attention Mechanism
DeBERTa V3 maintains the disentangled attention mechanism, which differs from the traditional BERT approach. In BERT, each word in the input layer is represented by a single vector that combines its word (content) embedding and position embedding. DeBERTa V3, however, uses two separate vectors to encode a word's content and its position. The attention interactions between words are determined by separate matrices that account for both the contents of the words and their relative positional information. This nuanced representation acknowledges that the attention weight between a pair of words depends on both their contents and their relative positions.

Mathematically, the disentangled attention can be represented as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + P\right)V \tag{2.10}$$

where Q, K, and V are the query, key, and value matrices, d_k is the dimension of the keys, and P is the positional encoding matrix.

ELECTRA-Style Pre-Training
One of the significant improvements in DeBERTa V3 over its predecessor is the incorporation of ELECTRA-style pre-training with the Replaced Token Detection (RTD) task. This approach involves a generator-discriminator setup, where the generator replaces some input tokens with plausible alternatives, and the discriminator learns to distinguish between the original and replaced tokens. This method enhances the model's efficiency and effectiveness in learning representations, leading to better performance on downstream tasks [18].

Pre-Training Data

DeBERTa V3 is pre-trained using a substantial text corpus similar to those used for ELECTRA, ensuring comprehensive language understanding. The primary objective for pre-training is the Replaced Token Detection (RTD) task, supplemented by an MLM task. Additionally, the authors propose a method of virtual adversarial training for fine-tuning pretrained language models, involving the normalization of word embeddings followed by slight perturbations to enhance model robustness.

The DeBERTa V3 model leverages these datasets and the RTD task to learn robust language representations, demonstrating a shift from generative to discriminative pre-training tasks for improved performance and efficiency.

2.2.6 Supervised Fine-Tuning

Supervised fine-tuning is crucial for adapting pre-trained language models to specific downstream tasks in natural language processing (NLP) [27]. This process involves further training a pre-trained model on a labeled dataset specific to the task, enabling it to learn task-specific features and enhance performance [32].

In supervised learning, models are trained on datasets with input-output label pairs. Fine-tuning, a form of transfer learning, involves training a model pre-trained on a large corpus of general data on a smaller, task-specific dataset. Pre-trained models like BERT and GPT, initially trained on extensive corpora such as Wikipedia and BooksCorpus, learn general language representations. However, fine-tuning is necessary for tasks like sentiment analysis, question answering, or legal text entailment to achieve optimal performance.

The steps involved in fine-tuning include preparing a relevant labeled dataset, initializing the pre-trained model with pre-learned weights, and potentially modifying the model architecture for the specific task. The model is then trained on the task-specific dataset, with loss computed based on predictions and true labels, updating weights through backpropagation. Hyperparameters like learning rate and batch size are tuned to avoid overfitting.

Performance is evaluated on a validation set to ensure good generalization, using techniques such as cross-validation, early stopping, and regularization. For domain-specific tasks like legal text entailment, fine-tuning on specialized datasets allows the model to capture domain-specific knowledge, improving task performance.

Supervised fine-tuning offers improved performance, reduced need for large labeled datasets, and better adaptation to task-specific nuances. By leveraging large-scale pre-trained models and fine-tuning them on domain-specific datasets allows for state-of-the-art performance on various NLP downstream tasks. In summary, supervised fine-tuning is essential step for adapting pre-trained language models to specific tasks.

2.3 Model Interpretability

Model interpretability refers to the ability to understand and explain the predictions made by machine learning models. As these models, especially deep learning networks, become more complex, the need for interpretability grows. This is particularly crucial in sensitive applications like healthcare, finance, and law, where understanding the rationale behind model predictions is necessary for trust and accountability.

Transformers, a class of models known for their state-of-the-art performance in natural language processing (NLP) tasks, are inherently complex due to their deep architecture and use of attention mechanisms. Despite their effectiveness, the opacity of these models can be a significant barrier to their adoption in critical applications. Interpretability techniques help bridge this gap by providing insights into how and why these models make certain predictions.

Transformers utilize self-attention mechanisms to weigh the importance of different parts of the input sequence when making predictions. This mechanism allows the model to consider the entire sequence simultaneously, capturing dependencies between distant tokens more effectively than traditional recurrent neural networks (RNNs) or long short-term memory networks (LSTMs). However, the multi-layer and multi-head attention structure of Transformers adds to their complexity, making it challenging to directly interpret their behavior.

To make Transformers more interpretable, several techniques have been developed. These techniques can broadly be categorized into intrinsic interpretability, where the model structure itself is made interpretable, and post-hoc interpretability, where external methods are used to explain an already trained model [55]. One of the most prominent post-hoc interpretability methods to provide insights on a model's prediction is Integrated Gradients (IG), introduced by Sundararajan et al. [47].

Integrated Gradients (IG) for Feature Attribution

Integrated Gradients (IG) [47] is a powerful technique for attributing the prediction of a neural network to its input features. IG addresses some of the limitations of simpler gradient-based methods, such as saliency maps, which can suffer from issues like saturation and noise [55].

Conceptual Foundation

The core idea behind IG is to attribute the difference between the model output for a given input and a baseline (usually an input with all features zeroed out) to the individual input features. This is achieved by integrating the gradients of the model output with respect to the input along a path from the baseline to the input. The resulting attribution scores indicate the contribution of each input feature to the prediction of the model.

Mathematical Formulation

Let F be a neural network with input $x \in \mathbb{R}^n$ and baseline input $x' \in \mathbb{R}^n$. The attribution score for the i-th feature of input x is defined as:

$$\mathrm{IG}_i(x) = (x_i - x'_i) \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \tag{2.11}$$

where α is a scaling factor that interpolates between the baseline x' and the input x. The term $\frac{\partial F(x'+\alpha(x-x'))}{\partial x_i}$ represents the gradient of the model's output with respect to the i-th input feature, evaluated at points along the straight line path from x' to x.

Steps to Compute Integrated Gradients

1. **Baseline Selection**: Choose a baseline input x'. This is often a zero vector or an average input.
2. **Interpolate**: Generate a set of interpolated inputs between the baseline x' and the actual input x by varying α from 0 to 1.
3. **Compute Gradients**: For each interpolated input, compute the gradient of the model's output with respect to the input features.
4. **Integrate**: Integrate these gradients along the interpolation path to obtain the attribution scores.

In the context of Transformers, IG can be used to interpret the contributions of individual tokens to the model's predictions. For example, in a sentiment analysis task, IG can highlight which tokens in a sentence are most responsible for the model's positive or negative sentiment prediction. Similarly, in natural language inference (NLI) tasks, IG can identify which parts of the premise and hypothesis the model relies on to determine the entailment relationship. Additionally, in the context of language artefacts, this feature attribution method can help us perform a qualitative analysis of the model's predictions by examining whether artefacts contribute to those predictions.

Visualizing IG attributions often involves heatmaps or color-coding the text to show the importance of each token. These visualizations can potentially help the users understand the model's decisions by providing clear and intuitive explanations of which parts of the input are most influential.

2.4 Summary

This chapter provides a comprehensive overview of the definition and challenges posed by language artefacts in Natural Language Inference (NLI) datasets, as well as the existing methods for their detection, evaluation, and mitigation. It also includes a brief overview of Transformer models, with a focus on BERT-based encoder-only variants used in this research. Furthermore, the chapter explores model interpretability, emphasizing the application of Integrated Gradients to visualize the feature attribution within Transformer models.

3. Dataset

3.1 Overview of COLIEE

The Competition on Legal Information Extraction and Entailment (COLIEE)[1] is an annual competition that aims to advance the development of artificial intelligence techniques in the legal domain. The competition comprises several tasks designed to tackle challenges in legal domain related to information retrieval and natural language inference. Among these tasks, Task 4, known as "Statute Law Textual Entailment," focuses on the ability of systems to determine whether a given legal statement (hypothesis) can be inferred (entailed) from a set of legal statutes (premises). This task is crucial for automating legal reasoning and providing decision support in legal contexts.

The following are the tasks involved in COLIEE.

1. **Task 1**: Case law retrieval - entails analyzing a novel legal case and identifying pertinent supporting cases from a comprehensive database of Federal Court of Canada case laws. The objective is to find cases that substantiate the decision of the new case.
2. **Task 2**: Case law textual entailment - involves discerning specific paragraphs within relevant case law that substantiate the legal conclusion of a new case. This task requires a detailed semantic comparison to accurately identify the supporting text.
3. **Task 3**: Statute law retrieval - involves examining a legal bar exam question and extracting the relevant articles from the Japanese Civil Code that are pertinent to resolving the question.
4. **Task 4**: Statute law textual entailment - entails determining whether the identified articles from the Japanese Civil Code entail the given legal question or its negation. The task is focused on establishing a binary (yes/no) entailment relationship.

This thesis focuses on Task 4, which involves the complex challenge of textual entailment within the legal domain.

[1]https://sites.ualberta.ca/ rabelo/COLIEE2024/

3.2 Statute Law Textual Entailment Task Objective

The objective of Task 4, “Statute Law Textual Entailment,” is to develop a system that can predict whether a set of relevant statute law articles entails (“YES”) or contradicts (“NO”) a given legal bar examination question (Q). The corpus for Task 4 consists of legal questions drawn from Japanese Legal Bar exams, and all relevant Japanese statute law articles are provided [56]. The training data includes pairs of queries and relevant articles, each labeled with a correct answer “Y” or “N.” The test data comprises pairs of queries and relevant articles without the “Y/N” labels. In this task, the relevant statute articles are considered as the premises, and the legal bar exam questions are considered as the hypotheses [56].

Task 4 Dataset

The dataset utilized for Task 4 comprises legal questions extracted from Japanese Legal Bar examinations and corresponding articles from statute law. In the training set, each legal question is paired with relevant articles and a label indicating whether the question is entailed by the articles. The test set follows the same structure, omitting the entailment labels to assess model performance in inferring relationships between complex legal texts.

Example Training Instance

```
<pair id="H18-1-2" label="N">
  <t1>
    Article 565
    The provisions of the preceding three Articles apply mutatis
       mutandis if the right transferred by the seller to the
       buyer does not conform to the terms of the contract (
       including the case in which the seller fails to transfer
       part of a right that belongs to another person).
    Article 566
    If the subject matter delivered by the seller to the buyer
       does not conform to the terms of the contract with respect
        to the kind or quality, and the buyer fails to notify the
        seller of the non-conformity within one year from the
       time when the buyer becomes aware of it, the buyer may not
        demand cure of the non-conformity of performance, demand
       a reduction of the price, claim compensation for loss or
       damage, or cancel the contract, on the grounds of the non-
       conformity; provided, however, that this does not apply if
        the seller knew or did not know due to gross negligence
       the non-conformity at the time of the delivery.
  </t1>
  <t2>
    There is a limitation period on pursuance of warranty if
       there is restriction due to superficies on the subject
       matter, but there is no restriction on pursuance of
       warranty if the seller's rights were revoked due to
       execution of the mortgage.
  </t2>
</pair>
```

Example Test Instance

```
<pair id='R03-01-E'>
  <t1>
    Article 32 (1) Having received proof that a missing person is
        alive or that a missing person died at a time different
        from the time set forth in the preceding Article, the
        family court, at the request of the missing person or an
        interested person, must rescind the declaration of that
        person's disappearance. In this case, the rescission does
        not affect the validity of any act performed in good faith
        after the declaration of disappearance but before the
        rescission thereof.
     (2) A person who has acquired property due to a declaration
        of disappearance loses the rights in question due to its
        rescission; provided, however, that the person has the
        obligation to return that property only to the extent
        currently enriched.
  </t1>
  <t2>
    In the event that a declaration of disappearance is rescinded,
        a person who has acquired property due to the declaration
        of disappearance must return to the missing person the
        full amount of the benefits received from the acquisition
        of the property, even if the person acted in good faith
        regarding the survival of the missing person.
  </t2>
</pair>
```

The dataset is provided in XML format. Each data instance is enclosed within a <pair> tag, containing a unique identifier (e.g., "H18-1-2") and, for training instances, an entailment label ("Y" for entailment, "N" for not entailment). Within the pair, the <t1> tag encompasses the premise, representing the Japanese statute law article(s), while the <t2> tag contains the hypothesis, corresponding to the Japanese Legal Bar examination question.

The COLIEE training dataset for each year is constructed by adding the test dataset from the previous year. This method results in an incremental increase in the size of the training dataset each year, ensuring the inclusion of a diverse range of legal questions and statutes. For example, the 2022 dataset contains training instances from all previous years up to 2021, plus the 2021 test dataset.

The provided dataset statistics in table 3.1 illustrates that for each year, the training and test datasets are nearly balanced, with a similar number of instances for both entailed and not-entailed labels. Despite this balance, the number of training instances per year is relatively small to train a neural network model. The domain-specific nature of the dataset can lead to models becoming overfitted and biased towards certain weak features, such as language artefacts, which are further discussed in this thesis.

3.3 Dataset Statistics

To investigate the presence and impact of language artefacts in COLIEE Task 4, a comprehensive analysis spanning five years of data from 2018 to 2022 was conducted. This longitudinal approach aimed to provide a robust understanding of the evolution and influence of language artefacts within the task's dataset.

Table 3.1: Training and Test Dataset Statistics

Year	Training Instances	Entailed Instances	Not-Entailed Instances	Test Instances	Entailed Instances	Not-Entailed Instances
2018	567	287	280	58	27	31
2019	625	314	311	70	36	34
2020	695	350	345	111	59	52
2021	806	409	397	81	38	43
2022	887	447	440	109	36	34

The COLIEE training dataset is built incrementally each year, incorporating the test dataset from the previous year. This methodology results in a progressively expanding training corpus, enriching the diversity of legal questions and statutes included. For instance, the 2022 training dataset encompasses all instances from prior years up to 2021, supplemented by the 2021 test set. However, this approach also allows for the potential propagation of language artefacts through each yearly iteration if not properly mitigated.

3.4 Summary

Thus the COLIEE Task 4 dataset provides a robust framework for developing and evaluating models for statute law textual entailment. By using legal questions from Japanese Legal Bar exams and corresponding statute law articles, the dataset offers a realistic and challenging environment for testing the inference capabilities of AI models [56]. The incremental growth and balanced nature of the dataset make it a valuable resource for training and testing NLI models. However, the limited size of the dataset requires careful consideration to avoid overfitting and bias. A thorough understanding of the dataset is crucial for the subsequent analysis and development of robust models capable of handling legal textual entailment tasks effectively.

4. Methodology

This section outlines the methodologies used in this research. It begins with a description of the techniques for detecting language artefacts in the COLIEE legal statute law entailment dataset. This is followed by an evaluation of the robustness of various BERT-based models. Finally, it discusses the mitigation strategies employed to reduce word overlap and contradiction word artefacts [56].

4.1 Language Artefacts Detection

In supervised learning, neural network models may inadvertently generalize from weak features that do not accurately represent the task at hand [56]. These features introduce biases related to specific labels within the dataset, affecting the reliability of model predictions. Unlike societal biases, which stem from broader social behaviors, dataset biases or artefacts are typically implicit and unintentional. Ensuring the detection and mitigation of these minor yet significant traits is important for enhancing the accuracy and effectiveness of neural network models in supervised learning settings. [56].

In the field of Natural Language Processing (NLP), language artefacts refer to these subtle biases within a language dataset [56]. Particularly in the context of Natural Language Inference (NLI), as discussed in the background chapter, datasets often contain spurious correlations or weak heuristics. These can be exploited by statistical learning models to predict textual entailment relationships without fully considering the context or stronger features.

This research aims to detect various lexical and syntactic artefacts, including annotation artefacts, word overlap artefacts, contradiction word artefacts, sentence length artefacts, and subsequence heuristics. The following section details the methods used to identify these language artefacts in the COLIEE Task 4 dataset.

4.1.1 Annotation Artefacts

Definition: Annotation artefacts emerge when annotators, while labeling or curating sentences for specific classes, follow a consistent pattern. This pattern, when generalized across a dataset, leads to implicit biases. For example, if a unigram feature ‘A’ is strongly correlated with a label ‘X’, the model might predict ‘X’ whenever it encounters ‘A’, neglecting other relevant features or the overall semantic context [56].

Detection Method: Previous research has employed methods such as Pointwise Mutual Information (PMI) [15] and Z-statistics [12] to detect annotation artefacts in datasets like SNLI and MNLI. However, these methods may not be as effective with smaller datasets due to their reliance on larger sample sizes for accurate estimation of statistical measures. In this research, the binomial significance test is chosen to identify statistically significant annotation artefacts because it is well-suited for small sample sizes, providing a robust measure of significance even when data points are limited. Given that the premises are derived from Japanese statute law and the hypotheses are based on Japanese bar examination questions, it is assumed that distinct patterns for entailment and non-entailment labels will be found in the questions. Bar exam questions vary to simulate real-world scenarios, introducing potential artefacts, whereas statute laws are consistently written, minimizing such variations. Consequently, identifying artefacts in the bar exam questions (hypotheses) is more practical, and modifying these questions is easier than altering statute laws (premises). Therefore, concentrating on bar examination questions (hypotheses) affords a more adaptable and practical methodology for the detection of annotation artefacts. Additionally, all unigram features, inclusive of stopwords, are incorporated into the analysis, as even an asymmetrical distribution of such features may influence model performance. The detection method involves the steps described below.

1. Calculate the total number of times a particular token occurs in the training corpus only in hypothesis sentences and not in the premise.
2. Calculate the number of times a particular token occurs in entailment label "Y" and non-entailment label "N".
3. Calculate the conditional probability of a particular label occurring given the token, $P(\text{label}|\text{token})$.
4. **Calculate the p-value of a token using the binomial significance test.** If a particular token for a given label has a conditional probability greater than 0.5 and a p-value less than 0.1, the token can be considered an annotation artefact for that particular class label. The significance level is a threshold set by the researcher to determine whether to reject the null hypothesis. It represents the probability of observing the data, or something more extreme, assuming the null hypothesis is true. In this context, a significance level of 0.1 means there is a 10% chance of incorrectly identifying a token as an annotation artefact when it is not. This relatively high significance level is chosen due to the small dataset size and to ensure that potential artefacts are not overlooked.

The binomial significance test formula is:

$$P(X = k|n, p) = \binom{n}{k} p^k (1 - p)^{n-k} \tag{4.1}$$

Where:

- X is the number of successes (here, occurrences in entailment).
- $k = 70$ (number of occurrences in entailment).
- $n = 100$ (total occurrences).
- p is the hypothesized probability of success (if there is no artefact, we might assume $p = 0.5$ for equal likelihood).

Example Calculation: Consider a token "A" that occurs 100 times in hypothesis sentences. Out of these, it occurs 70 times in instances labeled as entailment ("Y") and 30 times in instances labeled as non-entailment ("N").

- Total occurrences of token "A" in hypotheses: 100
- Occurrences in entailment ("Y"): 70
- Occurrences in non-entailment ("N"): 30

Conditional Probability

- **Conditional probability for entailment:** $P(Y|A) = \frac{70}{100} = 0.7$
- **Conditional probability for non-entailment:** $P(N|A) = \frac{30}{100} = 0.3$

Binomial Significance Test

The binomial coefficient $\binom{100}{70}$ is calculated as:

$$\binom{100}{70} = \frac{100!}{70!(100-70)!} = \frac{100!}{70! \cdot 30!}$$

Let's calculate the p-value:

- **n (total trials):** 100
- **k (successes):** 70
- **p (probability of success):** 0.5

So, the p-value for the token "A" using the binomial significance test is approximately 0.0002674.

Since the p-value 0.0002674 is much less than 0.1, token "A" can be considered an annotation artefact for the entailment class label.

The procedure mentioned above is carried out for each dataset throughout 2018 to 2022 to detect the annotation artefacts associated with entailment label "Y" and non-entailment label "N".

4.1.2 Sentence Length Artefacts

Definition: Sentence length artefacts emerge from irregularities in the distribution of the input text's sentence lengths. Gururangan et al. (2018) discussed how this irregularity in hypothesis length is present among the different classes in the SNLI dataset. This research considers the irregularities in the hypotheses (bar examination questions) sentence lengths in the statute law entailment task for detecting sentence length artefacts.

Detection Method: To detect irregularities in the distribution of hypothesis sentence lengths, the sentence lengths within both the entailment and non-entailment labels are analyzed, with a focus on instances that exceed the average sentence length. This analysis is conducted for each dataset from 2018 to 2022. If one of the labels has a higher number of instances exceeding the average sentence length of the hypothesis, this indicates the presence of a sentence length artefact for that particular label.

4.1.3 Word Overlap Artefacts

Definition: Word overlap artefacts in Natural Language Inference (NLI) occur when there is a significant overlap of words between the premise and the hypothesis [56]. The guiding presumption behind this artefact is that entailment classes typically exhibit greater overlaps than non-entailment classes [56].

Detection Method: To detect word overlap artefacts, the number of overlapping words between the premise (statute law) and the hypothesis (bar examination question) is counted. For this research, an instance is considered as word overlap artefact if more than 50% of the words in the hypothesis are present in the premise. However, hypotheses that are entirely contained within the premise which is a continuous subsequence are not considered word overlap artefacts.

4.1.4 Contradiction Word Artefacts

Definition: "Contradiction word artefacts occur when there is a strong correlation between the presence of negation words in a training instance and the non-entailment (N) class label" [56]. In this research, a high co-occurrence of negation words in a hypothesis and the non-entailment (N) label in the training dataset is considered indicative of a contradiction word artefact [56].

Detection Method: The correlation between the presence of negation words in the hypotheses and the assigned class labels were examined. The negation words used to identify the contradiction word artefacts are: "not", "no", "n't", "none", "neither", "never", "nobody", "nothing", "nowhere", "hardly", "scarcely", "barely", "rarely", "seldom" [56]. Instances classified as non-entailment exhibited a higher frequency of negation words than those classified as entailment, indicating contradiction word artefacts. [56].

4.1.5 Subsequence Heuristic Artefacts

Definition Subsequence heuristic artefacts in Natural Language Inference (NLI) occur when the hypothesis is constructed entirely from a continuous subsequence of the premise [56]. According to McCoy et al. [26], each segment of a hypothesis should be considered "a continuous subsequence of the premise". They hypothesized that a higher number of subsequence heuristics are typically found in entailment instances.

Detection Method To detect these artefacts, instances where the entire hypothesis (bar examination question) appears as a subsequence within the premise (statute laws) were analyzed. If the hypothesis is entirely embedded within the premise, this indicates a specific bias called subsequence heuristics artefacts.

The aforementioned methods outline the detection techniques utilized for identifying various language artefacts within the statute law textual entailment task training dataset in COLIEE.

4.2 NLI Models

4.2.1 Model Selection

In the COLIEE competition, a variety of neural networks and traditional machine learning algorithms have been employed to address Task 4, which focuses on statute law textual entailment. Table 4.1 provides a summary of the unique algorithms and models used in solving the statute law textual entailment task from 2018 to 2022. Since the introduction of BERT in 2018, a significant number of teams have adopted transformer models for this task, highlighting their transformative impact on natural language processing (NLP) tasks.

Table 4.1: Summary of machine learning algorithms used in COLIEE from 2018-2022

Year	Deep Learning Models	Traditional Machine Learning Models	Other Models
2018	Deep Neural Network, Word2Vec	None	None
2019	BERT, LSTM, GRU, CNN, FastText	None	None
2020	BERT, RoBERTa, LSTM, GRU, GloVe	Support Vector Machine (SVM)	None
2021	BERT, T5, ELECTRA, RoBERTa, GPT-3	None	Graph Neural Networks
2022	BERT, ELECTRA, RoBERTa, LegalBERT	Support Vector Machine (SVM), Ensemble Methods	Graph Neural Networks, Longest Uncommon Subsequence Similarity Comparison, Predicate-Argument Structures

The various models were sourced from the COLIEE proceedings and summaries of the respective years [20, 22, 35, 36, 37]. Based on their historical performance and relevance to this research, several BERT-based encoder-only models were selected to evaluate robustness against language artefacts. The chosen models include BERT, LegalBERT, RoBERTa, ELECTRA, and DeBERTa. These models were selected due to their proven effectiveness in the COLIEE tasks for handling statute law textual entailment.

Since the models used in COLIEE for Task 4 are not publicly available and implementing them from scratch is beyond the scope of this research, open-source models available from Huggingface[2] were chosen. A critical issue identified with the base pre-trained BERT-based language models is overfitting and poor performance when fine-tuned solely on the limited COLIEE training dataset. To mitigate this, models already fine-tuned on larger, more diverse NLI datasets were selected. This pre-finetuning helps the models generalize better and reduces the risk of overfitting, as these models were already exposed to NLI tasks with relatively larger training data.

For BERT, RoBERTa, and ELECTRA, models fine-tuned on Multi-Genre Natural Language Inference (MNLI) dataset [57], a large and diverse NLI dataset, were chosen. As there were no existing LegalBERT models fine-tuned on NLI datasets, the base LegalBERT model continually pre-trained on legal texts, was selected, making it inherently suitable for the legal domain. For DeBERTa, a model fine-tuned on multiple NLI datasets such as MNLI [57], Fact Extraction and VERification (FEVER) [50], and Adversarial-NLI (ANLI) [28] was chosen. This diverse fine-tuning makes DeBERTa base well-equipped to handle various types of entailment relationships.

The selected models include those extensively fine-tuned on single and multiple NLI datasets, as well as LegalBERT, which, while not fine-tuned on any textual entailment task, possesses domain-specific knowledge. This mixture of models offers a comprehensive approach to analyzing the impact of language artefacts, allowing for a thorough investigation of their influence on model performance in determining entailment relationships. Therefore, this diverse set of models, each with unique strengths and architectures, offers a robust basis to explore how language artefacts found in the COLIEE Task 4 training dataset influence BERT-based NLI models during textual entailment prediction.

Before moving on to the robustness evaluation, it is essential to understand the dynamics of fine-tuning the chosen NLI models on COLIEE task 4 dataset. The following section provides an overview of constructing language artefact-aware feature representation for fine-tuning the models, with the primary objective being to understand the impact of these language artefacts on model performance.

4.2.2 Model Fine-tuning

To perform robustness evaluation against language artefacts, the selected models were fine-tuned on the COLIEE task 4 training dataset. The input structure for these models was specifically designed to study the effects of various language artefacts, both when they were explicitly provided and when they were not. This approach aims to assess how models interact with these artefacts during inference.

The input to the above selected NLI models consisted of three elements: the premise, the hypothesis, and the language artefact features, each separated by a ‘SEP’ token [56]. The language artefacts were represented as either numeric or boolean features, depending on their type, except for annotation artefacts. Annotation artefacts, which are detected through high correlations between certain tokens and class labels, are challenging to represent explicitly as simple numeric or boolean values.

[2]https://huggingface.co/

The representation of language artefacts using simple numeric or boolean values rather than complex phrases or sentences was chosen to determine if these straightforward features create patterns that models might exploit during inference. This resulted in six different types of input configurations for each model:

- No language artefact features.
- All language artefact features (except annotation artefacts).
- Sentence length as a language artefact feature.
- Word overlap as a language artefact feature.
- Presence of contradiction words as a language artefact feature.
- Subsequence heuristics as a language artefact feature.

The pattern for appending all language artefact features to the hypothesis follows a structured sequence. First, the sentence length feature is represented by the number of tokens in the hypothesis, providing a simple numeric value. This is followed by the word overlap feature, which includes both the number of overlapping words between the hypothesis and the premise, as well as a boolean value indicating whether any word overlap exists. Next, the contradiction word feature is introduced with a boolean value that signals the presence of negation words, such as "not," within the hypothesis. Finally, the subsequence heuristic feature is represented by another boolean value that indicates whether the hypothesis is a subsequence of the premise.

The table 4.2 further summarizes the language artefact features and their representation types, along with example inputs based on an example textual entailment pair.

Each model variant was finetuned with these six different input configurations to assess the impact and interaction of language artefacts on these models. This diverse input structure allowed for a comprehensive evaluation of the model's susceptibility to language artefacts and their ability to handle them during inference.

The primary goal of this fine-tuning approach was to create language artefact-aware training. By explicitly including or excluding these features in the input, it is possible to observe the models' performance and determine whether they rely on these artefacts to make predictions. This setup provides insights into how models might exploit patterns created by these features during inference.

After finetuning, the model variants were evaluated for their robustness against language artefacts using various evaluation methods. These methods provide a comprehensive understanding of the robustness and capability of the models, particularly concerning language artefacts.

4.3 Robustness Evaluation for NLI Models

Evaluating the robustness of Natural Language Inference (NLI) models is a crucial aspect of ensuring their reliability and applicability in real-world scenarios. Robustness refers to the model's ability to maintain high performance and accurate predictions when faced with challenging, diverse, or previously unseen data without exploiting weak features. This

Table 4.2: Features and Model Inputs for Textual Entailment

Feature	Description	Example Model Input
None	No language artefact feature is appended to the premise-hypothesis pair.	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [EOS]
All	All language artefact features are appended to the premise-hypothesis pair (except annotation artefact).	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [SEP] 9 [SEP] 8 [SEP] True [SEP] False [SEP] False
Sentence length	The number of words in the hypothesis sentence.	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [SEP] 9 [EOS]
Word overlap	The count of words shared between the premise and the hypothesis, and a true/false value indicating if any overlap exists.	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [SEP] 8 [SEP] True [EOS]
Contradiction words	True/false value indicating the presence of negation words in the hypothesis.	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [SEP] False [EOS]
subsequence	True/false value showing if the hypothesis is a part of the premise.	[BOS] All students in the class passed the final exam. [SEP] Some students in the class passed the final exam. [SEP] False [EOS]

evaluation is particularly important for NLI models, which are tasked with understanding and reasoning complex language inputs to determine logical entailment relationships between two statements. Robust models are less likely to be fooled by adversarial attacks or spurious correlations, which can undermine their credibility and effectiveness. By systematically evaluating the robustness of NLI models, researchers and practitioners can identify and mitigate potential vulnerabilities, leading to the development of more reliable NLI systems.

Robustness evaluation encompasses various methodologies designed to stress-test NLI models under different conditions. This includes hypothesis-only baselines, which check if the model can make predictions only based on the hypothesis; evaluation on challenging datasets that expose model weaknesses often by adversarial attacks that introduce subtle perturbations to test model resilience and interpretability techniques that shed light on the model's decision-making process to qualitatively assess the important features considered during inference.

The NLI models are evaluated with the standard test set provided by COLIEE for task 4, an adversarial test that is created to evaluate the robustness against the language artefacts and finally a baseline evaluation on hypothesis-only model. For the quantitative evaluation accuracy is used officially as the performance metric in COLIEE.

This section will thoroughly examine the various methodologies employed to assess the robustness of Natural Language Inference (NLI) models, offering an in-depth analysis of their strengths and potential areas for enhancement by focusing on language artefacts.

4.3.1 Standard Test Set

In COLIEE, evaluating the performance of Natural Language Inference (NLI) models for the statute law textual entailment task is necessary to assess their robustness and effectiveness. Each selected BERT-based model, with its six input variants, is fine-tuned on the COLIEE-provided training dataset. The performance of these models is then evaluated using the test dataset provided by COLIEE. The primary metric for evaluation is accuracy, which measures the proportion of correctly predicted instances to the total number of instances in the test set.

The test dataset consists of pairs of legal statutes and corresponding bar examination questions, where the task is to determine whether the statute entails the question ("Y") or contradicts it ("N"). For each model variant, the predictions are compared against the ground truth labels provided in the test set.

To ensure a comprehensive evaluation, each model's accuracy is calculated for six different input configurations. These configurations include the model's performance without explicit language artefact features, as well as with individual and combined artefact features. By comparing the accuracy across these configurations, insights can be gained into the impact of language artefacts on the model's predictive capabilities. However, a generally prepared test dataset may not adequately assess the robustness of the models against language artefacts. This limitation necessitates the creation of a challenging adversarial test set specifically designed to evaluate the models' resilience to language artefacts.

4.3.2 Adversarial Test Set

Adversarial evaluation is a fundamental method for testing the robustness of Natural Language Inference (NLI) models. This evaluation involves using specially crafted test instances designed to challenge the models' generalization capabilities. By exposing models to these challenging scenarios, it is possible to assess whether the models exploit spurious correlations or consider the relevant features for the task.

In the context of COLIEE statute law textual entailment task, adversarial test sets are constructed to evaluate the robustness of the fine-tuned models against various language artefacts. These test sets include instances with and without the presence of specific artefacts, such as word overlap, contradiction words, and a few annotation artefact tokens. For the creation of the adversarial test set only these mentioned artefacts were considered due to the limited domain-specific knowledge and also considering the scope of this research; the remaining artefact types were not considered during adversarial test set generation. Evaluating model performance on these adversarial instances helps identify vulnerabilities and areas where models may fail to generalize beyond their training data.

Adversarial Test Set Generation

The adversarial test set for COLIEE Task 4 is designed to challenge the models' reliance on specific language artefacts. The artefact types chosen for this test set include word overlap, contradiction words, and specific tokens associated with annotation artefacts. For annotation artefacts, the tokens "building" and "person" were selected due to their strong association with the entailment ("Y") label, and "rescind" and "property" were selected for their strong association with the non-entailment ("N") label. Further, the instances are categorized as either "challenging" or "supporting" based on their alignment with the assumptions underlying these artefacts.

- **Challenging Instances**: These instances are intentionally designed to contradict the typical patterns associated with specific language artefacts. For example, a challenging instance for the word overlap artefact would involve a hypothesis with high word overlap that is labeled as non-entailment. Similarly, a challenging instance for the contradiction word artefact would involve a hypothesis with negation words labeled as entailment. Such instances are selected from the 2018 training set and subsequently modified to ensure that models are tested on adversarial examples derived from known training data. These artefact instances are manually edited with minimal modifications to change the ground truth label, transforming them into challenging adversarial instances.
- **Supporting Instances**: These instances exhibit the typical patterns associated with specific language artefacts. For instance, a supporting instance for word overlap would involve a hypothesis with high word overlap labeled as entailment. Similarly, a supporting instance for the contradiction word artefact would involve a hypothesis with negation words labeled as non-entailment. These instances are chosen from the 2022 test set, ensuring that they are novel and have not been encountered during training. Some instances are directly taken from the 2022 test set without modification, while others are selected and altered to serve as adversarial examples for specific artefacts. These artefact instances are selected with their ground truth labels intact to act as supporting adversarial instances.

Challenging Instances

The table 4.3 summarizes the distribution of adversarial instances across different types of language artefacts.

The total number of adversarial instances is 44, with 22 challenging and 22 supporting instances. This balanced approach ensures a thorough evaluation of the robustness of the models against language artefacts.

During the evaluation, each fine-tuned model variant is tested on this adversarial set. The primary metric used for this evaluation is accuracy, measuring the proportion of correctly

Table 4.3: Distribution of Adversarial Instances

Artefact Type	**Challenging Instances**	**Supporting Instances**
Word Overlap	5	5
Contradiction Words	5	5
Annotation Artefact (Y label)		
Token "Building"	3	3
Token "Person"	3	3
Annotation Artefact (N label)		
Token "Rescind"	3	3
Token "Property"	3	3
Total Instances	22	22

predicted instances out of the total number of instances. This rigorous testing helps identify the extent to which language artefacts impact the models' predictions and their ability to generalize to adversarial scenarios. The insights gained from this evaluation can guide further refinement of the models and the development of strategies to mitigate the effects of language artefacts.

4.3.3 Hypothesis-only Baseline

The hypothesis-only baseline is a method used in evaluating the robustness of Natural Language Inference (NLI) models [34]. This baseline involves training and evaluating a model using only the hypothesis component of the dataset excluding the premise. This approach aims to assess whether the hypothesis alone contains enough signal to predict the entailment label, thereby identifying potential weak features or artefacts in the hypothesis that the model might exploit.

Importance Over Majority Baseline

The hypothesis-only baseline is considered a more stringent and informative evaluation metric compared to the majority baseline. The majority baseline simply predicts the most frequent class in the training data for all instances, providing a basic benchmark for model performance. However, the majority baseline does not account for any specific patterns or biases in the data. In contrast, the hypothesis-only baseline tests the model's reliance on the hypothesis alone, revealing whether superficial patterns or spurious correlations in the hypothesis are driving the model's predictions.

From a scientific perspective, using the majority baseline has significant limitations. The majority baseline is inherently simplistic and fails to capture any meaningful patterns in the data, as it does not utilize any of the input features to make predictions. As a result, it sets a very low bar for model performance, and achieving higher accuracy than the majority baseline does not necessarily indicate a model's robustness or true understanding of the task in NLI models.

In contrast, the hypothesis-only baseline provides a deeper insight into the model's behavior. If a model performs significantly better than the majority baseline using only the hypothesis, it indicates that there are identifiable patterns or artefacts in the hypothesis that the model can exploit to make predictions. This suggests that the model might be over-relying on these weak features rather than understanding the nuanced relationship between the premise and the hypothesis. Thus, the hypothesis-only baseline serves as a more rigorous test of the model's ability to generalize and its susceptibility to dataset artefacts.

In this research, each selected NLI model was fine-tuned on the COLIEE task 4 training dataset using six different input configurations with only the hypothesis. The goal was to determine the extent to which the weak features in the hypothesis influenced the model's performance.

After fine-tuning, the models were evaluated on the standard test set, considering only the hypothesis. The primary evaluation metric used was accuracy. If a model's performance using only the hypothesis exceeded the majority baseline, it would indicate that the hypothesis contains weak features or artefacts that contribute significantly to the prediction accuracy. Such a result would suggest that the model is not solely relying on capturing the premise-hypothesis relationship, but is influenced by artefacts within the hypothesis.

By evaluating these hypothesis-only configurations, the research aims to uncover the extent to which language artefacts within the hypothesis contribute to model predictions.

4.3.4 Qualitative Evaluation

In addition to the quantitative evaluation methods mentioned above, model interpretability techniques are used to qualitatively understand the features that the models focus on during prediction. Feature attribution techniques, such as Integrated Gradients introduced by Sundararajan et al. [47], are employed for this purpose. For a given input text, Integrated Gradients calculates the attribution score for each token, indicating its contribution to the predicted label. A high attribution score for a particular token suggests a significant influence, while a low negative attribution score suggests minimal influence on the predicted label. This approach highlights the patterns or weak features exploited by the model during prediction, allowing for an interpretation of the model's capabilities and robustness.

These evaluations help to identify the vulnerabilities of NLI models, thereby improving the robustness and reliability of these models in statute law textual entailment tasks.

4.4 Language Artefacts Mitigation

In this study, the focus is on mitigating two types of language artefacts: word overlap and contradiction words. These artefacts were chosen due to their significant impact on model performance and their prevalence in the dataset. Additionally, given the limited scope of this research and the constraints related to legal domain knowledge, other artefact types were not considered for mitigation. Data augmentation was selected as the primary method to address these artefacts within the training dataset, aiming to enhance the robustness and generalization capabilities of the models.

"Instances exhibiting high word overlap between the premise and hypothesis with a class label 'Y' (entailment) are indicative of word overlap artefacts" [56]. To counter this, instances with high word overlap but labeled 'N' (non-entailment) are required. Similarly, "instances with negation words in the hypothesis and a class label 'N' (non-entailment) exhibit contradiction word artefacts" [56]. To counter these, instances with negation words in the hypothesis but labeled 'Y' (entailment) are needed [56]. These instances, needed for balancing, are referred to as counter-artefact instances. Counter-artefact instances help balance the artefacts in the training dataset. These instances can be filtered and used to generate an augmented dataset to mitigate word overlap and contradiction word artefacts. The following steps were implemented to create synthetic counter-artefact instances and balance the artefact instances.

4.4.1 Data Augmentation Pipeline

Step 1: Identify and Filter Counter-Artefact Instances

The first step involves identifying and filtering existing counter-artefact instances in the training dataset for the word overlap and contradiction word artefacts . The required number of counter-artefact instances to balance the dataset is calculated as the difference between the number of artefact instances and counter-artefact instances in the current training dataset [56].

For the contradiction word artefact, the filtering criterion for a counter-artefact instance is that the hypothesis must contain negation words and the label must be "Y" (entailment). Similarly, for the word overlap counter-artefact instances, the criterion is that there must be a high overlap of words (more than 50%) between the premise and hypothesis, without the presence of negation words or subsequence heuristic patterns, and the label must be "N" (non-entailment). Based on these criteria, the counter-artefact instances were filtered and used for data augmentation.

Step 2: Generate Paraphrased Hypotheses

Next, the hypotheses from the filtered counter-artefact instances were paraphrased using GPT-4, a large language model [29]. This step aimed to generate synthetic counter-artefact instances, balancing the dataset by providing clear, specific prompts to GPT-4. Importantly, only the hypotheses in the counter-artefact instances were paraphrased to create the augmented dataset. GPT-4 was chosen for paraphrasing due to its ability to handle complex legal language and generate high-quality text [29]. Clear instructions were provided to the model, including the premise and the original hypothesis as input for paraphrasing. For mitigating contradiction word artefacts, the paraphrased hypotheses were required to contain at least one negation word. Similarly, for mitigating word overlap artefacts, the paraphrased hypotheses were required to maintain at least 60% word overlap with the premise. The prompts used for generating the paraphrased hypotheses are provided in the Appendix.

Step 3: Validation of Paraphrased Hypotheses

Once the paraphrased hypotheses were generated, they underwent a three-fold validation process to ensure their quality and relevance.

- **GPT-4 Based Similarity Validation**: The first step involves using GPT-4 to validate whether the original and paraphrased hypotheses have the same meaning. The output is either 'YES' or 'NO'.

- **Embedding-Based Semantic Similarity**: The second fold calculates the cosine similarity between the original and paraphrased hypotheses using a sentence transformer [41]. A threshold of 75% or above is used for validation to accept the paraphrased hypothesis.
- **Manual Validation**: The final fold involves manual validation to check whether the paraphrased hypothesis is similar to the original hypothesis and holds the same entailment relationship with the premise.

If a paraphrased hypothesis fails any of these validations, it is discarded, and steps 2 and 3 are repeated until a valid paraphrased hypothesis is generated [56]. The Figure 4.1 illustrates the data augmentation pipeline used for generating synthetic counter-artefact instances.

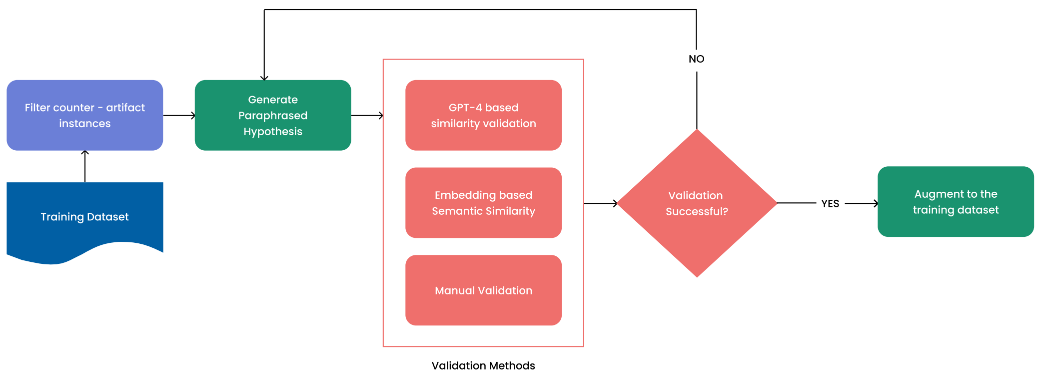

Figure 4.1: Data Augmentation Process to Mitigate Language Artefacts

Contradiction Word Artefacts

The table 4.4 shows the dataset imbalance and the balance achieved after augmentation for contradiction word artefacts.

Table 4.4: Balancing Contradiction Word Artefacts in Entailment instances

Year	Before Augmentation		After Augmentation	
	Entailment Instances	Non-Entailment Instances	Entailment Instances	Balanced Non-Entailment Instances
2018	138	150	150	150
2019	152	172	172	172
2020	171	194	194	194
2021	195	223	223	223
2022	218	247	247	247

An example instance for augmented contradiction word artefact.

Premise: Article 698 If a manager engages in benevolent intervention in another's business in order to allow a principal to escape imminent danger to the principal's person, reputation, or property, the manager is not liable to compensate for damage resulting from this unless the manager has acted in bad faith or with gross negligence.

Original Hypothesis: In cases where an individual rescues another person from getting hit by a car by pushing that person out of the way, causing the person's luxury kimono to get dirty, the rescuer does not have to compensate damages for the kimono

Paraphrased Hypothesis: If someone steps in to prevent someone else from being struck by a vehicle, resulting in that person's expensive kimono getting soiled, the intervener is not liable for the cost of the kimono.

Label: "Y" Entailment

As mentioned above this new paraphrased hypothesis will be combined with the Premise to form a new training instance which will act as a counter-artefact in balancing the contradiction word artefacts.

Word Overlap Artefact

The table 4.5 shows the dataset imbalance and the balance achieved after augmentation for word overlap artefacts.

Table 4.5: Balancing Word Overlap Artefacts in Non-Entailment instances

Year	Before Balancing		After Balancing	
	Entailment Instances	Non-Entailment Instances	Entailment Instances	Non-Entailment Instances
2018	109	70	109	109
2019	117	79	117	117
2020	129	87	129	129
2021	150	101	150	150
2022	157	118	157	157

An example instance for augmented word overlap artefact.

Premise: Article 506 (1) Set-offs are effected through the manifestation of one party's intention to the other. In such a case, no condition or time limit may be added to the manifestation of intention. (2) A manifestation of intention as referred to in the preceding paragraph is effective retroactive to the time when the obligations of both parties became eligible to be set-offs.

Original Hypothesis: A condition may be added to manifestation of set-offs.

Synthetic Hypothesis: It's permissible to append a condition to the expression of set-offs.

Label: "N" Non-entailment

As mentioned above this new synthetic hypothesis will be combined with the Premise to form a new training instance which will act as a counter-artefact in balancing the word overlap artefact.

4.4.2 Model Fine-Tuning with Augmented Dataset

Once the necessary number of counter-artefact instances are generated, they are appended to the original training dataset to balance the artefacts. This new augmented dataset is then used to fine-tune the selected NLI models [56]. The model inputs for fine-tuning remain consistent with the previous section of NLI model fine-tuning. These fine-tuned models on augmented data undergo the same robustness evaluation methods to understand the impact of artefact mitigation on the performance of the models [56].

By implementing this structured approach to mitigate language artefacts, the study aims to enhance the robustness and reliability of the NLI models in legal textual entailment tasks. The generated synthetic counter-artefact instances and the balanced dataset play a critical role in achieving more robust and generalizable model performance.

4.5 Summary

The methodology used in this study involved systematic detection and analysis of language artefacts within legal Natural Language Inference (NLI) dataset in COLIEE. Five BERT-based encoder-only models (BERT, LegalBERT, RoBERTa, ELECTRA, and DeBERTa) were strategically selected and fine-tuned on the dataset, incorporating identified artefacts as features to evaluate their impact on model robustness against the language artefacts. Robustness assessment included evaluation on standard and adversarial test sets, as well as a hypothesis-only baseline. Additionally, a data augmentation approach was implemented to mitigate specific artefacts (word overlap and contradiction words), aiming to enhance model generalizability and overall performance in statute law textual entailment task in COLIEE.

5. Implementation

This chapter discusses in detail the experimental setup and implementation strategies used to address the research questions concerning the impact and mitigation of language artefacts on selected BERT-based NLI models in the COLIEE statute law textual entailment task.

5.1 Experiments

The primary objective of the experiments is to evaluate the influence of language artefacts on model performance and to assess the effectiveness of mitigation strategies.

Design

The experiments were designed and carried out using the COLIEE Task 4 dataset from 2018 to 2022 on the selected BERT-based models which were fine-tuned using two setups: full-context (including both premise and hypothesis) and hypothesis-only. Each model setup was tested with six different input configurations including language artefact features appended to the hypothesis. Evaluations were performed on both standard and adversarial datasets before and after mitigating the language artefacts.

The experimental design aimed to analyse the impact of language artefacts on BERT-based models and to evaluate the effectiveness of mitigation strategies [56]. The design consisted of three main components:

1. **Full Context vs. Hypothesis-Only**: Models were trained and evaluated in two categories which are Full Context (including both premise and hypothesis) and Hypothesis-Only. Each category was further subdivided based on six different input configurations.
2. **Yearly Analysis**: Experiments were conducted separately for each year from 2018 to 2022 to assess model performance across multiple datasets.
3. **Mitigation Analysis**: Experiments were repeated with an augmented dataset after mitigating language artefacts to compare model performance before and after the mitigation process.

5.1.1 Setup

The chosen BERT-based NLI models (BERT, LegalBERT, RoBERTa, ELECTRA, and DeBERTa) were fine-tuned using six different input configurations for each year's dataset.

For each year, the following steps were performed:

1. **Dataset Preparation**: The training dataset for each year was prepared with and without the inclusion of language artefact features. The input configurations included:
    - No language artefact features
    - All language artefact features
    - Individual language artefact features (sentence length, word overlap, contradiction words, and subsequence heuristics)
2. **Model Fine-Tuning**: Each model was fine-tuned using the respective year's training dataset with and without data augmentation for both the full context (premise and hypothesis) and the hypothesis-only baseline. The hypothesis-only baseline involved fine-tuning the models using only the hypothesis part of the input to understand the models' reliance on superficial features within the hypothesis.
3. **Evaluation**: After fine-tuning, the models were evaluated for robustness using the standard test set and the adversarial test set, as described in chapter 4. This evaluation was conducted for both the full context models and the hypothesis-only baseline models.

Similarly, the same experimental setup was applied to assess the impact of language artefact mitigation. The mitigation process involved augmenting the original Task 4 dataset to balance the artefact instances, specifically targeting word overlap and contradiction word artefacts. The augmented dataset was then used to repeat the fine-tuning and evaluation process [56].

Model Variants

The experiment involves a comprehensive evaluation of BERT-based models using various input configurations and mitigation strategies. Each model is tested under the following conditions:

- **Input Configurations**: Six different input configurations are used for each model.
- **Evaluation Period**: Experiments are conducted over five years (2018 to 2022).
- **Setup Types**: Two setups are considered—Full Context (including both premise and hypothesis) and Hypothesis-Only.

For each model, this results in a total of 60 model variants:

- 6 configurations $\times$ 5 years $\times$ 2 setups $= 60$ variants

These experiments are conducted both before and after applying mitigation strategies for language artefacts, resulting in:

- 60 variants (before mitigation)
- 60 variants (after mitigation)

Thus, for each model, the total number of variants is:

- 60 variants (before mitigation) $+ 60$ variants (after mitigation) $= 120$ total variants

Considering five different BERT-based models, the overall number of model variants is:

- 120 total variants per model $\times 5$ models $= 600$ model variants

To ensure the reliability and robustness of the results, the experiments are repeated using three different seed values (42, 66, and 5). This results in:

- 600 model variants $\times 3$ seed values $= 1800$ total experiments

This detailed experimental setup, encompassing 1800 experiments across various years, input configurations, and models, provides a robust framework to evaluate the impact of language artefacts and the effectiveness of mitigation strategies. This systematic approach ensures a thorough understanding of the performance and limitations of BERT-based models in the legal NLI task of COLIEE.

5.2 Model Fine-tuning and Evaluation

This section details the methodological implementation of model fine-tuning and evaluation for the task of statute law textual entailment using the COLIEE dataset. The implementation leveraged a range of software tools, pre-trained language models, and custom data processing techniques to ensure robust and reproducible results.

Framework

The Hugging Face Transformers library[3], a widely adopted toolkit for deep learning models, formed the foundation for model fine-tuning and evaluation. This library provides pre-trained models and essential utilities for adapting them to specific tasks. The model fine-tuning and evaluation was done on Tesla T4 GPU with 16GB VRAM compute resource. Standard Python libraries such as NumPy [17], pandas [10], and scikit-learn [30] were utilized for numerical operations, data manipulation, and evaluation metrics, respectively. The Hugging Face datasets[4] library was employed for dataset loading and preprocessing, while the Weights & Biases (WandB)[5] platform was integrated to monitor fine-tuning progress and log performance metrics.

[3] https://github.com/huggingface/transformers
[4] https://huggingface.co/docs/datasets/en/index
[5] https://wandb.ai/site

Pre-trained Language Models

A diverse set of pre-trained language models, publicly available and pre-finetuned on large Natural Language Inference (NLI) tasks, was utilized. Due to the limited compute resource, the base version of all the models was selected. The architecture of all these base version of the BERT-base models contained 12 layers of transformer blocks containing 12 attentions heads with 768 hidden dimension. Also, all these models have a 512 token maximum sequence length. The open-sourced NLI models chosen for this experiment are as follows:

- BERT-Base-Cased-Finetuned-MNLI[6]
- Legal-BERT-Base-Uncased[7]
- RoBERTa-Base-MNLI[8]
- ELECTRA-Base-MNLI[9]
- DeBERTa-v3-base-mnli-fever-anli[10]

These models were selected from the Hugging Face Model Hub.

Dataset Preparation

The COLIEE dataset, spanning multiple years (2018-2022), was preprocessed to meet the input requirements of the chosen models. For each year, COLIEE provides separate training and test datasets. During data preparation, the provided training data was split into 90% train set and 10% validation set.

Tokenization was performed using model-specific tokenizers to ensure consistent input formatting. Input sequences were structured to include the premise, hypothesis, and, when applicable, extracted features designed to capture specific language artefacts. These features included sentence length, word overlap, the presence of contradiction words, and subsequence heuristics. Various feature configurations were accommodated, enabling ablation studies to assess the impact of each feature on model performance.

Model Fine-tuning

Fine-tuning was conducted independently for each model variant and input configuration. The fine-tuning process involved several key steps:

1. **Model Initialization**: The pre-trained models were initialized using 'AutoModelForSequenceClassification' class from the HuggingFace Transformers[11] library, which includes a classifier layer on top of the encoder. The encoder's sequence representation is extracted using the first [CLS] token, which is then fed into the classifier for sequence prediction.

[6] https://huggingface.co/gchhablani/bert-base-cased-finetuned-mnli
[7] https://huggingface.co/nlpaueb/legal-bert-base-uncased
[8] https://huggingface.co/textattack/roberta-base-MNLI
[9] https://huggingface.co/howey/electra-base-mnli
[10] https://huggingface.co/MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli [56]
[11] https://github.com/huggingface/transformers

2. **Data Preparation**: Custom PyTorch 'Dataset' classes were defined to streamline data loading and preprocessing. A data collator was employed to dynamically pad input sequences within each batch to a maximum length (512 tokens for full-context and 180 tokens for hypothesis-only), ensuring efficient batch processing during training.
3. **Training Configuration**: The training process was configured with the following parameters:
   - Learning rate: 3×10^{-5}
   - Batch size: 8
   - Number of epochs: 15
   - Gradient accumulation steps: 4 (to effectively increase the batch size without excessive memory usage)

   The AdamW optimizer, along with a linear learning rate scheduler, was utilized to update the model parameters. The optimizer adjusted the weights based on the computed cross-entropy loss during each epoch.
4. **Evaluation**: Model performance was continuously monitored on the validation set using accuracy as the evaluation metric. This approach ensured that the model's learning process was properly tracked and adjustments could be made if necessary.

This comprehensive fine-tuning process ensured that each model variant was thoroughly evaluated and optimized for the sequence classification tasks in the COLIEE statute law textual entailment task.

Evaluation

The evaluation of the fine-tuned models consisted of two types: quantitative and qualitative analyses.

Quantitative evaluation was performed by calculating the accuracy metric to assess model performance on two distinct test sets: the standard COLIEE test set and a custom-built language artefact-focused adversarial test set. For full-context models, both the standard and adversarial test sets were used. In contrast, for hypothesis-only models, only the hypothesis from the standard and adversarial test sets were used.

For each year, a standard test set was provided by COLIEE, while the adversarial test set remained consistent across all years (2018-2022). Within the adversarial test set, accuracy metrics were calculated for each model variant on both challenging instances (adversarial instances without artefacts) and supporting instances (adversarial instances with artefacts). This approach helps to identify whether the fine-tuned models exploit language artefacts, thereby providing a quantitative measure of model performance and revealing if the models' capabilities are overestimated or accurately assessed.

For qualitative analysis, model interpretability techniques such as Integrated Gradients were used to understand feature attribution during the prediction of a model variant. The implementation of Integrated Gradients on the fine-tuned transformer models was facilitated by the open-source library 'transformers-interpret'[12]. This library loads the

[12]https://github.com/cdpierse/transformers-interpret

fine-tuned model and, for a given input text, provides the attribution score for each token, indicating its influence on the predicted label.

Tokens with high positive attribution scores, often indicated by green color, were significant in the model's prediction, while tokens with low negative attribution scores, often indicated by red color, were less influential. This method allows for the interpretation of whether the models focused on language artefact features or meaningful patterns in the input text, thereby aiding in understanding model behavior.

This ensures a thorough assessment of model performance, robustness, and interpretability for COLIEE statute law textual entailment task.

5.3 Language Artefact Mitigation

To mitigate the impact of contradiction word and word overlap artefacts, a data augmentation pipeline was implemented. This pipeline leveraged OpenAI's GPT-4 language model [29] for generating paraphrased hypothesis instances, the Sentence Transformers library [41] for evaluating semantic similarity, and additional libraries for text processing and analysis.

Libraries and Tools

The implementation utilized the OpenAI library[13] to interface with the GPT-4 model for text generation and the sentence-transformers library[14] to employ the BAAI/bge-large-en-v1.5 [58] embedding model for semantic similarity calculations. Standard data manipulation and analysis were conducted using pandas [10], while natural language processing tasks like tokenization were performed using the nltk (Natural Language Toolkit) library [4].

Identification of Counter-Artefact Instances

The pipeline started by identifying instances in the training data exhibiting a high prevalence of contradiction words and word overlap biases. This was achieved through a multi-faceted analysis. First, negation words (e.g., "not," "no," "never") were extracted from the hypotheses with the nltk library's tokenizer using a predefined list of negation words. Instances with a high count of negations were flagged as potential candidates for contradiction word artefact augmentation. For word overlap bias, the total number of word overlaps were calculated between the premise and hypothesis token sets, representing the proportion of shared tokens. Instances with a high word overlap exceeding a predefined threshold were marked for word overlap artefact augmentation. Using these criteria, counter-artefact instances were identified and filtered.

Hypotheses Paraphrasing

A crucial component of the pipeline involved paraphrasing the hypotheses of the identified counter-artefact instances. OpenAI's GPT-4 [29] language model was used as the paraphrasing engine, guided by meticulously crafted prompts. The specific prompts used are provided in the Appendix.

[13]https://github.com/openai/openai-python

[14]https://github.com/UKPLab/sentence-transformers

For contradiction word artefacts, the prompt instructed the model to preserve negation words in the hypothesis while minimizing lexical overlap with the premise. This approach aimed to generate different versions of the hypothesis that maintained the original entailment relationship.

For word overlap artefacts, the prompt directed the model to either maintain or increase the word overlap between the premise and hypothesis while preserving the ground truth of the counter-artefact instance. The goal was to produce paraphrases that reinforced the existing semantic connections between the two sentences, thereby mitigating the potential for spurious correlations based solely on lexical overlap.

Validation and Quality Control

The quality of the generated paraphrased hypotheses was rigorously assessed. Semantic similarity between the original and paraphrased hypotheses was computed using the Sentence Transformers library's pre-trained BAAI/bge-large-en-v1.5 model, which embeds sentences in a high-dimensional space where semantic similarity can be quantified using cosine similarity. The embedding dimension of this model is 1024 and the context length is 512. Hypotheses with a cosine similarity score below a certain threshold of 75% were deemed to have deviated significantly in meaning and were discarded.

Further, a validation step was performed using GPT-4 to determine whether the paraphrased hypothesis preserved the entailment label of the original instance. The model was tasked with answering a simple yes/no question regarding the equivalence of the two hypotheses in terms of their entailment relationship with the premise. This ensured that the paraphrasing process did not inadvertently alter the ground truth label.

Iterative Refinement and Integration

In instances where the generated paraphrased hypotheses failed to meet the above described quality criteria, an iterative refinement process was initiated. The language model was provided with modified prompts incorporating feedback from the validation step, aiming to generate paraphrases that were both semantically similar and label-preserving. The validated, high-quality paraphrased hypotheses were subsequently integrated into the original COLIEE dataset, augmenting it with instances specifically designed to counteract the word overlap and contradiction word artefacts.

5.4 Summary

This chapter provides implementation details to study the impact of language artefacts on BERT-based NLI models in the COLIEE statute law textual entailment task [56]. It presents an experimental setup, evaluating various model variants over multiple years with diverse input configurations, both with and without language artefact mitigation. Model fine-tuning and evaluation implementation are detailed, encompassing pre-processing, training, and analysis of results using both quantitative and qualitative methods. Finally, a data augmentation pipeline is outlined for mitigating contradiction word and word overlap artefacts, using large language models for paraphrasing, semantic similarity assessment, and iterative refinement to enhance model robustness.

6. Results

This chapter presents the experimental results and discusses the findings related to the three research questions posed in this thesis. Each section corresponds to a research question, providing a detailed analysis of the experimental results and discussing their implications.

6.1 Detection of Language Artefacts

The first research question investigates the presence of language artefacts within the COLIEE Task 4 statute law entailment datasets. This section presents the results of the detection methods applied to identify these artefacts, followed by a discussion of the findings.

6.1.1 Detection Results

The detection of language artefacts involved analysing the COLIEE datasets for specific biases, such as annotation artefacts, sentence length artefacts, word overlap artefacts, contradiction word artefacts, and subsequence heuristics artefacts. The results for each type of artefact are summarized below.

Annotation Artefacts

Analysis of the token associations in hypotheses with entailment and non-entailment labels over the years 2018 to 2022 reveals interesting trends, potentially indicative of annotation artefacts within the datasets. The binomial significance test was used to identify tokens with significant bias towards either the entailment or non-entailment labels. The results are shown in Table 6.1. There are more tokens biased towards the entailment than the non-entailment label. Definite articles ("the", "a") and prepositions ("in", "for", "with") consistently emerge as strong indicators of entailment, underscoring the role of clear grammatical structures in hypotheses. Additionally, the frequent association of general nouns ("person", "cases") and pronouns with entailment suggests that statements involving specific entities or scenarios are more likely to exhibit entailment relationship. There is an increasing prevalence of "or" in recent years that may reflect a shift in the types of entailment tasks, potentially encompassing more scenarios involving choices or alternative possibilities. The appearance of the token "b", likely a dataset-specific placeholder term that often denotes relationships and situations among persons, which warrants further investigation into its contextual role in influencing entailment judgments. Similarly, the presence of words indicating negation or limitation, such as "not," "cannot," and "unless," often signals a non-entailment relationship between statements, underscoring how restrictions and conditions can impede straightforward inference. Similarly, there is a prevalence of legal and transactional terms like "sale," "property," and "pledge" more in cases of non-entailment. Furthermore, there is also strong association of dynamic verbs such as "was," "return," and "effect" with non-entailment labels.

Table 6.1: Number of tokens biased towards entailment and non-entailment labels (2018-2022)

Label	**2018**	**2019**	**2020**	**2021**	**2022**
Entailment	**70**	**63**	**74**	**92**	**93**
Non-entailment	36	44	47	48	52

The presence of annotation artefacts, evidenced by the significant bias of certain tokens toward specific class labels, raises concerns about the potential for models to exploit these superficial cues for prediction rather than engaging in genuine semantic understanding. The list of the top ten tokens demonstrating bias towards each specific label for every year is detailed in the Appendix.

Sentence Length Artefacts

Sentence length artefacts were identified by comparing the lengths of hypothesis sentences in the entailment and non-entailment classes. Instances where the hypothesis length was longer than the average were particularly analyzed. The entailment label has more instances where the hypothesis is longer than the average hypothesis length. This indicates a potential sentence length artefact. The results indicate a notable difference in sentence lengths between the two classes, as shown in Table 6.2.

Table 6.2: Number of instances longer than average sentence length (2018-2022)

Label	**2018**	**2019**	**2020**	**2021**	**2022**
Entailment	**145**	**161**	**177**	**199**	**219**
Non-entailment	126	144	152	171	190

The observed difference in sentence lengths between entailment and non-entailment instances indicates that models might inadvertently learn to associate longer hypotheses with entailment.

Word Overlap Artefacts

Word overlap artefacts were detected by calculating the percentage of word overlap between the premise and the hypothesis. Instances with more than 50% overlap were considered significant. The analysis showed a higher frequency of word overlaps in entailment instances compared to non-entailment instances, indicating the presence of word overlap artefacts [56]. The results are summarized in Table 6.3.

Table 6.3: Number of instances with word overlap above 50% (2018-2022)

Label	2018	2019	2020	2021	2022
Entailment	**109**	**117**	**129**	**150**	**157**
Non-entailment	70	79	87	101	118

Contradiction Word Artefacts

Contradiction word artefacts were identified by examining the presence of negation words in the hypothesis sentences. Instances with negation words were compared across the entailment and non-entailment labels [56]. The results, shown in Table 6.4, indicate a higher occurrence of negation words in non-entailment instances.

Table 6.4: Number of instances with negation words (2018-2022)

Label	2018	2019	2020	2021	2022
Entailment	138	152	171	195	218
Non-entailment	**150**	**172**	**194**	**223**	**247**

Subsequence Heuristic Artefacts

Subsequence heuristic artefacts were detected by identifying instances where the hypothesis appeared as a subsequence within the premise [56]. This type of artefact was found to be predominantly present in entailment instances, as shown in Table 6.5.

Table 6.5: Number of instances with subsequence heuristics (2018-2022)

Label	2018	2019	2020	2021	2022
Entailment	**2**	**3**	**3**	**4**	**10**
Non-Entailment	0	0	0	0	0

6.1.2 Discussion

The detection results provide strong evidence of language artefacts within the COLIEE Task 4 statute law entailment datasets. The presence of annotation artefacts indicates that certain tokens are significantly biased towards specific class labels, which could mislead the models. The sentence length artefacts suggest that hypothesis length is not uniformly distributed across labels, potentially causing models to rely on length as a cue for prediction. The word overlap artefacts reveal that high overlap between premise and hypothesis is more common in entailment instances, which could lead models to erroneously predict entailment based on overlap alone. The contradiction word artefacts highlight a bias where negation words in the hypothesis correlate with non-entailment labels, suggesting that models might use the presence of negation as a heuristic. Lastly, the significant subsequence heuristic artefacts demonstrate that models might predict entailment simply because the hypothesis is a subsequence of the premise. One of the common patterns observed in the artefact detection results is that, except for the contradiction word artefact, the remaining artefacts are highly correlated with the entailment label.

Overall, these findings confirm that language artefacts are present in the COLIEE Task 4 dataset and have the potential to impact the performance and reliability of NLI models. Addressing these artefacts is key for developing more robust and generalizable models.

6.2 Robustness Evaluation of Models

This section addresses the second research question: To what extent do language artefacts influence the performance of BERT-based models in statute law entailment tasks? The analysis includes results from various evaluations, such as standard test sets, adversarial test sets, and hypothesis-only baseline evaluations.The accuracy reported for each test is the average of three runs, each conducted with a different seed value (42, 66, and 5), along with the corresponding standard error.

The performance of selected BERT-based models (BERT base, LegalBERT, RoBERTa base, ELECTRA base, and DeBERTa base) was assessed using different strategies to understand the impact of language artefacts. This section presents the results of three major performance comparisons:

1. The comparison between standard and adversarial test sets on Full Context models.
2. The performance comparison within adversarial groups, specifically between challenging and supporting instances on Full Context models.
3. The baseline comparison between hypothesis-only and majority classifier evaluations.

6.2.1 Standard vs Adversarial Test Set

This evaluation strategy compares the performance of the full context models on the standard and the adversarial test set. The full context models are those trained with both the premise and the hypothesis. The standard test set represents the official test set from COLIEE for taks 4, while the adversarial test set includes instances specifically designed to challenge the model by introducing variations that test the model's robustness against language artefacts. By comparing the performance on these two sets, the goal is to assess how well the models can generalize beyond the patterns present in the training data and how vulnerable they are to manipulations that exploit language artefacts. The scatter plots in Figures 6.31 to 6.35 illustrate the performance comparison of various BERT-based model variants over the years 2018 to 2022. For the sake of brevity, the artefacts are denoted in abbreviated forms: 'N' represents no artefact features, 'A' denotes all artefact features, 'SL' corresponds to sentence length artefacts, 'CW' refers to contradiction word artefacts, 'WO' stands for word overlap, and 'SH' indicates subsequence heuristics.

Tables 6.6 to 6.10 present a comparative analysis of model performance on standard and adversarial test sets, over the years 2018 to 2022. BERT-Base model variants consistently exhibit lower performance on adversarial test sets compared to standard test sets across all years, as illustrated in Figures 6.1 to 6.6. LegalBERT generally underperforms, achieving approximately 50% accuracy on standard test sets, regardless of the presence of artefact features (Table 6.7). However, in 2022, LegalBERT shows improved performance on the standard test set, exceeding 60% accuracy when all features, sentence length, and word overlap are considered (Figures 6.8 to 6.10). Despite this improvement, LegalBERT's performance on adversarial test sets remains poor across all settings.

Table 6.8 presents the results for the RoBERTa Base model. RoBERTa demonstrates consistent performance around 60% on standard test sets in most settings, except in 2018 when all features are included (Figure 6.14). Its performance on adversarial test sets, while higher than BERT Base and LegalBERT at around 50% as shown in Figures 6.13 to 6.18, still reveals a significant gap compared to the standard test set.

The ELECTRA Base model exhibits a notable improvement in adversarial robustness compared to previous models, particularly from 2020 to 2022 as presented in Table 6.9. It maintains consistent performance on standard test sets, ranging from 60% to 65% across all years and feature settings. However, a performance boost is observed on adversarial test sets from 2020 to 2022, specifically when considering all features, contradiction words, and subsequence heuristic features (Figures 6.20, 6.23, and 6.24). In these later years, ELECTRA effectively reduces the performance gap between standard and adversarial test sets when utilizing these specific artefact features.

Finally, the DeBERTa Base model outperforms all previous models on both standard and adversarial test sets (Table 6.10). Figures 6.25 to 6.30 clearly demonstrate that its performance on adversarial test sets is comparable to that on standard test sets. In many cases, when incorporating sentence length, word overlap, contradiction words, and subsequence heuristic features, the model even surpasses its performance on the standard test set. Furthermore, DeBERTa is the only model to achieve accuracy above 70% on adversarial test sets (Figures 6.25 and 6.30).

Thus, most models exhibit a significant performance gap between standard and adversarial test sets, the DeBERTa Base model effectively closes this gap, demonstrating enhanced robustness against adversarial attacks. This trend is further corroborated by Figures 6.31 to 6.35, which illustrate the varying robustness of the models over time. These figures highlight the consistent underperformance of BERT-Base and LegalBERT, the moderate performance of RoBERTa Base and ELECTRA Base, and the superior performance of DeBERTa Base across all years.

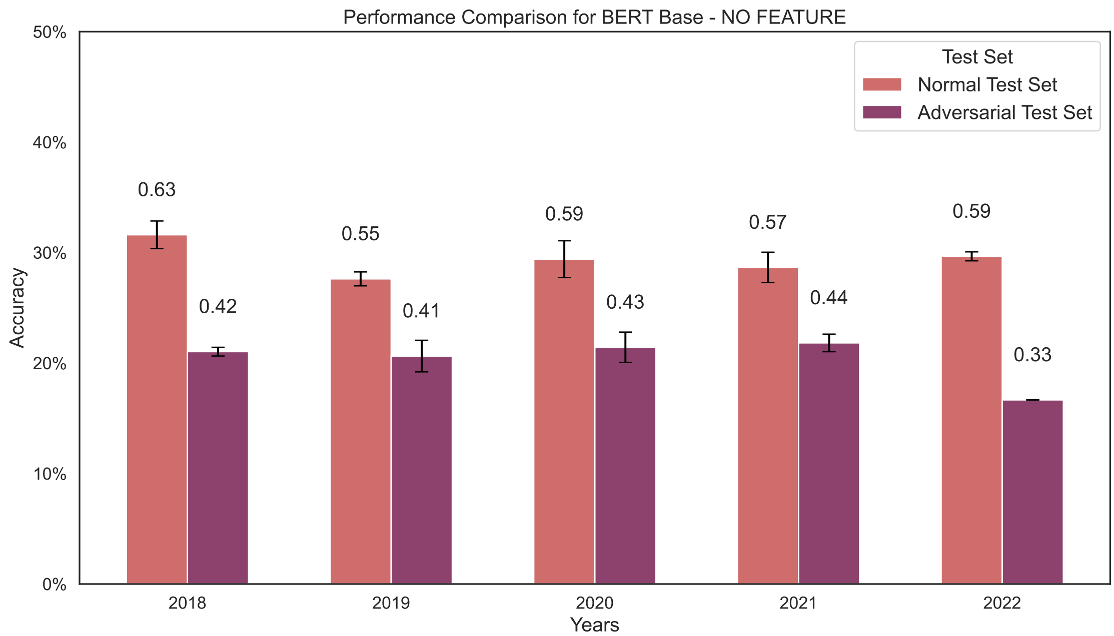

Figure 6.1: BERT Base performance on standard vs adversarial test sets with no artefacts.

Table 6.6: Performance comparison of BERT Base on Standard and Adversarial test sets

Year	Feature	Standard	Adversarial
2018	None	0.6322 $\pm$ 0.03	0.4207 $\pm$ 0.01
	All	0.6034 $\pm$ 0.03	0.3968 $\pm$ 0.04
	Sentence Length	0.6379 $\pm$ 0.03	0.3809 $\pm$ 0.01
	Word Overlap	0.6149 $\pm$ 0.03	0.3968 $\pm$ 0.03
	Contradiction Word	0.6322 $\pm$ 0.03	0.4207 $\pm$ 0.01
	Subsequence Heuristic	0.6207 $\pm$ 0.03	0.3968 $\pm$ 0.02
2019	None	0.5524 $\pm$ 0.01	0.4127 $\pm$ 0.03
	All	0.5333 $\pm$ 0.03	0.3730 $\pm$ 0.04
	Sentence Length	0.5714 $\pm$ 0.01	0.4286 $\pm$ 0.01
	Word Overlap	0.5762 $\pm$ 0.03	0.4524 $\pm$ 0.01
	Contradiction Word	0.5476 $\pm$ 0.02	0.3889 $\pm$ 0.02
	Subsequence Heuristic	0.5667 $\pm$ 0.02	0.3730 $\pm$ 0.03
2020	None	0.5882 $\pm$ 0.03	0.4286 $\pm$ 0.03
	All	0.5705 $\pm$ 0.03	0.4445 $\pm$ 0.01
	Sentence Length	0.5555 $\pm$ 0.02	0.4286 $\pm$ 0.04
	Word Overlap	0.5706 $\pm$ 0.02	0.4286 $\pm$ 0.01
	Contradiction Word	0.5555 $\pm$ 0.03	0.4286 $\pm$ 0.03
	Subsequence Heuristic	0.5916 $\pm$ 0.03	0.4286 $\pm$ 0.02
2021	None	0.5732 $\pm$ 0.03	0.4365 $\pm$ 0.02
	All	0.5679 $\pm$ 0.02	0.4127 $\pm$ 0.04
	Sentence Length	0.5309 $\pm$ 0.03	0.3968 $\pm$ 0.03
	Word Overlap	0.5720 $\pm$ 0.02	0.3968 $\pm$ 0.03
	Contradiction Word	0.5843 $\pm$ 0.03	0.4365 $\pm$ 0.02
	Subsequence Heuristic	0.5597 $\pm$ 0.03	0.4127 $\pm$ 0.04
2022	None	0.5932 $\pm$ 0.01	0.3333 $\pm$ 0.00
	All	0.5688 $\pm$ 0.03	0.4048 $\pm$ 0.05
	Sentence Length	0.6208 $\pm$ 0.02	0.4286 $\pm$ 0.00
	Word Overlap	0.5963 $\pm$ 0.03	0.3809 $\pm$ 0.01
	Contradiction Word	0.5932 $\pm$ 0.01	0.3333 $\pm$ 0.00
	Subsequence Heuristic	0.5963 $\pm$ 0.04	0.4365 $\pm$ 0.04

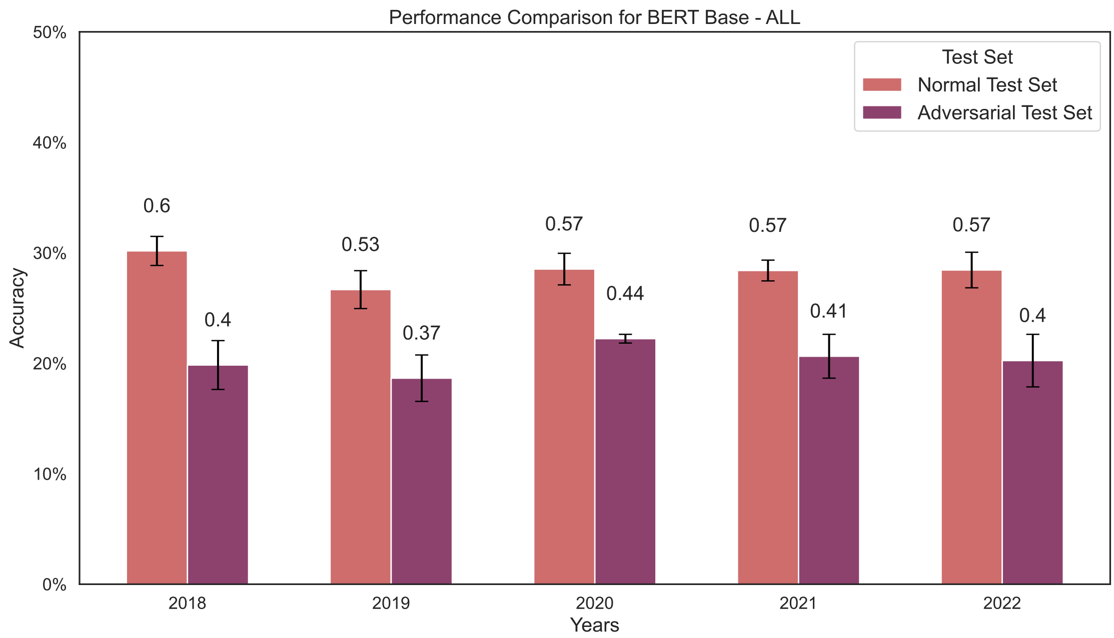

Figure 6.2: BERT Base performance on standard vs adversarial test sets with all artefacts.

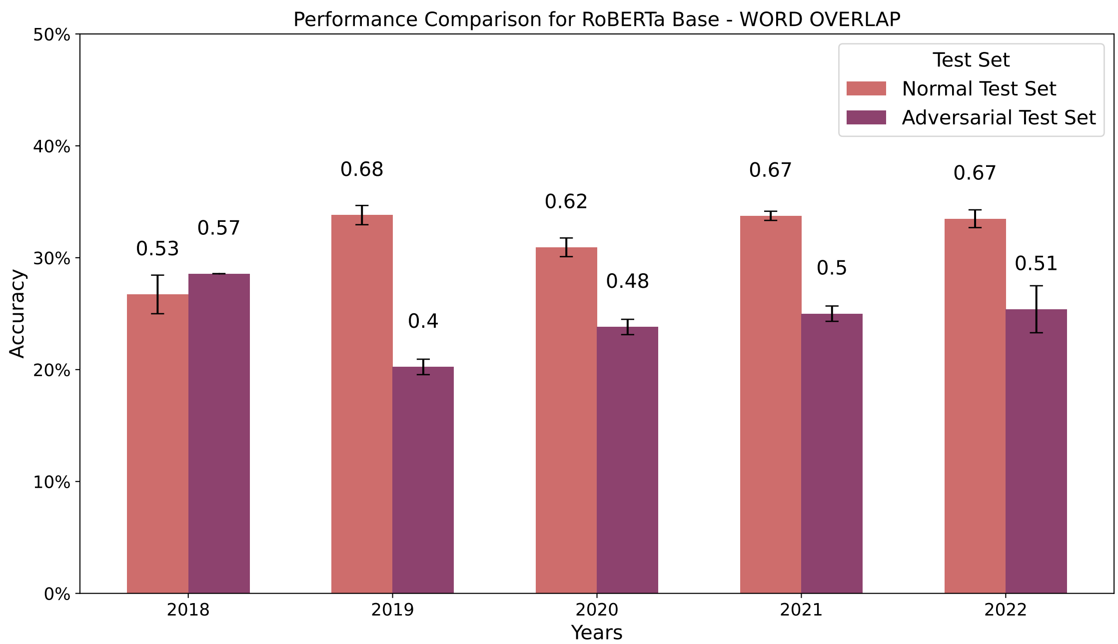

Figure 6.16: RoBERTa Base performance on standard vs adversarial test sets with word overlap artefact.

6.2.2 Adversarial Challenging vs Supporting Instances

This evaluation strategy examines the performance of the full context models on adversarial instances that either comply with or violate the assumptions of specific language artefacts. Here, adversarial instances are crafted to highlight the model's dependency on these artefacts. Challenging instances do not contain artefacts whereas the supporting instances

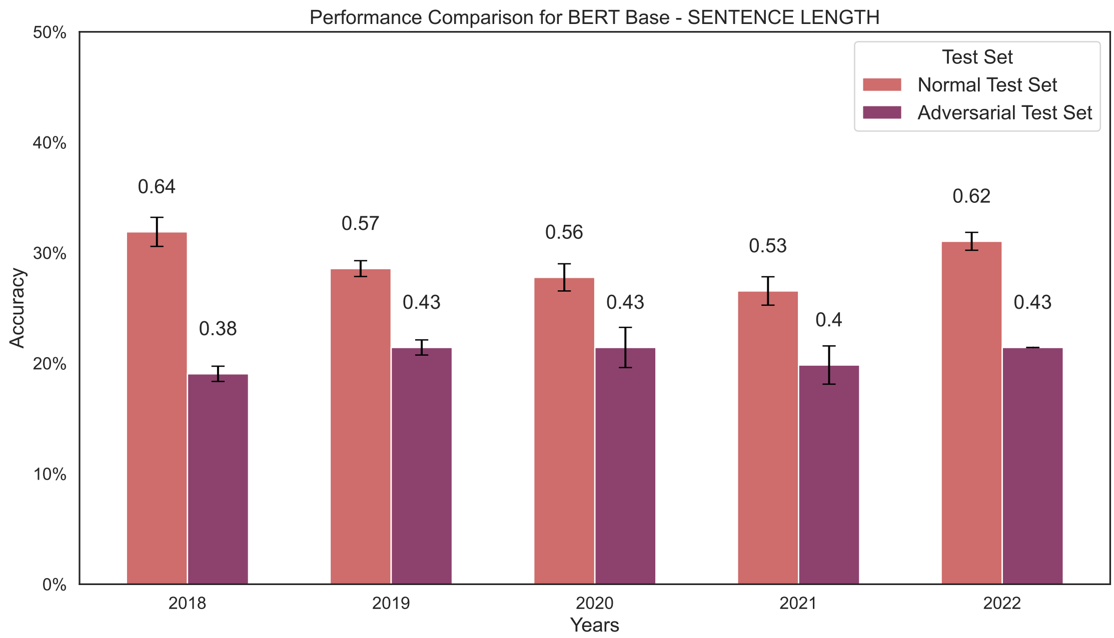

Figure 6.3: BERT Base performance on standard vs adversarial test sets with sentence length artefact.

contain artefacts. This comparison helps to identify the extent to which the models rely on artefact-related patterns to make predictions.

The goal is to analyse the performance gap between the adversarial challenging and supporting instances. Table 6.11 provides the performance result of BERT Base model on challenging and supporting instances. This clearly indicates a good performance on supporting instances and poor performance on challenging instances, as it can be seen on figures 6.36 to 6.41. This performance gap is consistent throughout the years. Similarly, Table 6.12 presents the LegalBERT adversarial results indicating the similar large performance gap on all artefact features over all the years as illustrated in Figure 6.42 to Figure 6.47. Comparatively RoBERTa Base model tries to shorten the performance gap as presented in Table 6.13. The illustrations of the model with Sentence length feature (Figure 6.50), contradiction word feature (Figure 6.52) and word overlap feature (Figure 6.51) narrows the performance gap. Unfortunately, ELECTRA Base model as presented in Table 6.14 does not shorten the performance gap. From the illustrations (Figure 6.54 to 6.59) it is evident that the model performs poorly on challenging instances over the years on all artefact features. Finally DeBERTa Base model provides a clear trend of narrowing the performance gap over the years especially from 2020 to 2022 as mentioned in Table 6.15. As illustrated in Figure 6.63, the word overlap feature provides the least performance gap between the challenging and supporting adversarial instances. These observations on the BERT-based models provide a clear indication that models rely on language artefacts during prediction, thus performing better on artefacting supporting instances and poorly on challenging instances.

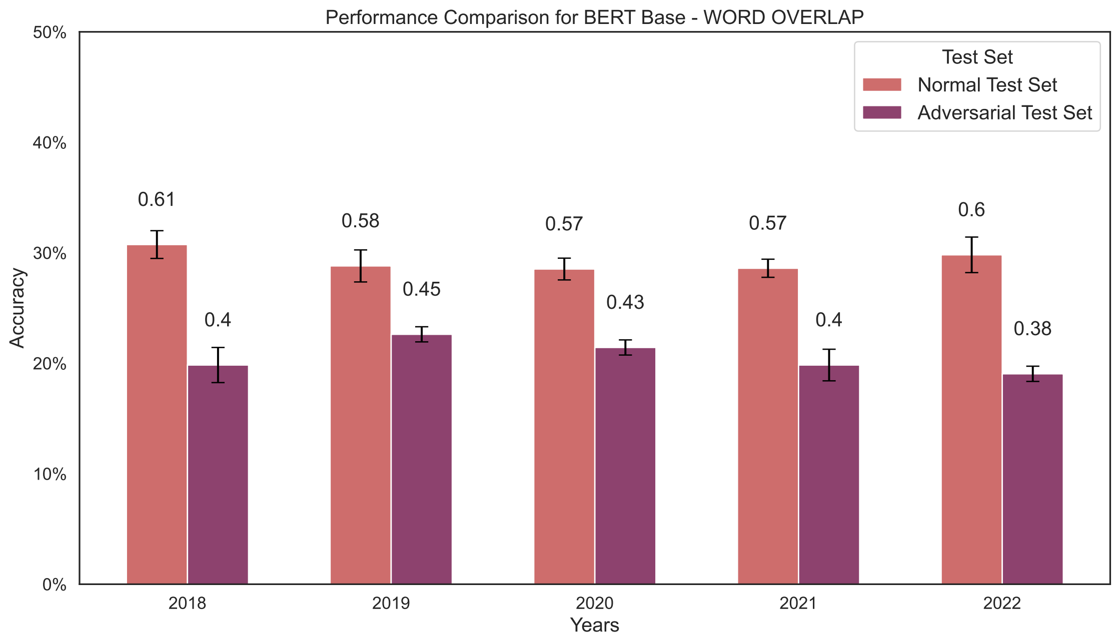

Figure 6.4: BERT Base performance on standard vs adversarial test sets with word overlap artefact.

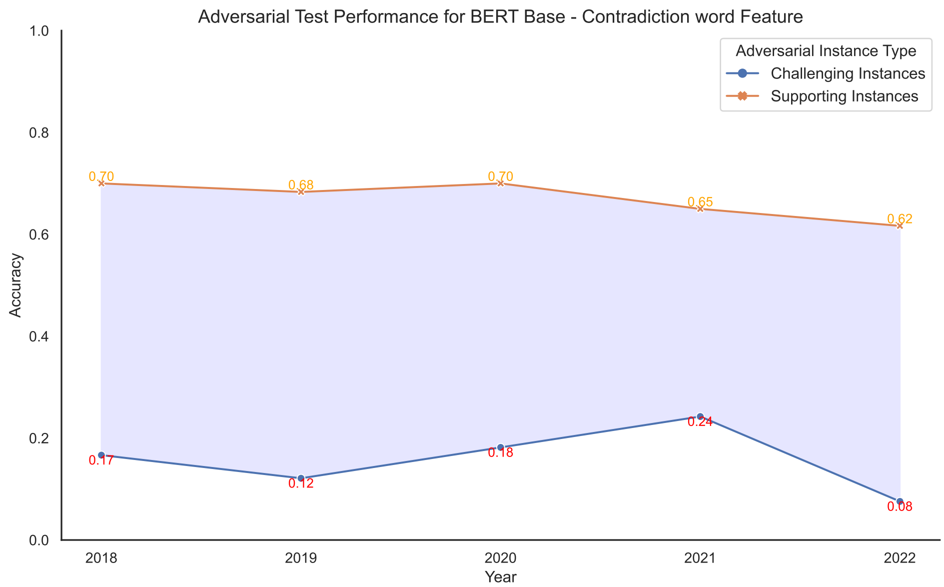

Figure 6.40: BERT Base performance on Challenging and Supporting adversarial instances with contradiction word artefact.

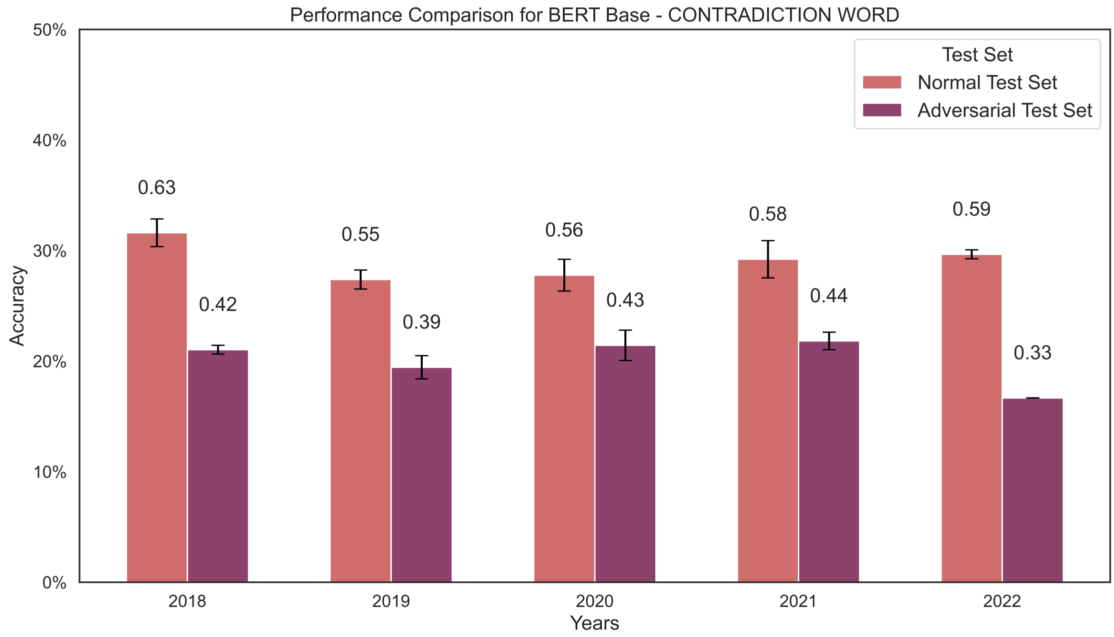

Figure 6.5: BERT Base performance on standard vs adversarial test sets with contradiction word artefact.

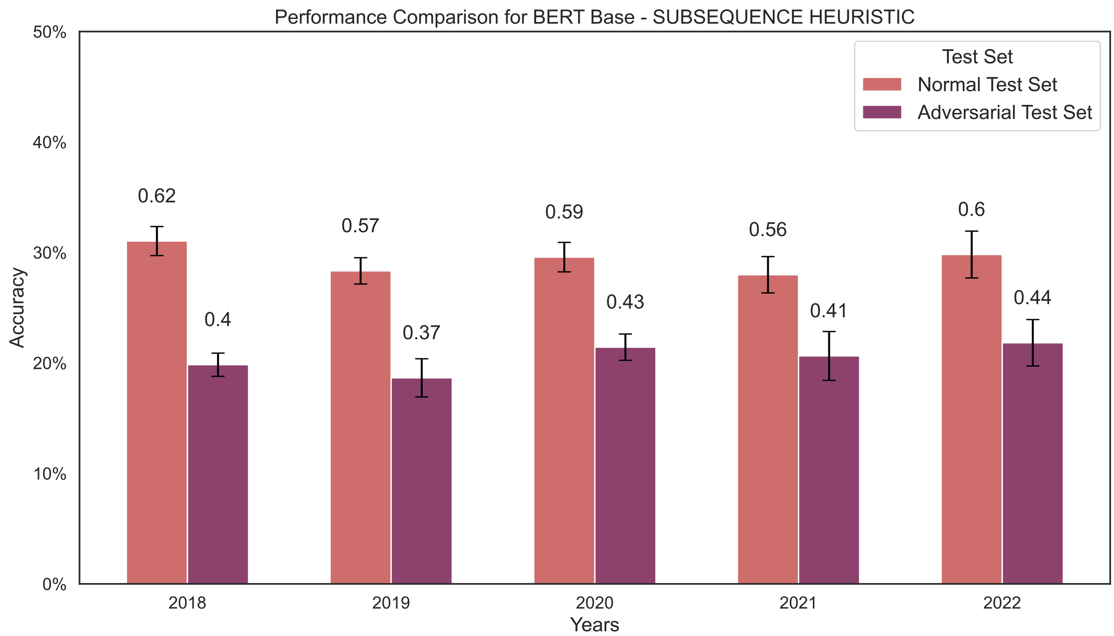

Figure 6.6: BERT Base performance on standard vs adversarial test sets with subsequence heuristic artefact.

Table 6.7: Performance comparison of LegalBERT on Standard and Adversarial test sets

Year	Feature	Standard	Adversarial
2018	None	0.5632 ± 0.02	0.4206 ± 0.04
	All	0.5287 ± 0.04	0.4206 ± 0.06
	Sentence Length	0.5689 ± 0.01	0.3571 ± 0.03
	Word Overlap	0.4885 ± 0.04	0.4841 ± 0.03
	Contradiction Word	0.5632 ± 0.02	0.4206 ± 0.04
	Subsequence Heuristic	0.5345 ± 0.00	0.4921 ± 0.01
2019	None	0.5190 ± 0.02	0.4365 ± 0.06
	All	0.5048 ± 0.00	0.5000 ± 0.08
	Sentence Length	0.5218 ± 0.03	0.3651 ± 0.04
	Word Overlap	0.4619 ± 0.04	0.4444 ± 0.03
	Contradiction Word	0.5190 ± 0.02	0.4206 ± 0.05
	Subsequence Heuristic	0.5191 ± 0.01	0.3730 ± 0.03
2020	None	0.5015 ± 0.01	0.4048 ± 0.06
	All	0.4895 ± 0.01	0.4445 ± 0.03
	Sentence Length	0.4865 ± 0.00	0.3889 ± 0.03
	Word Overlap	0.5195 ± 0.03	0.3730 ± 0.02
	Contradiction Word	0.4955 ± 0.00	0.3968 ± 0.06
	Subsequence Heuristic	0.4835 ± 0.02	0.4127 ± 0.03
2021	None	0.5720 ± 0.00	0.3571 ± 0.03
	All	0.5432 ± 0.04	0.4127 ± 0.02
	Sentence Length	0.5144 ± 0.03	0.4841 ± 0.06
	Word Overlap	0.5515 ± 0.01	0.4365 ± 0.03
	Contradiction Word	0.5720 ± 0.00	0.3571 ± 0.03
	Subsequence Heuristic	0.4815 ± 0.04	0.4206 ± 0.07
2022	None	0.5596 ± 0.06	0.4445 ± 0.03
	All	0.6300 ± 0.02	0.4047 ± 0.02
	Sentence Length	0.6208 ± 0.01	0.3968 ± 0.04
	Word Overlap	0.6238 ± 0.02	0.4127 ± 0.03
	Contradiction Word	0.5596 ± 0.06	0.4445 ± 0.03
	Subsequence Heuristic	0.5077 ± 0.04	0.4365 ± 0.06

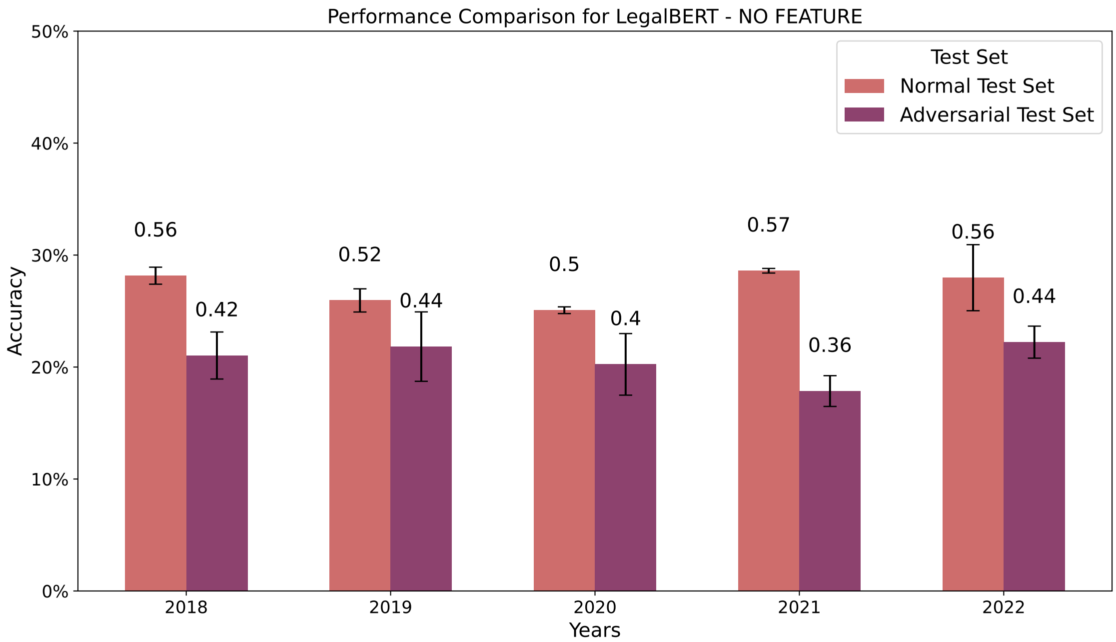

Figure 6.7: LegalBERT performance on standard vs adversarial test sets with no artefacts.

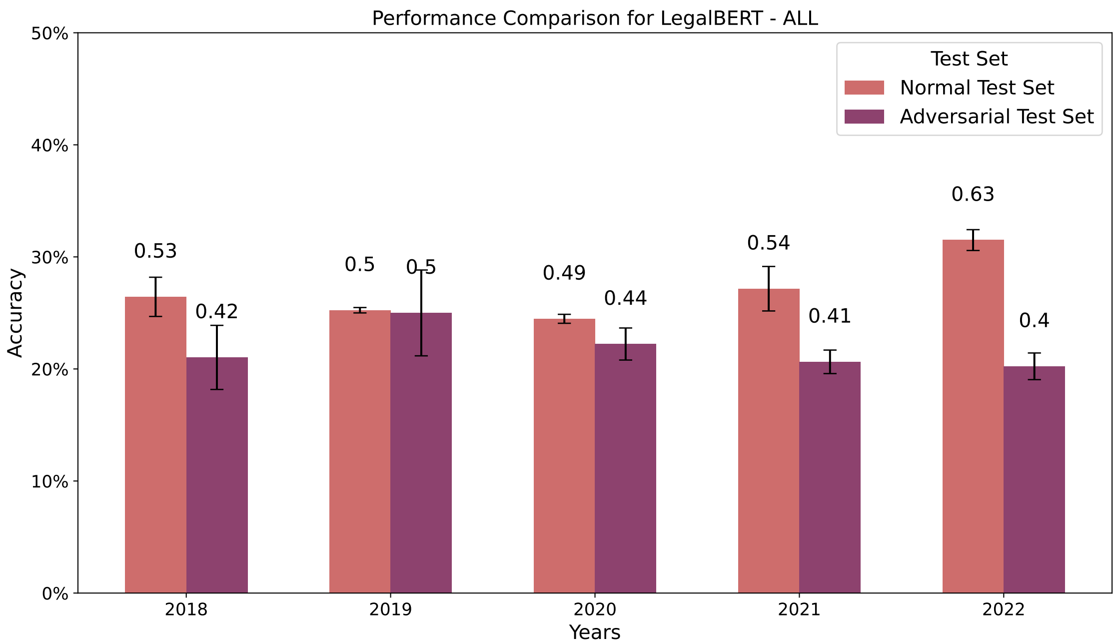

Figure 6.8: LegalBERT performance on standard vs adversarial test sets with all artefacts.

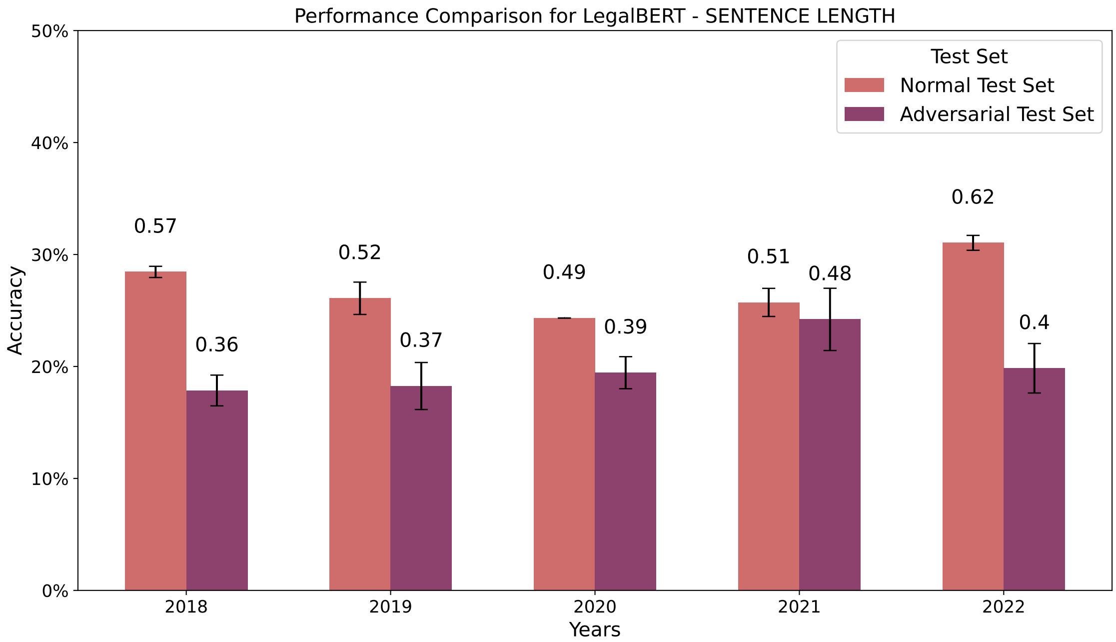

Figure 6.9: LegalBERT performance on standard vs adversarial test sets with sentence length artefact.

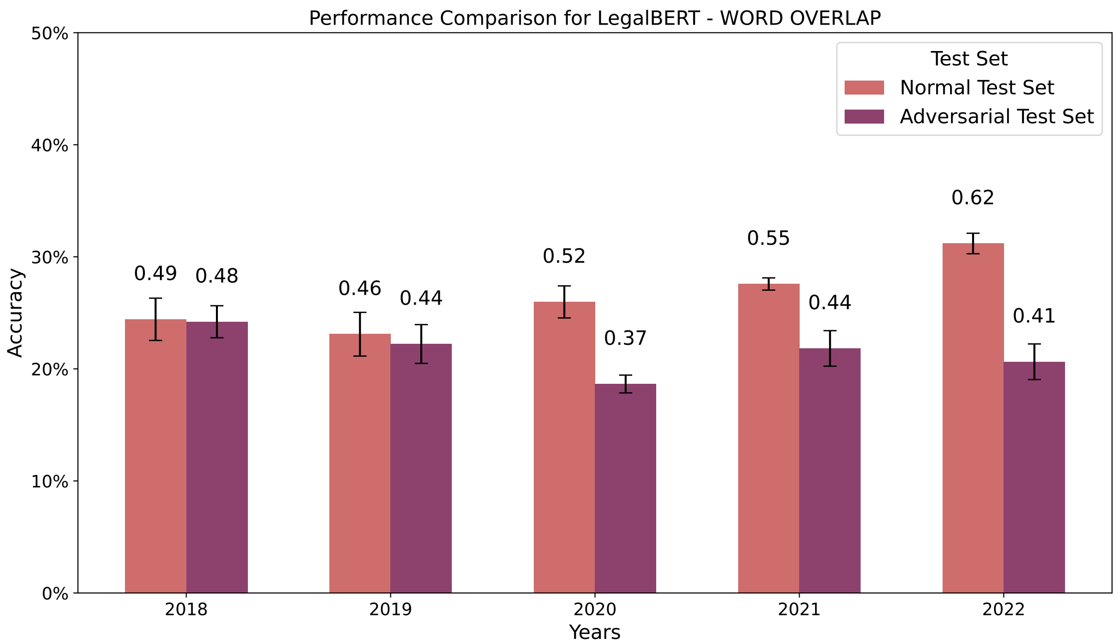

Figure 6.10: LegalBERT performance on standard vs adversarial test sets with word overlap artefact.

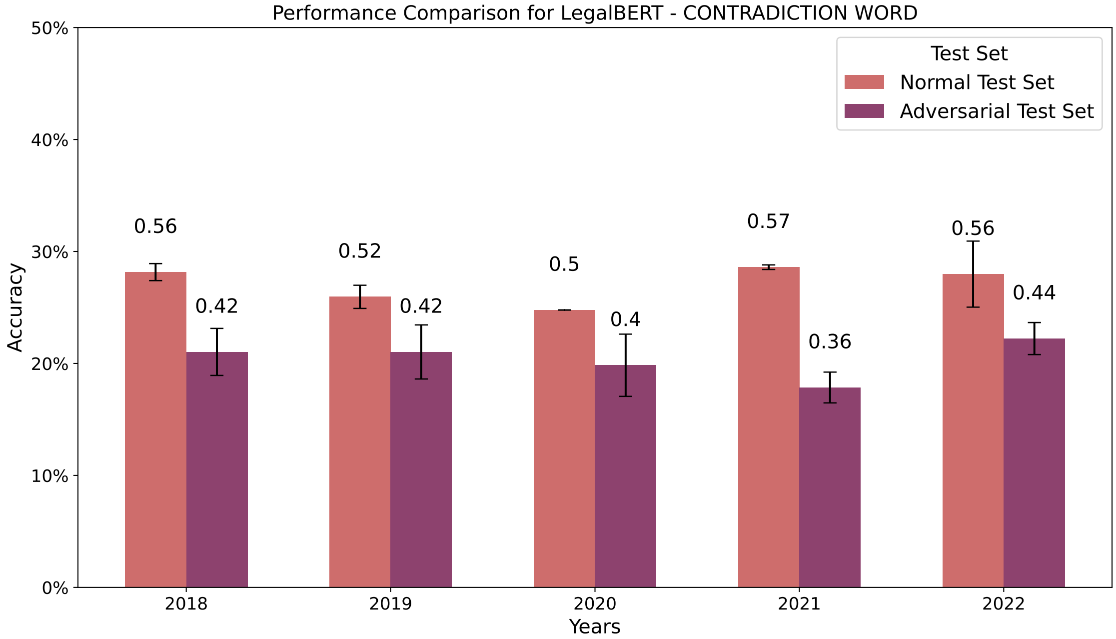

Figure 6.11: LegalBERT performance on standard vs adversarial test sets with contradiction word artefact.

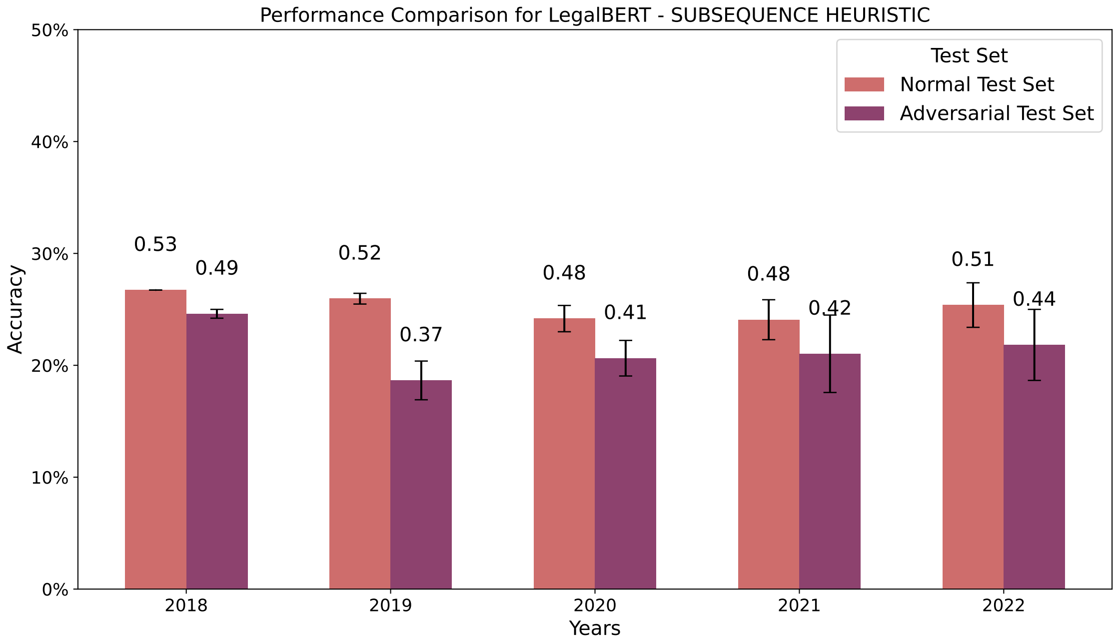

Figure 6.12: LegalBERT performance on standard vs adversarial test sets with subsequence heuristic artefact.

Table 6.8: Performance comparison of RoBERTa Base on Standard and Adversarial test sets

Year	Feature	Standard	Adversarial
2018	None	0.6092 ± 0.02	0.5397 ± 0.03
	All	0.5115 ± 0.03	0.5397 ± 0.02
	Sentence Length	0.5689 ± 0.02	0.5556 ± 0.06
	Word Overlap	0.5344 ± 0.03	0.5714 ± 0.00
	Contradiction Word	0.6092 ± 0.02	0.5397 ± 0.03
	Subsequence Heuristic	0.5919 ± 0.02	0.5317 ± 0.01
2019	None	0.6429 ± 0.02	0.5238 ± 0.03
	All	0.5857 ± 0.05	0.4603 ± 0.04
	Sentence Length	0.6191 ± 0.01	0.5079 ± 0.01
	Word Overlap	0.6762 ± 0.02	0.4048 ± 0.01
	Contradiction Word	0.6429 ± 0.02	0.5238 ± 0.03
	Subsequence Heuristic	0.6048 ± 0.02	0.4683 ± 0.02
2020	None	0.6336 ± 0.01	0.4921 ± 0.02
	All	0.6396 ± 0.02	0.5476 ± 0.01
	Sentence Length	0.6216 ± 0.01	0.5079 ± 0.02
	Word Overlap	0.6186 ± 0.02	0.4762 ± 0.01
	Contradiction Word	0.6336 ± 0.01	0.4921 ± 0.02
	Subsequence Heuristic	0.6156 ± 0.02	0.5317 ± 0.01
2021	None	0.6502 ± 0.02	0.4921 ± 0.04
	All	0.6872 ± 0.02	0.4920 ± 0.03
	Sentence Length	0.6831 ± 0.03	0.4841 ± 0.02
	Word Overlap	0.6749 ± 0.01	0.5000 ± 0.01
	Contradiction Word	0.6502 ± 0.02	0.4921 ± 0.04
	Subsequence Heuristic	0.6872 ± 0.02	0.5317 ± 0.01
2022	None	0.6697 ± 0.01	0.5000 ± 0.01
	All	0.6881 ± 0.01	0.5238 ± 0.02
	Sentence Length	0.6727 ± 0.01	0.5079 ± 0.02
	Word Overlap	0.6697 ± 0.02	0.5079 ± 0.04
	Contradiction Word	0.6697 ± 0.01	0.4921 ± 0.02
	Subsequence Heuristic	0.5933 ± 0.06	0.5159 ± 0.01

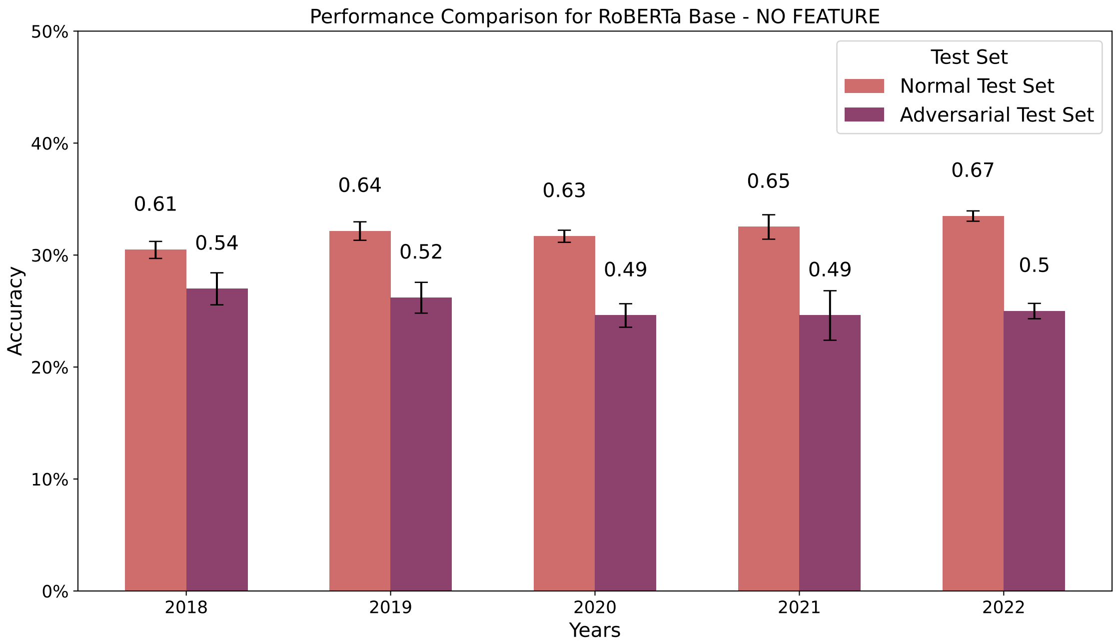

Figure 6.13: RoBERTa Base performance on standard vs adversarial test sets with no artefacts.

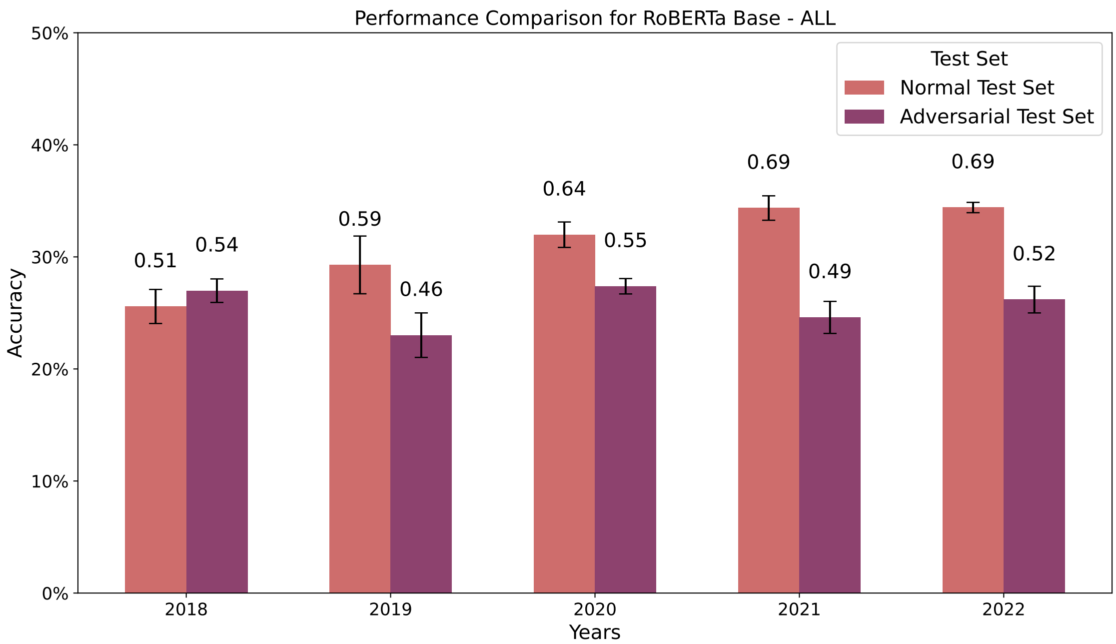

Figure 6.14: RoBERTa Base performance on standard vs adversarial test sets with all artefacts.

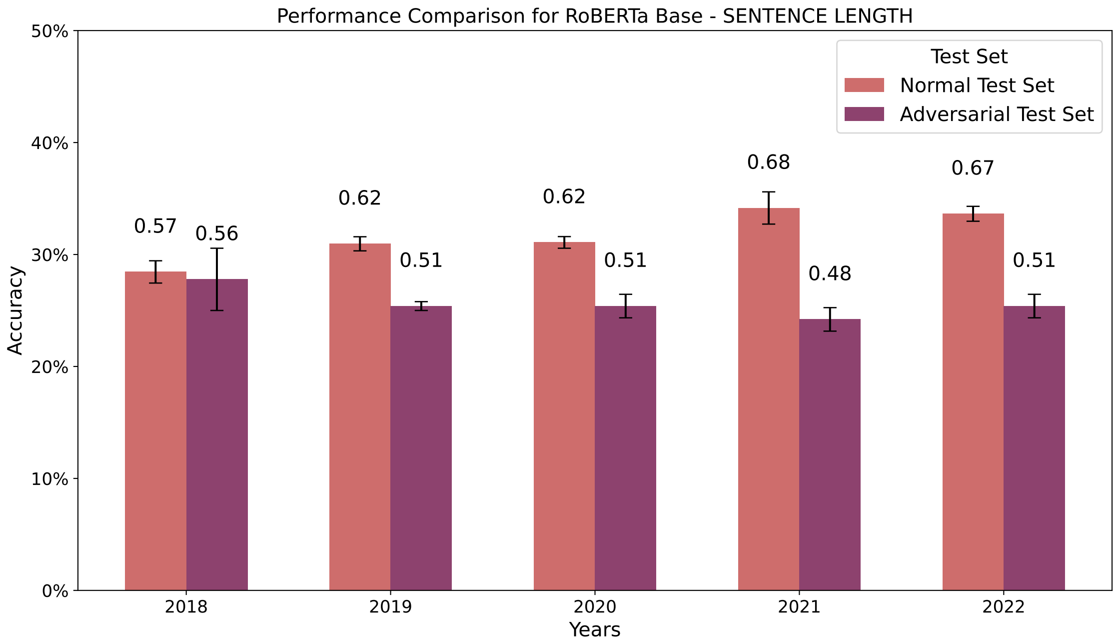

Figure 6.15: RoBERTa Base performance on standard vs adversarial test sets with sentence length artefact.

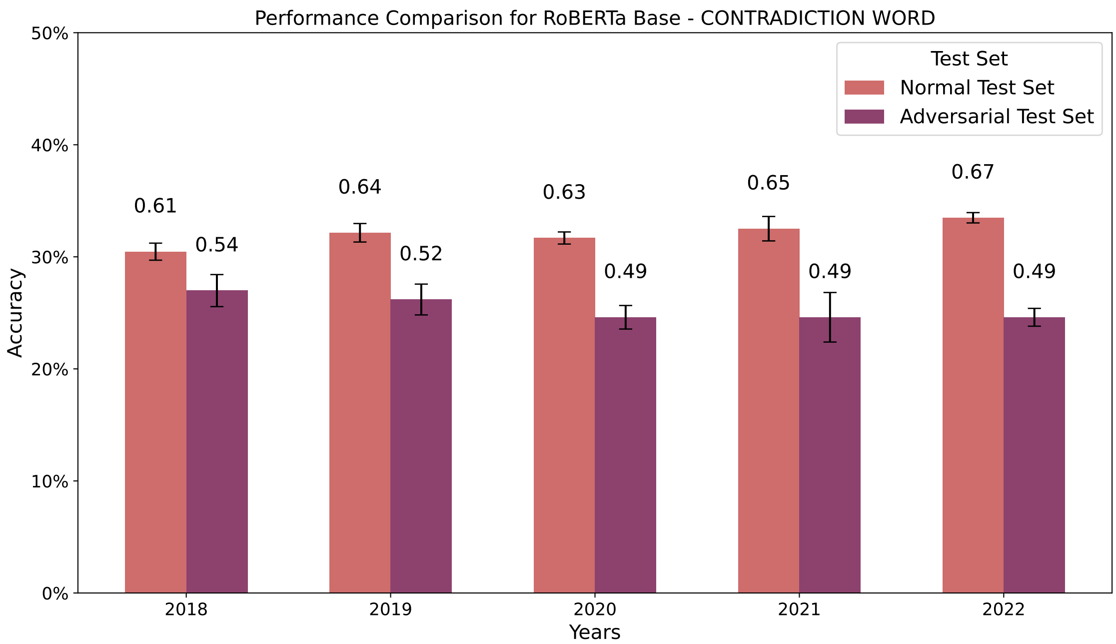

Figure 6.17: RoBERTa Base performance on standard vs adversarial test sets with contradiction word artefact.

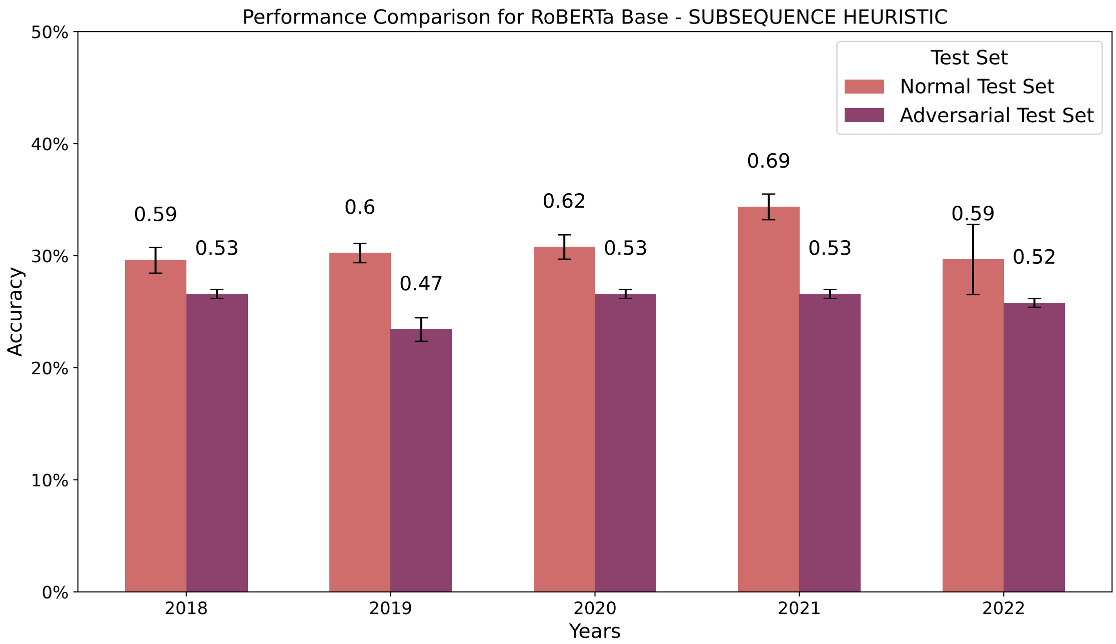

Figure 6.18: RoBERTa Base performance on standard vs adversarial test sets with subsequence heuristic artefact.

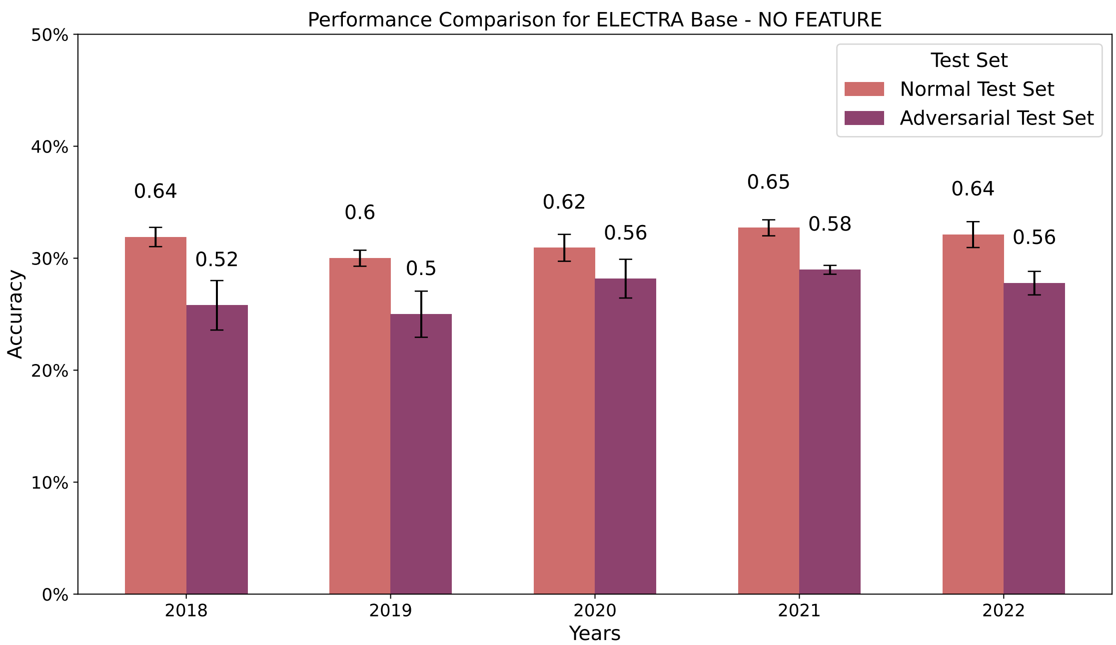

Figure 6.19: ELECTRA Base performance on standard vs adversarial test sets with no artefacts.

Table 6.9: Performance comparison of ELECTRA Base on Standard and Adversarial test sets

Year	Feature	Standard	Adversarial
2018	None	0.6379 $\pm$ 0.02	0.5159 $\pm$ 0.04
	All	0.6437 $\pm$ 0.01	0.5000 $\pm$ 0.02
	Sentence Length	0.6494 $\pm$ 0.03	0.4762 $\pm$ 0.01
	Word Overlap	0.6092 $\pm$ 0.01	0.5000 $\pm$ 0.00
	Contradiction Word	0.6379 $\pm$ 0.02	0.5159 $\pm$ 0.04
	Subsequence Heuristic	0.6264 $\pm$ 0.03	0.5079 $\pm$ 0.02
2019	None	0.6000 $\pm$ 0.01	0.5000 $\pm$ 0.04
	All	0.5952 $\pm$ 0.01	0.4683 $\pm$ 0.01
	Sentence Length	0.5714 $\pm$ 0.01	0.5159 $\pm$ 0.06
	Word Overlap	0.5905 $\pm$ 0.01	0.4603 $\pm$ 0.04
	Contradiction Word	0.6000 $\pm$ 0.01	0.5000 $\pm$ 0.04
	Subsequence Heuristic	0.5714 $\pm$ 0.01	0.4920 $\pm$ 0.04
2020	None	0.6186 $\pm$ 0.02	0.5635 $\pm$ 0.03
	All	0.6216 $\pm$ 0.01	0.5635 $\pm$ 0.03
	Sentence Length	0.6246 $\pm$ 0.01	0.5238 $\pm$ 0.02
	Word Overlap	0.5916 $\pm$ 0.02	0.5476 $\pm$ 0.04
	Contradiction Word	0.6186 $\pm$ 0.02	0.5397 $\pm$ 0.02
	Subsequence Heuristic	0.6096 $\pm$ 0.02	0.5714 $\pm$ 0.02
2021	None	0.6543 $\pm$ 0.01	0.5793 $\pm$ 0.01
	All	0.6749 $\pm$ 0.03	0.5397 $\pm$ 0.04
	Sentence Length	0.6996 $\pm$ 0.02	0.5555 $\pm$ 0.02
	Word Overlap	0.7078 $\pm$ 0.01	0.5238 $\pm$ 0.01
	Contradiction Word	0.6543 $\pm$ 0.01	0.5793 $\pm$ 0.01
	Subsequence Heuristic	0.6831 $\pm$ 0.02	0.5159 $\pm$ 0.02
2022	None	0.6422 $\pm$ 0.02	0.5555 $\pm$ 0.02
	All	0.6667 $\pm$ 0.01	0.5555 $\pm$ 0.02
	Sentence Length	0.6728 $\pm$ 0.01	0.5317 $\pm$ 0.02
	Word Overlap	0.6697 $\pm$ 0.01	0.5079 $\pm$ 0.02
	Contradiction Word	0.6422 $\pm$ 0.02	0.5555 $\pm$ 0.02
	Subsequence Heuristic	0.6820 $\pm$ 0.01	0.5714 $\pm$ 0.01

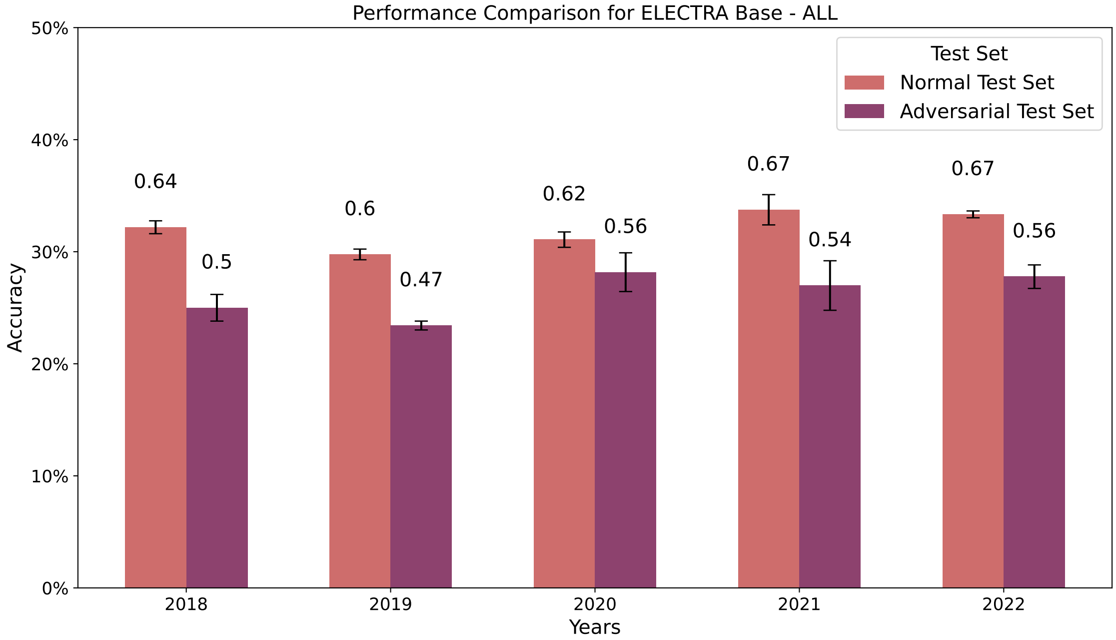

Figure 6.20: ELECTRA Base performance on standard vs adversarial test sets with all artefacts.

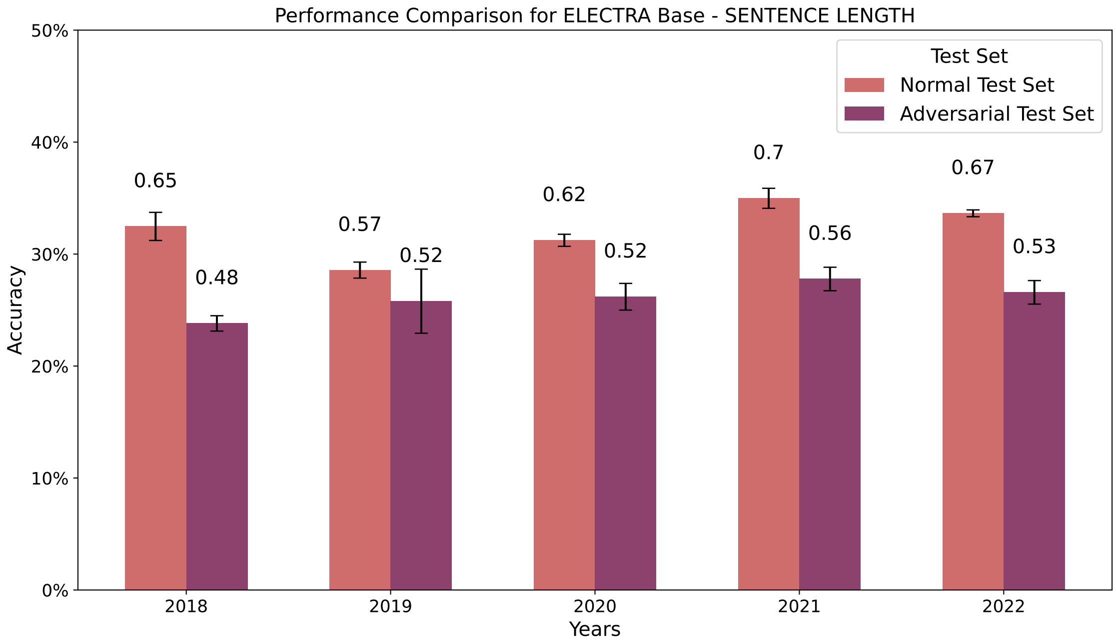

Figure 6.21: ELECTRA Base performance on standard vs adversarial test sets with sentence length artefact.

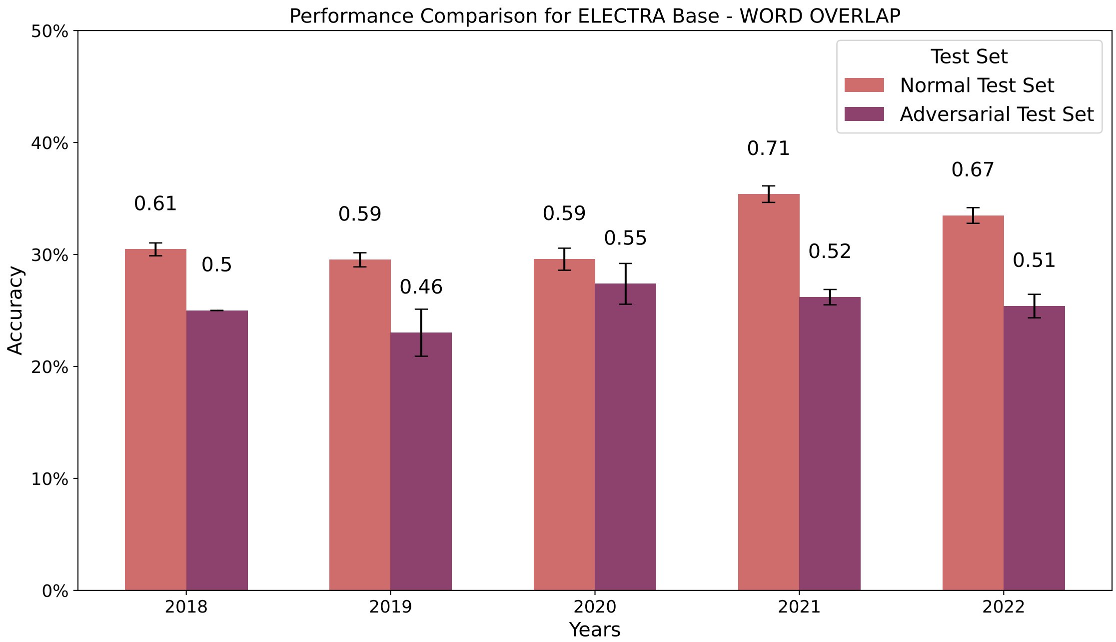

Figure 6.22: ELECTRA Base performance on standard vs adversarial test sets with word overlap artefact.

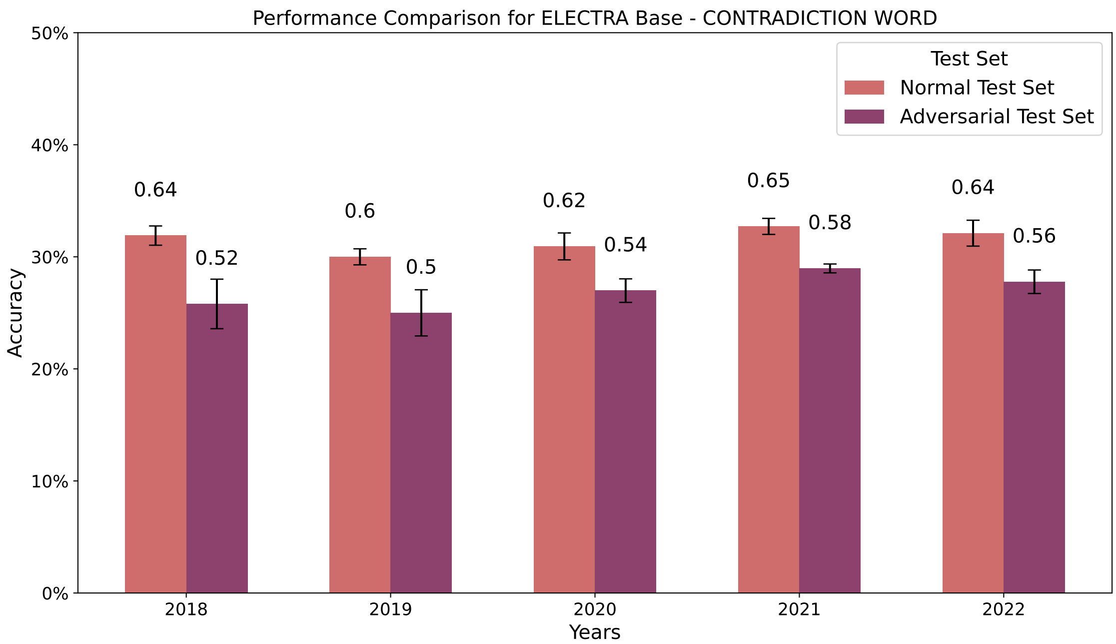

Figure 6.23: ELECTRA Base performance on standard vs adversarial test sets with contradiction word artefact.

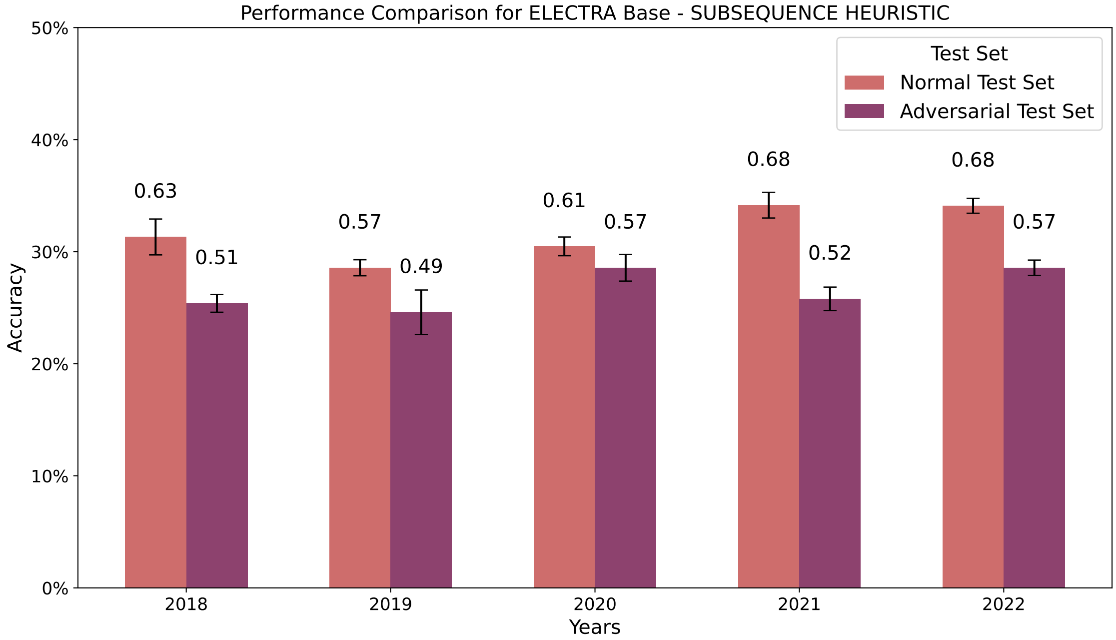

Figure 6.24: ELECTRA Base performance on standard vs adversarial test sets with subsequence heuristic artefact.

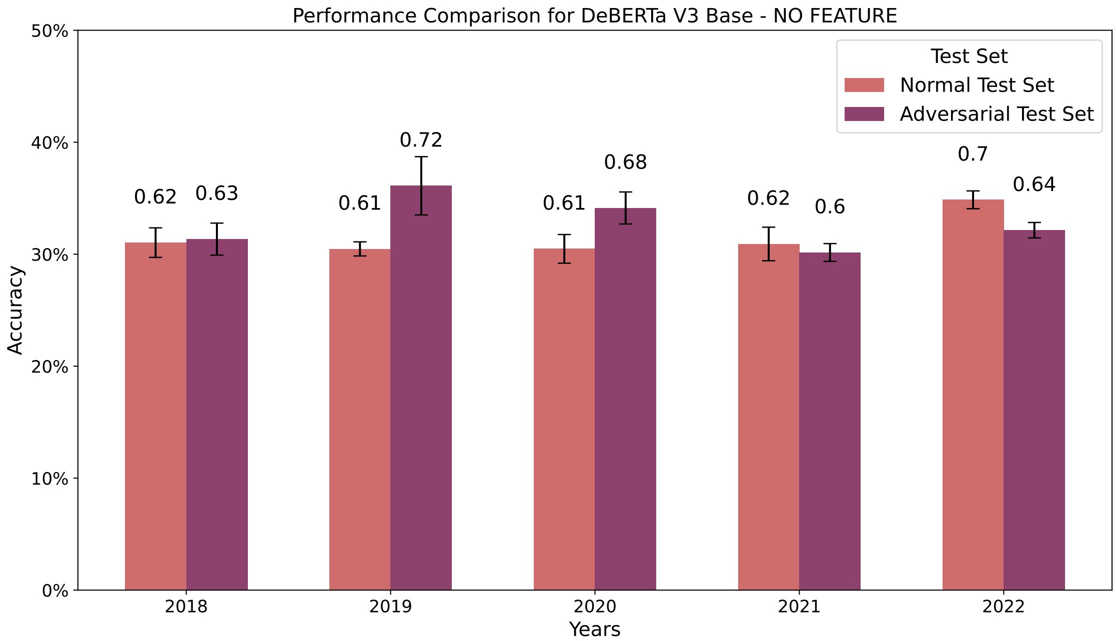

Figure 6.25: DeBERTa V3 Base performance on standard vs adversarial test sets with no artefacts.

Table 6.10: Performance comparison of DeBERTa Base on Standard and Adversarial test sets

Year	Feature	Standard	Adversarial
2018	None	0.6207 ± 0.03	0.6270 ± 0.03
	All	0.6839 ± 0.02	0.5793 ± 0.01
	Sentence Length	0.6494 ± 0.02	0.5635 ± 0.01
	Word Overlap	0.6092 ± 0.02	0.6111 ± 0.04
	Contradiction Word	0.6207 ± 0.03	0.5794 ± 0.03
	Subsequence Heuristic	0.6437 ± 0.02	0.6032 ± 0.03
2019	None	0.6095 ± 0.01	0.7222 ± 0.05
	All	0.6238 ± 0.01	0.6349 ± 0.02
	Sentence Length	0.6476 ± 0.02	0.6269 ± 0.03
	Word Overlap	0.6238 ± 0.03	0.6746 ± 0.02
	Contradiction Word	0.6143 ± 0.01	0.7302 ± 0.04
	Subsequence Heuristic	0.6048 ± 0.01	0.6508 ± 0.01
2020	None	0.6096 ± 0.03	0.6826 ± 0.03
	All	0.6006 ± 0.02	0.6429 ± 0.01
	Sentence Length	0.6186 ± 0.00	0.6905 ± 0.04
	Word Overlap	0.5886 ± 0.03	0.6508 ± 0.07
	Contradiction Word	0.6006 ± 0.03	0.6826 ± 0.02
	Subsequence Heuristic	0.6396 ± 0.02	0.6508 ± 0.03
2021	None	0.6183 ± 0.03	0.6031 ± 0.02
	All	0.6954 ± 0.07	0.6270 ± 0.02
	Sentence Length	0.6269 ± 0.01	0.6667 ± 0.04
	Word Overlap	0.6134 ± 0.04	0.6587 ± 0.02
	Contradiction Word	0.5999 ± 0.03	0.6429 ± 0.01
	Subsequence Heuristic	0.6790 ± 0.01	0.7143 ± 0.04
2022	None	0.6972 ± 0.02	0.6429 ± 0.01
	All	0.6606 ± 0.02	0.6588 ± 0.02
	Sentence Length	0.6657 ± 0.00	0.6270 ± 0.03
	Word Overlap	0.6544 ± 0.01	0.6429 ± 0.00
	Contradiction Word	0.6902 ± 0.04	0.5952 ± 0.03
	Subsequence Heuristic	0.6954 ± 0.03	0.7064 ± 0.01

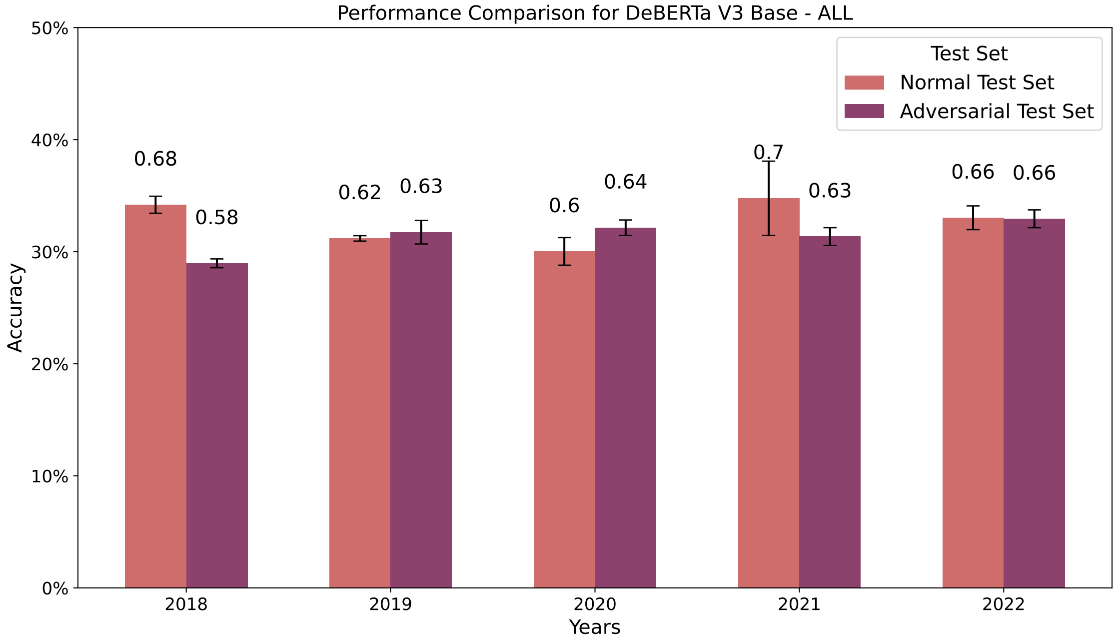

Figure 6.26: DeBERTa V3 Base performance on standard vs adversarial test sets with all artefacts.

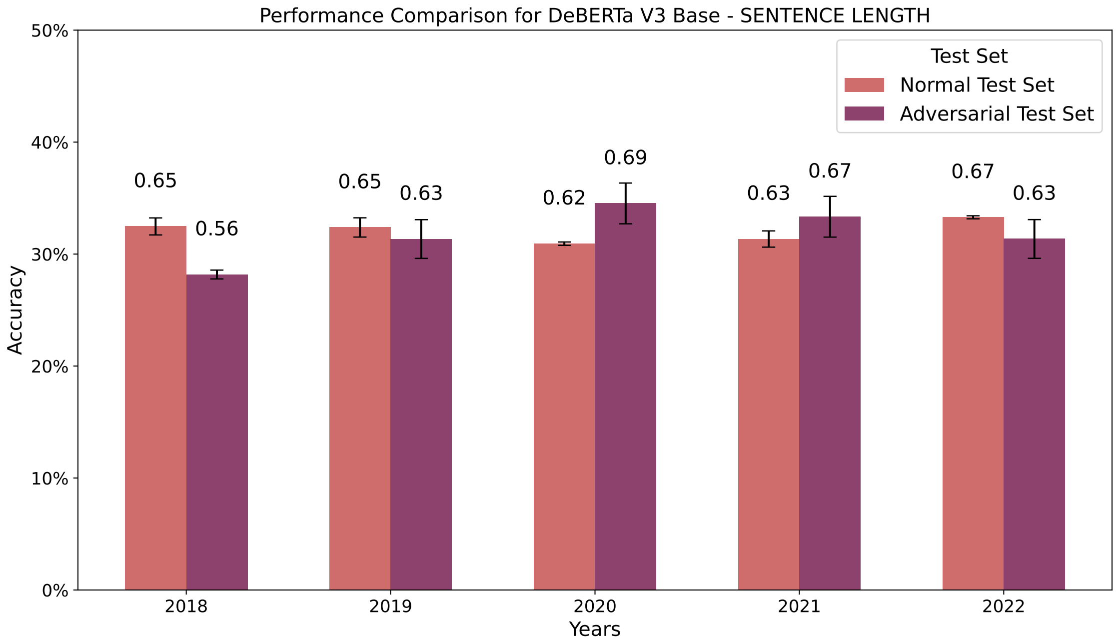

Figure 6.27: DeBERTa V3 Base performance on standard vs adversarial test sets with sentence length artefact.

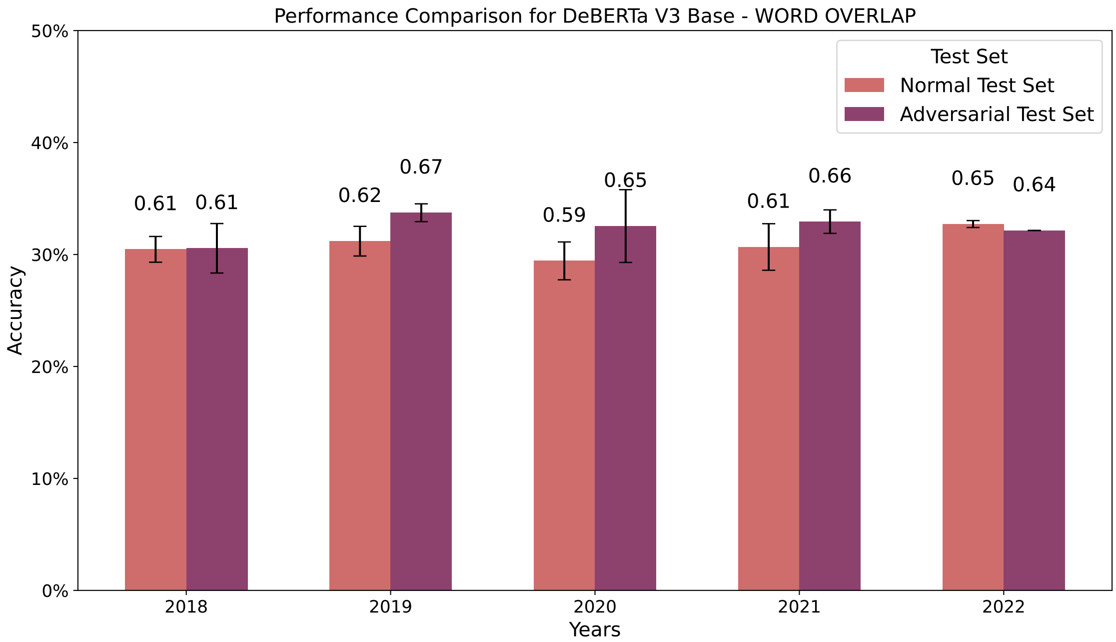

Figure 6.28: DeBERTa V3 Base performance on standard vs adversarial test sets with word overlap artefact.

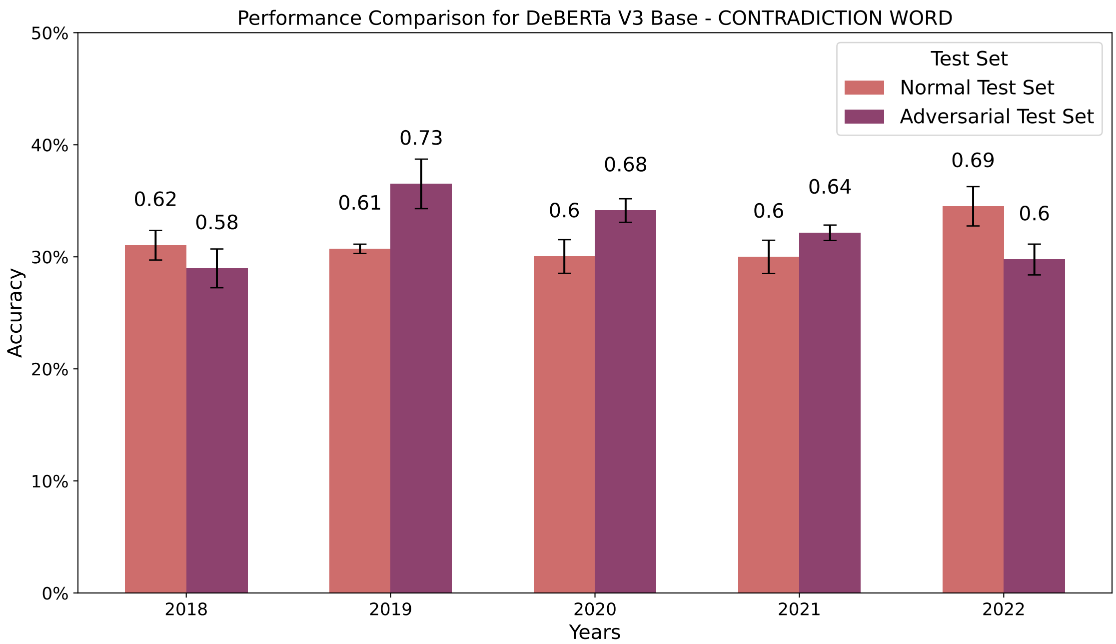

Figure 6.29: DeBERTa V3 Base performance on standard vs adversarial test sets with contradiction word artefact.

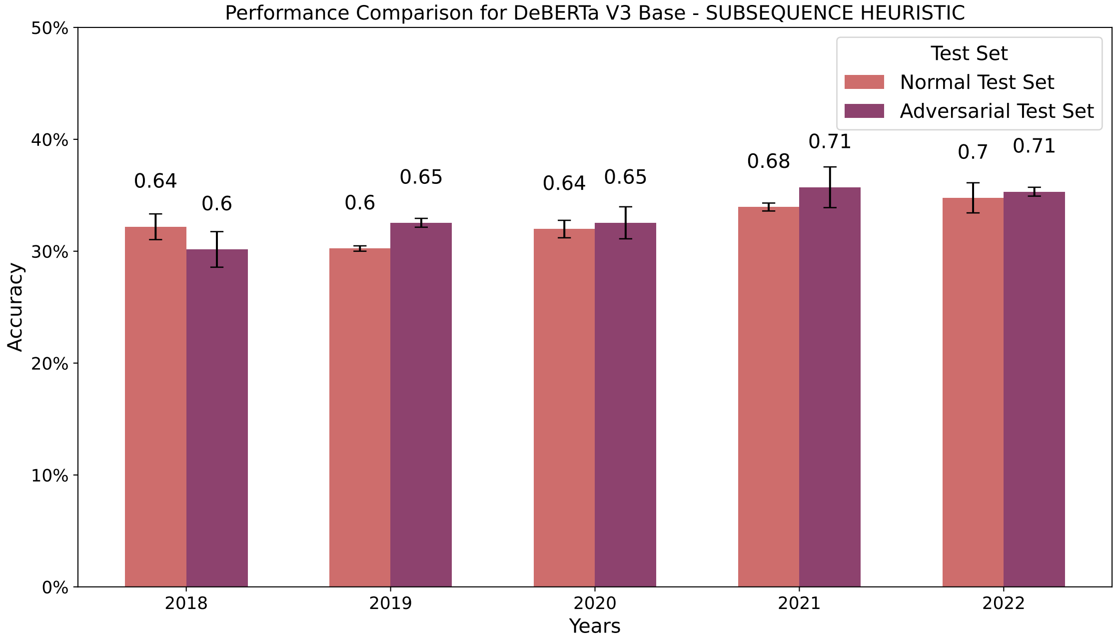

Figure 6.30: DeBERTa V3 Base performance on standard vs adversarial test sets with subsequence heuristic artefact.

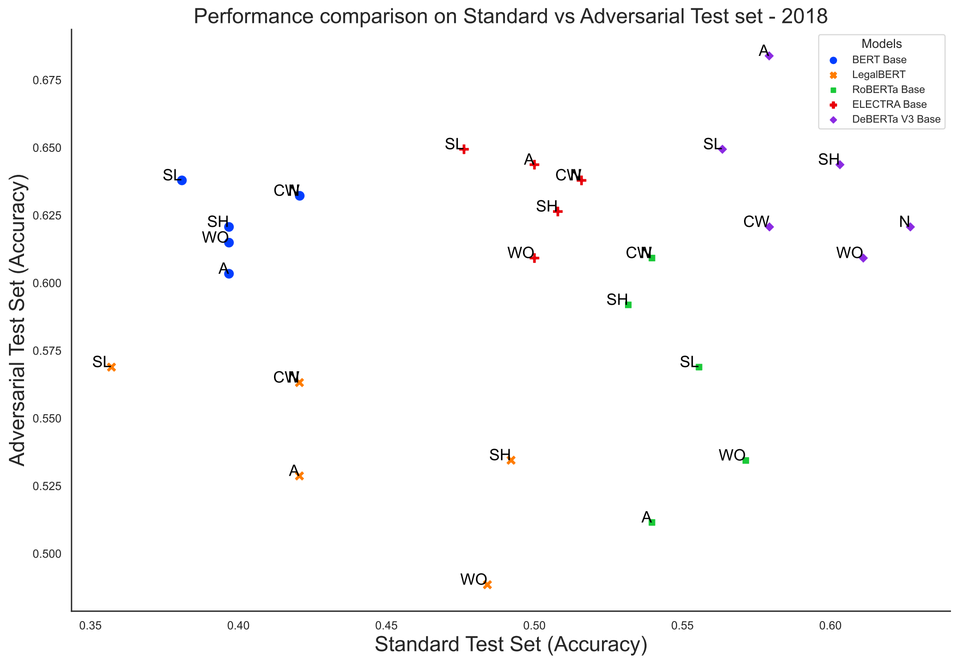

Figure 6.31: Performance comparison of Standard vs Adversarial test sets for 2018

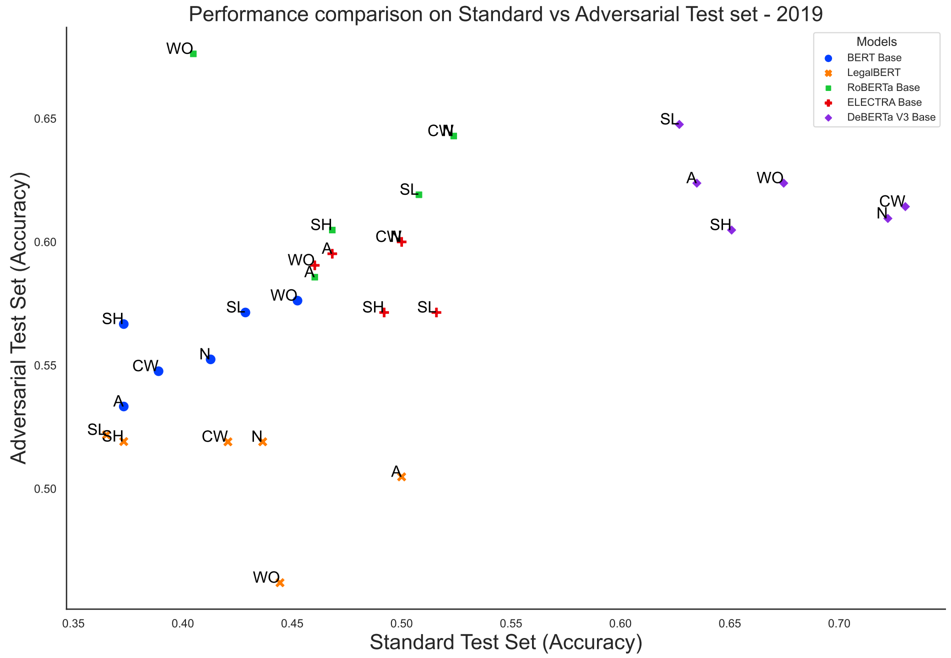

Figure 6.32: Performance comparison of Standard vs Adversarial test sets for 2019

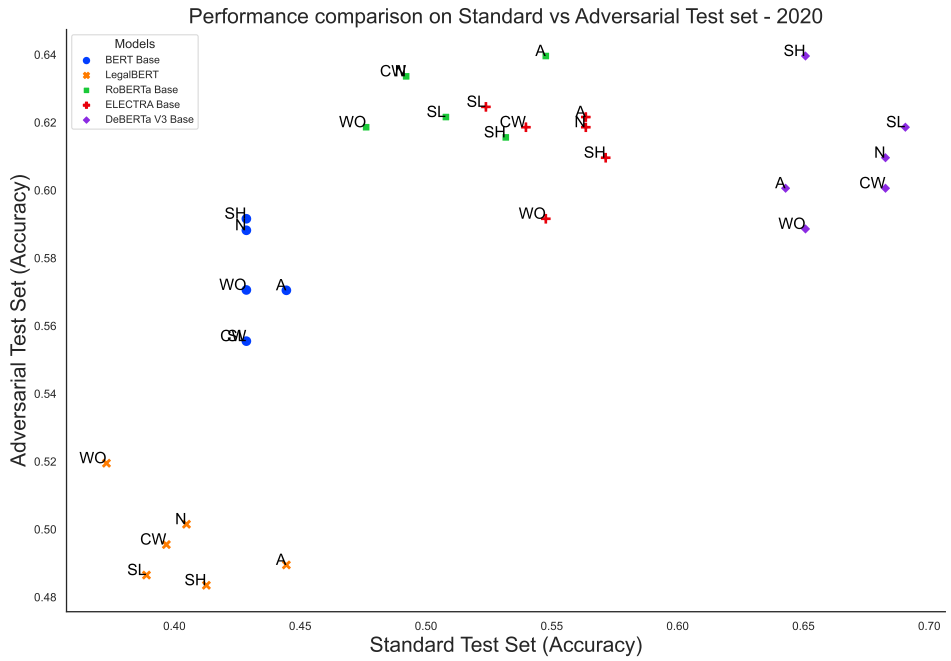

Figure 6.33: Performance comparison of Standard vs Adversarial test sets for 2020

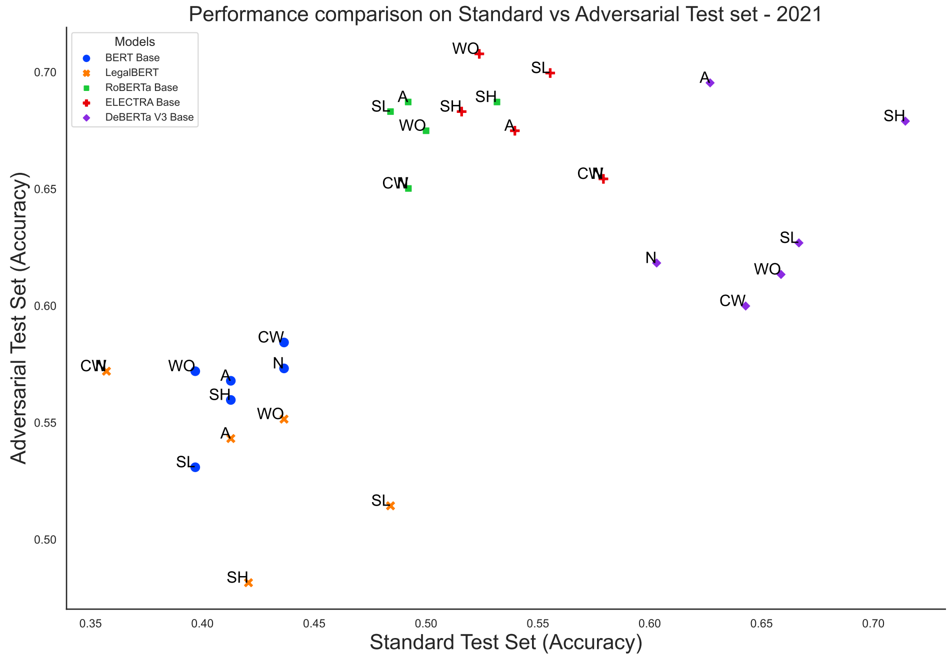

Figure 6.34: Performance comparison of Standard vs Adversarial test sets for 2021

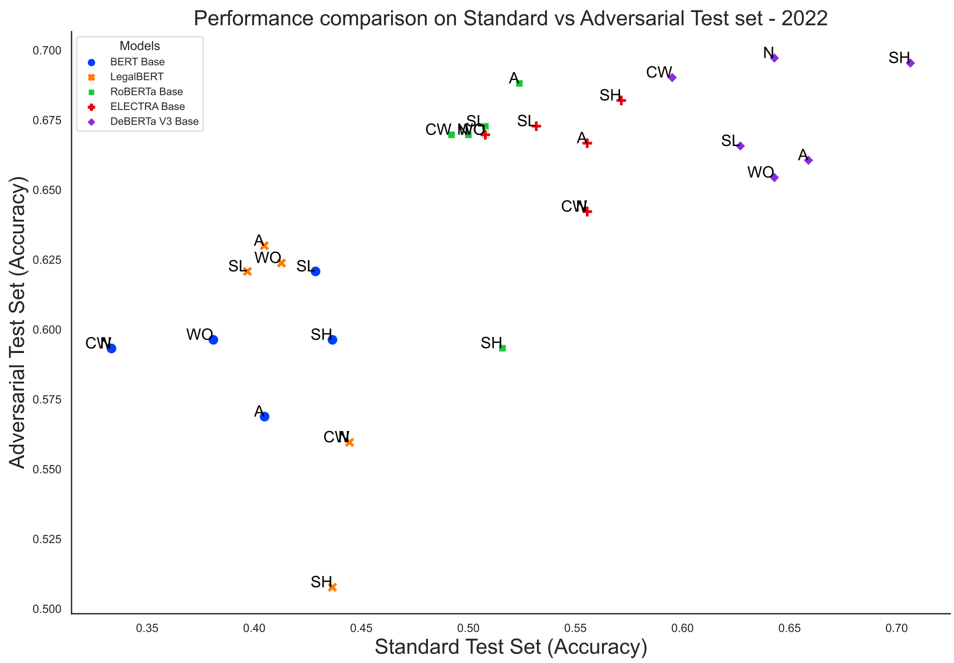

Figure 6.35: Performance comparison of Standard vs Adversarial test sets for 2022

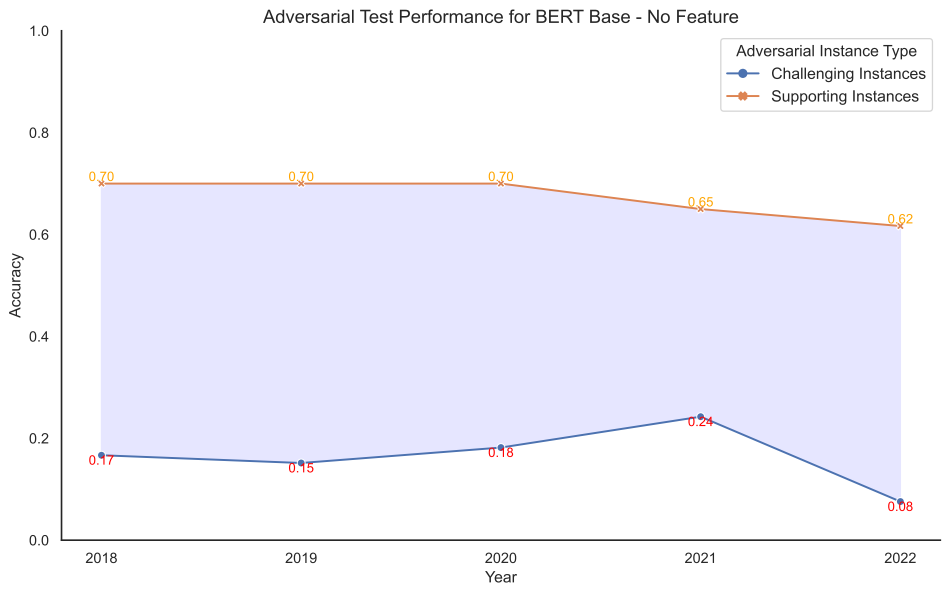

Figure 6.36: BERT Base performance on Challenging and Supporting adversarial instances with no artefacts.

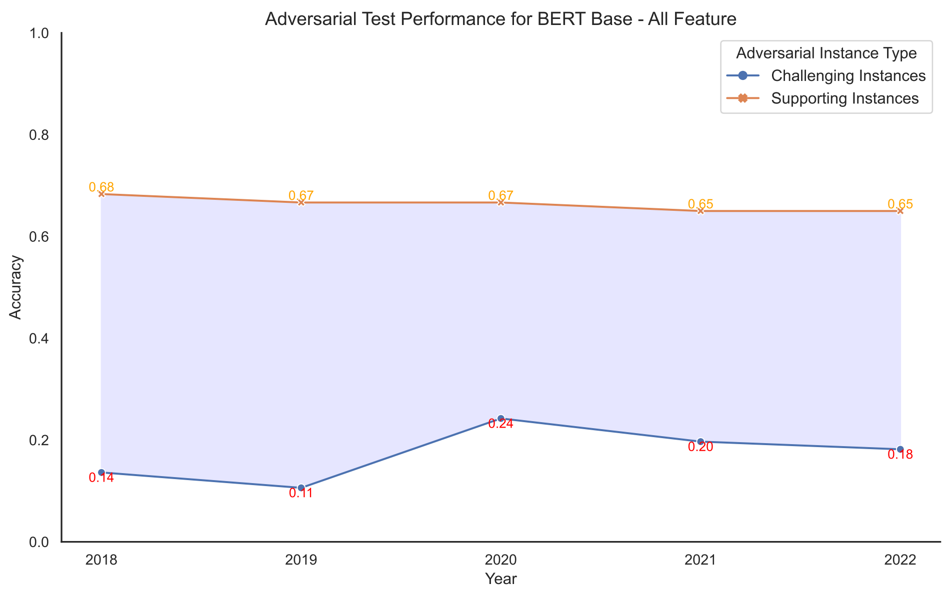

Figure 6.37: BERT Base performance on Challenging and Supporting adversarial instances with all artefacts.

Table 6.11: Performance comparison of BERT Base on Adversarial Challenging and Supporting instances

Year	Feature	Challenging	Supporting
2018	None	0.1667 ± 0.04	0.7000 ± 0.05
	All	0.1363 ± 0.03	0.6833 ± 0.07
	Sentence Length	0.1363 ± 0.03	0.6500 ± 0.03
	Word Overlap	0.1212 ± 0.04	0.7000 ± 0.03
	Contradiction Word	0.1666 ± 0.04	0.7000 ± 0.05
	Subsequence Heuristic	0.1515 ± 0.05	0.6667 ± 0.02
2019	None	0.1515 ± 0.06	0.7000 ± 0.05
	All	0.1061 ± 0.04	0.6667 ± 0.06
	Sentence Length	0.1212 ± 0.04	0.7667 ± 0.03
	Word Overlap	0.2272 ± 0.03	0.7000 ± 0.06
	Contradiction Word	0.1212 ± 0.03	0.6833 ± 0.06
	Subsequence Heuristic	0.1515 ± 0.05	0.6167 ± 0.02
2020	None	0.1818 ± 0.05	0.7000 ± 0.03
	All	0.2424 ± 0.02	0.6667 ± 0.02
	Sentence Length	0.1363 ± 0.03	0.7500 ± 0.09
	Word Overlap	0.2576 ± 0.02	0.6167 ± 0.03
	Contradiction Word	0.1818 ± 0.05	0.7000 ± 0.03
	Subsequence Heuristic	0.2424 ± 0.06	0.6333 ± 0.06
2021	None	0.2424 ± 0.02	0.6500 ± 0.03
	All	0.1969 ± 0.07	0.6500 ± 0.05
	Sentence Length	0.2727 ± 0.05	0.5333 ± 0.04
	Word Overlap	0.1666 ± 0.04	0.6500 ± 0.05
	Contradiction Word	0.2424 ± 0.02	0.6500 ± 0.03
	Subsequence Heuristic	0.1969 ± 0.07	0.6500 ± 0.03
2022	None	0.0758 ± 0.04	0.6167 ± 0.04
	All	0.1818 ± 0.11	0.6500 ± 0.06
	Sentence Length	0.2273 ± 0.05	0.6500 ± 0.06
	Word Overlap	0.1515 ± 0.06	0.6333 ± 0.04
	Contradiction Word	0.0758 ± 0.04	0.6167 ± 0.04
	Subsequence Heuristic	0.1818 ± 0.05	0.7167 ± 0.11

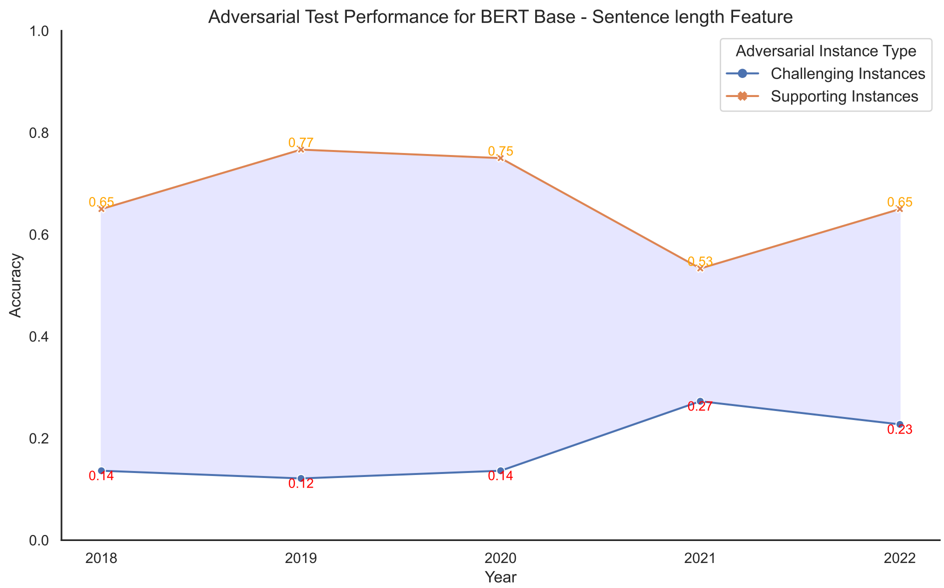

Figure 6.38: BERT Base performance on Challenging and Supporting adversarial instances with sentence length artefact.

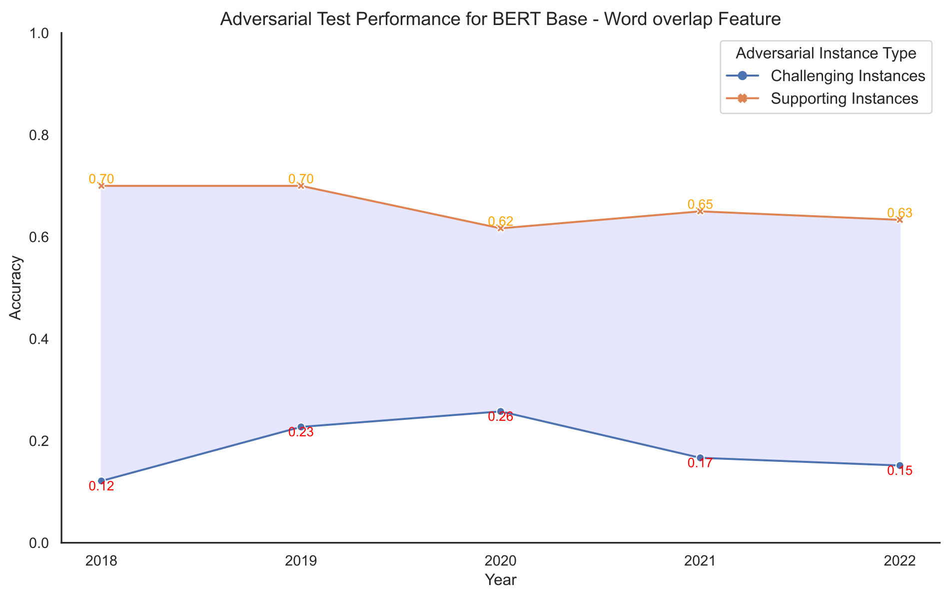

Figure 6.39: BERT Base performance on Challenging and Supporting adversarial instances with word overlap artefact.

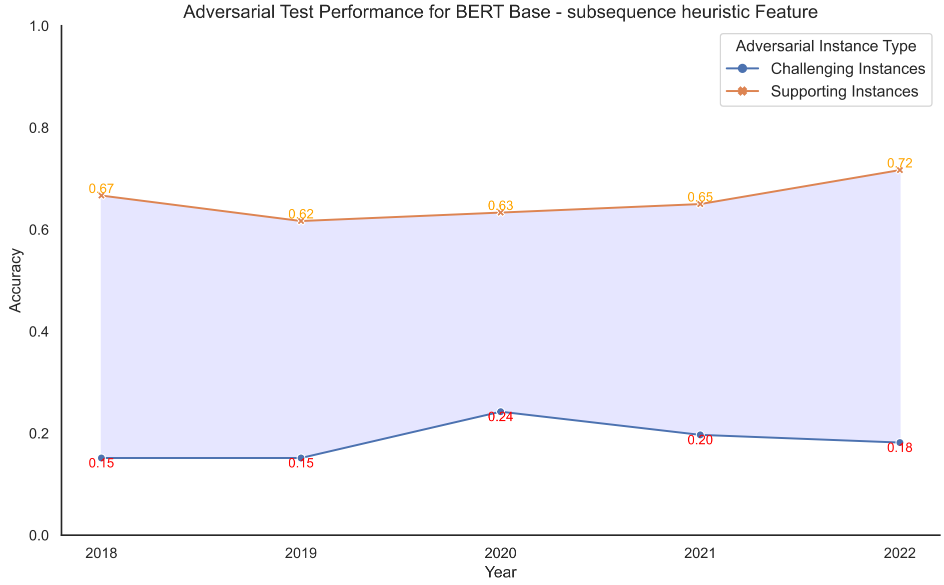

Figure 6.41: BERT Base performance on Challenging and Supporting adversarial instances with subsequence heuristic artefact.

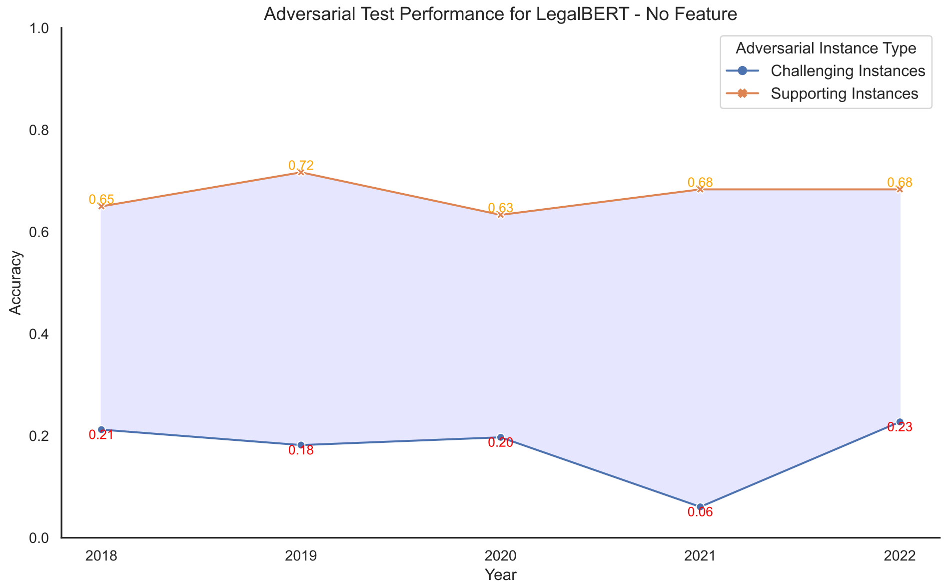

Figure 6.42: LegalBERT performance on Challenging and Supporting adversarial instances with no artefacts.

Table 6.12: Performance comparison of LegalBERT on Adversarial Challenging and Supporting instances

Year	Feature	Challenging	Supporting
2018	None	0.2121 $\pm$ 0.14	0.6500 $\pm$ 0.08
	All	0.3030 $\pm$ 0.12	0.5500 $\pm$ 0.03
	Sentence Length	0.0909 $\pm$ 0.05	0.6500 $\pm$ 0.03
	Word Overlap	0.3939 $\pm$ 0.11	0.5833 $\pm$ 0.06
	Contradiction Word	0.2121 $\pm$ 0.14	0.6500 $\pm$ 0.08
	Subsequence Heuristic	0.4545 $\pm$ 0.05	0.5333 $\pm$ 0.03
2019	None	0.1818 $\pm$ 0.09	0.7167 $\pm$ 0.06
	All	0.3030 $\pm$ 0.17	0.7167 $\pm$ 0.04
	Sentence Length	0.1363 $\pm$ 0.05	0.6167 $\pm$ 0.03
	Word Overlap	0.2424 $\pm$ 0.08	0.6667 $\pm$ 0.04
	Contradiction Word	0.2273 $\pm$ 0.14	0.6333 $\pm$ 0.09
	Subsequence Heuristic	0.0909 $\pm$ 0.03	0.6833 $\pm$ 0.04
2020	None	0.1970 $\pm$ 0.15	0.6333 $\pm$ 0.07
	All	0.1818 $\pm$ 0.07	0.7333 $\pm$ 0.02
	Sentence Length	0.0758 $\pm$ 0.02	0.7333 $\pm$ 0.04
	Word Overlap	0.0758 $\pm$ 0.02	0.7000 $\pm$ 0.03
	Contradiction Word	0.1970 $\pm$ 0.15	0.6167 $\pm$ 0.06
	Subsequence Heuristic	0.0758 $\pm$ 0.03	0.7833 $\pm$ 0.03
2021	None	0.0606 $\pm$ 0.02	0.6833 $\pm$ 0.06
	All	0.3335 $\pm$ 0.12	0.5000 $\pm$ 0.10
	Sentence Length	0.4242 $\pm$ 0.12	0.5500 $\pm$ 0.03
	Word Overlap	0.2576 $\pm$ 0.12	0.6333 $\pm$ 0.07
	Contradiction Word	0.0606 $\pm$ 0.02	0.6833 $\pm$ 0.06
	Subsequence Heuristic	0.2273 $\pm$ 0.09	0.6333 $\pm$ 0.06
2022	None	0.2273 $\pm$ 0.14	0.6833 $\pm$ 0.09
	All	0.1363 $\pm$ 0.07	0.7000 $\pm$ 0.08
	Sentence Length	0.1060 $\pm$ 0.05	0.7167 $\pm$ 0.04
	Word Overlap	0.1061 $\pm$ 0.04	0.7500 $\pm$ 0.03
	Contradiction Word	0.2273 $\pm$ 0.14	0.6833 $\pm$ 0.09
	Subsequence Heuristic	0.3485 $\pm$ 0.15	0.5333 $\pm$ 0.03

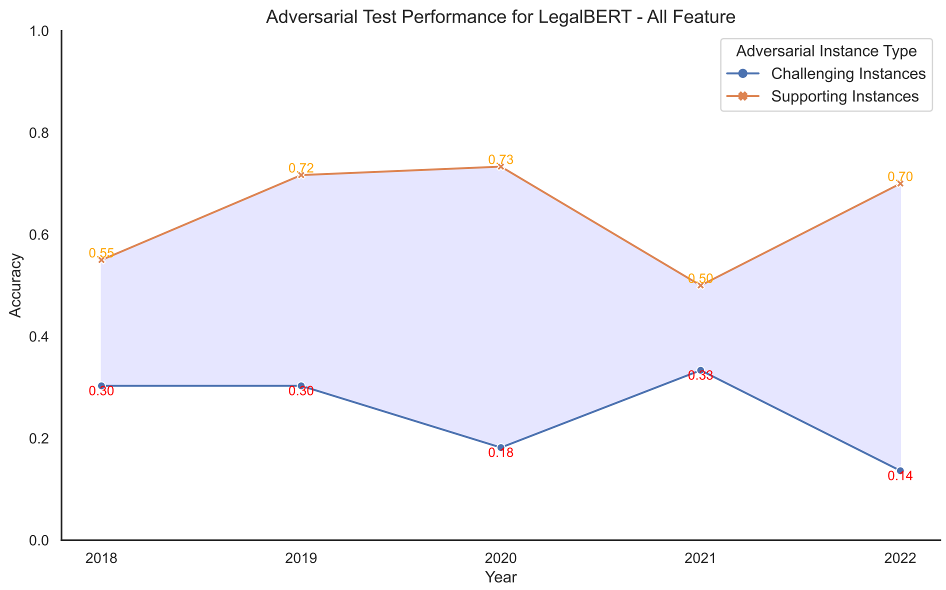

Figure 6.43: LegalBERT performance on Challenging and Supporting adversarial instances with all artefacts.

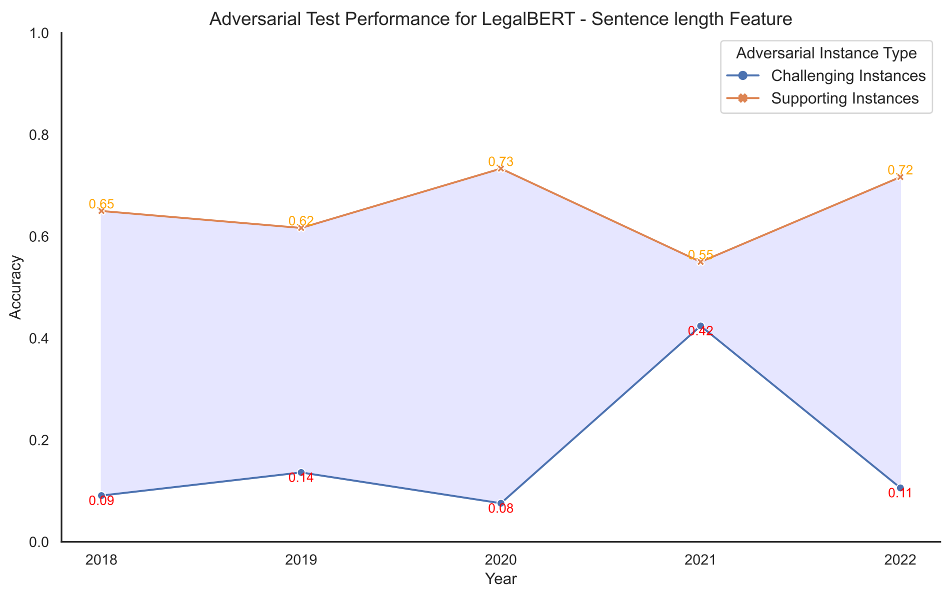

Figure 6.44: LegalBERT performance on Challenging and Supporting adversarial instances with sentence length artefact.

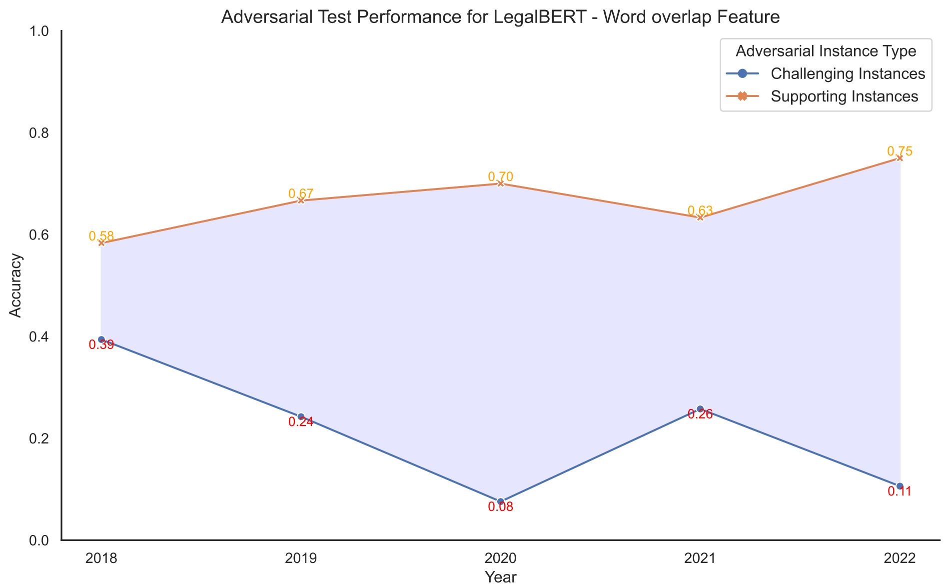

Figure 6.45: LegalBERT performance on Challenging and Supporting adversarial instances with word overlap artefact.

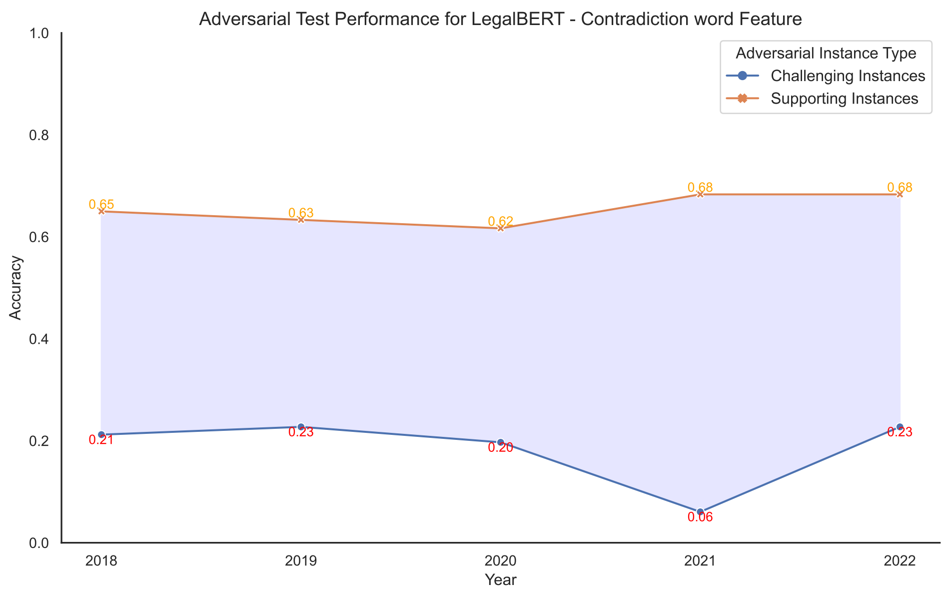

Figure 6.46: LegalBERT performance on Challenging and Supporting adversarial instances with contradiction word artefact.

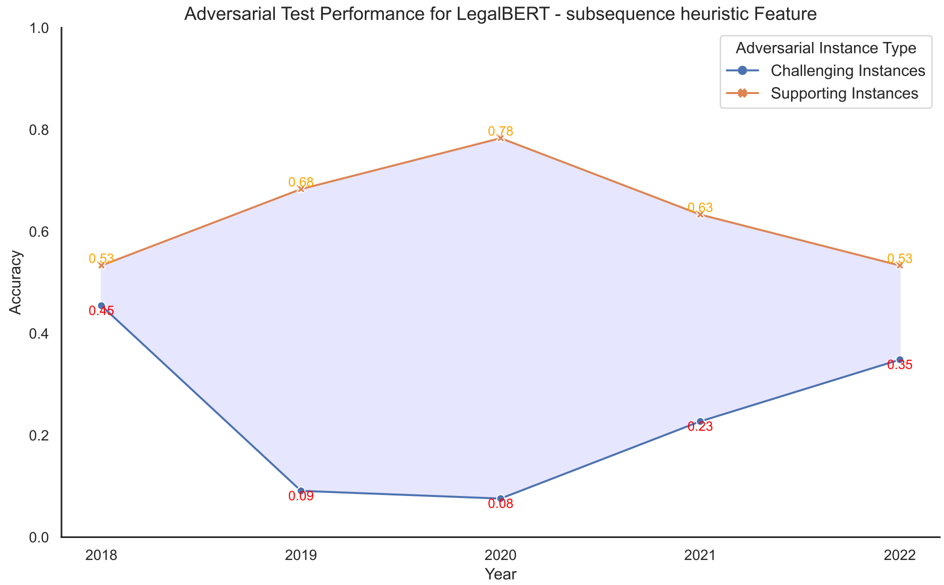

Figure 6.47: LegalBERT performance on Challenging and Supporting adversarial instances with subsequence heuristic artefact.

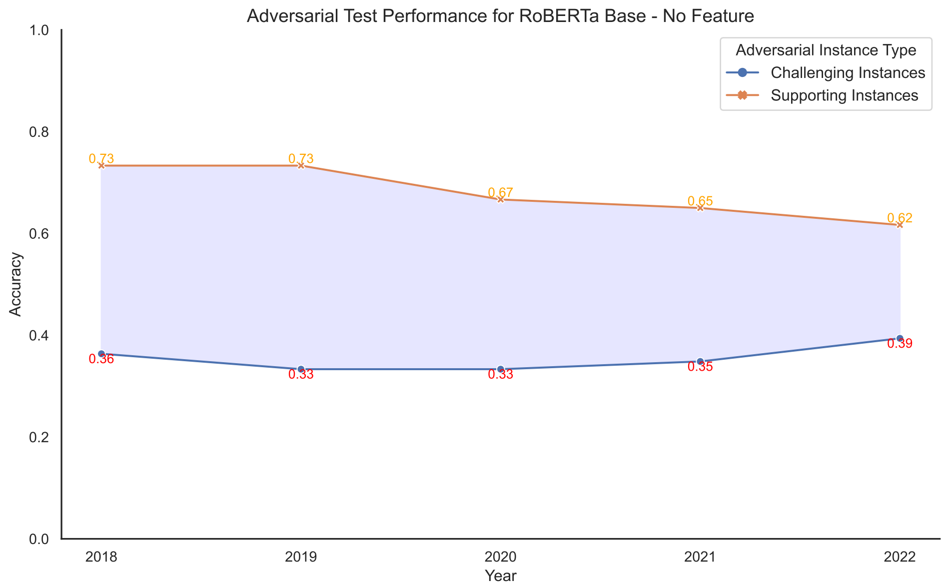

Figure 6.48: RoBERTa Base performance on Challenging and Supporting adversarial instances with no artefacts.

Table 6.13: Performance comparison of RoBERTa Base on Adversarial Challenging and Supporting instances

Year	Feature	Challenging	Supporting
2018	None	0.3636 ± 0.03	0.7333 ± 0.07
	All	0.3788 ± 0.03	0.7167 ± 0.02
	Sentence Length	0.4091 ± 0.09	0.7167 ± 0.04
	Word Overlap	0.4091 ± 0.03	0.7500 ± 0.03
	Contradiction Word	0.3636 ± 0.03	0.7333 ± 0.07
	Subsequence Heuristic	0.3333 ± 0.04	0.7500 ± 0.03
2019	None	0.3333 ± 0.04	0.7333 ± 0.02
	All	0.3333 ± 0.08	0.6000 ± 0.08
	Sentence Length	0.4242 ± 0.03	0.6000 ± 0.05
	Word Overlap	0.3182 ± 0.00	0.5000 ± 0.03
	Contradiction Word	0.3333 ± 0.04	0.7333 ± 0.02
	Subsequence Heuristic	0.3030 ± 0.02	0.6500 ± 0.06
2020	None	0.3333 ± 0.03	0.6667 ± 0.02
	All	0.3939 ± 0.04	0.7167 ± 0.02
	Sentence Length	0.3636 ± 0.03	0.6667 ± 0.02
	Word Overlap	0.3636 ± 0.08	0.6000 ± 0.06
	Contradiction Word	0.3333 ± 0.03	0.6667 ± 0.02
	Subsequence Heuristic	0.3485 ± 0.03	0.7333 ± 0.02
2021	None	0.3485 ± 0.03	0.6500 ± 0.09
	All	0.3182 ± 0.05	0.6833 ± 0.04
	Sentence Length	0.4091 ± 0.05	0.5667 ± 0.07
	Word Overlap	0.3636 ± 0.05	0.6500 ± 0.03
	Contradiction Word	0.3485 ± 0.03	0.6500 ± 0.09
	Subsequence Heuristic	0.3333 ± 0.03	0.7500 ± 0.03
2022	None	0.3939 ± 0.02	0.6167 ± 0.03
	All	0.3788 ± 0.02	0.6833 ± 0.03
	Sentence Length	0.3485 ± 0.02	0.6833 ± 0.03
	Word Overlap	0.4394 ± 0.05	0.5833 ± 0.04
	Contradiction Word	0.3939 ± 0.02	0.6000 ± 0.03
	Subsequence Heuristic	0.4091 ± 0.05	0.6333 ± 0.07

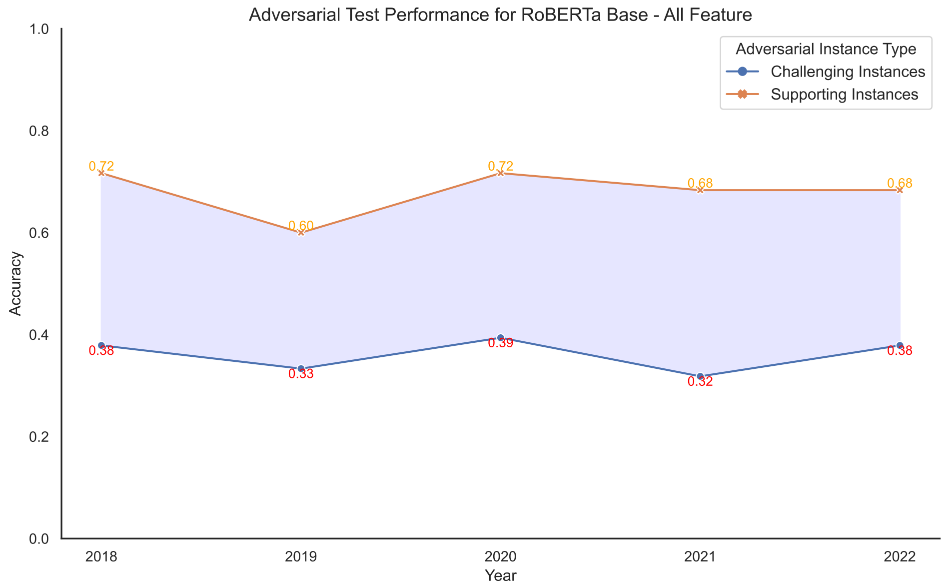

Figure 6.49: RoBERTa Base performance on Challenging and Supporting adversarial instances with all artefacts.

6.2.3 Hypothesis-Only Model vs Majority Baseline

This evaluation strategy compares the performance of hypothesis-only models with the majority baseline. Hypothesis-only models are trained using only the hypothesis part of the input, excluding the premise. The majority baseline represents a simple heuristic where the model always predicts the majority class from the training data. By comparing these, the aim is to determine whether the hypothesis-only models can achieve better performance than the majority baseline, indicating the presence of strong artefacts within the hypothesis alone. This comparison is fundamental for understanding if models can exploit superficial features in the hypothesis to make predictions.

Tables 6.16 to 6.20 present a comparative analysis of hypothesis-only models against the majority baseline. For the BERT Base hypothesis-only model, as illustrated in Figure 6.66, the hypothesis-only model consistently outperforms the majority classifier across all years and artefact features. Similarly, Table 6.17 and Figure 6.67 demonstrate a similar trend for the LegalBERT hypothesis-only model, where it surpasses the majority classifier.

Table 6.18 presents the results for the RoBERTa Base hypothesis-only model, with observations largely consistent with BERT Base and LegalBERT. However, a notable deviation occurs in 2020, where all RoBERTa Base model variants underperform the majority classifier (Figure 6.68). This pattern also holds true for the ELECTRA Base hypothesis-only model (Table 6.19 and Figure 6.69), with model variants in 2020 and 2022 performing worse than the majority classifier.

For DeBERTa, the trend of poor performance in 2020 persists (Table 6.20 and Figure 6.70). However, in other years, the model variants outperform the majority classifier.

Thus, for RoBERTa Base, ELECTRA Base, and DeBERTa Base, the 2020 dataset appears to be an outlier where hypothesis-only models underperform. Excluding this outlier,

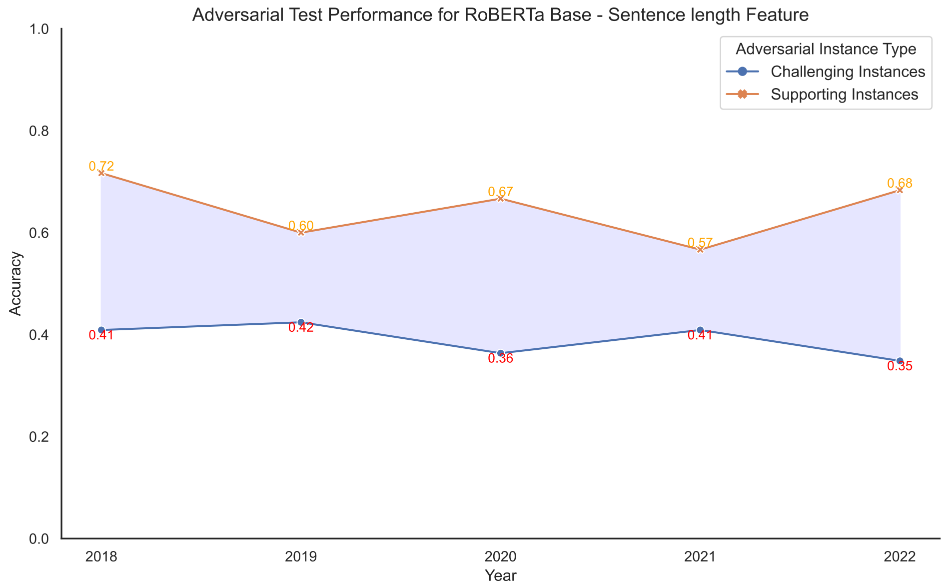

Figure 6.50: RoBERTa Base performance on Challenging and Supporting adversarial instances with sentence length artefact.

the remaining model variants generally outperform the majority classifier, suggesting a potential reliance on language artefacts in these models.

6.2.4 Discussion

The results from the robustness evaluation reveal significant insights into how language artefacts impact the performance of BERT-based models in the COLIEE statute law entailment task. This discussion section summarises the findings from the various evaluation strategies to provide a fundamental understanding of the influence of language artefacts on model performance.

The evaluation of various BERT-based models on the standard and adversarial test sets reveals a complex interplay between model architectures, pre-training strategies, and the impact of language artefacts on natural language inference (NLI) tasks. The consistent performance gap observed between the standard and adversarial test sets underscores the limitations of current NLI models in generalizing beyond the biases and superficial patterns present in the training data. The varying degrees of vulnerability to language artefacts across different models shed light on the strengths and weaknesses of specific architectures and pre-training strategies. BERT Base and LegalBERT, despite the latter's legal domain adaptation, exhibit significant performance drops on the adversarial set as presented in table 6.7, indicating their susceptibility to exploiting shallow cues like sentence length and word overlap. This suggests that domain-specific pre-training alone might not be sufficient to overcome the challenges posed by language artefacts.

In contrast, DeBERTa V3 Base consistently demonstrates greater robustness, achieving comparable or even superior performance on the adversarial set compared to the standard set as illustrated in Figure 6.25 to 6.30. This suggests that its diverse pre-training on various NLI tasks, including adversarial examples, contributes to its ability to capture

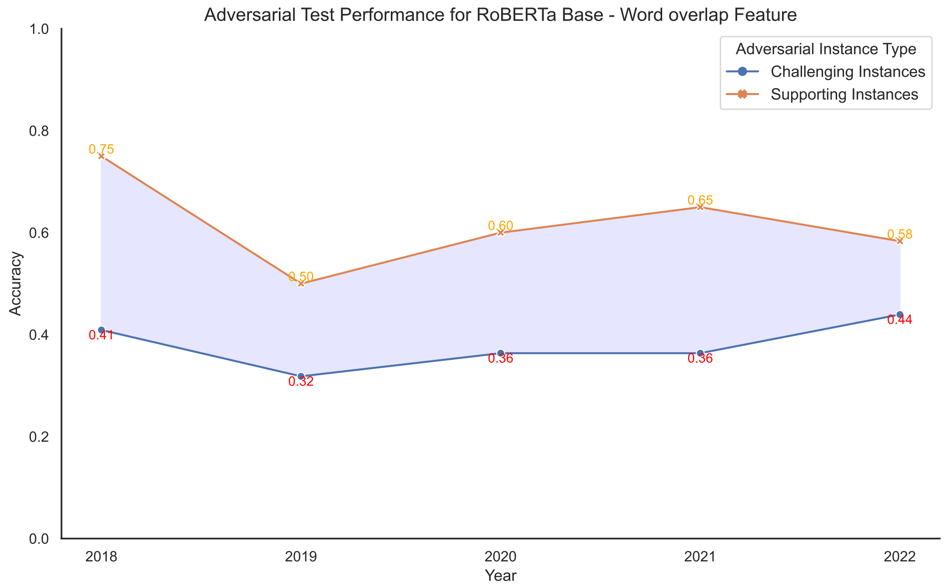

Figure 6.51: RoBERTa Base performance on Challenging and Supporting adversarial instances with word overlap artefact.

deeper semantic relationships and rely less on superficial patterns. The impact of specific language artefacts also varies across models. While sentence length and word overlap seem to affect most models to some extent, the effect of contradiction words and subsequence heuristics is more pronounced in models like BERT Base and LegalBERT. This suggests that these models might be more prone to relying on simple heuristics based on negation or subsequence matching, highlighting the need for architectural or training modifications to address these vulnerabilities.

The comparison of model performance on adversarial challenging and supporting instances provides even deeper insights into the extent to which models are relying on language artefacts for their predictions. A striking disparity in performance is observed between challenging and supporting instances across almost all models and artefact types. Models generally excel on supporting instances (containing artefacts), whereas their performance significantly drops on challenging instances (Table 6.11 to 6.15). This gap serves as compelling evidence of models leveraging these artefacts for prediction, rather than focusing solely on the underlying semantic relationships. DeBERTa V3 Base again distinguishes itself by showcasing a less pronounced performance difference between challenging and supporting instances, particularly in later years as shown in Figure 6.65. While it still exhibits some reliance on artefacts, its ability to perform reasonably well even on challenging instances indicates a greater capacity for genuine semantic understanding compared to other models.

The hypothesis-only baseline evaluations presented in Table 6.16 to 6.20 provide further evidence of the impact of language artefacts. The fact that hypothesis-only models, which are trained without the premise, often outperform the majority baseline is indicative of strong artefacts within the hypothesis alone. This suggests that models can make predic-

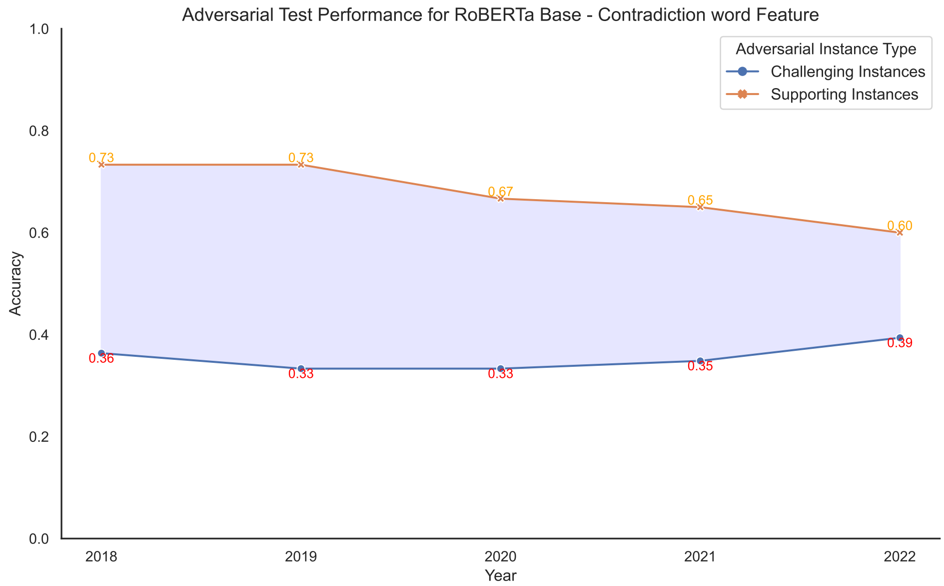

Figure 6.52: RoBERTa Base performance on Challenging and Supporting adversarial instances with contradiction word artefact.

tions based solely on superficial features present in the hypothesis, without considering the full context provided by the premise.

This finding is critical because it highlights that the models are not truly understanding the legal entailment task but rather exploiting patterns in the hypothesis that correlate with the labels. This undermines the validity of the models' predictions and highlights the necessity for robust training data that does not contain such artefacts.

The comprehensive analysis of the performance of BERT-based models on standard, adversarial, and hypothesis-only baselines reveals a widespread issue with language artefacts. These artefacts can significantly skew model training and evaluation, leading to unreliable and non-generalizable models. The findings underscore the importance of addressing language artefacts through targeted mitigation strategies, such as data augmentation, to improve model performance.

Based on the robustness evaluation results, it is evident that language artefacts significantly impact the performance of BERT-based models in statute law entailment tasks. The models' performance on adversarial test sets and hypothesis-only baselines clearly indicates that these artefacts introduce biases that can lead to incorrect predictions.

In conclusion, the detailed examination of model performance across various evaluation strategies reveals that language artefacts are a critical factor influencing the effectiveness of BERT-based models in the COLIEE statute law entailment task. The findings highlight the need for robust mitigation techniques to ensure that these models perform without exploiting the language artefacts hidden in the dataset.

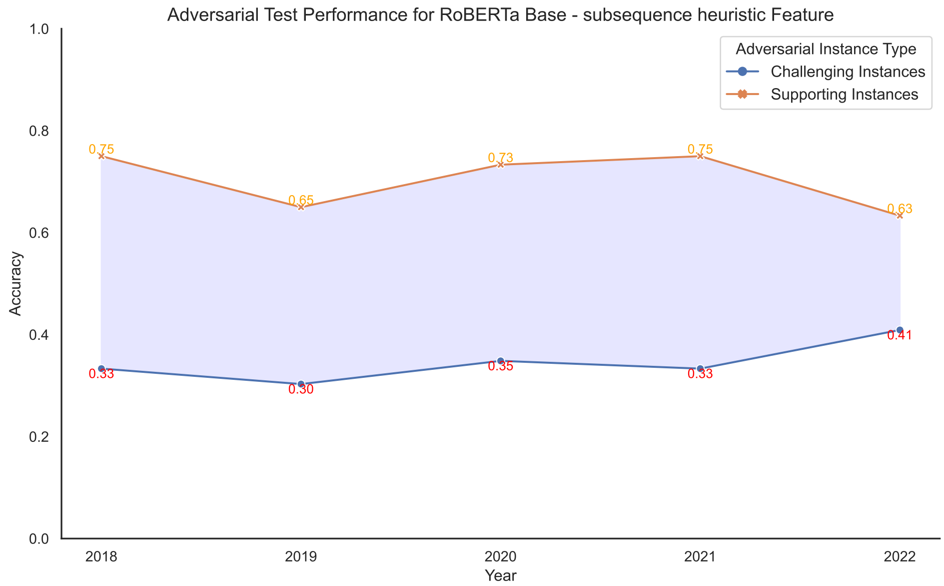

Figure 6.53: RoBERTa Base performance on Challenging and Supporting adversarial instances with subsequence heuristic artefact.

6.3 Effectiveness of Artefact Mitigation

This section addresses the third research question: Can data-level debiasing through data augmentation techniques effectively reduce the contradiction word and word overlap artefacts within the dataset?

Data-level mitigation efforts were focused on word overlap and contradiction word artefacts. Experiments involved training models with various input configurations on the mitigated dataset to assess the overall impact of these efforts, even though mitigation was applied to only two types of language artefacts.

Four main performance comparisons were conducted to evaluate the effectiveness of artefact mitigation on full-context BERT-based models:

- On standard test set before and after mitigation.
- On the adversarial test set before and after mitigation.
- On supporting adversarial instances both before and after mitigation.
- On challenging adversarial instances both before and after mitigation.

This evaluation strategy aims to provide a thorough understanding of how data augmentation techniques affect model behavior in the presence and absence of language artefacts. The accuracy reported for each test is the average of three runs, each conducted with a different seed value (42, 66, and 5), along with the corresponding standard error.

Table 6.14: Performance comparison of ELECTRA Base on Adversarial Challenging and Supporting instances

Year	Feature	Challenging	Supporting
2018	None	0.3333 ± 0.04	0.7167 ± 0.07
	All	0.3030 ± 0.04	0.7167 ± 0.02
	Sentence Length	0.3485 ± 0.02	0.6167 ± 0.02
	Word Overlap	0.3939 ± 0.02	0.6167 ± 0.02
	Contradiction Word	0.3333 ± 0.04	0.7167 ± 0.07
	Subsequence Heuristic	0.3788 ± 0.02	0.6500 ± 0.03
2019	None	0.3485 ± 0.05	0.6667 ± 0.03
	All	0.1970 ± 0.04	0.7667 ± 0.04
	Sentence Length	0.3636 ± 0.05	0.6833 ± 0.07
	Word Overlap	0.3030 ± 0.06	0.6333 ± 0.03
	Contradiction Word	0.3485 ± 0.05	0.6667 ± 0.03
	Subsequence Heuristic	0.3333 ± 0.04	0.6667 ± 0.04
2020	None	0.3939 ± 0.02	0.7500 ± 0.09
	All	0.3788 ± 0.02	0.7667 ± 0.09
	Sentence Length	0.4091 ± 0.07	0.6500 ± 0.03
	Word Overlap	0.3939 ± 0.05	0.7167 ± 0.02
	Contradiction Word	0.3939 ± 0.02	0.7000 ± 0.05
	Subsequence Heuristic	0.3939 ± 0.02	0.7667 ± 0.07
2021	None	0.4242 ± 0.04	0.7500 ± 0.03
	All	0.3788 ± 0.04	0.7167 ± 0.09
	Sentence Length	0.3788 ± 0.03	0.7500 ± 0.03
	Word Overlap	0.3788 ± 0.02	0.6833 ± 0.02
	Contradiction Word	0.4242 ± 0.04	0.7500 ± 0.03
	Subsequence Heuristic	0.3636 ± 0.00	0.6833 ± 0.04
2022	None	0.4242 ± 0.04	0.7000 ± 0.00
	All	0.3788 ± 0.03	0.7500 ± 0.05
	Sentence Length	0.4394 ± 0.02	0.6333 ± 0.03
	Word Overlap	0.4091 ± 0.03	0.6167 ± 0.03
	Contradiction Word	0.4242 ± 0.04	0.7000 ± 0.00
	Subsequence Heuristic	0.4091 ± 0.05	0.7500 ± 0.08

6.3.1 Standard Test set

This section compares the performance of BERT-based models on the standard test set before and after mitigation. The detailed results are presented in Tables 6.21 to 6.25.

The results indicate that mitigation efforts had a varied impact on the models' performance on standard test sets. For BERT Base, there is a slight decrease in performance across most features after mitigation. This suggests that while the mitigation strategies effectively reduced the reliance on spurious artefacts, they might also have removed some genuinely useful patterns that the model had learned to exploit. The slight drop in performance might reflect a loss in the model's ability to leverage these patterns, leading to a more challenging learning environment.

LegalBERT and RoBERTa Base models exhibit a similar trend, though with some improvements in handling specific artefacts like contradiction words. The mixed results indicate that the mitigation strategies are not uniformly effective across different artefacts and

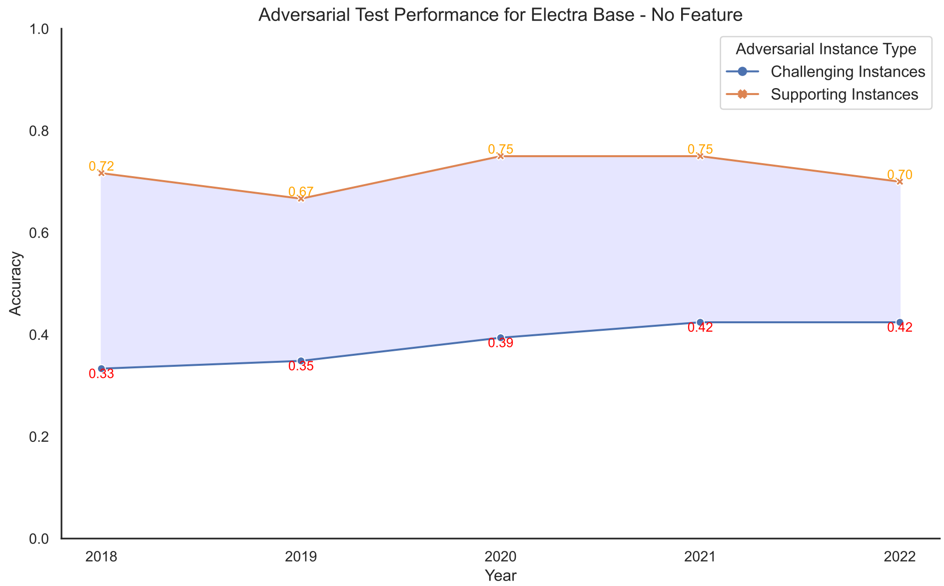

Figure 6.54: ELECTRA Base performance on Challenging and Supporting adversarial instances with no artefacts.

models. This highlights the complexity of fully addressing artefact reliance and suggests that further refinement of mitigation techniques is necessary.

In some cases, such as with the ELECTRA Base, there are improvements in handling word overlap artefacts. This indicates that the mitigation strategies were partially successful in addressing specific biases, even if the overall performance did not significantly improve. These insights underscore the need for targeted mitigation approaches tailored to the specific weaknesses of each model.

6.3.2 Adversarial Test set

Tables 6.26 to 6.30 present the performance of models on adversarial test sets before and after mitigation.

The results show a general improvement in model performance on adversarial test sets after mitigation. For instance, the BERT BASE exhibits a notable increase in performance with the 'ALL' feature configuration. This suggests that the mitigation efforts were effective in reducing the models' susceptibility to adversarial manipulations.

The LegalBERT and ELECTRA Base models also show similar trends, with improved performance on adversarial test sets post-mitigation. This improvement highlights the effectiveness of data augmentation techniques in enhancing the models' ability to generalize and handle more challenging inputs. The consistent gains across different models suggest that these mitigation strategies can be broadly applicable and beneficial.

However, it is important to note that the improvements are not uniform across all configurations and artefacts. Some configurations showed marginal gains or even slight declines in performance, indicating that while the mitigation strategies are beneficial, they are not a magic bullet. This variability underscores the need for continuous refinement and evaluation of mitigation techniques to achieve optimal robustness.

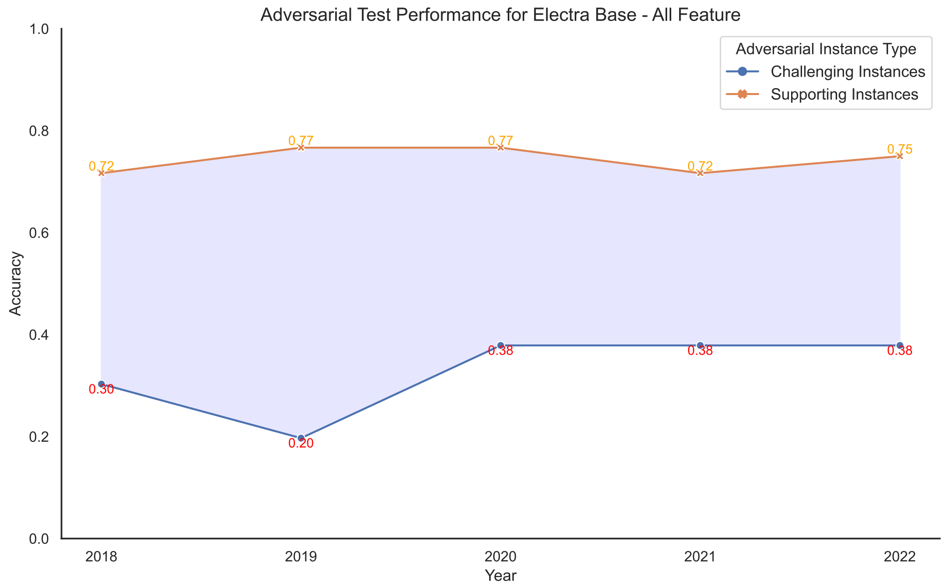

Figure 6.55: ELECTRA Base performance on Challenging and Supporting adversarial instances with all artefacts.

6.3.3 Adversarial Supporting Instances

Tables 6.31 to 6.35 detail the performance comparison on adversarial instances where artefacts were present.

The performance of models generally improved post-mitigation, though with some fluctuations. For example, the BERT BASE shows improved performance on instances with contradiction word artefacts, indicating a reduced reliance on these artefacts. This improvement suggests that the model has become more adept at handling inputs without over-relying on specific features, leading to more accurate predictions in adversarial scenarios.

However, some models, such as LegalBERT, showed inconsistent results, with performance improvements in some artefacts but not others. This inconsistency reflects the inherent difficulty in fully mitigating artefact reliance. The varying results highlight that while mitigation strategies can reduce artefact exploitation, achieving a complete and uniform mitigation across different types of artefacts remains challenging.

The nuanced performance changes indicate that different artefacts impact models in various ways, and mitigation strategies must be tailored accordingly. For instance, while word overlap mitigation might be effective for one model, it might not significantly impact another. These insights emphasize the need for a more granular approach to artefact mitigation, where strategies are customized based on the specific vulnerabilities of each model.

6.3.4 Adversarial Challenging Instances

The performance on adversarial instances designed to test models' resilience against non-artefact-based manipulations is shown in Tables 6.36 to 6.40.

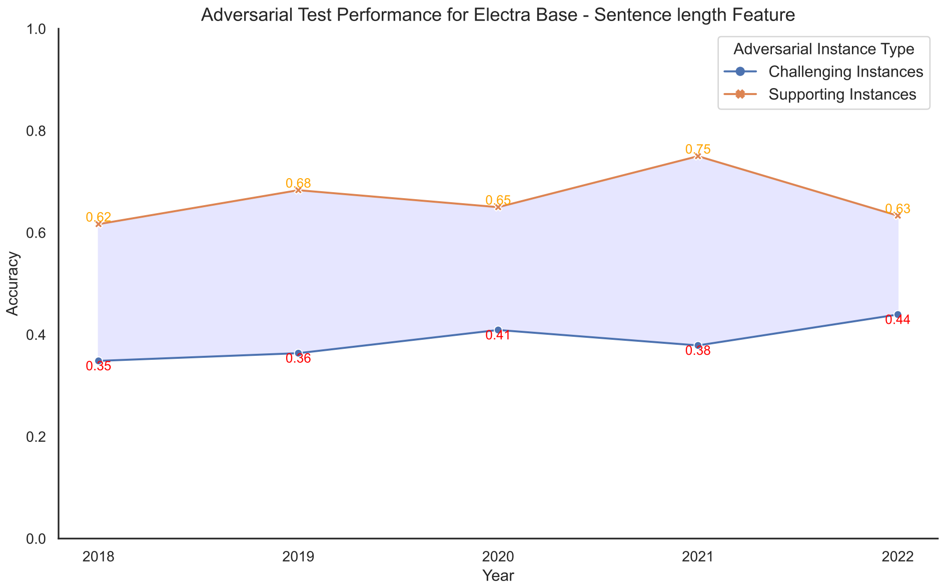

Figure 6.56: ELECTRA Base performance on Challenging and Supporting adversarial instances with sentence length artefact.

The results exhibit substantial improvement across models post-mitigation. For instance, the BERT Base model's performance increased significantly for the 'ALL' feature configuration. This substantial improvement highlights that the mitigation efforts not only reduced artefact reliance but also enhanced the models' general robustness.

This trend is observed across other models like RoBERTa Base and ELECTRA Base, further supporting the effectiveness of data augmentation techniques. The consistent improvements across various models and configurations indicate that these mitigation strategies are effective in promoting a more genuine understanding of the inputs, leading to better performance even in the absence of artefacts.

The positive results suggest that mitigation efforts help models become more adaptable and less reliant on specific patterns, which is essential for real-world applications where inputs can vary widely. These findings underscore the importance of developing and implementing robust mitigation strategies to enhance model generalization and reliability.

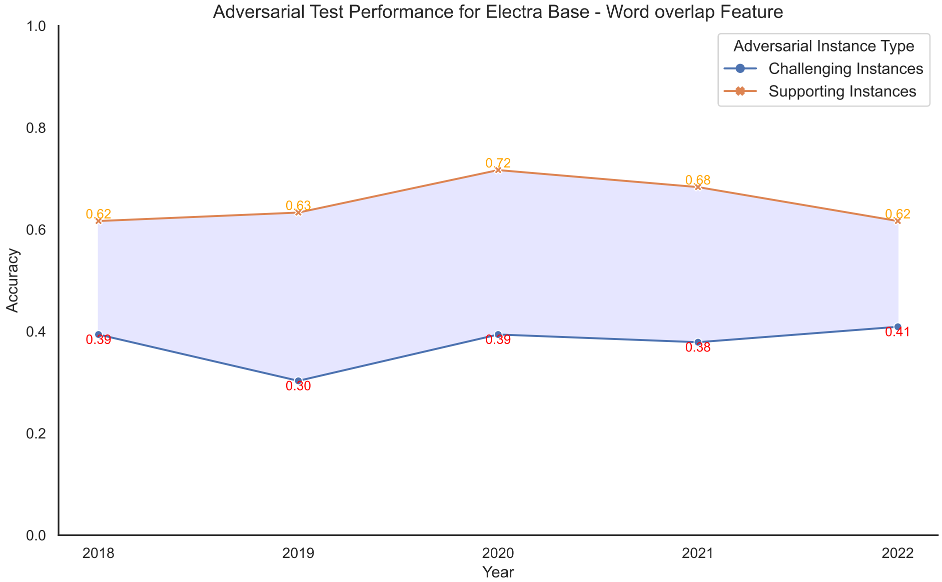

Figure 6.57: ELECTRA Base performance on Challenging and Supporting adversarial instances with word overlap artefact.

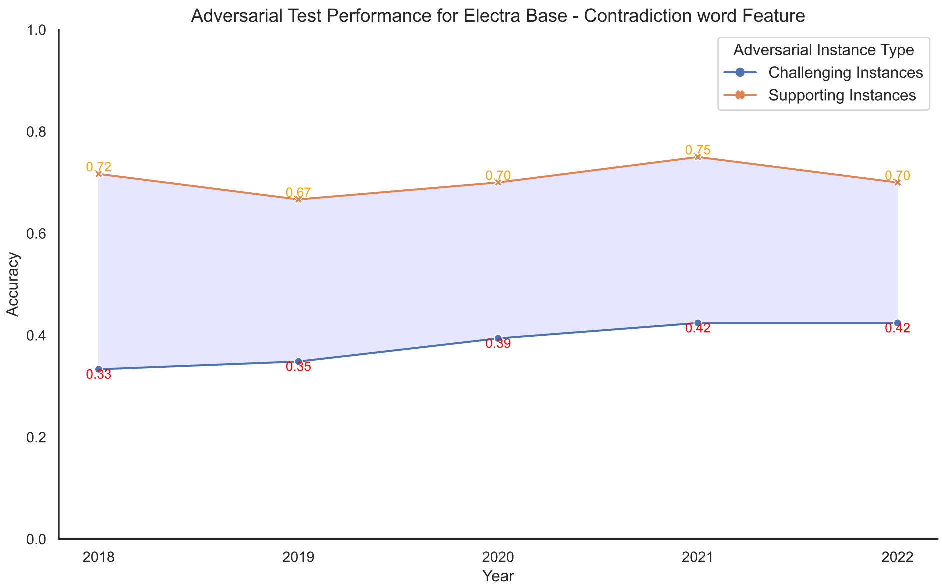

Figure 6.58: ELECTRA Base performance on Challenging and Supporting adversarial instances with contradiction word artefact.

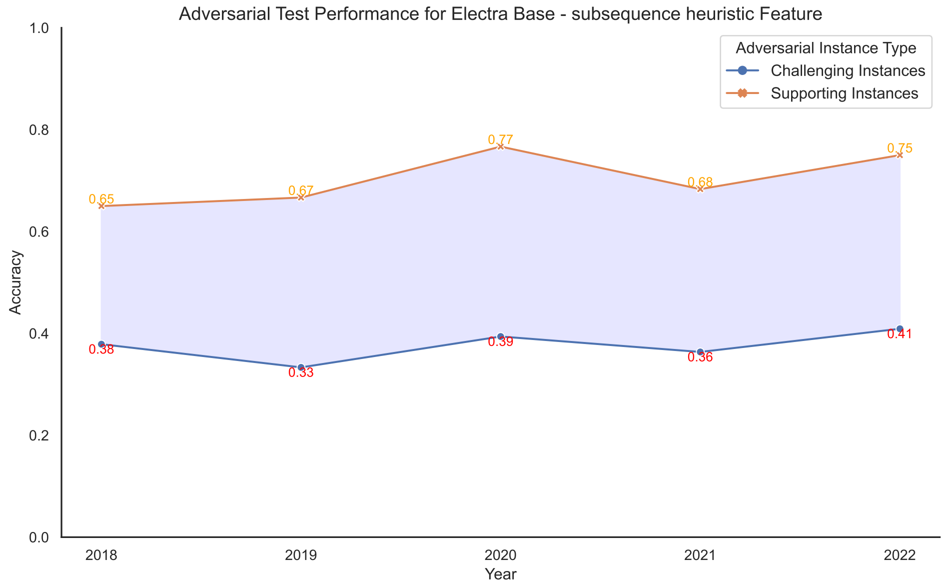

Figure 6.59: ELECTRA Base performance on Challenging and Supporting adversarial instances with subsequence heuristic artefact.

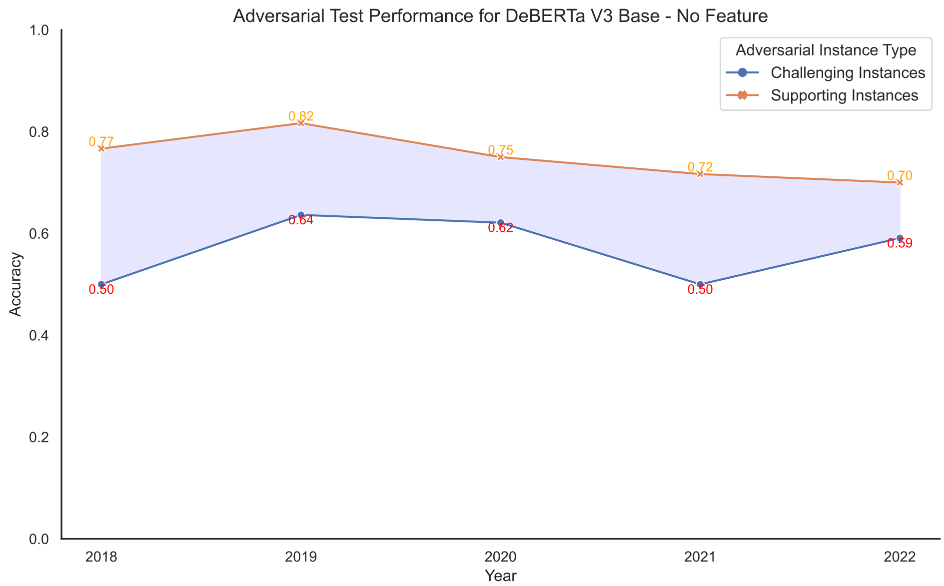

Figure 6.60: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with no artefacts.

Table 6.15: Performance comparison of DeBERTa Base on Adversarial Challenging and Supporting instances

Year	Feature	Challenging Instances	Supporting Instances
2018	None	0.5000 $\pm$ 0.04	0.7667 $\pm$ 0.01
	All	0.4394 $\pm$ 0.01	0.7333 $\pm$ 0.01
	Sentence Length	0.4091 $\pm$ 0.02	0.7333 $\pm$ 0.03
	Word Overlap	0.4697 $\pm$ 0.04	0.7667 $\pm$ 0.06
	Contradiction Word	0.4849 $\pm$ 0.06	0.6833 $\pm$ 0.04
	Subsequence Heuristic	0.4091 $\pm$ 0.07	0.8167 $\pm$ 0.04
2019	None	0.6364 $\pm$ 0.04	0.8167 $\pm$ 0.06
	All	0.5152 $\pm$ 0.03	0.7667 $\pm$ 0.01
	Sentence Length	0.5303 $\pm$ 0.05	0.7333 $\pm$ 0.01
	Word Overlap	0.6515 $\pm$ 0.04	0.7000 $\pm$ 0.05
	Contradiction Word	0.6060 $\pm$ 0.05	0.8667 $\pm$ 0.03
	Subsequence Heuristic	0.5757 $\pm$ 0.04	0.7333 $\pm$ 0.03
2020	None	0.6212 $\pm$ 0.05	0.7500 $\pm$ 0
	All	0.5151 $\pm$ 0.04	0.7833 $\pm$ 0.07
	Sentence Length	0.5909 $\pm$ 0.04	0.8000 $\pm$ 0.05
	Word Overlap	0.5606 $\pm$ 0.06	0.7500 $\pm$ 0.07
	Contradiction Word	0.5757 $\pm$ 0.05	0.8000 $\pm$ 0.05
	Subsequence Heuristic	0.5455 $\pm$ 0.02	0.7667 $\pm$ 0.07
2021	None	0.5000 $\pm$ 0.02	0.7167 $\pm$ 0.01
	All	0.5151 $\pm$ 0.08	0.7500 $\pm$ 0.05
	Sentence Length	0.6363 $\pm$ 0	0.7000 $\pm$ 0.07
	Word Overlap	0.6212 $\pm$ 0.03	0.7000 $\pm$ 0.05
	Contradiction Word	0.6212 $\pm$ 0.05	0.6667 $\pm$ 0.04
	Subsequence Heuristic	0.6515 $\pm$ 0.01	0.7833 $\pm$ 0.06
2022	None	0.5909 $\pm$ 0	0.7000 $\pm$ 0.02
	All	0.5606 $\pm$ 0.03	0.7667 $\pm$ 0.06
	Sentence Length	0.6061 $\pm$ 0.08	0.6500 $\pm$ 0.02
	Word Overlap	0.6515 $\pm$ 0.04	0.6333 $\pm$ 0.04
	Contradiction Word	0.5455 $\pm$ 0.02	0.6500 $\pm$ 0.05
	Subsequence Heuristic	0.6061 $\pm$ 0.01	0.8167 $\pm$ 0.01

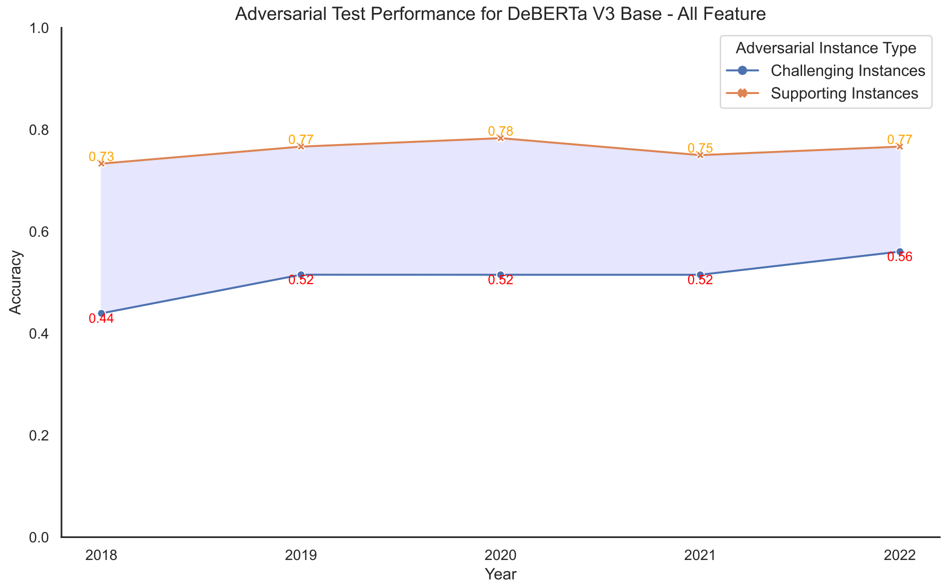

Figure 6.61: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with all artefacts.

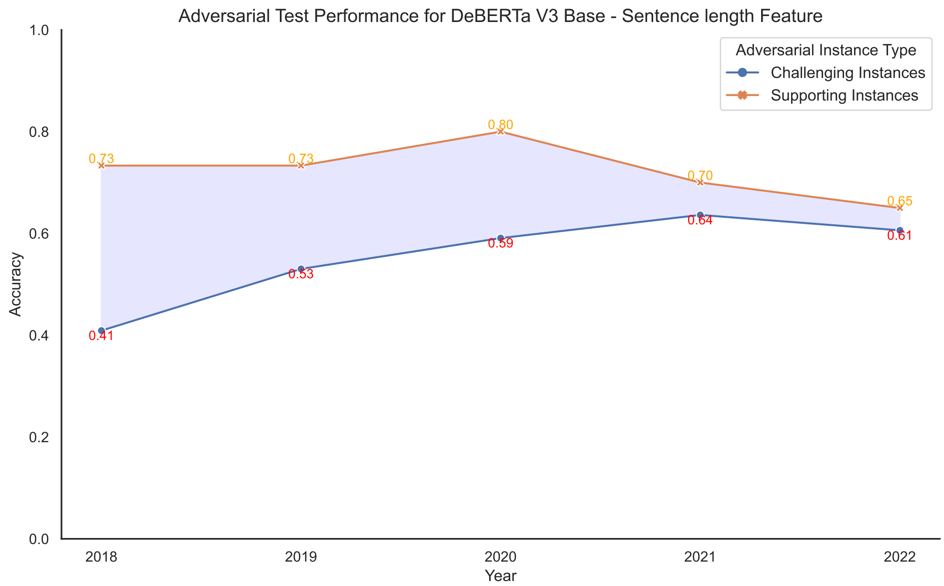

Figure 6.62: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with sentence length artefact.

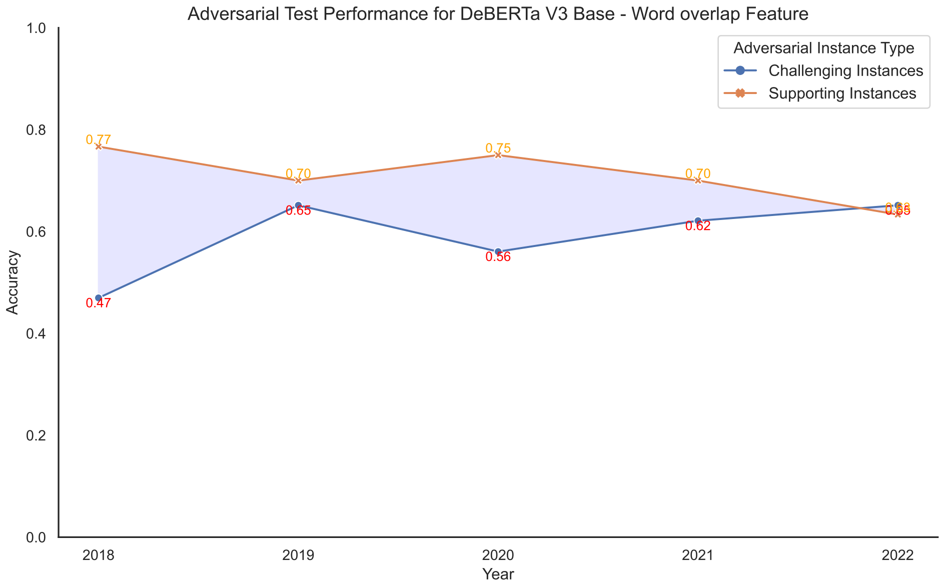

Figure 6.63: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with word overlap artefact.

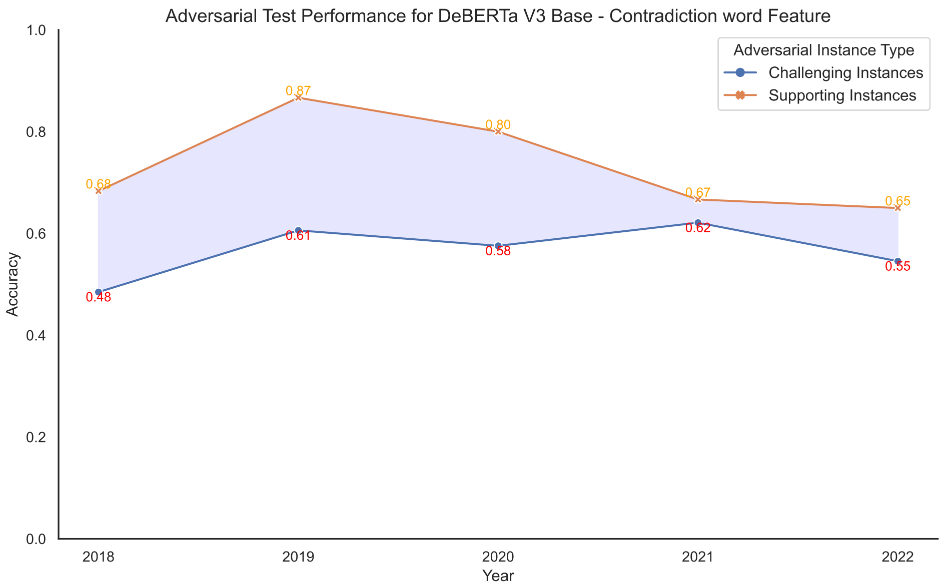

Figure 6.64: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with contradiction word artefact.

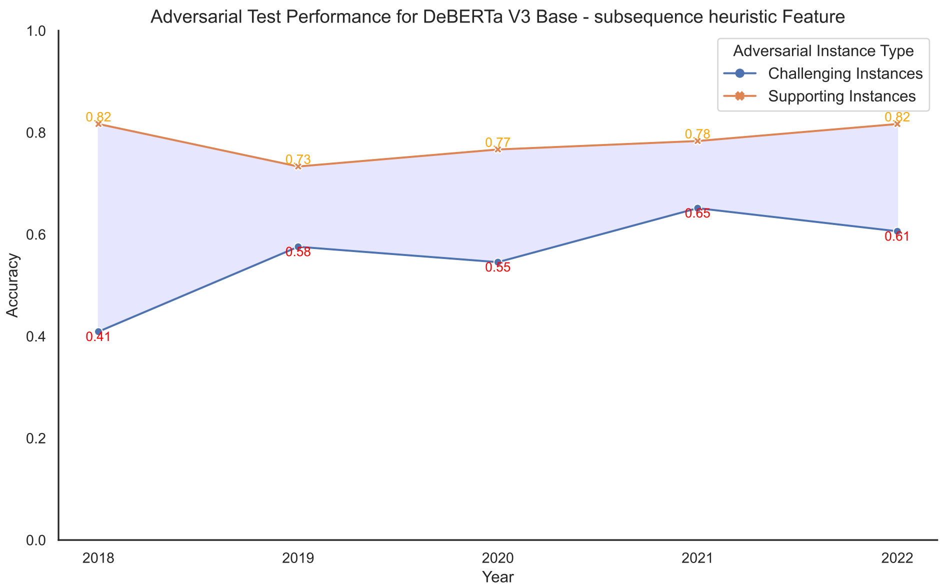

Figure 6.65: DeBERTa V3 Base performance on Challenging and Supporting adversarial instances with subsequence heuristic artefact.

Table 6.16: Performance comparison of BERT Base Hypothesis-only vs Majority classifier Baseline

Feature	2018	2019	2020	2021	2022
Majority Baseline	0.4655 ± 0	0.5143 ± 0	0.5315 ± 0	0.4691 ± 0	0.5321 ± 0
None	0.5862 ± 0.02	0.5141 ± 0.03	0.5285 ± 0	0.5596 ± 0.02	0.5688 ± 0.02
All	0.5402 ± 0.02	0.5571 ± 0.02	0.5946 ± 0.01	0.5308 ± 0.03	0.5535 ± 0.02
Sentence length	0.5230 ± 0.02	0.5667 ± 0.02	0.5706 ± 0	0.5021 ± 0.02	0.5443 ± 0.02
Word Overlap	0.5517 ± 0.02	0.5714 ± 0.01	0.5736 ± 0.01	0.5432 ± 0.01	0.5413 ± 0.01
Contradiction word	0.5862 ± 0.02	0.5143 ± 0.03	0.5345 ± 0	0.5596 ± 0.02	0.5688 ± 0.02
Subsequence Heuristic	0.5345 ± 0.03	0.5762 ± 0	0.5465 ± 0.02	0.5597 ± 0.02	0.5321 ± 0.01

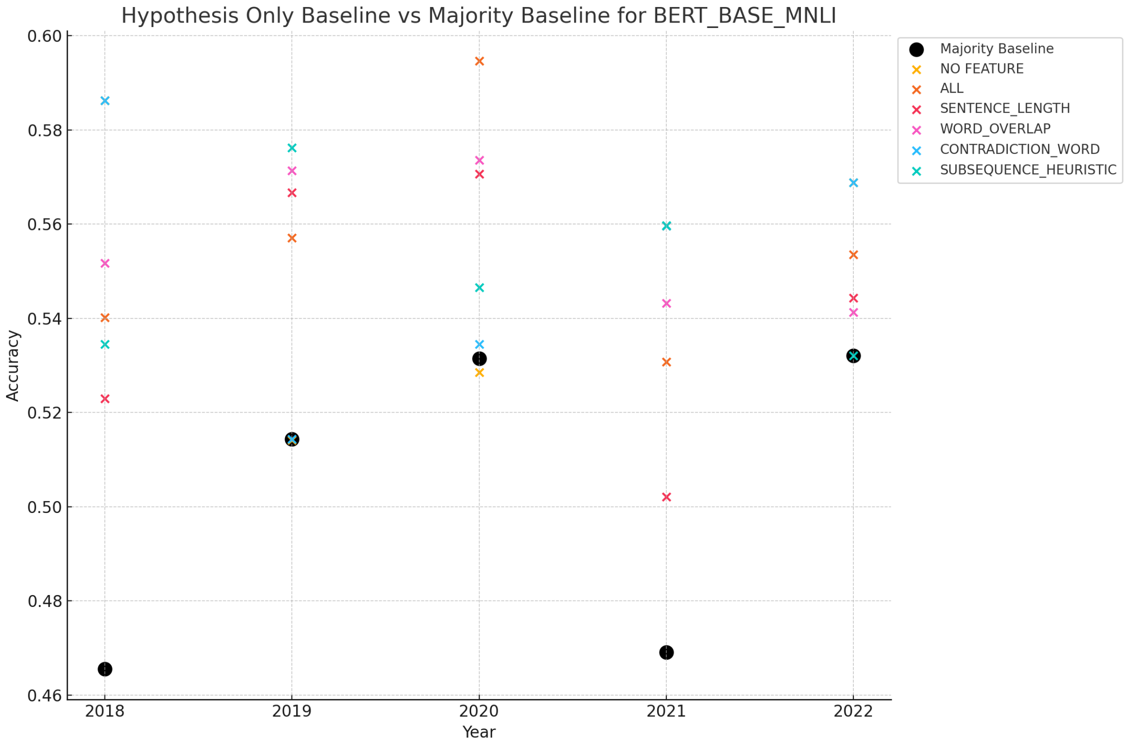

Figure 6.66: Performance comparison of BERT Base Hypothesis-only vs Majority classifier Baseline over multiple years on different model variants

Table 6.17: Performance comparison of LegalBERT Hypothesis-only vs Majority classifier Baseline

Feature	2018	2019	2020	2021	2022
Majority Baseline	0.4655 $\pm$ 0	0.5143 $\pm$ 0	0.5315 $\pm$ 0	0.4691 $\pm$ 0	0.5321 $\pm$ 0
None	0.5115 $\pm$ 0.02	0.5333 $\pm$ 0.03	0.5255 $\pm$ 0.53	0.5267 $\pm$ 0.03	0.5382 $\pm$ 0.02
All	0.5287 $\pm$ 0.01	0.5286 $\pm$ 0.03	0.5585 $\pm$ 0.56	0.5350 $\pm$ 0.03	0.5474 $\pm$ 0.03
Sentence length	0.5058 $\pm$ 0.04	0.5809 $\pm$ 0.01	0.5435 $\pm$ 0.54	0.5885 $\pm$ 0.02	0.5535 $\pm$ 0.05
Word overlap	0.5172 $\pm$ 0.01	0.5190 $\pm$ 0.03	0.5285 $\pm$ 0.53	0.5226 $\pm$ 0.03	0.4954 $\pm$ 0.03
Contradiction word	0.5000 $\pm$ 0.02	0.5333 $\pm$ 0.03	0.3634 $\pm$ 0.36	0.5267 $\pm$ 0.03	0.5382 $\pm$ 0.02
Subsequence Heuristic	0.5345 $\pm$ 0	0.5474 $\pm$ 0.02	0.4925 $\pm$ 0.49	0.5432 $\pm$ 0.04	0.5414 $\pm$ 0.02

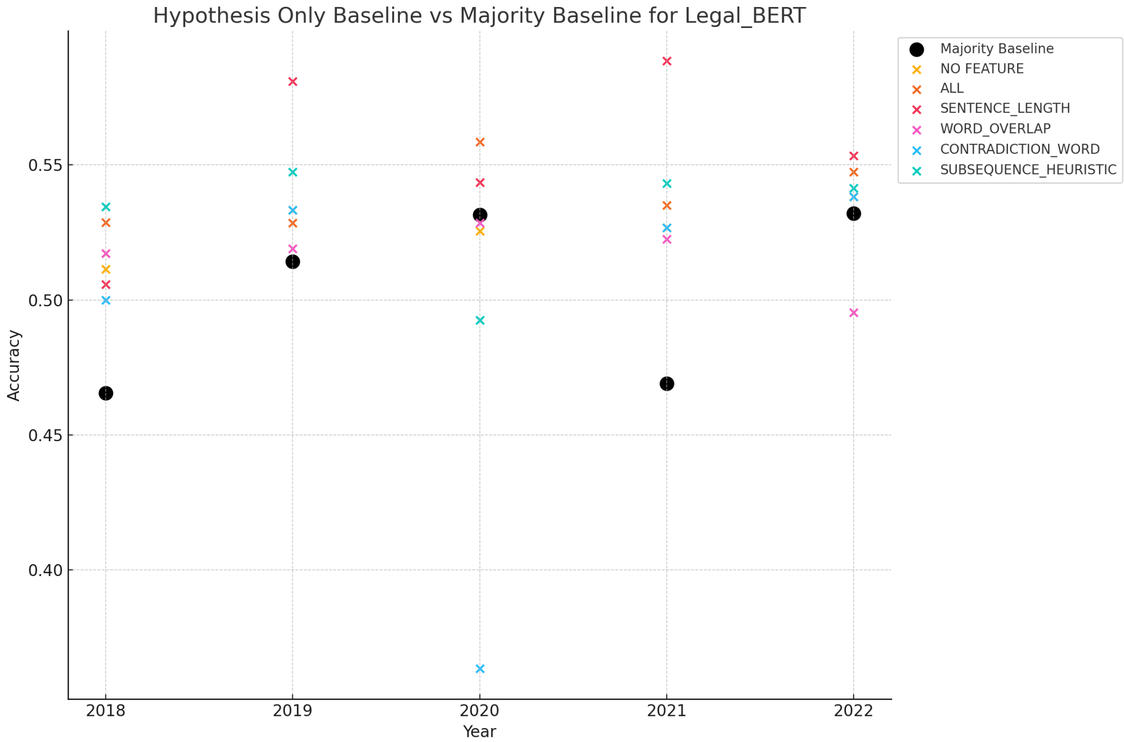

Figure 6.67: Performance comparison of LegalBERT Hypothesis-only vs Majority classifier Baseline over multiple years on different model variants

Table 6.18: Performance comparison of RoBERTa Base Hypothesis-only vs Majority classifier Baseline

Feature	2018	2019	2020	2021	2022
Majority Baseline	0.4655 $\pm$ 0	0.5143 $\pm$ 0	0.5315 $\pm$ 0	0.4691 $\pm$ 0	0.5321 $\pm$ 0
None	0.5765 $\pm$ 0.03	0.6000 $\pm$ 0.01	0.5075 $\pm$ 0.01	0.4938 $\pm$ 0.01	0.5870 $\pm$ 0.02
All	0.5390 $\pm$ 0.02	0.5952 $\pm$ 0.01	0.4955 $\pm$ 0.01	0.5556 $\pm$ 0.01	0.5688 $\pm$ 0.06
Sentence length	0.5742 $\pm$ 0.01	0.5714 $\pm$ 0.02	0.5285 $\pm$ 0.03	0.5103 $\pm$ 0.04	0.6422 $\pm$ 0.01
Word Overlap	0.4890 $\pm$ 0.04	0.5428 $\pm$ 0.01	0.4565 $\pm$ 0.02	0.5062 $\pm$ 0.03	0.5474 $\pm$ 0.02
Contradiction word	0.5765 $\pm$ 0.03	0.5952 $\pm$ 0.01	0.5075 $\pm$ 0.01	0.4815 $\pm$ 0.01	0.5872 $\pm$ 0.02
Subsequence Heuristic	0.5300 $\pm$ 0.01	0.5857 $\pm$ 0.02	0.4775 $\pm$ 0.03	0.5185 $\pm$ 0.03	0.6086 $\pm$ 0.02

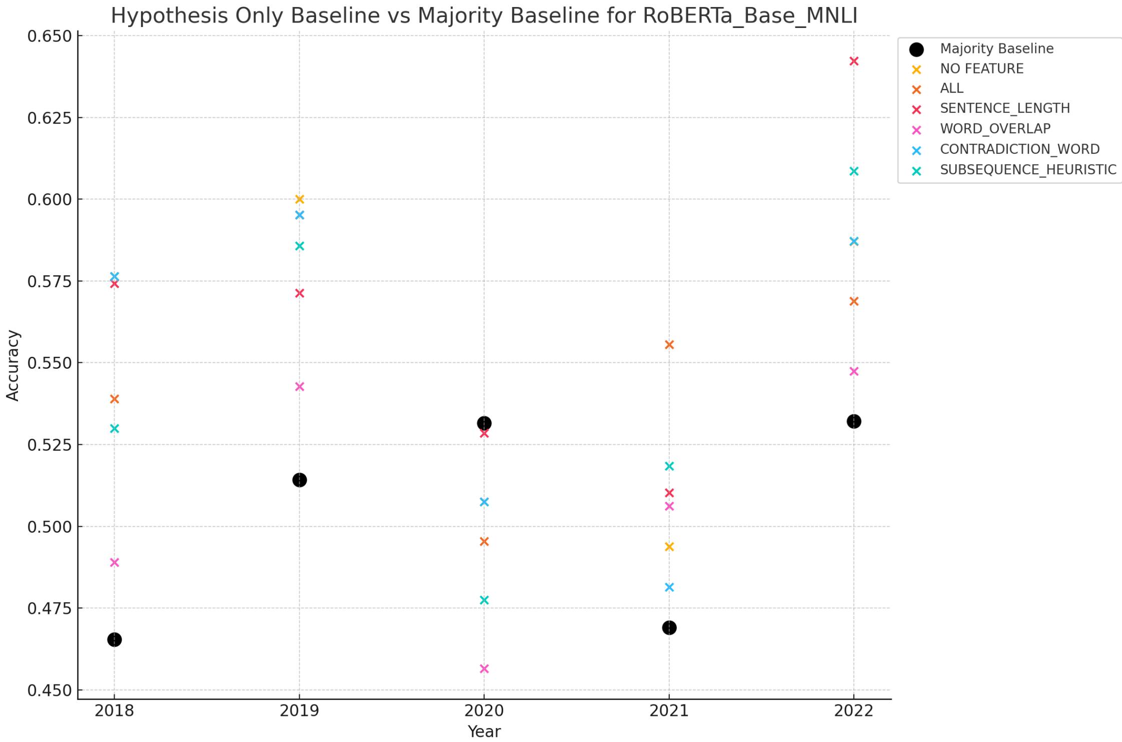

Figure 6.68: Performance comparison of RoBERTa Base Hypothesis-only vs Majority classifier Baseline over multiple years on different model variants

Table 6.19: Performance comparison of ELECTRA Base Hypothesis-only vs Majority classifier Baseline

Feature	2018	2019	2020	2021	2022
Majority Baseline	0.4655 $\pm$ 0	0.5143 $\pm$ 0	0.5315 $\pm$ 0	0.4691 $\pm$ 0	0.5321 $\pm$ 0
None	0.5747 $\pm$ 0.01	0.5905 $\pm$ 0.03	0.4865 $\pm$ 0.02	0.4814 $\pm$ 0.02	0.3730 $\pm$ 0.02
All	0.5632 $\pm$ 0.01	0.5619 $\pm$ 0.04	0.4925 $\pm$ 0.04	0.5679 $\pm$ 0.01	0.4286 $\pm$ 0.02
Sentence length	0.5517 $\pm$ 0.01	0.5762 $\pm$ 0.04	0.5045 $\pm$ 0.03	0.5412 $\pm$ 0.03	0.3412 $\pm$ 0.01
Word Overlap	0.5804 $\pm$ 0.02	0.5810 $\pm$ 0.04	0.4805 $\pm$ 0.01	0.5309 $\pm$ 0.01	0.3571 $\pm$ 0.02
Contradiction word	0.5747 $\pm$ 0.01	0.5905 $\pm$ 0.03	0.4865 $\pm$ 0.02	0.4815 $\pm$ 0.02	0.4048 $\pm$ 0.01
Subsequence Heuristic	0.5632 $\pm$ 0.01	0.5714 $\pm$ 0.01	0.5045 $\pm$ 0.03	0.4938 $\pm$ 0.02	0.3412 $\pm$ 0.01

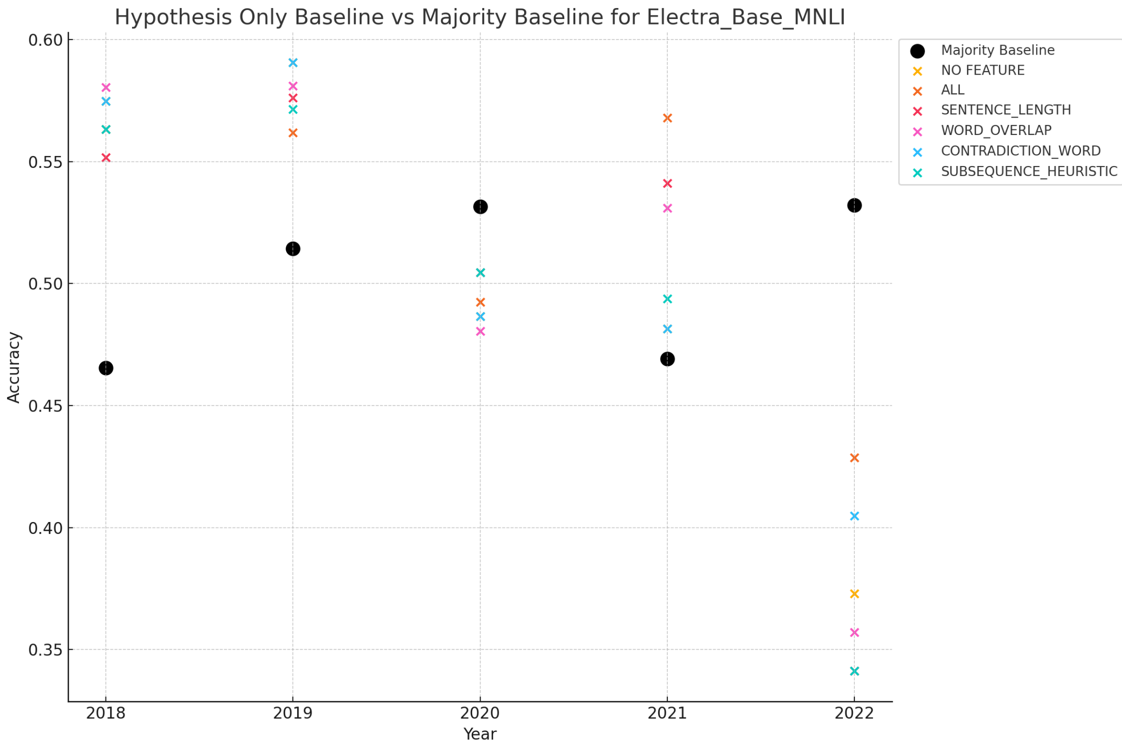

Figure 6.69: Performance comparison of ELECTRA Base Hypothesis-only vs Majority classifier Baseline over multiple years on different model variants

Table 6.20: Performance comparison of DeBERTa base Hypothesis-only vs Majority classifier Baseline

Feature	**2018**	**2019**	**2020**	**2021**	**2022**
Majority Baseline	0.4655 ± 0	0.5143 ± 0	0.5315 ± 0	0.4691 ± 0	0.5321 ± 0
None	0.5632 ± 0.05	0.5381 ± 0.05	0.4805 ± 0.01	0.4897 ± 0.01	0.5260 ± 0.03
All	0.5402 ± 0.04	0.6238 ± 0.01	0.5255 ± 0.02	0.4938 ± 0.01	0.5963 ± 0.02
Sentence length	0.5402 ± 0.01	0.5333 ± 0.04	0.4775 ± 0.02	0.5103 ± 0.03	0.5841 ± 0.02
Word overlap	0.5459 ± 0.03	0.5857 ± 0.01	0.4625 ± 0.00	0.5227 ± 0.02	0.5994 ± 0.03
Contradiction word	0.5632 ± 0.05	0.5381 ± 0.05	0.4805 ± 0.01	0.4897 ± 0.01	0.5260 ± 0.03
Subsequence Heuristic	0.4942 ± 0.04	0.5381 ± 0.02	0.5075 ± 0.02	0.4856 ± 0.02	0.5413 ± 0.01

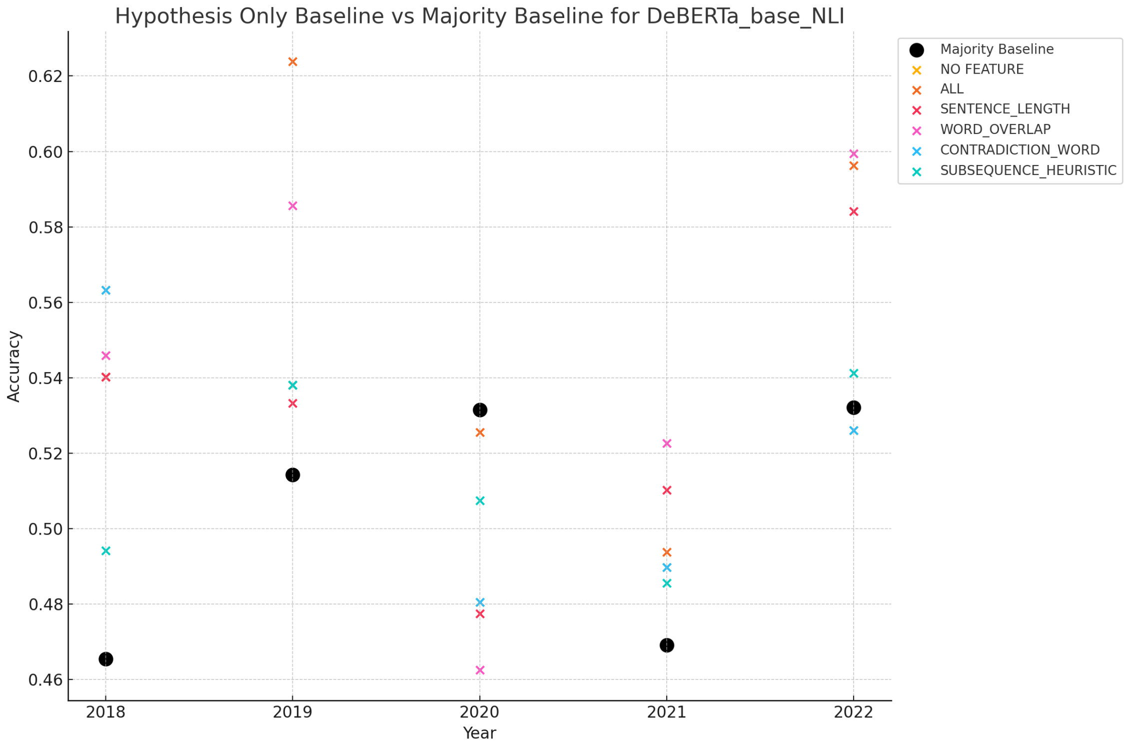

Figure 6.70: Performance comparison of DeBERTa Base Hypothesis-only vs Majority classifier Baseline over multiple years on different model variants

Table 6.21: Performance comparison of BERT Base on Standard Test Set before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6322 ± 0.03	0.6034 ± 0.03	0.6379 ± 0.03	0.6149 ± 0.03	0.6322 ± 0.03	0.6207 ± 0.03
	After	0.5977 ± 0.01	0.5977 ± 0.02	0.5747 ± 0.06	0.5804 ± 0.04	0.5977 ± 0.01	0.5862 ± 0.03
2019	Before	0.5524 ± 0.01	0.5333 ± 0.03	0.5714 ± 0.01	0.5762 ± 0.03	0.5476 ± 0.02	0.5667 ± 0.02
	After	0.5333 ± 0.03	0.5143 ± 0.02	0.5762 ± 0.03	0.5571 ± 0.01	0.5333 ± 0.03	0.4953 ± 0.02
2020	Before	0.5882 ± 0.04	0.5705 ± 0.06	0.5555 ± 0.04	0.5706 ± 0.04	0.5555 ± 0.04	0.5916 ± 0.07
	After	0.5315 ± 0.02	0.5315 ± 0.02	0.5766 ± 0.02	0.5285 ± 0.01	0.5315 ± 0.02	0.5225 ± 0.01
2021	Before	0.5732 ± 0.03	0.5679 ± 0.02	0.5309 ± 0.03	0.5720 ± 0.02	0.5843 ± 0.03	0.5597 ± 0.03
	After	0.5761 ± 0.03	0.4979 ± 0.03	0.4815 ± 0.02	0.5926 ± 0.04	0.5761 ± 0.03	0.5597 ± 0.06
2022	Before	0.5932 ± 0.01	0.5688 ± 0.03	0.6208 ± 0.02	0.5963 ± 0.03	0.5932 ± 0.01	0.5963 ± 0.04
	After	0.5810 ± 0.01	0.5810 ± 0.04	0.5596 ± 0.03	0.6025 ± 0.01	0.5810 ± 0.01	0.5810 ± 0.02

Table 6.22: Performance comparison of LegalBERT on Standard Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.5632 ± 0.02	0.5287 ± 0.04	0.5689 ± 0.01	0.4885 ± 0.04	0.5632 ± 0.02	0.5345 ± 0.00
	After	0.5287 ± 0.04	0.5057 ± 0.02	0.5115 ± 0.01	0.5402 ± 0.02	0.5287 ± 0.04	0.5459 ± 0.03
2019	Before	0.5190 ± 0.02	0.5048 ± 0.00	0.5218 ± 0.03	0.4619 ± 0.04	0.5190 ± 0.02	0.5191 ± 0.01
	After	0.5476 ± 0.00	0.5476 ± 0.03	0.4476 ± 0.03	0.5000 ± 0.03	0.5476 ± 0.00	0.5143 ± 0.03
2020	Before	0.5015 ± 0.01	0.4895 ± 0.01	0.4865 ± 0.00	0.5195 ± 0.03	0.4955 ± 0.00	0.4835 ± 0.02
	After	0.5015 ± 0.03	0.5285 ± 0.01	0.5225 ± 0.03	0.5225 ± 0.01	0.5015 ± 0.03	0.5255 ± 0.03
2021	Before	0.5720 ± 0.00	0.5432 ± 0.04	0.5144 ± 0.03	0.5515 ± 0.01	0.5720 ± 0.00	0.4815 ± 0.04
	After	0.5432 ± 0.04	0.4691 ± 0.00	0.4691 ± 0.00	0.4691 ± 0.00	0.5432 ± 0.04	0.4979 ± 0.06
2022	Before	0.5596 ± 0.06	0.6300 ± 0.02	0.6208 ± 0.01	0.6238 ± 0.02	0.5596 ± 0.06	0.5077 ± 0.04
	After	0.6055 ± 0.01	0.6178 ± 0.03	0.5719 ± 0.01	0.6147 ± 0.05	0.6055 ± 0.01	0.5504 ± 0.03

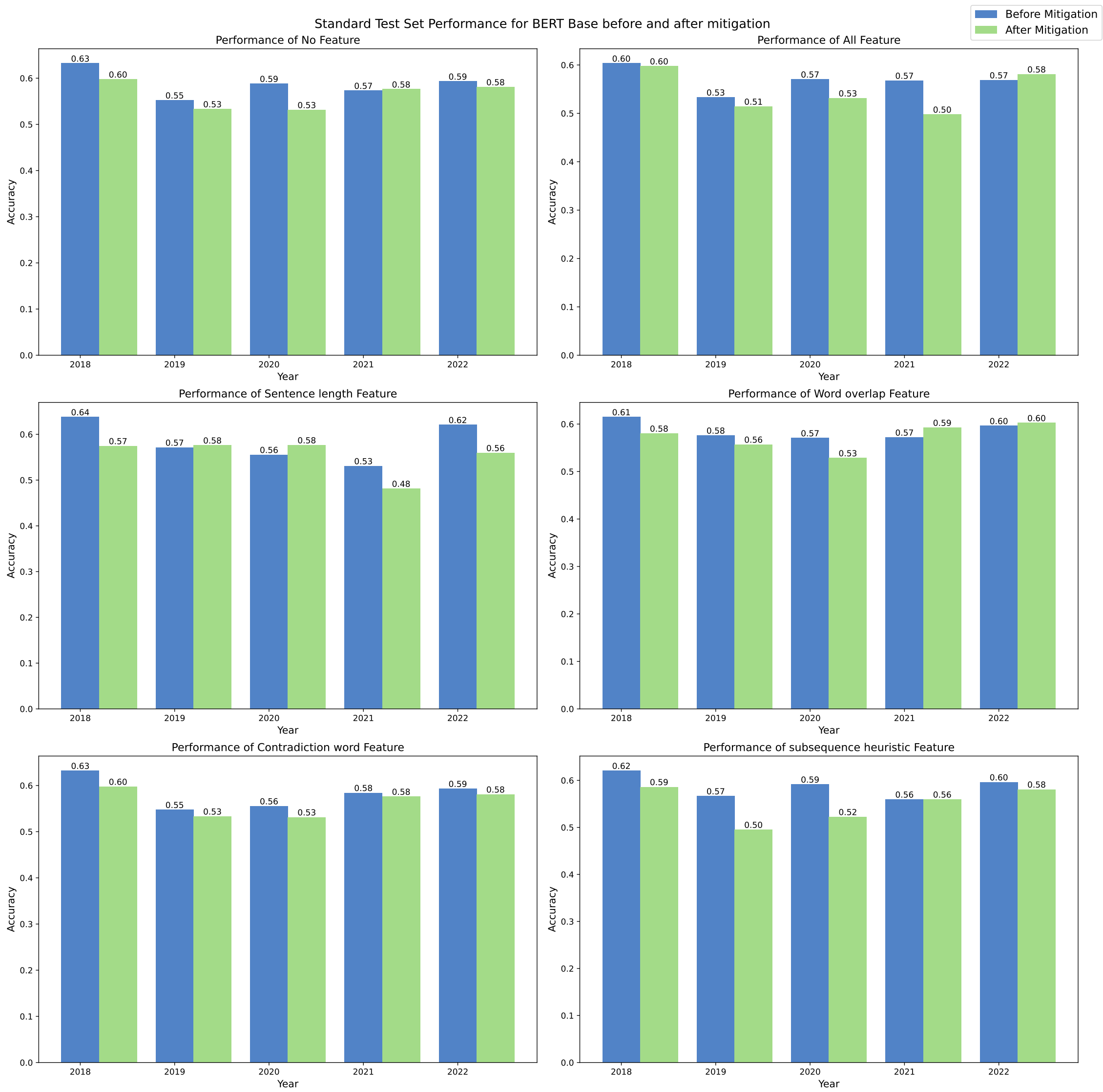

Figure 6.71: Performance comparison of BERT Base on Standard Test Set Before and After Mitigation

Table 6.23: Performance comparison of RoBERTa Base on Standard Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6025 ± 0.01	0.5738 ± 0.03	0.5738 ± 0.01	0.5632 ± 0.02	0.5882 ± 0.04	0.6025 ± 0.01
	After	0.5690 ± 0.04	0.5402 ± 0.02	0.5115 ± 0.01	0.5402 ± 0.02	0.5596 ± 0.06	0.5690 ± 0.04
2019	Before	0.5476 ± 0.02	0.5476 ± 0.02	0.5476 ± 0.00	0.5619 ± 0.04	0.5476 ± 0.02	0.5191 ± 0.01
	After	0.5571 ± 0.01	0.5571 ± 0.01	0.5476 ± 0.00	0.5571 ± 0.01	0.5476 ± 0.00	0.5476 ± 0.00
2020	Before	0.5135 ± 0.04	0.5135 ± 0.04	0.4925 ± 0.03	0.5175 ± 0.04	0.4895 ± 0.01	0.4895 ± 0.01
	After	0.5150 ± 0.03	0.5071 ± 0.03	0.5015 ± 0.03	0.5285 ± 0.01	0.5015 ± 0.03	0.5071 ± 0.03
2021	Before	0.5294 ± 0.04	0.5144 ± 0.03	0.4865 ± 0.01	0.5285 ± 0.01	0.4914 ± 0.03	0.4815 ± 0.04
	After	0.5371 ± 0.01	0.5144 ± 0.03	0.5071 ± 0.03	0.5225 ± 0.01	0.5071 ± 0.03	0.4979 ± 0.06
2022	Before	0.5596 ± 0.05	0.5110 ± 0.04	0.5037 ± 0.04	0.5486 ± 0.04	0.4856 ± 0.05	0.4920 ± 0.04
	After	0.5556 ± 0.01	0.5225 ± 0.01	0.5077 ± 0.04	0.5077 ± 0.04	0.5077 ± 0.04	0.5077 ± 0.04

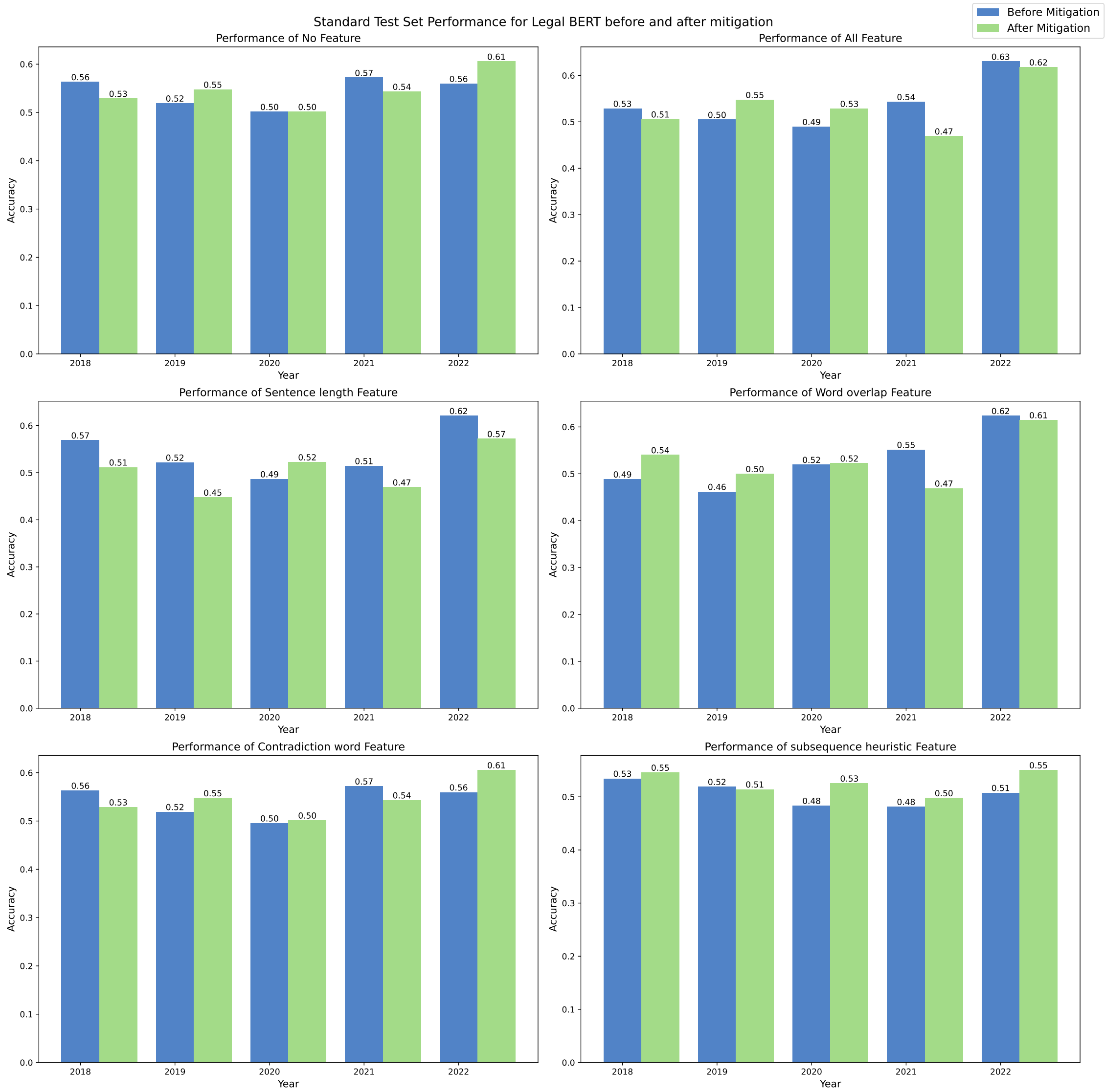

Figure 6.72: Performance comparison of LegalBERT on Standard Test Set Before and After Mitigation

Table 6.24: Performance comparison of ELECTRA Base on Standard Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6379 ± 0.02	0.6437 ± 0.01	0.6494 ± 0.03	0.6092 ± 0.01	0.6379 ± 0.02	0.6264 ± 0.03
	After	0.6667 ± 0.02	0.6839 ± 0.01	0.7011 ± 0.02	0.7241 ± 0.03	0.6667 ± 0.02	0.7069 ± 0.00
2019	Before	0.6000 ± 0.01	0.5952 ± 0.01	0.5714 ± 0.01	0.5905 ± 0.01	0.6000 ± 0.01	0.5714 ± 0.01
	After	0.6334 ± 0.00	0.6428 ± 0.03	0.6000 ± 0.01	0.6191 ± 0.01	0.6334 ± 0.00	0.5952 ± 0.02
2020	Before	0.6186 ± 0.02	0.6216 ± 0.01	0.6246 ± 0.01	0.5916 ± 0.02	0.6186 ± 0.02	0.6096 ± 0.02
	After	0.6366 ± 0.02	0.6006 ± 0.01	0.6216 ± 0.00	0.6096 ± 0.02	0.6366 ± 0.02	0.6336 ± 0.01
2021	Before	0.6543 ± 0.01	0.6749 ± 0.03	0.6996 ± 0.02	0.7078 ± 0.01	0.6543 ± 0.01	0.6831 ± 0.02
	After	0.7243 ± 0.04	0.6913 ± 0.03	0.6955 ± 0.05	0.7490 ± 0.01	0.7243 ± 0.04	0.7201 ± 0.04
2022	Before	0.6422 ± 0.02	0.6667 ± 0.01	0.6728 ± 0.01	0.6697 ± 0.01	0.6422 ± 0.02	0.6820 ± 0.01
	After	0.6361 ± 0.02	0.6453 ± 0.02	0.6758 ± 0.01	0.6697 ± 0.01	0.6361 ± 0.02	0.6269 ± 0.01

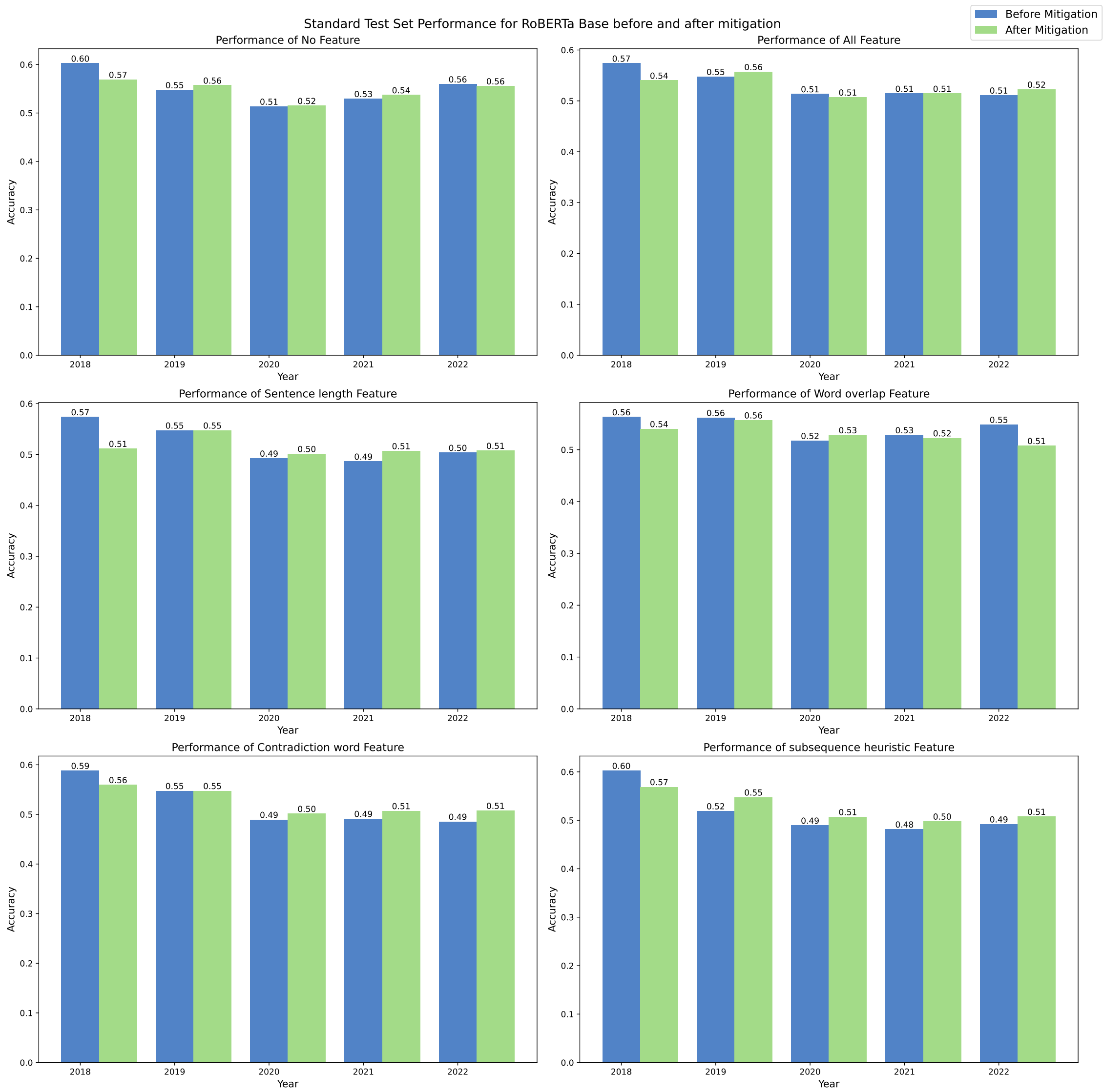

Figure 6.73: Performance comparison of RoBERTa Base on Standard Test Set Before and After Mitigation

Table 6.25: Performance comparison of DeBERTa Base on Standard Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6207 $\pm$ 0.01	0.5932 $\pm$ 0.01	0.5738 $\pm$ 0.01	0.5632 $\pm$ 0.02	0.5882 $\pm$ 0.04	0.6025 $\pm$ 0.01
	After	0.5804 $\pm$ 0.04	0.5556 $\pm$ 0.01	0.5402 $\pm$ 0.02	0.5402 $\pm$ 0.02	0.5596 $\pm$ 0.06	0.5690 $\pm$ 0.04
2019	Before	0.5714 $\pm$ 0.01	0.5476 $\pm$ 0.00	0.5714 $\pm$ 0.01	0.5619 $\pm$ 0.04	0.5476 $\pm$ 0.02	0.5191 $\pm$ 0.01
	After	0.5619 $\pm$ 0.04	0.5619 $\pm$ 0.04	0.5476 $\pm$ 0.00	0.5571 $\pm$ 0.01	0.5476 $\pm$ 0.00	0.5476 $\pm$ 0.00
2020	Before	0.5432 $\pm$ 0.04	0.5175 $\pm$ 0.04	0.5135 $\pm$ 0.04	0.5175 $\pm$ 0.04	0.4895 $\pm$ 0.01	0.4895 $\pm$ 0.01
	After	0.5285 $\pm$ 0.01	0.5015 $\pm$ 0.03	0.5225 $\pm$ 0.01	0.5285 $\pm$ 0.01	0.5015 $\pm$ 0.03	0.5071 $\pm$ 0.03
2021	Before	0.5144 $\pm$ 0.03	0.5432 $\pm$ 0.04	0.5015 $\pm$ 0.03	0.5285 $\pm$ 0.01	0.4914 $\pm$ 0.03	0.4815 $\pm$ 0.04
	After	0.5225 $\pm$ 0.01	0.4691 $\pm$ 0.00	0.5144 $\pm$ 0.03	0.5225 $\pm$ 0.01	0.5071 $\pm$ 0.03	0.4979 $\pm$ 0.06
2022	Before	0.5432 $\pm$ 0.04	0.5810 $\pm$ 0.01	0.4715 $\pm$ 0.04	0.5486 $\pm$ 0.04	0.4856 $\pm$ 0.05	0.4920 $\pm$ 0.04
	After	0.5077 $\pm$ 0.04	0.5077 $\pm$ 0.04	0.5077 $\pm$ 0.04	0.5077 $\pm$ 0.04	0.5077 $\pm$ 0.04	0.5077 $\pm$ 0.04

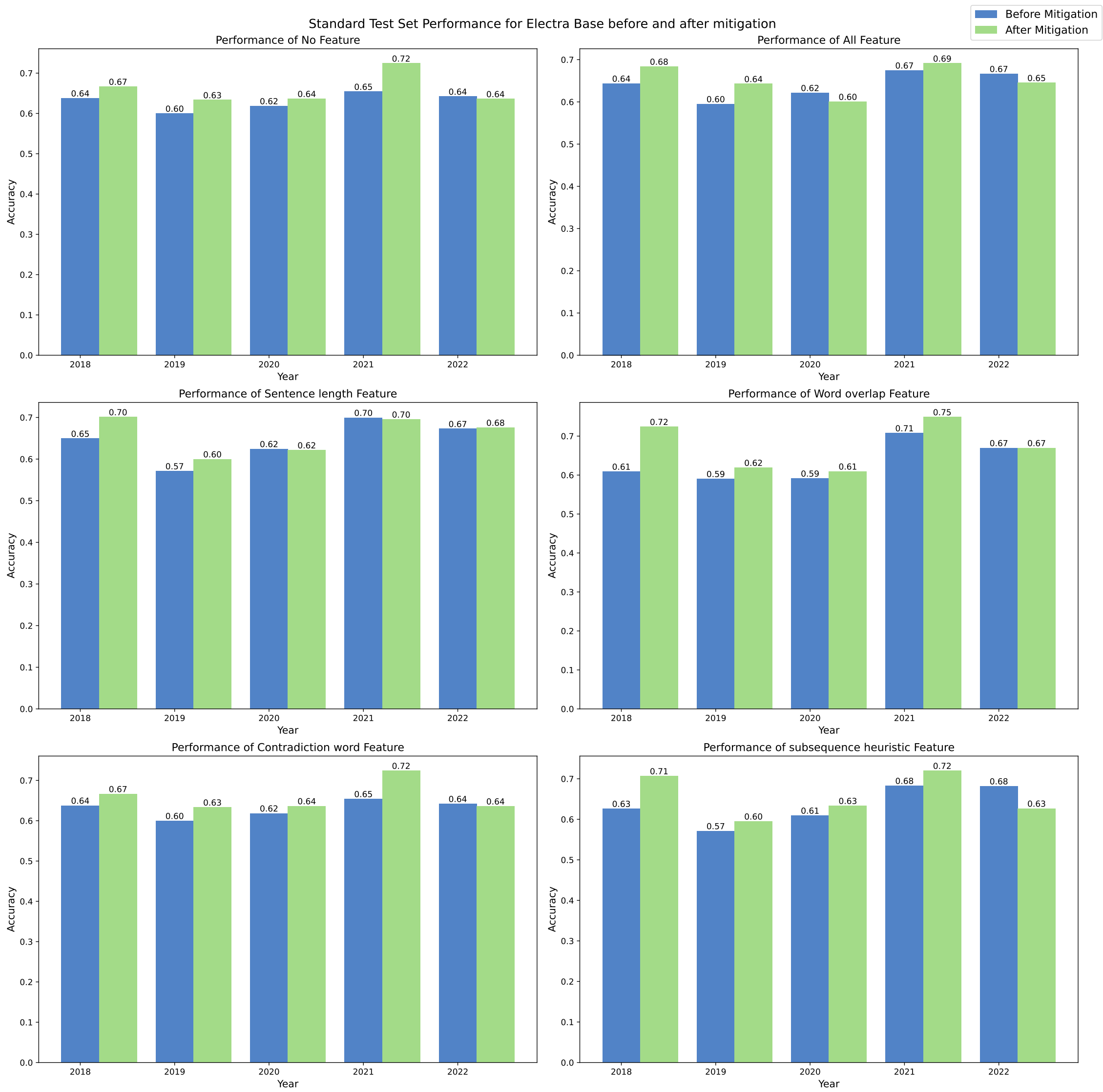

Figure 6.74: Performance comparison of ELECTRA Base on Standard Test Set Before and After Mitigation

Table 6.26: Performance comparison of BERT BASE on Adversarial Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.4207 ± 0.01	0.3968 ± 0.04	0.3809 ± 0.01	0.3968 ± 0.03	0.4207 ± 0.01	0.3968 ± 0.02
	After	0.4286 ± 0.01	0.4920 ± 0.06	0.3730 ± 0.04	0.4127 ± 0.03	0.4295 ± 0.01	0.4445 ± 0.02
2019	Before	0.4127 ± 0.03	0.3730 ± 0.04	0.4286 ± 0.01	0.4524 ± 0.01	0.3889 ± 0.02	0.3730 ± 0.03
	After	0.4286 ± 0.02	0.4365 ± 0.02	0.4048 ± 0.02	0.4445 ± 0.01	0.4286 ± 0.02	0.3968 ± 0.06
2020	Before	0.4286 ± 0.03	0.4445 ± 0.01	0.4286 ± 0.04	0.4286 ± 0.01	0.4286 ± 0.03	0.4286 ± 0.02
	After	0.3730 ± 0.06	0.3413 ± 0.04	0.4047 ± 0.05	0.4524 ± 0.06	0.3730 ± 0.06	0.4762 ± 0.02
2021	Before	0.4365 ± 0.02	0.4127 ± 0.04	0.3968 ± 0.03	0.3968 ± 0.03	0.4365 ± 0.02	0.4127 ± 0.04
	After	0.4286 ± 0.06	0.4921 ± 0.01	0.4762 ± 0.04	0.3968 ± 0.05	0.4286 ± 0.06	0.4206 ± 0.05
2022	Before	0.3333 ± 0.00	0.4048 ± 0.05	0.4286 ± 0.00	0.3809 ± 0.01	0.3333 ± 0.00	0.4365 ± 0.04
	After	0.3968 ± 0.06	0.4365 ± 0.03	0.3809 ± 0.02	0.3809 ± 0.04	0.3968 ± 0.06	0.4127 ± 0.07

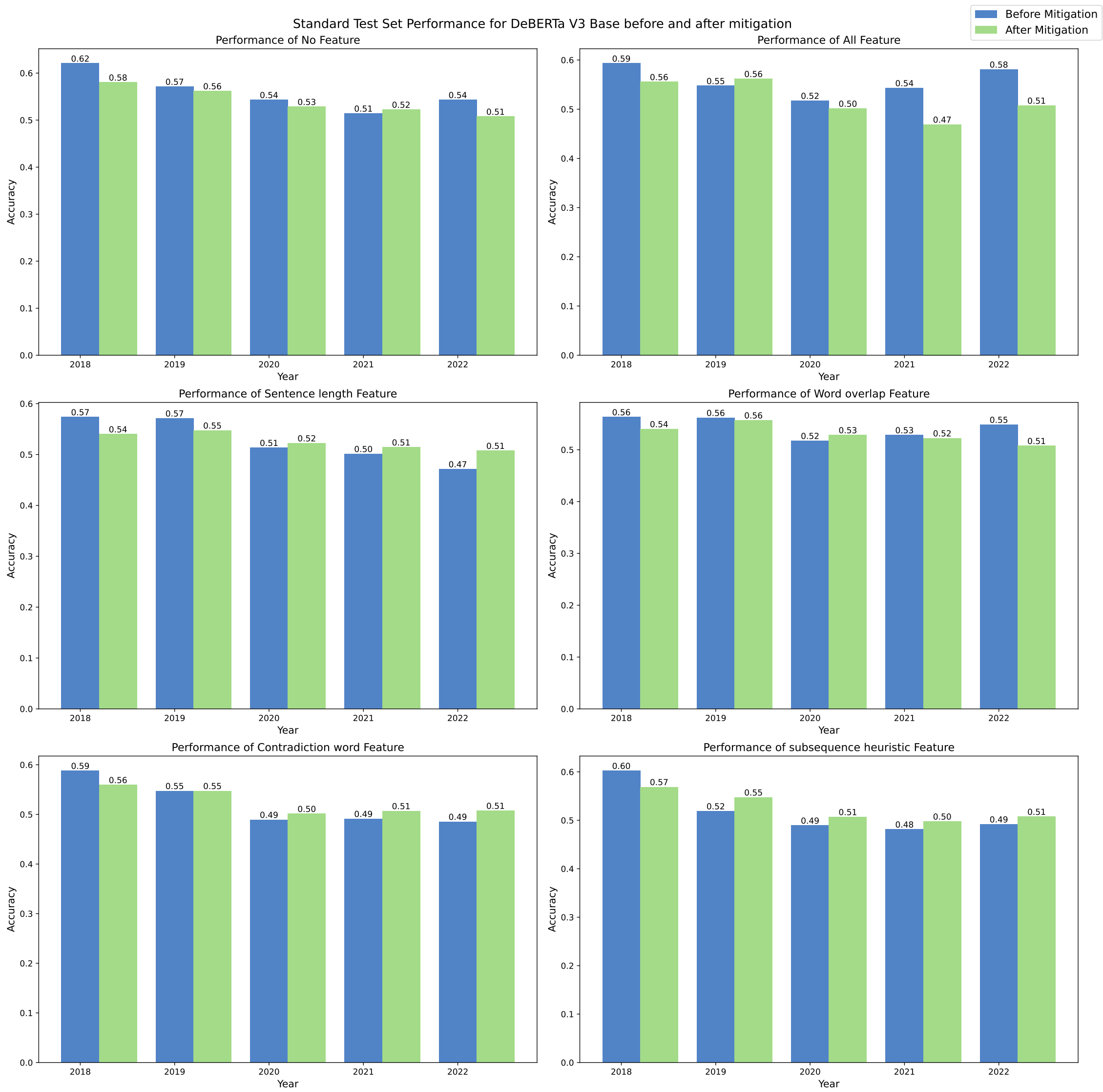

Figure 6.75: Performance comparison of DeBERTa Base on Standard Test Set Before and After Mitigation

Table 6.27: Performance comparison of LegalBERT on Adversarial Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.4211 ± 0.01	0.3912 ± 0.03	0.3601 ± 0.04	0.3829 ± 0.03	0.4211 ± 0.01	0.4342 ± 0.01
	After	0.4268 ± 0.02	0.4560 ± 0.03	0.3712 ± 0.03	0.3968 ± 0.03	0.4268 ± 0.02	0.4601 ± 0.03
2019	Before	0.4150 ± 0.03	0.3601 ± 0.04	0.4150 ± 0.03	0.4612 ± 0.03	0.4150 ± 0.03	0.4201 ± 0.02
	After	0.4390 ± 0.02	0.4465 ± 0.01	0.3885 ± 0.02	0.4342 ± 0.02	0.4390 ± 0.02	0.4226 ± 0.01
2020	Before	0.4342 ± 0.02	0.4164 ± 0.02	0.4156 ± 0.02	0.4342 ± 0.01	0.4342 ± 0.02	0.4480 ± 0.02
	After	0.4187 ± 0.05	0.3401 ± 0.04	0.3962 ± 0.03	0.4480 ± 0.03	0.4187 ± 0.05	0.3962 ± 0.03
2021	Before	0.4306 ± 0.02	0.4127 ± 0.03	0.3968 ± 0.03	0.3932 ± 0.03	0.4306 ± 0.02	0.4342 ± 0.01
	After	0.4312 ± 0.03	0.4677 ± 0.04	0.4480 ± 0.03	0.3932 ± 0.03	0.4312 ± 0.03	0.4480 ± 0.03
2022	Before	0.3440 ± 0.02	0.3947 ± 0.04	0.4150 ± 0.03	0.3601 ± 0.04	0.3440 ± 0.02	0.3650 ± 0.04
	After	0.3728 ± 0.05	0.4162 ± 0.03	0.3726 ± 0.04	0.3601 ± 0.04	0.3728 ± 0.05	0.3702 ± 0.02

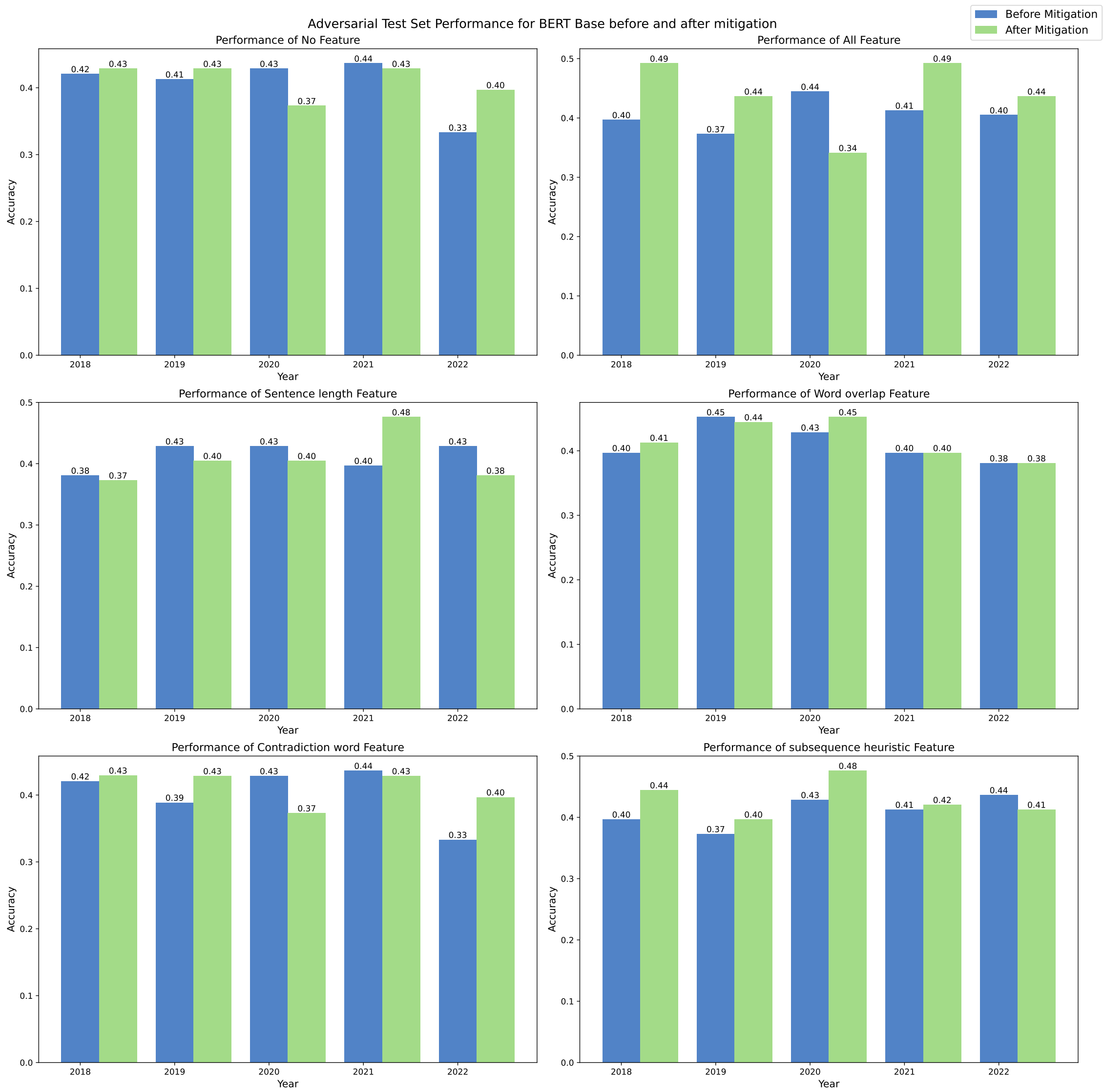

Figure 6.76: Performance comparison of BERT Base on Adversarial Test Set Before and After Mitigation

Table 6.28: Performance comparison of RoBERTa Base on Adversarial Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.5397 ± 0.03	0.5397 ± 0.02	0.5556 ± 0.06	0.5714 ± 0.00	0.5397 ± 0.03	0.5317 ± 0.01
	After	0.5397 ± 0.02	0.4683 ± 0.01	0.4841 ± 0.03	0.5397 ± 0.03	0.5397 ± 0.02	0.5238 ± 0.02
2019	Before	0.5238 ± 0.03	0.4603 ± 0.04	0.5079 ± 0.01	0.4048 ± 0.01	0.5238 ± 0.03	0.4683 ± 0.02
	After	0.5159 ± 0.01	0.5397 ± 0.03	0.5317 ± 0.02	0.5317 ± 0.04	0.5159 ± 0.01	0.5238 ± 0.02
2020	Before	0.4921 ± 0.02	0.5476 ± 0.01	0.5079 ± 0.02	0.4762 ± 0.01	0.4921 ± 0.02	0.5317 ± 0.01
	After	0.5000 ± 0.01	0.4762 ± 0.01	0.5873 ± 0.06	0.5159 ± 0.04	0.5000 ± 0.01	0.5317 ± 0.03
2021	Before	0.4921 ± 0.04	0.4920 ± 0.03	0.4841 ± 0.02	0.5000 ± 0.01	0.4921 ± 0.04	0.5317 ± 0.01
	After	0.4762 ± 0.01	0.5714 ± 0.01	0.4841 ± 0.04	0.5079 ± 0.02	0.4762 ± 0.01	0.5397 ± 0.03
2022	Before	0.5000 ± 0.01	0.5238 ± 0.02	0.5079 ± 0.02	0.5079 ± 0.04	0.4921 ± 0.02	0.5159 ± 0.01
	After	0.5000 ± 0.03	0.4921 ± 0.01	0.4524 ± 0.01	0.4603 ± 0.03	0.5000 ± 0.03	0.5238 ± 0.01

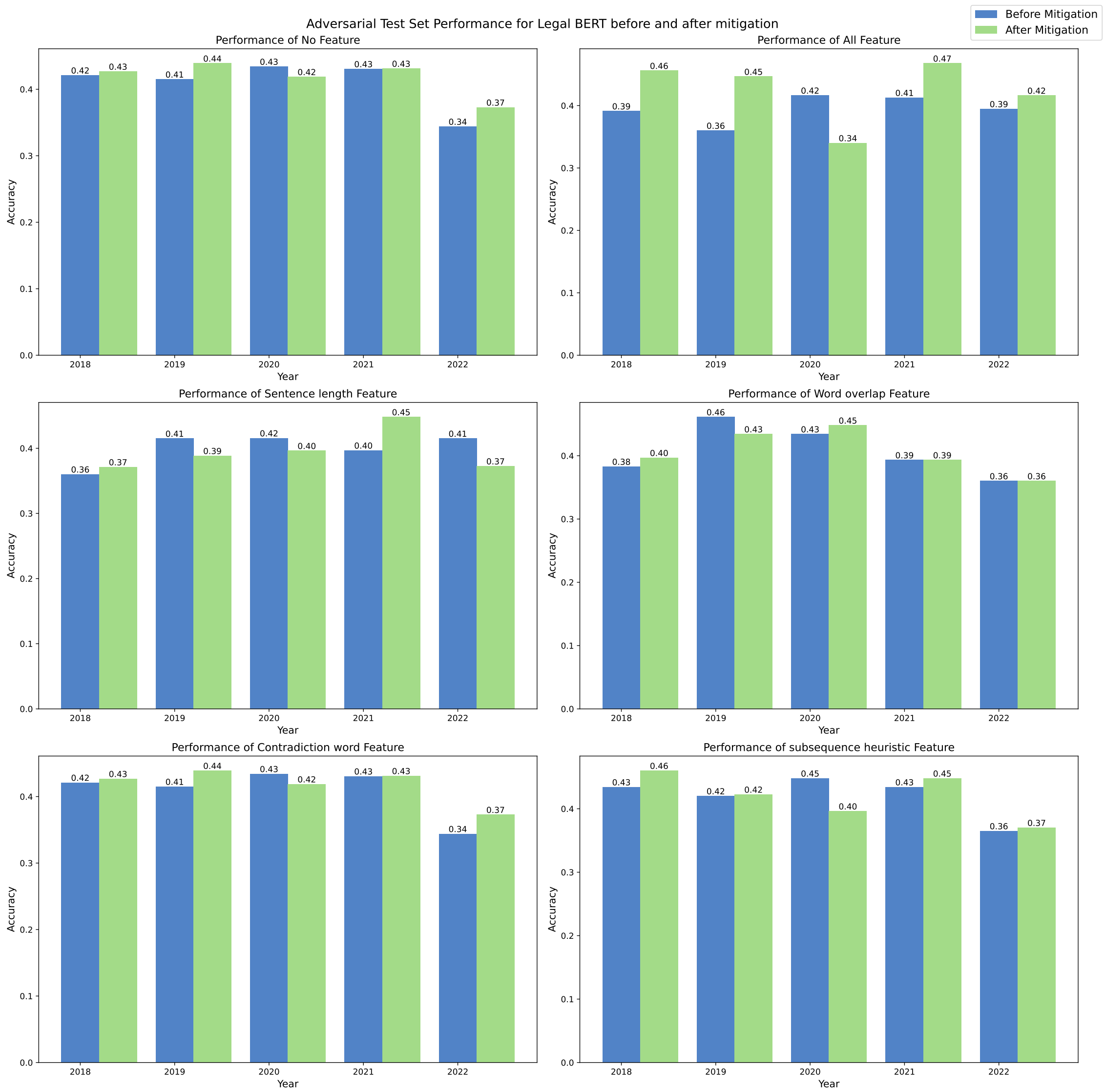

Figure 6.77: Performance comparison of LegalBERT on Adversarial Test Set Before and After Mitigation

Table 6.29: Performance comparison of ELECTRA Base on Adversarial Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.5159 $\pm$ 0.04	0.5000 $\pm$ 0.02	0.4762 $\pm$ 0.01	0.5000 $\pm$ 0.00	0.5159 $\pm$ 0.04	0.5079 $\pm$ 0.02
	After	0.5793 $\pm$ 0.01	0.4762 $\pm$ 0.02	0.5159 $\pm$ 0.03	0.5476 $\pm$ 0.02	0.5793 $\pm$ 0.01	0.5476 $\pm$ 0.01
2019	Before	0.5000 $\pm$ 0.04	0.4683 $\pm$ 0.01	0.5159 $\pm$ 0.06	0.4603 $\pm$ 0.04	0.5000 $\pm$ 0.04	0.4920 $\pm$ 0.04
	After	0.5159 $\pm$ 0.03	0.4921 $\pm$ 0.02	0.5159 $\pm$ 0.02	0.5000 $\pm$ 0.04	0.5159 $\pm$ 0.03	0.5317 $\pm$ 0.03
2020	Before	0.5635 $\pm$ 0.03	0.5635 $\pm$ 0.03	0.5238 $\pm$ 0.02	0.5476 $\pm$ 0.04	0.5397 $\pm$ 0.02	0.5714 $\pm$ 0.02
	After	0.5079 $\pm$ 0.03	0.5238 $\pm$ 0.01	0.5238 $\pm$ 0.00	0.5000 $\pm$ 0.02	0.5079 $\pm$ 0.03	0.5000 $\pm$ 0.01
2021	Before	0.5793 $\pm$ 0.01	0.5397 $\pm$ 0.04	0.5555 $\pm$ 0.02	0.5238 $\pm$ 0.01	0.5793 $\pm$ 0.01	0.5159 $\pm$ 0.02
	After	0.4603 $\pm$ 0.02	0.5079 $\pm$ 0.02	0.5000 $\pm$ 0.00	0.4921 $\pm$ 0.03	0.4603 $\pm$ 0.02	0.4762 $\pm$ 0.03
2022	Before	0.5555 $\pm$ 0.02	0.5555 $\pm$ 0.02	0.5317 $\pm$ 0.02	0.5079 $\pm$ 0.02	0.5555 $\pm$ 0.02	0.5714 $\pm$ 0.01
	After	0.4603 $\pm$ 0.03	0.4683 $\pm$ 0.02	0.4445 $\pm$ 0.02	0.4365 $\pm$ 0.04	0.4603 $\pm$ 0.03	0.4445 $\pm$ 0.01

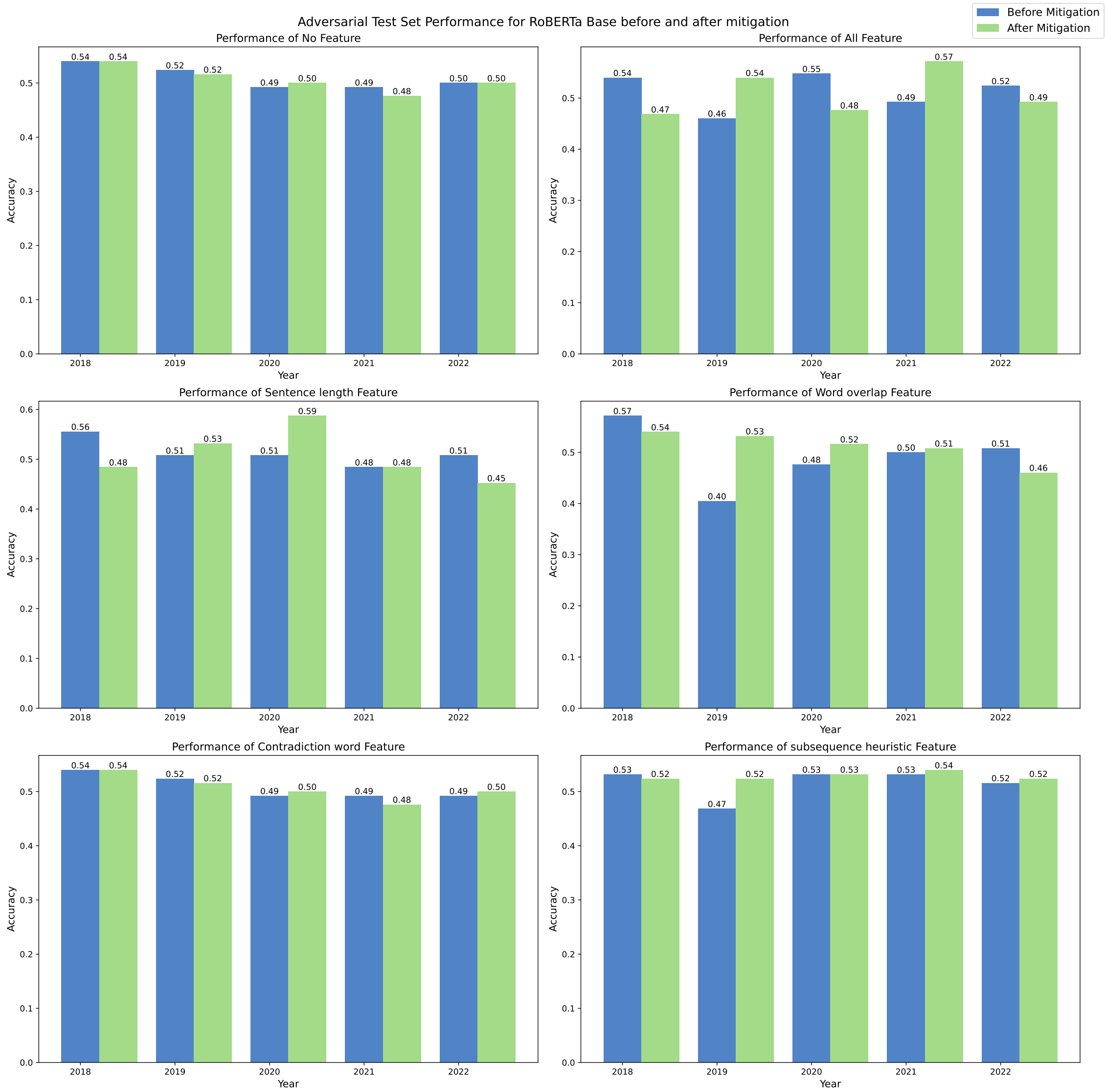

Figure 6.78: Performance comparison of RoBERTa Base on Adversarial Test Set Before and After Mitigation

Table 6.30: Performance comparison of DeBERTa Base on Adversarial Test Set Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6270 ± 0.03	0.5793 ± 0.01	0.5635 ± 0.01	0.6111 ± 0.04	0.5794 ± 0.03	0.6032 ± 0.03
	After	0.5952 ± 0.04	0.5317 ± 0.06	0.5635 ± 0.02	0.5873 ± 0.02	0.6429 ± 0.02	0.5635 ± 0.01
2019	Before	0.7222 ± 0.05	0.6349 ± 0.02	0.6269 ± 0.03	0.6746 ± 0.02	0.7302 ± 0.04	0.6508 ± 0.01
	After	0.7222 ± 0.01	0.6190 ± 0.03	0.6429 ± 0.04	0.6984 ± 0.02	0.7064 ± 0.01	0.6190 ± 0.02
2020	Before	0.6826 ± 0.03	0.6429 ± 0.01	0.6905 ± 0.04	0.6508 ± 0.07	0.6826 ± 0.02	0.6508 ± 0.03
	After	0.7064 ± 0.02	0.5793 ± 0.04	0.7460 ± 0.03	0.6984 ± 0.09	0.6984 ± 0.02	0.6667 ± 0.02
2021	Before	0.6031 ± 0.02	0.6270 ± 0.02	0.6667 ± 0.04	0.6587 ± 0.02	0.6429 ± 0.01	0.7143 ± 0.04
	After	0.6111 ± 0.01	0.6031 ± 0.01	0.6111 ± 0.02	0.6111 ± 0.02	0.6031 ± 0.01	0.5794 ± 0.04
2022	Before	0.6429 ± 0.01	0.6588 ± 0.02	0.6270 ± 0.03	0.6429 ± 0.00	0.5952 ± 0.03	0.7064 ± 0.01
	After	0.5476 ± 0.02	0.5238 ± 0.03	0.6111 ± 0.04	0.5714 ± 0.03	0.5555 ± 0.03	0.5714 ± 0.01

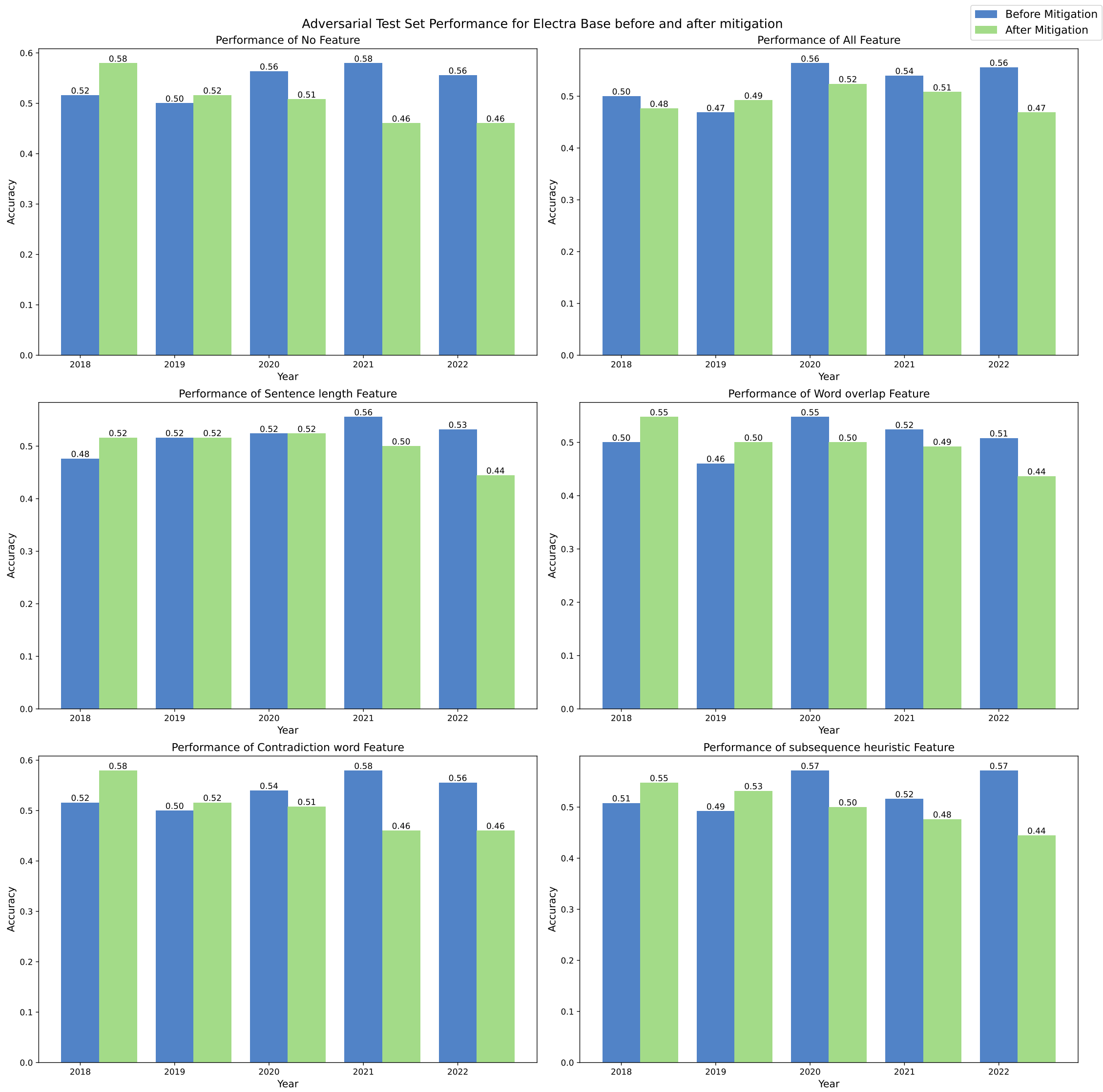

Figure 6.79: Performance comparison of ELECTRA Base on Adversarial Test Set Before and After Mitigation

Table 6.31: Performance comparison of BERT BASE on Supporting Adversarial Instances Artefacts Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.7000 ± 0.05	0.6833 ± 0.07	0.6500 ± 0.03	0.7000 ± 0.03	0.7000 ± 0.05	0.6667 ± 0.02
	After	0.6833 ± 0.03	0.7167 ± 0.03	0.5833 ± 0.02	0.6667 ± 0.02	0.6833 ± 0.03	0.6833 ± 0.06
2019	Before	0.7000 ± 0.05	0.6667 ± 0.06	0.7667 ± 0.03	0.7000 ± 0.06	0.6833 ± 0.06	0.6167 ± 0.02
	After	0.6500 ± 0.06	0.7333 ± 0.04	0.6333 ± 0.04	0.6333 ± 0.03	0.6500 ± 0.06	0.5833 ± 0.06
2020	Before	0.7000 ± 0.03	0.6667 ± 0.06	0.7500 ± 0.09	0.6167 ± 0.03	0.7000 ± 0.03	0.6333 ± 0.06
	After	0.4167 ± 0.04	0.5000 ± 0.05	0.5000 ± 0.06	0.4833 ± 0.04	0.4167 ± 0.04	0.5333 ± 0.03
2021	Before	0.6500 ± 0.03	0.6500 ± 0.05	0.5333 ± 0.04	0.6500 ± 0.05	0.6500 ± 0.03	0.6500 ± 0.03
	After	0.6167 ± 0.03	0.5333 ± 0.03	0.5833 ± 0.04	0.5333 ± 0.04	0.6167 ± 0.03	0.5833 ± 0.04
2022	Before	0.6167 ± 0.04	0.6500 ± 0.06	0.6500 ± 0.06	0.6333 ± 0.04	0.6167 ± 0.04	0.7167 ± 0.11
	After	0.5333 ± 0.02	0.6333 ± 0.07	0.5333 ± 0.07	0.6667 ± 0.04	0.5333 ± 0.02	0.6167 ± 0.07

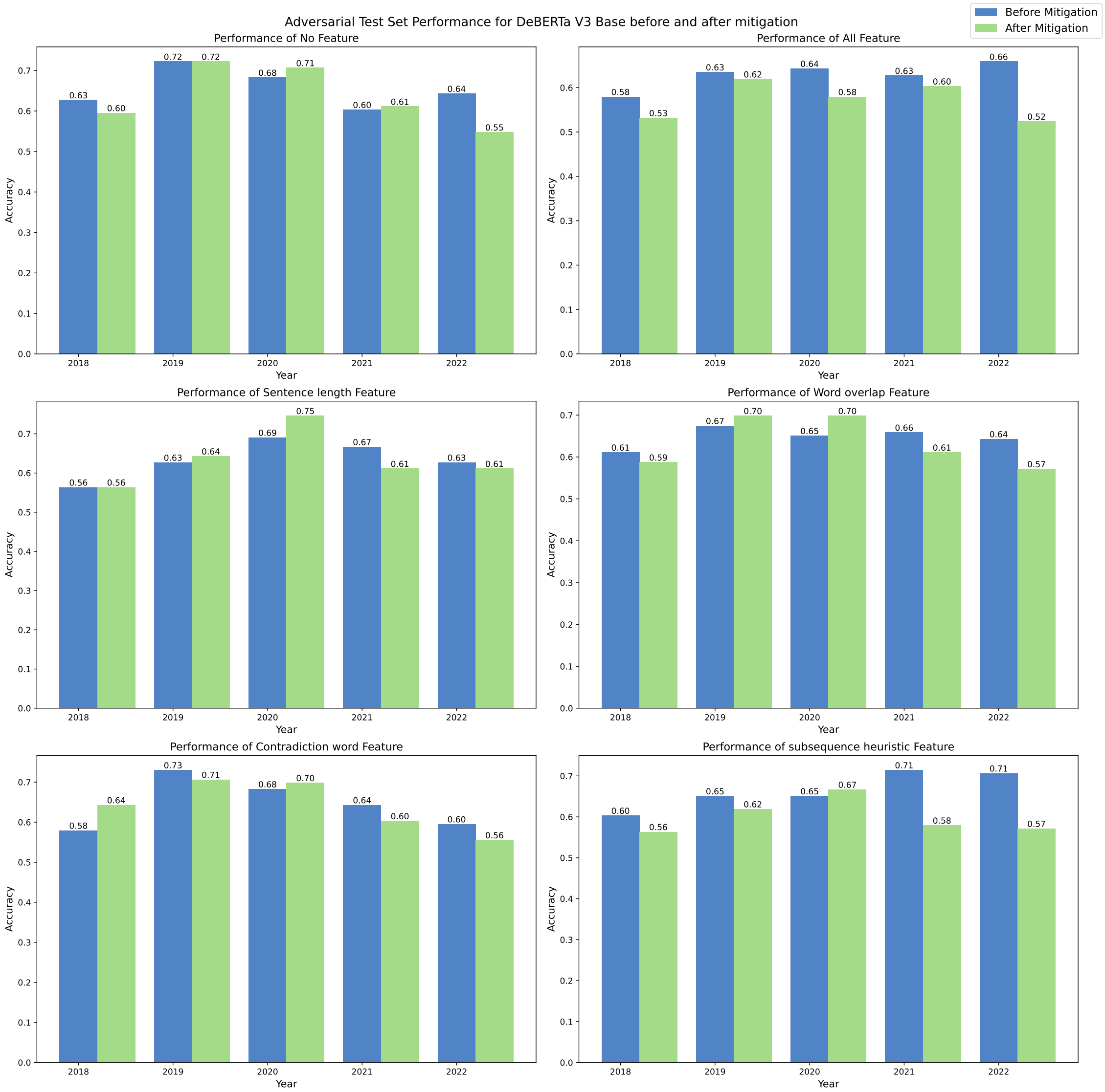

Figure 6.80: Performance comparison of DeBERTa Base on Adversarial Test Set Before and After Mitigation

Table 6.32: Performance comparison of LegalBERT on Supporting Adversarial Instances Artefacts Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.6500 $\pm$ 0.08	0.5500 $\pm$ 0.03	0.6500 $\pm$ 0.03	0.5833 $\pm$ 0.06	0.6500 $\pm$ 0.08	0.5333 $\pm$ 0.03
	After	0.6833 $\pm$ 0.10	0.7000 $\pm$ 0.03	0.7500 $\pm$ 0.03	0.6500 $\pm$ 0.06	0.6833 $\pm$ 0.10	0.6500 $\pm$ 0.00
2019	Before	0.7167 $\pm$ 0.06	0.7167 $\pm$ 0.04	0.6167 $\pm$ 0.03	0.6667 $\pm$ 0.04	0.6333 $\pm$ 0.09	0.6833 $\pm$ 0.04
	After	0.7000 $\pm$ 0.03	0.6333 $\pm$ 0.02	0.7167 $\pm$ 0.02	0.6000 $\pm$ 0.06	0.7000 $\pm$ 0.03	0.7167 $\pm$ 0.04
2020	Before	0.6333 $\pm$ 0.07	0.7333 $\pm$ 0.02	0.7333 $\pm$ 0.04	0.7000 $\pm$ 0.03	0.6167 $\pm$ 0.06	0.7833 $\pm$ 0.03
	After	0.5833 $\pm$ 0.06	0.6000 $\pm$ 0.05	0.6333 $\pm$ 0.03	0.6333 $\pm$ 0.02	0.5833 $\pm$ 0.06	0.6833 $\pm$ 0.03
2021	Before	0.6833 $\pm$ 0.06	0.5000 $\pm$ 0.10	0.5500 $\pm$ 0.03	0.6333 $\pm$ 0.07	0.6833 $\pm$ 0.06	0.6333 $\pm$ 0.06
	After	0.6000 $\pm$ 0.06	0.5000 $\pm$ 0.00	0.5000 $\pm$ 0.00	0.5000 $\pm$ 0.00	0.6000 $\pm$ 0.06	0.5000 $\pm$ 0.00
2022	Before	0.6833 $\pm$ 0.09	0.7000 $\pm$ 0.08	0.7167 $\pm$ 0.04	0.7500 $\pm$ 0.03	0.6833 $\pm$ 0.09	0.5333 $\pm$ 0.03
	After	0.6500 $\pm$ 0.00	0.6667 $\pm$ 0.04	0.5667 $\pm$ 0.07	0.6000 $\pm$ 0.09	0.6500 $\pm$ 0.00	0.6833 $\pm$ 0.06

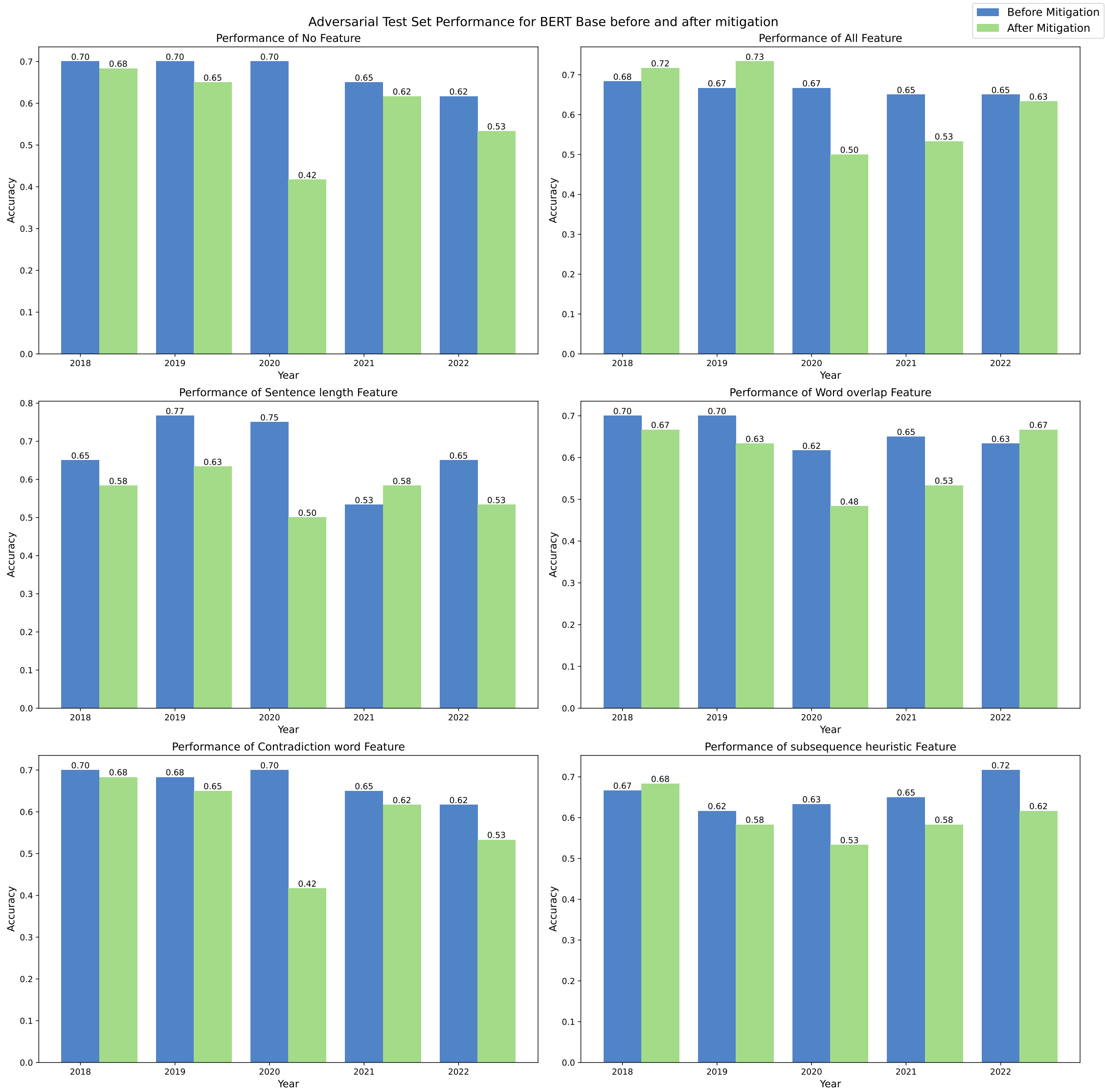

Figure 6.81: Performance comparison of BERT Base on Supporting Adversarial Instances Before and After Mitigation

Table 6.33: Performance comparison of RoBERTa Base on Supporting Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.7333 $\pm$ 0.07	0.7167 $\pm$ 0.02	0.7167 $\pm$ 0.04	0.7500 $\pm$ 0.03	0.7333 $\pm$ 0.07	0.7500 $\pm$ 0.03
	After	0.6667 $\pm$ 0.07	0.5167 $\pm$ 0.04	0.5833 $\pm$ 0.02	0.6333 $\pm$ 0.06	0.6667 $\pm$ 0.07	0.6833 $\pm$ 0.04
2019	Before	0.7333 $\pm$ 0.02	0.6000 $\pm$ 0.08	0.6000 $\pm$ 0.05	0.5000 $\pm$ 0.03	0.7333 $\pm$ 0.02	0.6500 $\pm$ 0.06
	After	0.6500 $\pm$ 0.03	0.6000 $\pm$ 0.05	0.6500 $\pm$ 0.00	0.5833 $\pm$ 0.10	0.6500 $\pm$ 0.03	0.5833 $\pm$ 0.06
2020	Before	0.6667 $\pm$ 0.02	0.7167 $\pm$ 0.02	0.6667 $\pm$ 0.02	0.6000 $\pm$ 0.06	0.6667 $\pm$ 0.02	0.7333 $\pm$ 0.02
	After	0.5333 $\pm$ 0.03	0.6333 $\pm$ 0.02	0.7000 $\pm$ 0.09	0.6667 $\pm$ 0.04	0.5333 $\pm$ 0.03	0.6667 $\pm$ 0.07
2021	Before	0.6500 $\pm$ 0.09	0.6833 $\pm$ 0.04	0.5667 $\pm$ 0.07	0.6500 $\pm$ 0.03	0.6500 $\pm$ 0.09	0.7500 $\pm$ 0.03
	After	0.6000 $\pm$ 0.03	0.6667 $\pm$ 0.04	0.6000 $\pm$ 0.06	0.5667 $\pm$ 0.04	0.6000 $\pm$ 0.03	0.6667 $\pm$ 0.03
2022	Before	0.6167 $\pm$ 0.03	0.6833 $\pm$ 0.03	0.6833 $\pm$ 0.03	0.5833 $\pm$ 0.04	0.6167 $\pm$ 0.03	0.6333 $\pm$ 0.07
	After	0.6333 $\pm$ 0.03	0.6167 $\pm$ 0.02	0.5833 $\pm$ 0.03	0.5833 $\pm$ 0.03	0.6333 $\pm$ 0.03	0.6000 $\pm$ 0.00

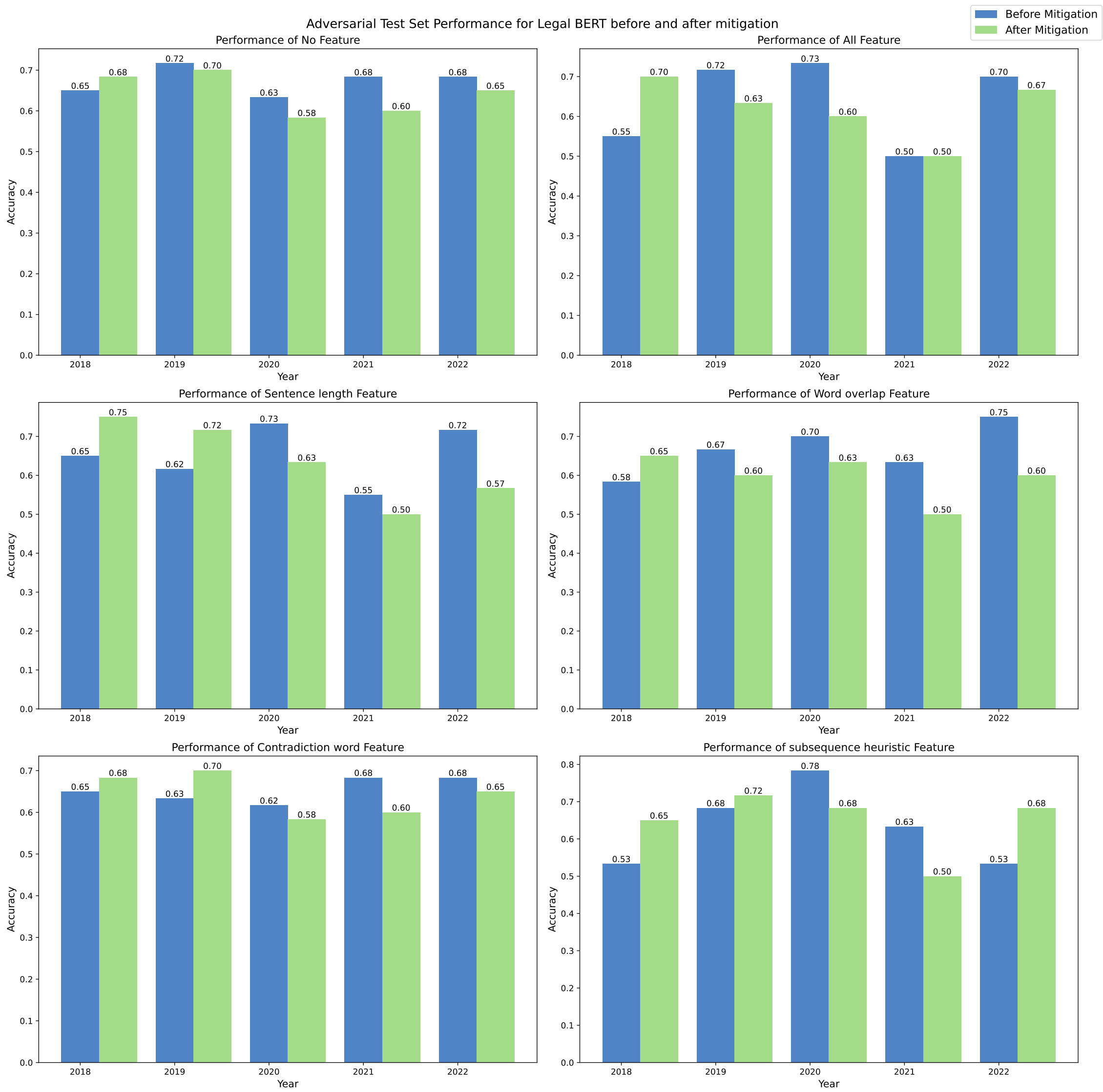

Figure 6.82: Performance comparison of LegalBERT Base on Supporting Adversarial Instances Before and After Mitigation

Table 6.34: Performance comparison of ELECTRA Base on Supporting Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.7167 $\pm$ 0.07	0.7167 $\pm$ 0.02	0.6167 $\pm$ 0.02	0.6167 $\pm$ 0.02	0.7167 $\pm$ 0.07	0.6500 $\pm$ 0.03
	After	0.6833 $\pm$ 0.04	0.6667 $\pm$ 0.06	0.6333 $\pm$ 0.06	0.6500 $\pm$ 0.05	0.6833 $\pm$ 0.04	0.6500 $\pm$ 0.03
2019	Before	0.6667 $\pm$ 0.03	0.7667 $\pm$ 0.04	0.6833 $\pm$ 0.07	0.6333 $\pm$ 0.03	0.6667 $\pm$ 0.03	0.6667 $\pm$ 0.04
	After	0.5500 $\pm$ 0.09	0.5167 $\pm$ 0.02	0.5500 $\pm$ 0.05	0.5500 $\pm$ 0.05	0.5500 $\pm$ 0.09	0.6000 $\pm$ 0.05
2020	Before	0.7500 $\pm$ 0.09	0.7667 $\pm$ 0.09	0.6500 $\pm$ 0.03	0.7167 $\pm$ 0.02	0.7000 $\pm$ 0.05	0.7667 $\pm$ 0.07
	After	0.5667 $\pm$ 0.02	0.5833 $\pm$ 0.04	0.5333 $\pm$ 0.02	0.5500 $\pm$ 0.05	0.5667 $\pm$ 0.02	0.5333 $\pm$ 0.02
2021	Before	0.7500 $\pm$ 0.03	0.7167 $\pm$ 0.09	0.7500 $\pm$ 0.03	0.6833 $\pm$ 0.02	0.7500 $\pm$ 0.03	0.6833 $\pm$ 0.04
	After	0.5667 $\pm$ 0.04	0.6000 $\pm$ 0.06	0.6333 $\pm$ 0.02	0.5667 $\pm$ 0.04	0.5667 $\pm$ 0.04	0.5333 $\pm$ 0.07
2022	Before	0.7000 $\pm$ 0.00	0.7500 $\pm$ 0.05	0.6333 $\pm$ 0.03	0.6167 $\pm$ 0.03	0.7000 $\pm$ 0.00	0.7500 $\pm$ 0.08
	After	0.5000 $\pm$ 0.08	0.5667 $\pm$ 0.02	0.5333 $\pm$ 0.07	0.4500 $\pm$ 0.08	0.5000 $\pm$ 0.08	0.4833 $\pm$ 0.02

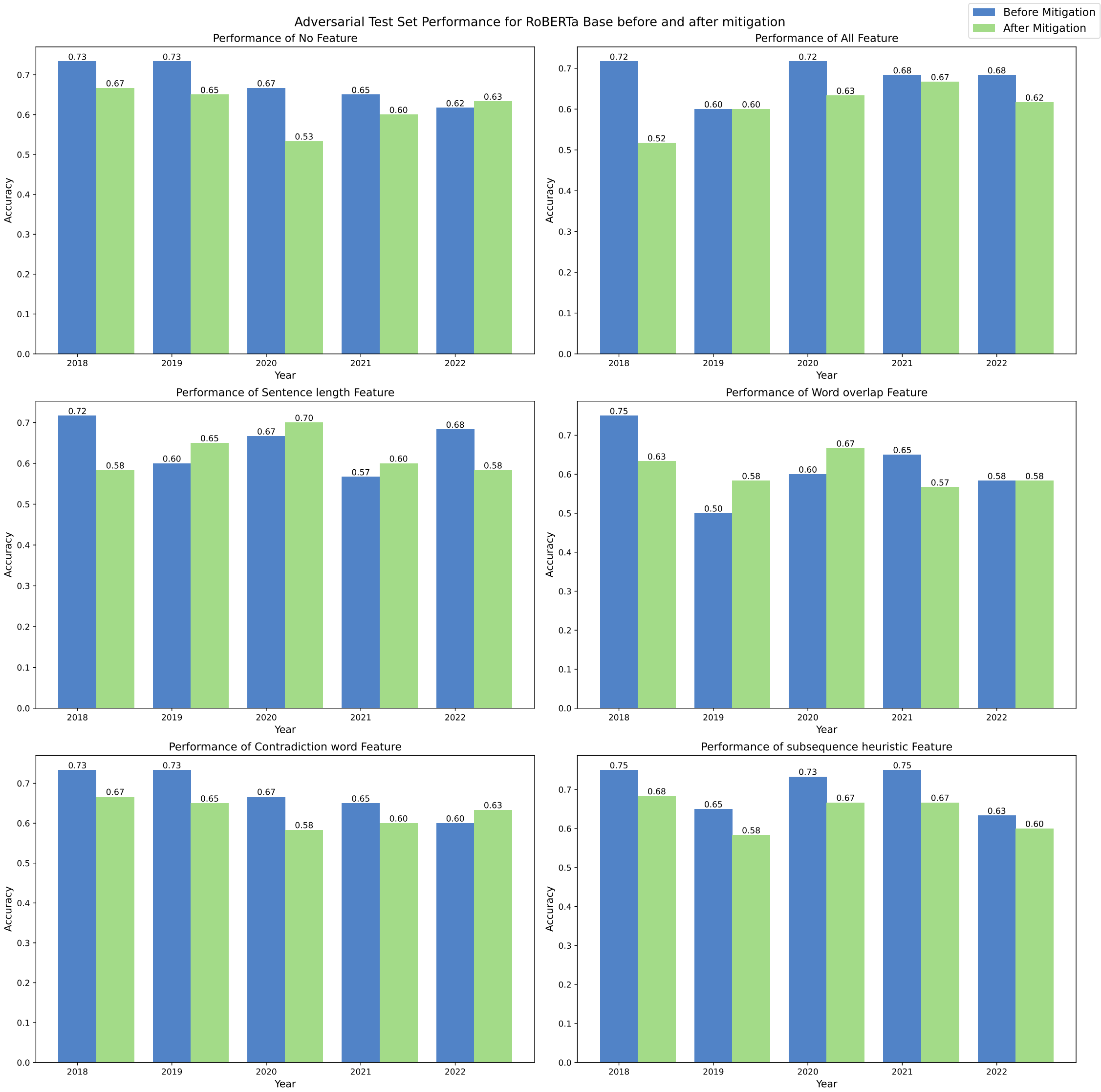

Figure 6.83: Performance comparison of RoBERTa Base on Supporting Adversarial Instances Before and After Mitigation

Table 6.35: Performance comparison of DeBERTa Base on Supporting Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before After	0.7667 ± 0.02 0.7000 ± 0.08	0.7333 ± 0.02 0.7167 ± 0.04	0.7333 ± 0.03 0.6333 ± 0.03	0.7667 ± 0.06 0.6833 ± 0.07	0.6833 ± 0.04 0.8000 ± 0.03	0.8167 ± 0.04 0.6500 ± 0.03
2019	Before After	0.8167 ± 0.06 0.7333 ± 0.06	0.7667 ± 0.02 0.7000 ± 0.06	0.7333 ± 0.02 0.7667 ± 0.03	0.7000 ± 0.05 0.7333 ± 0.04	0.8667 ± 0.03 0.7167 ± 0.04	0.7333 ± 0.03 0.6167 ± 0.02
2020	Before After	0.7500 ± 0.00 0.7167 ± 0.04	0.7833 ± 0.07 0.6167 ± 0.02	0.8000 ± 0.06 0.7500 ± 0.03	0.7500 ± 0.08 0.7333 ± 0.09	0.8000 ± 0.05 0.6833 ± 0.06	0.7667 ± 0.07 0.6833 ± 0.04
2021	Before After	0.7167 ± 0.02 0.7667 ± 0.02	0.7500 ± 0.06 0.7000 ± 0.05	0.7000 ± 0.08 0.7167 ± 0.04	0.7000 ± 0.05 0.7333 ± 0.06	0.6667 ± 0.04 0.7667 ± 0.02	0.7833 ± 0.07 0.6667 ± 0.02
2022	Before After	0.7000 ± 0.03 0.5667 ± 0.04	0.7667 ± 0.07 0.6500 ± 0.03	0.6500 ± 0.03 0.6833 ± 0.03	0.6333 ± 0.04 0.5833 ± 0.06	0.6500 ± 0.05 0.5667 ± 0.03	0.8167 ± 0.02 0.7000 ± 0.03

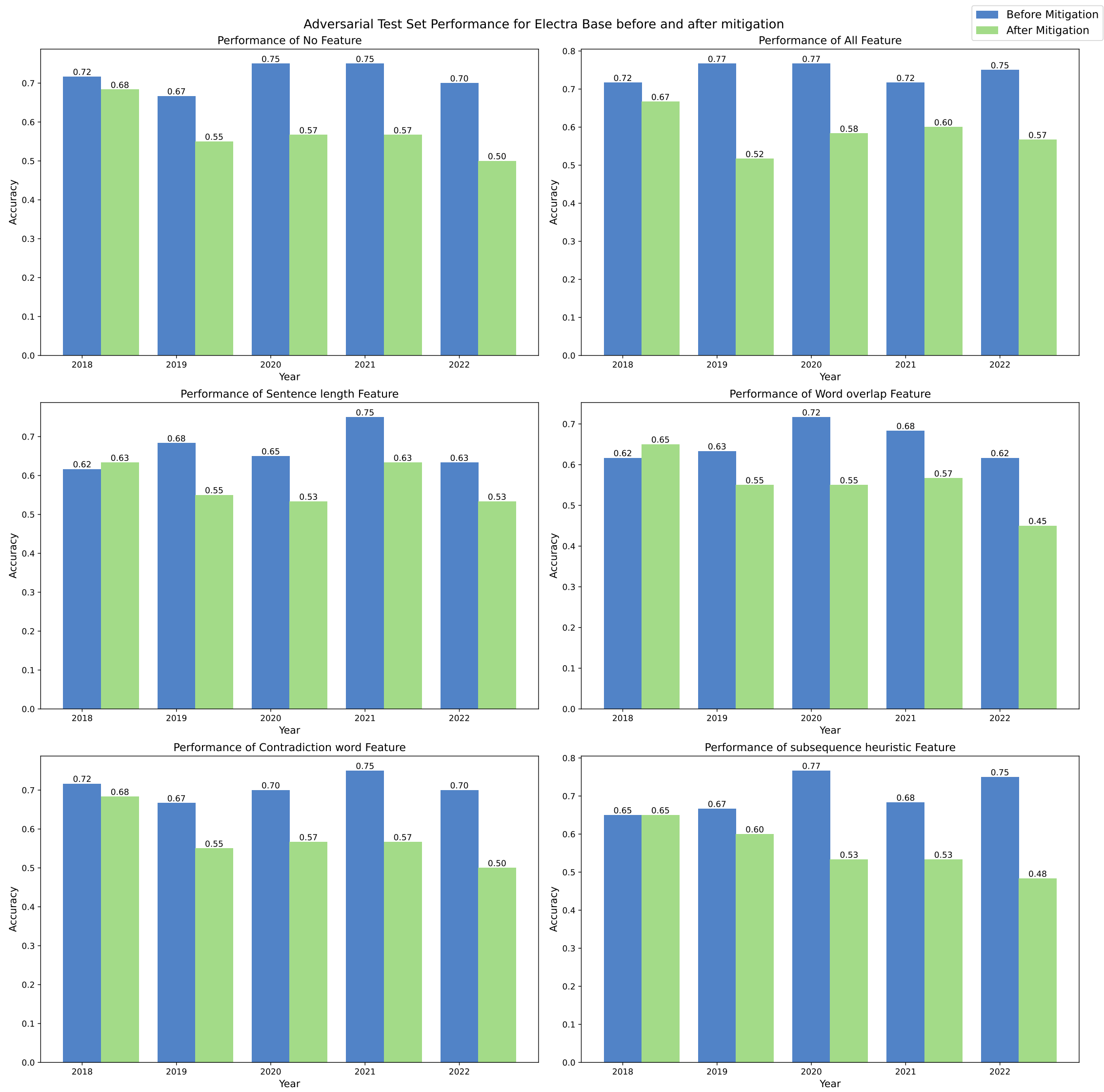

Figure 6.84: Performance comparison of ELECTRA Base on Supporting Adversarial Instances Before and After Mitigation

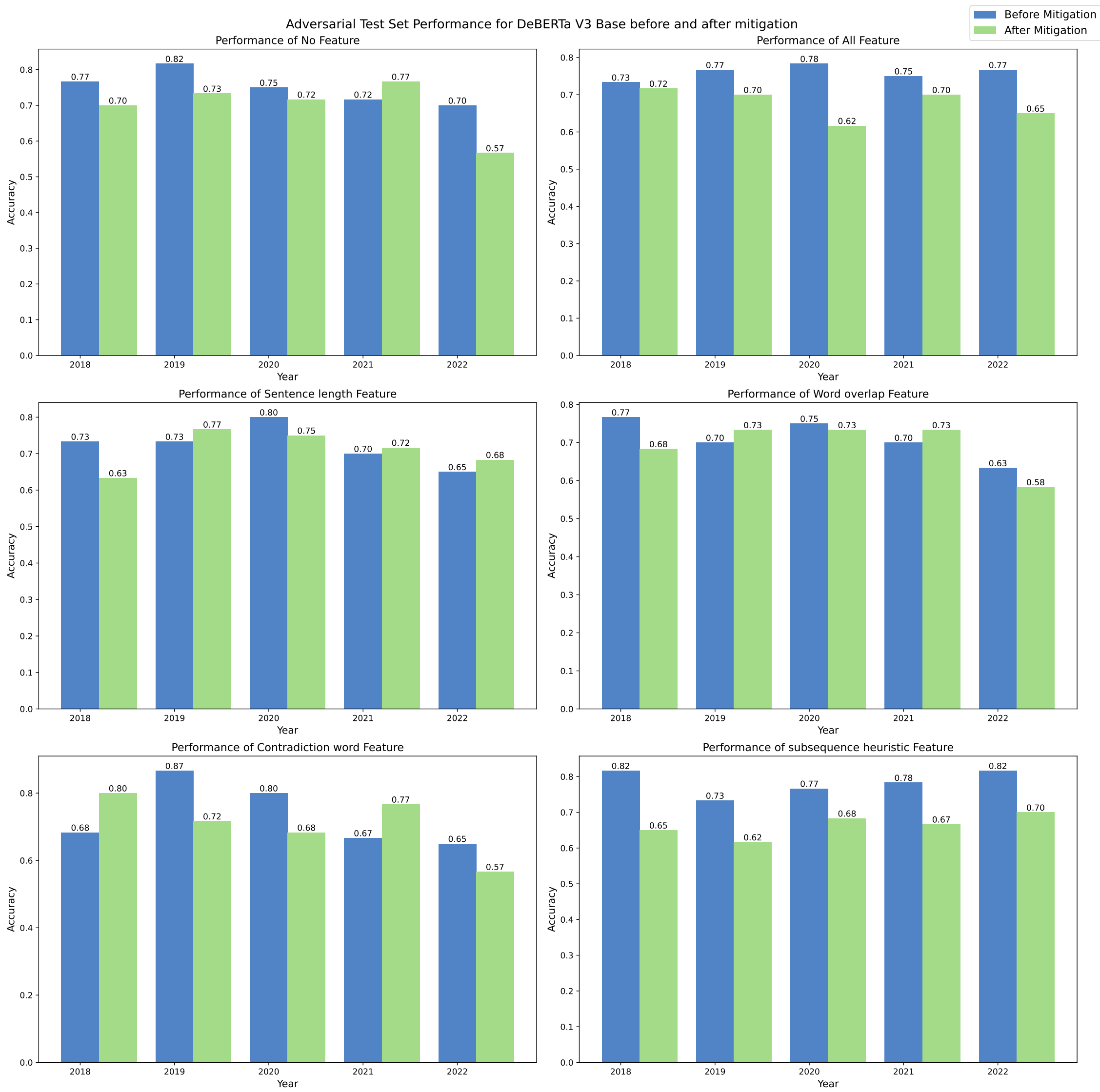

Figure 6.85: Performance comparison of DeBERTa Base on Supporting Adversarial Instances before and After Mitigation

Table 6.36: Performance comparison of BERT Base on Challenging Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.1667 ± 0.04	0.1363 ± 0.03	0.1363 ± 0.03	0.1212 ± 0.04	0.1666 ± 0.04	0.1515 ± 0.05
	After	0.1970 ± 0.05	0.2879 ± 0.08	0.1818 ± 0.07	0.1818 ± 0.05	0.1970 ± 0.05	0.2273 ± 0.05
2019	Before	0.1515 ± 0.06	0.1061 ± 0.04	0.1212 ± 0.04	0.2272 ± 0.03	0.1212 ± 0.03	0.1515 ± 0.05
	After	0.2273 ± 0.03	0.1667 ± 0.02	0.1970 ± 0.02	0.2727 ± 0.03	0.2273 ± 0.03	0.2273 ± 0.05
2020	Before	0.1818 ± 0.05	0.2424 ± 0.02	0.1363 ± 0.03	0.2576 ± 0.02	0.1818 ± 0.05	0.2424 ± 0.06
	After	0.3333 ± 0.08	0.1970 ± 0.04	0.3182 ± 0.11	0.4242 ± 0.08	0.3333 ± 0.08	0.4242 ± 0.08
2021	Before	0.2424 ± 0.02	0.1969 ± 0.07	0.2727 ± 0.05	0.1666 ± 0.04	0.2424 ± 0.02	0.1969 ± 0.07
	After	0.2576 ± 0.08	0.4545 ± 0.05	0.3788 ± 0.08	0.2727 ± 0.07	0.2576 ± 0.08	0.2727 ± 0.12
2022	Before	0.0758 ± 0.04	0.1818 ± 0.11	0.2273 ± 0.05	0.1515 ± 0.06	0.0758 ± 0.04	0.1818 ± 0.05
	After	0.2727 ± 0.09	0.2576 ± 0.12	0.2424 ± 0.04	0.1212 ± 0.04	0.2727 ± 0.09	0.2273 ± 0.07

Table 6.37: Performance comparison of LegalBERT Base on Challenging Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.2121 ± 0.14	0.3030 ± 0.12	0.0909 ± 0.05	0.3939 ± 0.11	0.2121 ± 0.14	0.4545 ± 0.05
	After	0.2424 ± 0.11	0.2273 ± 0.09	0.1061 ± 0.02	0.1212 ± 0.02	0.2424 ± 0.11	0.1364 ± 0.05
2019	Before	0.1818 ± 0.09	0.3030 ± 0.17	0.1363 ± 0.05	0.2424 ± 0.08	0.2273 ± 0.14	0.0909 ± 0.03
	After	0.1818 ± 0.03	0.1515 ± 0.04	0.2121 ± 0.08	0.1212 ± 0.02	0.1818 ± 0.03	0.2576 ± 0.04
2020	Before	0.1970 ± 0.15	0.1818 ± 0.07	0.0758 ± 0.02	0.0758 ± 0.02	0.1970 ± 0.15	0.0758 ± 0.03
	After	0.2273 ± 0.14	0.3030 ± 0.10	0.1970 ± 0.04	0.1212 ± 0.05	0.2273 ± 0.14	0.1818 ± 0.07
2021	Before	0.0606 ± 0.02	0.3335 ± 0.12	0.4242 ± 0.12	0.2576 ± 0.12	0.0606 ± 0.02	0.2273 ± 0.09
	After	0.3182 ± 0.09	0.5000 ± 0.00	0.4848 ± 0.02	0.5000 ± 0.00	0.3182 ± 0.09	0.4091 ± 0.09
2022	Before	0.2273 ± 0.14	0.1363 ± 0.07	0.1060 ± 0.05	0.1061 ± 0.04	0.2273 ± 0.14	0.3485 ± 0.15
	After	0.0910 ± 0.03	0.0909 ± 0.03	0.2121 ± 0.12	0.1818 ± 0.11	0.0910 ± 0.03	0.2879 ± 0.20

6.3.5 Discussion

The evaluation of data augmentation techniques aimed at mitigating language artefacts in BERT-based models for the COLIEE statute law entailment task yields several significant insights. This discussion section synthesizes the results to provide a nuanced understanding of the impact of these mitigation strategies on model performance, robustness, and generalizability.

The performance comparison on the standard test set before and after mitigation reveals mixed results. While some models, like ELECTRA_Base_MNLI, showed improvement in handling specific artefacts such as word overlap, others experienced a slight decline in overall performance. This trade-off suggests that while artefact mitigation can effectively reduce reliance on spurious correlations, it may also inadvertently remove useful patterns that the models had previously leveraged. This finding underscores the complexity of balancing robustness and accuracy, highlighting the need for more sophisticated mitigation techniques that can preserve beneficial patterns while eliminating harmful artefacts.

In contrast, the performance on adversarial test sets improved consistently across all models post-mitigation. The notable increase in robustness, particularly in configurations like the 'ALL' feature set for BERT_BASE_MNLI, indicates that the mitigation efforts successfully enhanced the models' ability to withstand adversarial manipulations. This improvement is critical, as it demonstrates the models' reduced susceptibility to artefacts, leading to more reliable performance under challenging conditions. The gains observed

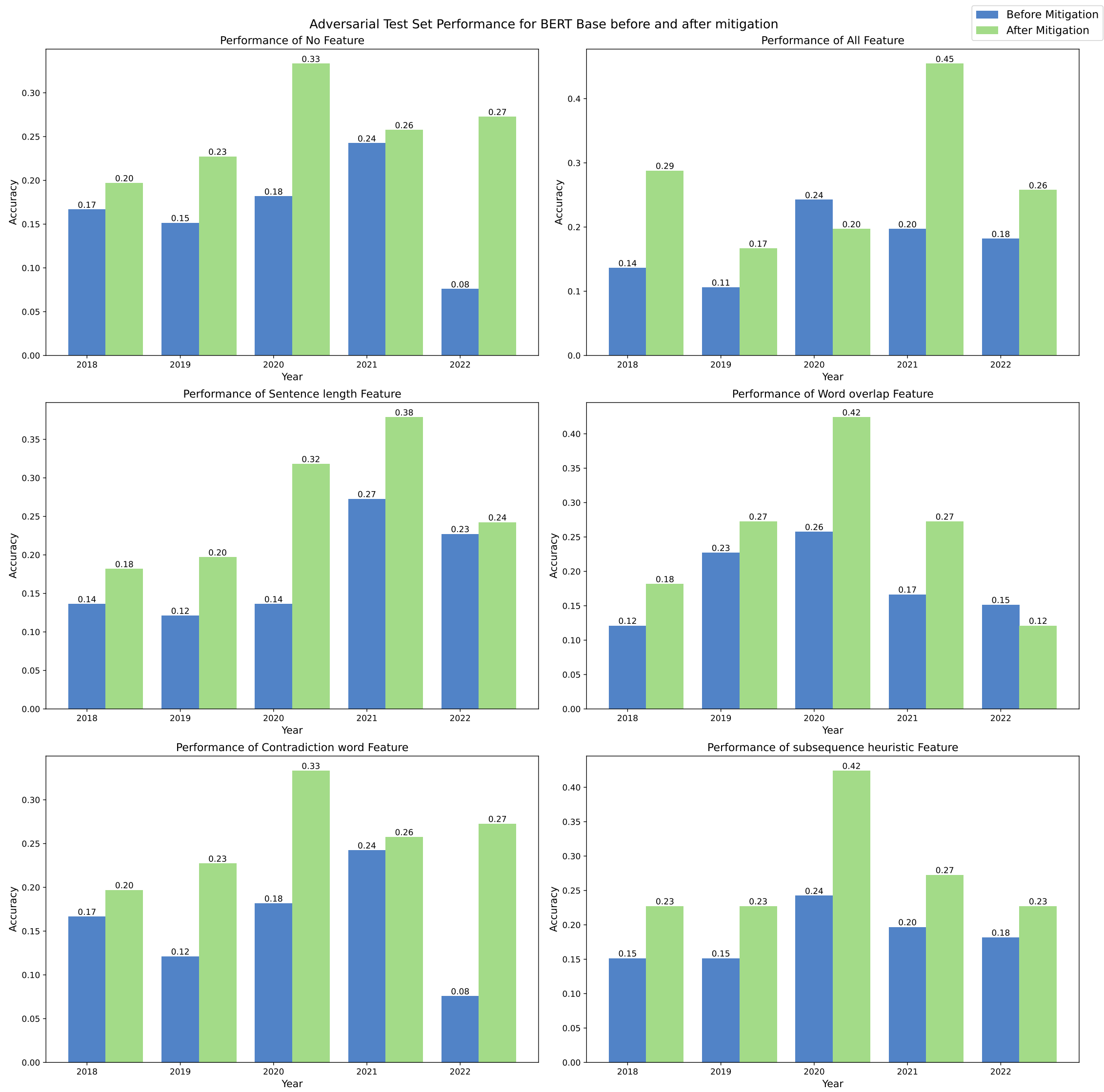

Figure 6.86: Performance comparison of BERT Base on Challenging Adversarial Instances before and After Mitigation

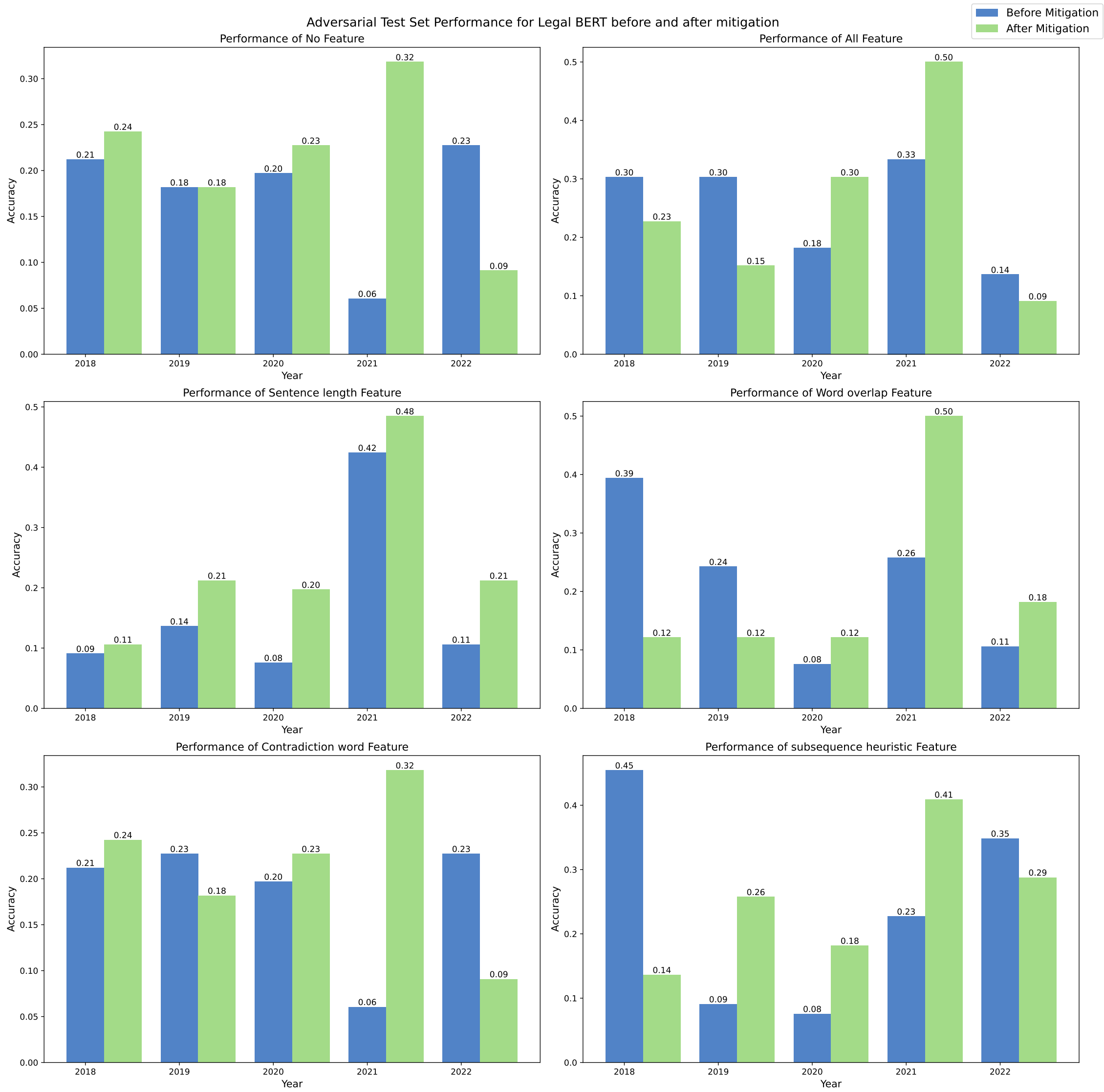

Figure 6.87: Performance comparison of LegalBERT Base on Challenging Adversarial Instances before and After Mitigation

in Legal_BERT and ELECTRA_Base_MNLI further validate the effectiveness of data augmentation strategies in promoting generalizability and resilience.

The results from adversarial instances with artefacts provide further evidence of the positive impact of mitigation. While there were some fluctuations, the overall trend indicates that models became less reliant on artefacts and more adept at handling inputs designed to exploit these weaknesses. This reduction in artefact dependence is crucial for developing models that can generalize well to diverse and unseen examples. However, the inconsistencies observed in certain models suggest that one particular method may not be sufficient. Tailoring mitigation strategies to address the specific vulnerabilities of each model type could yield more consistent improvements.

The most compelling evidence of the effectiveness of artefact mitigation comes from the performance on adversarial instances without artefacts. The substantial improvements across various models and configurations post-mitigation indicate that the models

Table 6.38: Performance comparison of RoBERTa Base on Challenging Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.3636 ± 0.03	0.3788 ± 0.03	0.4091 ± 0.09	0.4091 ± 0.03	0.3636 ± 0.03	0.3333 ± 0.04
	After	0.4242 ± 0.08	0.4242 ± 0.04	0.3939 ± 0.05	0.4545 ± 0.00	0.4242 ± 0.08	0.3788 ± 0.02
2019	Before	0.3333 ± 0.04	0.3333 ± 0.08	0.4242 ± 0.03	0.3182 ± 0.00	0.3333 ± 0.04	0.3030 ± 0.02
	After	0.3939 ± 0.03	0.4848 ± 0.02	0.4242 ± 0.04	0.4848 ± 0.07	0.3939 ± 0.03	0.4697 ± 0.02
2020	Before	0.3333 ± 0.03	0.3939 ± 0.04	0.3636 ± 0.03	0.3636 ± 0.08	0.3333 ± 0.03	0.3485 ± 0.03
	After	0.4697 ± 0.04	0.3333 ± 0.02	0.4848 ± 0.07	0.3788 ± 0.08	0.3333 ± 0.03	0.4091 ± 0.00
2021	Before	0.3485 ± 0.03	0.3182 ± 0.05	0.4091 ± 0.05	0.3636 ± 0.05	0.3485 ± 0.03	0.3333 ± 0.03
	After	0.3636 ± 0.00	0.4848 ± 0.04	0.3788 ± 0.03	0.4545 ± 0.03	0.3636 ± 0.00	0.4242 ± 0.03
2022	Before	0.3939 ± 0.02	0.3788 ± 0.02	0.3485 ± 0.02	0.4394 ± 0.05	0.3939 ± 0.02	0.4091 ± 0.05
	After	0.3788 ± 0.06	0.3788 ± 0.02	0.3333 ± 0.02	0.3485 ± 0.05	0.3788 ± 0.06	0.4394 ± 0.00

Table 6.39: Performance comparison of ELECTRA Base on Challenging Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.3333 ± 0.04	0.3030 ± 0.04	0.3485 ± 0.02	0.3939 ± 0.02	0.3333 ± 0.04	0.3788 ± 0.02
	After	0.4848 ± 0.03	0.3030 ± 0.04	0.4091 ± 0.00	0.4545 ± 0.00	0.4848 ± 0.03	0.4545 ± 0.00
2019	Before	0.3485 ± 0.05	0.1970 ± 0.04	0.3636 ± 0.05	0.3030 ± 0.06	0.3485 ± 0.05	0.3333 ± 0.04
	After	0.4848 ± 0.02	0.4697 ± 0.03	0.4848 ± 0.03	0.4545 ± 0.03	0.4848 ± 0.02	0.4697 ± 0.05
2020	Before	0.3939 ± 0.02	0.3788 ± 0.02	0.4091 ± 0.07	0.3939 ± 0.05	0.3939 ± 0.02	0.3939 ± 0.02
	After	0.4545 ± 0.05	0.4697 ± 0.03	0.5152 ± 0.02	0.4545 ± 0.03	0.4545 ± 0.05	0.4697 ± 0.02
2021	Before	0.4242 ± 0.04	0.3788 ± 0.04	0.3788 ± 0.03	0.3788 ± 0.02	0.4242 ± 0.04	0.3636 ± 0.00
	After	0.3636 ± 0.03	0.4242 ± 0.02	0.4091 ± 0.05	0.4242 ± 0.03	0.3636 ± 0.03	0.4242 ± 0.03
2022	Before	0.4242 ± 0.04	0.3788 ± 0.03	0.4394 ± 0.02	0.4091 ± 0.03	0.4242 ± 0.04	0.4091 ± 0.05
	After	0.4242 ± 0.02	0.3788 ± 0.03	0.3636 ± 0.05	0.4242 ± 0.03	0.4242 ± 0.02	0.4091 ± 0.00

have developed a deeper, more robust feature representation for the task. By moving beyond superficial patterns, the models demonstrated enhanced semantic comprehension, which is essential for real-world legal applications where inputs can vary widely and unpredictably. This deeper understanding and reduced artefact reliance suggest that the mitigation strategies have successfully addressed one of the primary challenges in natural language inference tasks.

In conclusion, the data augmentation techniques employed to mitigate word overlap and contradiction word artefacts have shown promising results in enhancing the robustness and generalizability of BERT-based models. While there are trade-offs in performance on standard test sets, the overall improvements in handling adversarial inputs and reducing artefact dependency are significant. These findings highlight the importance of continuous refinement and customization of mitigation strategies to balance robustness and accuracy effectively. Future research should focus on developing more sophisticated techniques that can achieve this balance, ensuring that legal NLI models perform reliably in real-world scenarios.

This study employed the Integrated Gradients technique [47], a model interpretability method, to identify and evaluate the features contributing to model predictions. This technique was applied to both the Hypothesis-only and Full-context models to visualize feature attribution before and after the implementation of mitigation strategies. The analysis specifically focused on instances from an adversarial test set, which comprised both challenging and supportive cases.

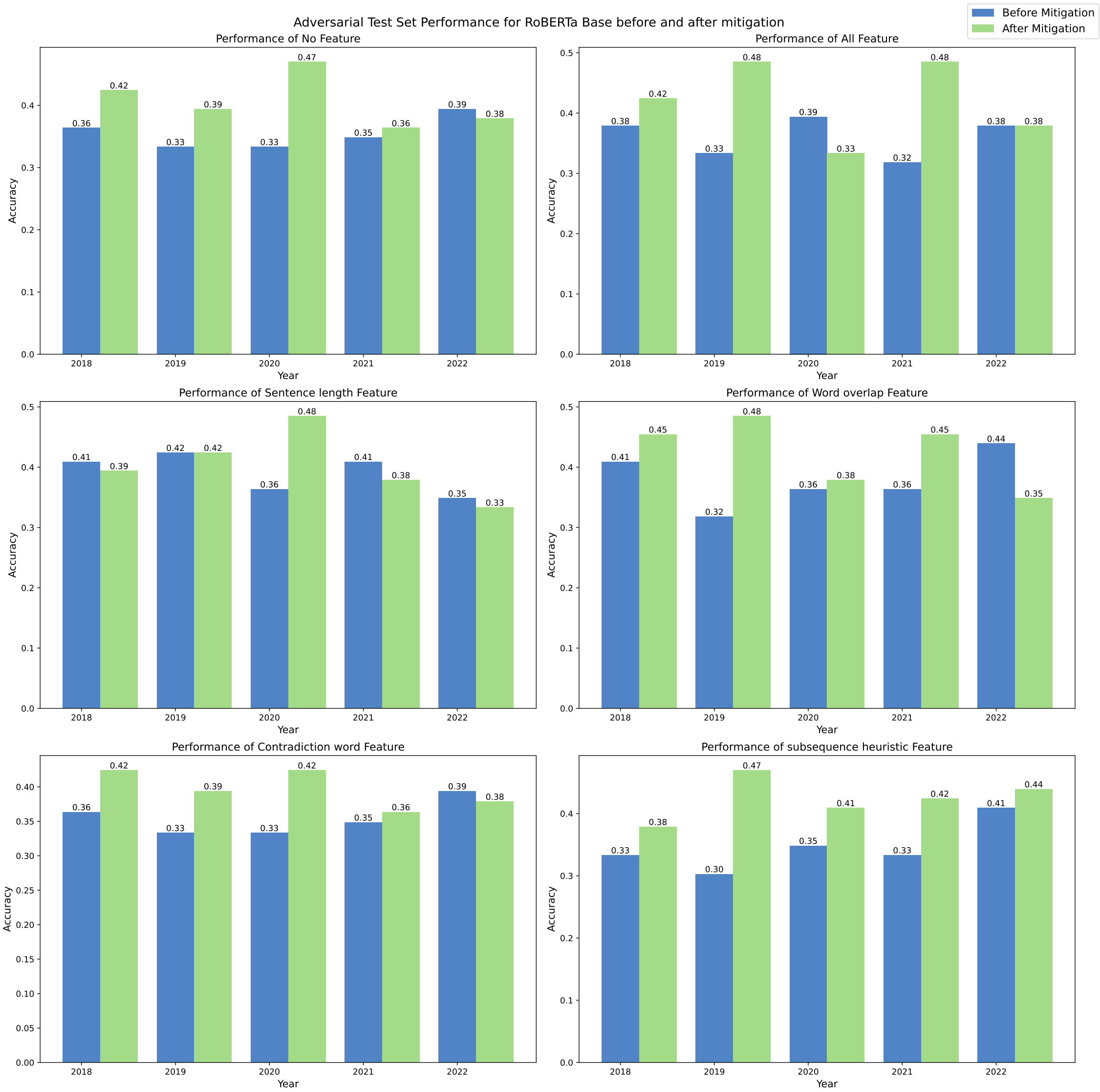

Figure 6.88: Performance comparison of RoBERTa Base on Challenging Adversarial Instances before and After Mitigation

The qualitative analysis did not reveal definitive evidence that the models relied explicitly on artefacts to make predictions. However, it did provide valuable insights into the extent to which models focus on language artefacts, such as contradiction words, during the prediction process. The analysis indicated that, although the models did not consistently rely on artefacts, there was a tendency to focus on certain artefact features during predictions.

Figures 6.31 to 6.35 compare the performance of different model variants over time. These figures demonstrate that DeBERTa V3 consistently outperformed its predecessor, BERT Base, on both standard and adversarial test sets, indicating its robustness across various test scenarios.

Figure 6.91 illustrates the feature attribution for the BERT Base model trained on the 2018 dataset, focusing on the contradiction word artefact feature before and after mitigation with the same random seed used in the model. Prior to mitigation, the Full-context model excessively emphasized (as indicated by dark green shading) the token “not take,” which

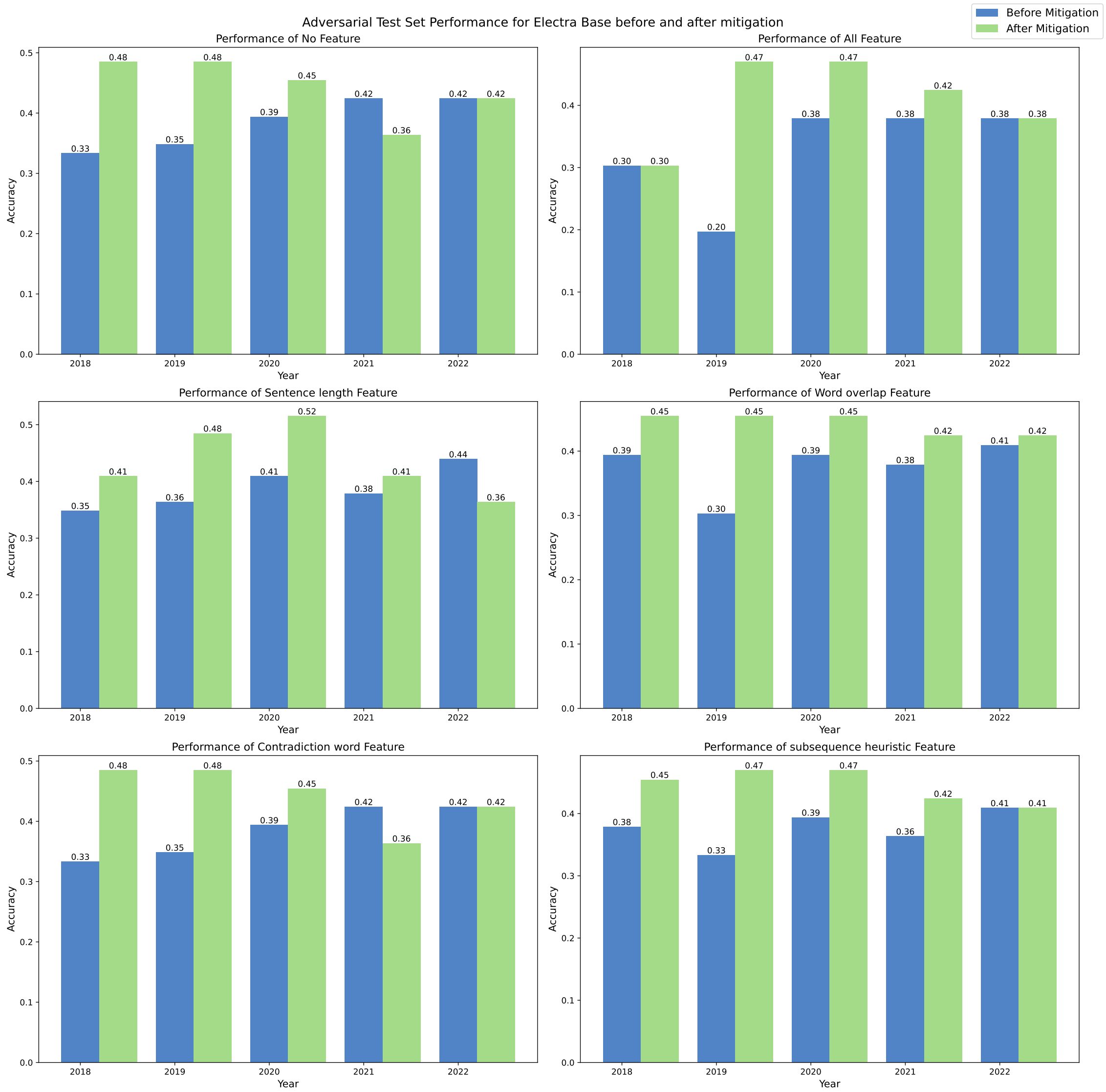

Figure 6.89: Performance comparison of ELECTRA Base on Challenging Adversarial Instances before and After Mitigation

contributed to incorrect predictions. After mitigation, the model's focus shifted away from these contradiction tokens to other, more relevant tokens. Although the model did not predict the correct label, the mitigation strategy effectively redirected the model's attention, reducing its reliance on misleading artefacts.

For the best-performing model, DeBERTa, the analysis revealed that prior to mitigation, the model similarly focused on weak features, such as contradiction words, leading to incorrect predictions of "non-entailment" in challenging adversarial instances. However, as depicted in Figure 6.92, following mitigation, DeBERTa successfully shifted its focus away from contradiction words and appropriately reduced the influence of the contradiction word artefact feature. This adjustment led to the correct prediction of "entailment."

In summary, while the qualitative analysis presented here is limited, it provides compelling evidence that mitigation efforts are effective in redirecting model attention from weaker features to more robust, relevant features during prediction. This shift not only enhances

Table 6.40: Performance comparison of DeBERTa Base on Challenging Adversarial Instances Before and After Mitigation

Year	Mitigation	None	All	Sentence Length	Word Overlap	Contradiction Word	Subsequence Heuristic
2018	Before	0.5000 $\pm$ 0.05	0.4394 $\pm$ 0.02	0.4091 $\pm$ 0.03	0.4697 $\pm$ 0.04	0.4849 $\pm$ 0.06	0.4091 $\pm$ 0.08
	After	0.5000 $\pm$ 0.03	0.3636 $\pm$ 0.09	0.5000 $\pm$ 0.05	0.5000 $\pm$ 0.03	0.5000 $\pm$ 0.03	0.4848 $\pm$ 0.02
2019	Before	0.6364 $\pm$ 0.05	0.5152 $\pm$ 0.03	0.5303 $\pm$ 0.05	0.6515 $\pm$ 0.04	0.6061 $\pm$ 0.05	0.5757 $\pm$ 0.04
	After	0.7121 $\pm$ 0.04	0.5455 $\pm$ 0.00	0.5303 $\pm$ 0.05	0.6667 $\pm$ 0.04	0.6970 $\pm$ 0.05	0.6212 $\pm$ 0.05
2020	Before	0.6212 $\pm$ 0.05	0.5152 $\pm$ 0.04	0.5909 $\pm$ 0.05	0.5606 $\pm$ 0.07	0.5757 $\pm$ 0.05	0.5455 $\pm$ 0.03
	After	0.6970 $\pm$ 0.02	0.5455 $\pm$ 0.07	0.7424 $\pm$ 0.04	0.6667 $\pm$ 0.08	0.7121 $\pm$ 0.02	0.6515 $\pm$ 0.05
2021	Before	0.5000 $\pm$ 0.03	0.5152 $\pm$ 0.05	0.6364 $\pm$ 0.00	0.6212 $\pm$ 0.03	0.6212 $\pm$ 0.05	0.6515 $\pm$ 0.02
	After	0.4697 $\pm$ 0.02	0.5152 $\pm$ 0.05	0.5152 $\pm$ 0.04	0.5000 $\pm$ 0.05	0.4545 $\pm$ 0.00	0.5000 $\pm$ 0.07
2022	Before	0.5909 $\pm$ 0.00	0.5606 $\pm$ 0.03	0.6061 $\pm$ 0.08	0.6515 $\pm$ 0.04	0.5455 $\pm$ 0.03	0.6061 $\pm$ 0.02
	After	0.5303 $\pm$ 0.03	0.4091 $\pm$ 0.03	0.5455 $\pm$ 0.07	0.5606 $\pm$ 0.04	0.5455 $\pm$ 0.05	0.4545 $\pm$ 0.03

the model's interpretability but also improves its predictive accuracy, particularly in the context of challenging adversarial examples.

6.4 Summary

This chapter confirmed the presence of language artefacts in the COLIEE Task 4 statute law entailment datasets, including annotation artefacts, sentence length, word overlap, contradiction words, and subsequence heuristics, which can mislead models and degrade performance. It then examined the impact of these artefacts on BERT-based models, revealing that the models often exploited language artefacts in the training data, leading to incorrect predictions, particularly on adversarial test sets and hypothesis-only baselines. The effectiveness of data augmentation for mitigating these artefacts was evaluated, showing mixed results on standard test sets but consistent improvements on adversarial test sets. This approach reduced the models' reliance on artefacts, enhancing robustness and generalization. A qualitative analysis using Integrated Gradients further explored feature attribution during predictions.

In summary, the chapter provided a detailed analysis of language artefacts in COLIEE statute law textual entailment task, demonstrating their impact on model performance and the potential of data augmentation in mitigating these language artefacts required for the development of robust NLI models.

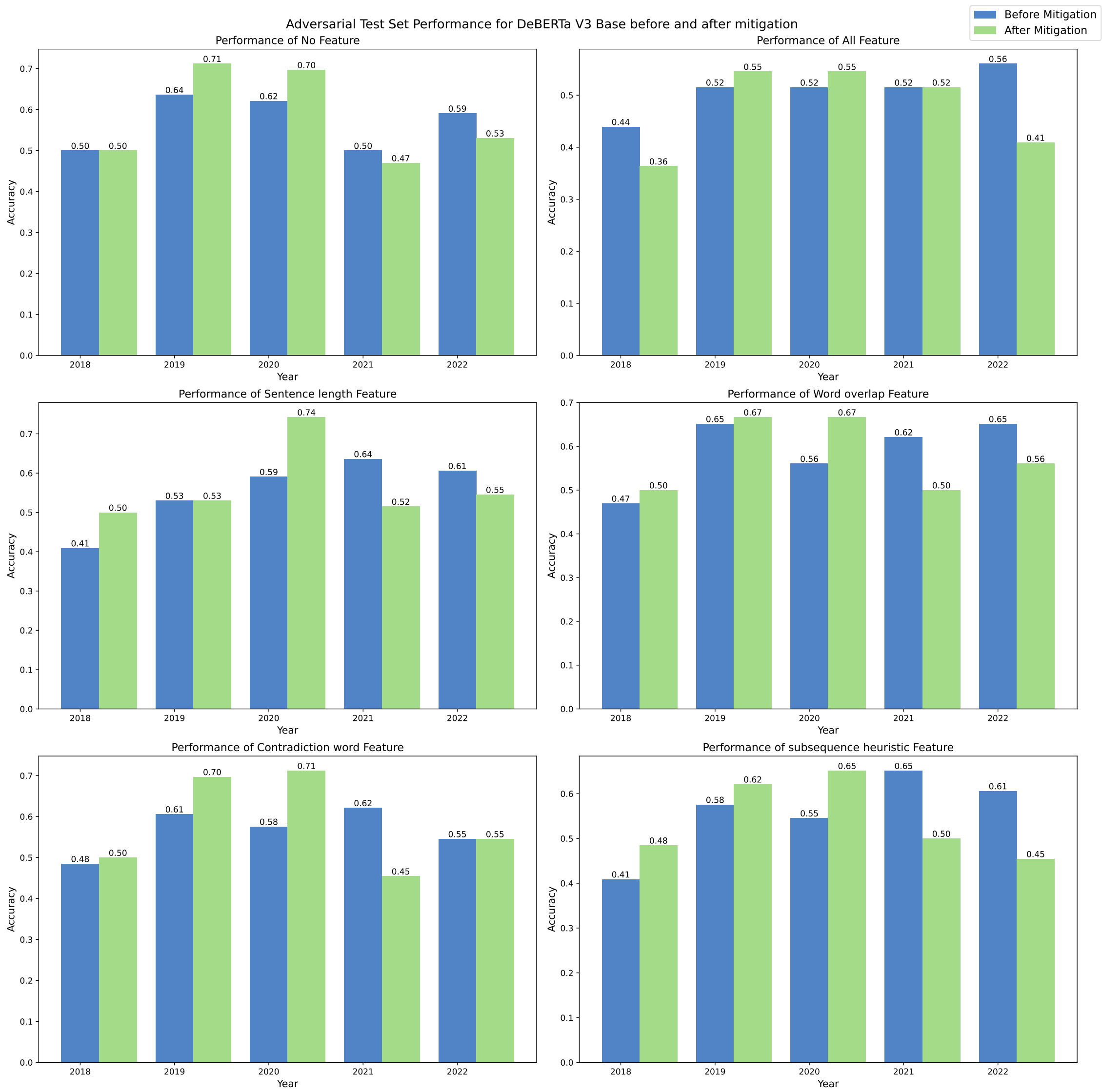

Figure 6.90: Performance comparison of DeBERTa Base on Challenging Adversarial Instances before and After Mitigation

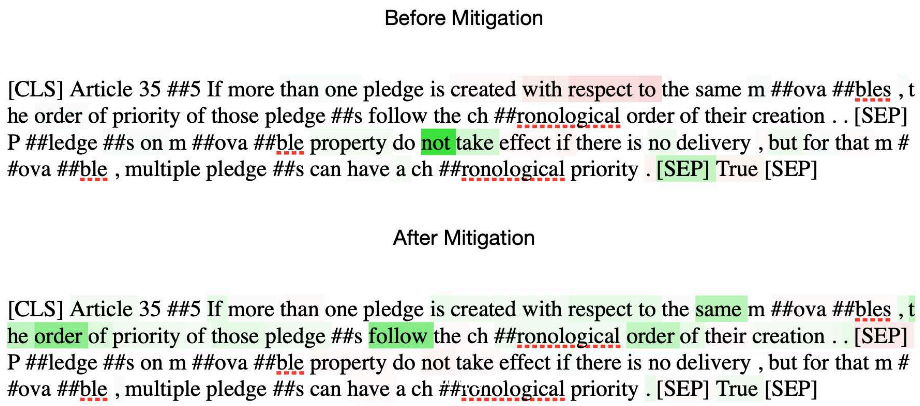

Figure 6.91: BERT Base with contradiction artefact feature model activation map before and after mitigation on a challenging adversarial instance

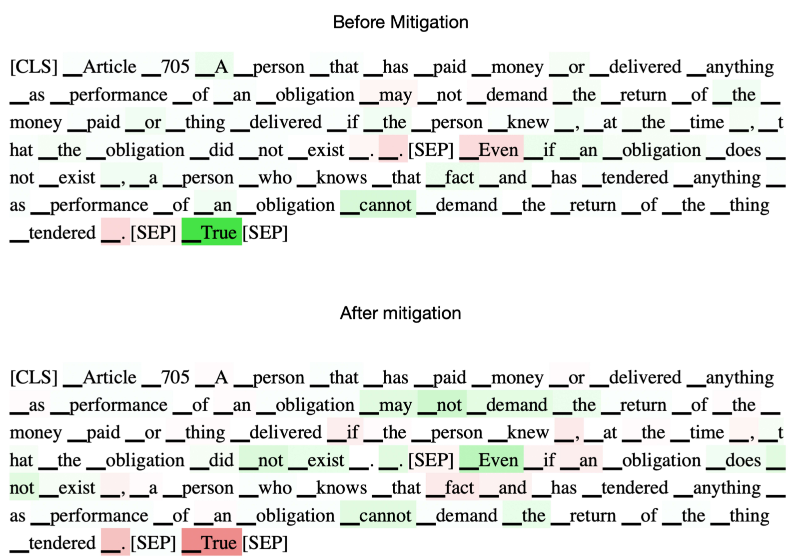

Figure 6.92: DeBERTa Base with contradiction word artefact feature activation map before and after mitigation on a challenging adversarial instance

7. Conclusion

This thesis presented an investigative study on language artefacts in the legal statute law entailment task dataset from COLIEE. This research focused on three main components.

Identification and Detection of Language Artefacts

The first component aimed to identify and detect the language artefacts within the dataset. Five types of language artefacts were identified: annotation, sentence length, word overlap, contradiction word, and subsequence heuristic artefacts [56]. The detection of these artefacts was conducted on five consecutive years of dataset releases from 2018 - 2022. Due to the nature of the dataset preparation in COLIEE, the prevalence of these artefacts increased over time.

Evaluation of Model Robustness

Once the artefacts were detected, popular BERT-based transformer models that had previously achieved good results in the statute law textual entailment task were selected. The selected models were BERT base, LegalBERT, RoBERTa Base, ELECTRA Base, and DeBERTa V3 Base. As pre-trained models often face challenges such as overfitting when fine-tuned on the limited COLIEE dataset, publicly available models already fine-tuned on NLI tasks using large NLI datasets were chosen. These models were then trained and evaluated using various strategies to understand the impact of the language artefacts on their performance.

The evaluations were done using:

- The standard test set from COLIEE task 4.
- An adversarial dataset containing challenging instances without the artefacts and supporting instances containing the artefacts.
- A hypothesis-only baseline method.

The evaluations revealed that most models performed poorly on the adversarial test set, except for the DeBERTa V3 base model. The hypothesis-only baseline evaluations also indicated that, over the years, most models performed better than a majority classifier, suggesting that the hypotheses contain generalizable weak features that aid in prediction. Furthermore, the performance gap between the challenging and supporting instances in the adversarial test set confirmed the models' reliance on language artefacts for prediction.

Data Augmentation for Mitigation

The third component of this research focused on mitigating the language artefacts by augmenting the original dataset with instances designed to balance the contradiction word and word overlap artefacts. After augmentation, the same evaluation methods were carried out.

The post-augmentation evaluation showed that the models' performance increased on the adversarial challenging instances and decreased slightly on the supporting instances. Additionally, there was not much degradation observed on the standard test set; this might be due to the level of augmentation applied on contradiction word and word overlap artefacts in the dataset. Models such as RoBERTa, ELECTRA, and DeBERTa performed slightly better on the standard dataset after mitigation. These results indicate that mitigation efforts push the model in the right direction, reducing its reliance on weak features.

Overall Findings

Overall, the study demonstrated that the DeBERTa V3 base model performed better and was more robust compared to the other models in most settings. This superior performance might be attributed to the diverse pre-fine-tuning of the model using various NLI tasks, including MNLI [57], FEVER [50], and Adversarial-NLI [28]. This diverse training likely contributes to the model's better performance on adversarial instances.

In conclusion, this thesis work successfully detected various language artefacts, evaluated the influence of these artefacts on the selected encoder-only NLI models using robustness evaluation methods, and mitigated a few language artefacts using data augmentation techniques within the context of the COLIEE statute law entailment task.

7.1 Limitations and Future Work

Limitations

- *Types of Artefacts:* This thesis work has only studied a small subset of the different types of language artefacts that may be present in a dataset. Other types of artefacts might exist that could influence model behavior and performance.
- *Limited Domain Knowledge:* Due to the limited amount of domain knowledge, it was challenging to create more challenging dataset instances for the artefacts that accurately reflect the characteristics of a legal dataset.
- *Model Availability:* Since the real models used in COLIEE were not available, only the base pre-trained models with the same architectures were chosen to conduct this analysis.

Future Work

Future research directions include the identification of additional nuanced language artefacts within legal datasets, expanding beyond statute law entailment to encompass case law textual entailment tasks. It is also important to evaluate the impact of language artefacts on a broader range of NLI models utilized in COLIEE, such as encoder-decoder models (e.g., T5 [40]) and decoder-only models (e.g., GPT-2 [39]). The creation of more challenging datasets that incorporate a wider variety of language artefacts is necessary to facilitate robustness evaluations.

Furthermore, future work on mitigating language artefacts should explore model-centric approaches like Adversarial Classifiers [1] or model-ensemble methods [8] to address dataset biases. Critically, enhancing model interpretability is essential for understanding why models are susceptible to specific artefacts, thereby enabling the development of targeted mitigation strategies. Research in this area can illuminate the internal decision-making processes that contribute to artefact exploitation.

Appendix

Annotation Artefacts

The top 10 tokens most associated with Entailment and Non-Entailment labels from 2018 to 2022 are shown below. The "Total Count" column lists each token's frequency in the training dataset hypotheses. The "Count" column shows the token's frequency within each label. The "Probability" column indicates the conditional probability of the label given the token, and the "p-value" column presents the binomial significance test result, with a significance level of 0.1 [56].

Top 10 Tokens Biased towards Entailment Label in 2018							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
the	2531	1343	0.5306	0.0011	1188	0.4694	0.9990
a	1009	530	0.5253	0.0577	479	0.4747	0.9492
to	664	356	0.5361	0.0340	308	0.4639	0.9714
in	451	251	0.5565	0.0092	200	0.4435	0.9929
for	287	160	0.5575	0.0294	127	0.4425	0.9777
that	275	157	0.5709	0.0109	118	0.4291	0.9921
person	248	140	0.5645	0.0244	108	0.4355	0.9820
with	176	103	0.5852	0.0143	73	0.4148	0.9904
cases	168	93	0.5536	0.0948	75	0.4464	0.9288
such	104	65	0.6250	0.0069	39	0.3750	0.9961
Top 10 Tokens Biased towards Not-Entailment Label in 2018							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
by	204	92	0.4510	0.9294	112	0.5490	0.0916
an	147	62	0.4218	0.9763	85	0.5782	0.0346
sale	53	19	0.3585	0.9865	34	0.6415	0.0267
statutory	53	20	0.3774	0.9733	33	0.6226	0.0492
property	49	17	0.3469	0.9894	32	0.6531	0.0222
cannot	33	9	0.2727	0.9977	24	0.7273	0.0068
real	29	9	0.3103	0.9879	20	0.6897	0.0307
pledge	28	9	0.3214	0.9822	19	0.6786	0.0436
movable	28	9	0.3214	0.9822	19	0.6786	0.0436
guarantor	27	9	0.3333	0.9739	18	0.6667	0.0610

Table 1.1: 2018 Top 10 Tokens Biased towards Entailment and Not-Entailment Labels.

Top 10 Tokens Biased towards Entailment Label in 2019							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
a	1139	592	0.5198	0.0961	547	0.4802	0.9136
in	498	274	0.5502	0.0140	224	0.4498	0.9889
for	333	181	0.5435	0.0624	152	0.4565	0.9500
that	320	177	0.5531	0.0325	143	0.4469	0.9749
person	265	152	0.5736	0.0097	113	0.4264	0.9931
with	201	115	0.5721	0.0240	86	0.4279	0.9830
such	121	77	0.6364	0.0017	44	0.3636	0.9991
or	119	69	0.5798	0.0493	50	0.4202	0.9669
any	108	64	0.5926	0.0335	44	0.4074	0.9786
hisher	108	62	0.5741	0.0743	46	0.4259	0.9493
Top 10 Tokens Biased towards Non-Entailment Label in 2019							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
an	156	68	0.4359	0.9538	88	0.5641	0.0640
was	77	29	0.3766	0.9890	48	0.6234	0.0198
sale	74	27	0.3649	0.9930	47	0.6351	0.0133
statutory	62	25	0.4032	0.9510	37	0.5968	0.0809
property	54	18	0.3333	0.9955	36	0.6667	0.0099
cannot	36	9	0.2500	0.9994	27	0.7500	0.0020
effect	39	13	0.3333	0.9881	26	0.6667	0.0266
return	40	15	0.3750	0.9597	25	0.6250	0.0769
unless	35	12	0.3429	0.9795	23	0.6571	0.0448
immovable	32	10	0.3125	0.9900	22	0.6875	0.0251

Table 1.2: 2019 - Top 10 Tokens Biased towards Entailment and Not-Entailment Labels.

Top 10 Tokens Biased towards Entailment Label in 2020							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
the	3101	1594	0.5140	0.0612	1507	0.4860	0.9430
a	1288	671	0.5210	0.0699	617	0.4790	0.9373
in	548	304	0.5547	0.0058	244	0.4453	0.9954
for	365	199	0.5452	0.0469	166	0.4548	0.9625
that	348	195	0.5603	0.0139	153	0.4397	0.9895
person	273	156	0.5714	0.0106	117	0.4286	0.9923
with	222	130	0.5856	0.0064	92	0.4144	0.9956
b	223	122	0.5471	0.0902	101	0.4529	0.9297
cases	199	110	0.5528	0.0780	89	0.4472	0.9407
such	129	82	0.6357	0.0013	47	0.3643	0.9993
Top 10 Tokens Biased towards Non-Entailment Label in 2020							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
obligor	127	55	0.4331	0.9451	72	0.5669	0.0777
was	97	37	0.3814	0.9928	60	0.6186	0.0125
sale	74	27	0.3649	0.9930	47	0.6351	0.0133
property	56	19	0.3393	0.9948	37	0.6607	0.0111
effect	42	14	0.3333	0.9902	28	0.6667	0.0218
cannot	41	13	0.3171	0.9942	28	0.6829	0.0138
unless	39	13	0.3333	0.9881	26	0.6667	0.0266
return	41	15	0.3659	0.9702	26	0.6341	0.0586
pledge	33	11	0.3333	0.9825	22	0.6667	0.0401
guarantor	35	13	0.3714	0.9552	22	0.6286	0.0877

Table 1.3: 2020 - Top 10 Tokens Biased towards Entailment and Non-Entailment Labels.

Top 10 Tokens Biased towards Entailment Label in 2021							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
the	3543	1840	0.5193	0.0112	1703	0.4807	0.9898
a	1491	775	0.5198	0.0665	716	0.4802	0.9399
in	624	349	0.5593	0.0017	275	0.4407	0.9987
for	404	218	0.5396	0.0614	186	0.4604	0.9497
that	386	217	0.5622	0.0083	169	0.4378	0.9937
person	284	161	0.5669	0.0140	123	0.4331	0.9898
b	295	161	0.5458	0.0650	134	0.4542	0.9486
with	254	145	0.5709	0.0139	109	0.4291	0.9900
cases	240	134	0.5583	0.0406	106	0.4417	0.9695
such	147	92	0.6259	0.0014	55	0.3741	0.9992
Top 10 Tokens Biased towards Non-Entailment Label in 2021							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
not	442	206	0.4661	0.9299	236	0.5339	0.0839
even	146	64	0.4384	0.9422	82	0.5616	0.0796
was	102	41	0.4020	0.9815	61	0.5980	0.0297
sale	82	32	0.3902	0.9824	50	0.6098	0.0299
property	66	23	0.3485	0.9954	43	0.6515	0.0093
must	57	23	0.4035	0.9444	34	0.5965	0.0924
effect	48	16	0.3333	0.9934	32	0.6667	0.0147
unless	44	15	0.3409	0.9887	29	0.6591	0.0244
cannot	43	15	0.3488	0.9842	28	0.6512	0.0330
return	45	17	0.3778	0.9638	28	0.6222	0.0676

Table 1.4: 2021 - Top 10 Tokens Biased towards Entailment and Non-Entailment Labels.

Top 10 Tokens Biased towards Entailment Label in 2022							
		Y (Entailment)			N (Not-Entailment)		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
the	3896	2012	0.5164	0.0209	1884	0.4836	0.9806
in	666	367	0.5511	0.0047	299	0.4489	0.9963
that	441	248	0.5624	0.0050	193	0.4376	0.9962
for	454	244	0.5374	0.0607	210	0.4626	0.9498
person	303	169	0.5578	0.0253	134	0.4422	0.9808
with	268	150	0.5597	0.0290	118	0.4403	0.9782
cases	240	134	0.5583	0.0406	106	0.4417	0.9695
or	161	95	0.5901	0.0135	66	0.4099	0.9911
such	149	94	0.6309	0.0009	55	0.3691	0.9995
it	158	90	0.5696	0.0472	68	0.4304	0.9665
Top 10 Tokens Biased towards Non-Entailment Label in 2022							
		Y (Entailment)			**N (Not-Entailment)**		
Token	Total Count	Count	Probability	p-value	Count	Probability	p-value
not	492	230	0.4675	0.9316	262	0.5325	0.0811
contract	287	131	0.4564	0.9376	156	0.5436	0.0782
an	237	107	0.4515	0.9406	130	0.5485	0.0764
even	160	68	0.4250	0.9761	92	0.5750	0.0343
was	112	47	0.4196	0.9639	65	0.5804	0.0539
sale	95	37	0.3895	0.9883	58	0.6105	0.0198
property	70	23	0.3286	0.9987	47	0.6714	0.0028
must	63	25	0.3968	0.9615	38	0.6032	0.0650
effect	50	16	0.3200	0.9967	34	0.6800	0.0077
authority	57	23	0.4035	0.9444	34	0.5965	0.0924

Table 1.5: 2022 - Top 10 Tokens Biased towards Entailment and Non-Entailment Labels.

GPT-4 Prompts

> You are a seasoned legal NLP Researcher. You are skilled in understanding complex legal texts and paraphrase them effectively based on the instructions provided.

Figure 1.1: System Prompt

> You are given two sentences. Do the following two sentences contain the same meaning?
> You must provide the answer only as YES or NO.
>
> **Sentence 1:** {hypothesis}
> **Sentence 2:** {paraphrased_hypothesis}

Figure 1.2: Validation Prompt

```
You will be provided with a Legal Natural Language Inference pair
containing premise and hypothesis.
  Your task is to paraphrase the provided original hypothesis by
following the provided instructions.
  INSTRUCTIONS:
  1. In the paraphrased hypothesis negation words MUST be preserved.
For example, words such as "no", "not",
  "cannot" etc. If needed, you can introduce more negation words,
but the original negation words must not
  be eliminated.
  2. There should not be more overlap between premise and
hypothesis.
  3. Provide only the paraphrased hypothesis and nothing else.
  Example Input:
  "Premise":"Article 705
  A person that has paid money or delivered anything as performance
of an obligation may not demand the
  return of the money paid or thing delivered if the person knew, at
the time, that the obligation did not exist..",
  "Hypothesis": "A person who has tendered anything as performance
of an obligation may not demand the return of the thing
  tendered if the person were negligent in not knowing that the
obligation did not exist."
  Example Output:
  "Paraphrased Hypothesis": "If an individual was careless and
failed to realize that an obligation was nonexistent
  at the time they fulfilled it, they are not entitled to ask for the
return of the items they provided."
  Real Input:
  "Premise":{premise},
  "Hypothesis":{hypothesis}
  Real Output:
  "Paraphrased Hypothesis":
```

Figure 1.3: Contradiction Word Artefact Paraphrase Prompt

```
You will be provided with a Legal Natural Language Inference pair
containing premise, hypothesis, and its respective label.
  Label "Y" indicates entailment and "N" indicates non-entailment.
  Your task is to paraphrase the provided original hypothesis by
following the provided instructions.
  INSTRUCTIONS:
  1. In the paraphrased hypothesis, the number of word overlaps
between premise and hypothesis MUST be the same or more,
  but it MUST NOT be reduced.
  2. There MUST be more overlap between premise and hypothesis while
preserving the label.
  3. There MUST not be any negation words such as "no", "n't", "not",
"cannot" etc., in the paraphrased hypothesis at all.
  4. Provide ONLY the paraphrased hypothesis and nothing else.
  Example Input:
  "Premise":"Article 299
  (1) If the holder of a right of retention incurs necessary expenses
with respect to the thing retained, that holder
  may have the owner reimburse the same.
  (2) If the holder of a right of retention incurs beneficial
expenses with respect to the thing retained, to the extent
  that there is currently an increase in value as a result of the
same,
  that holder may have the expenses incurred or the increase in value
reimbursed at the owner's choice;
  provided, however, that the court may, at the request of the owner,
  grant a reasonable period of time for the reimbursement of the
same..",
  "Hypothesis": "If a holder of a right of retention incurs ordinary
unnecessary expenses with respect to the Thing retained,
  he/she may have the owner reimburse the same."
  "Label": "N"
  Example Output:
  "Paraphrased Hypothesis": "If an individual was careless and
failed to realize that an obligation was nonexistent
  at the time they fulfilled it, they are deprived of the right to
demand the return of the items they provided."
  Real Input:
  "Premise":{premise},
  "Hypothesis":{hypothesis}
  Real Output:
  "Paraphrased Hypothesis":
```

Figure 1.4: Word Overlap Artefact Paraphrase Prompt

Bibliography

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